

BT302 - Immunology

Teacher: Dr Numan Javed (Phd Holder at SHanghai Institute of Sciences, China)

1 to 167 Lectures merged

Effort by AHMAD SAEED

BT302 - Immunology

Lesson 1 : Overview of Immune System:

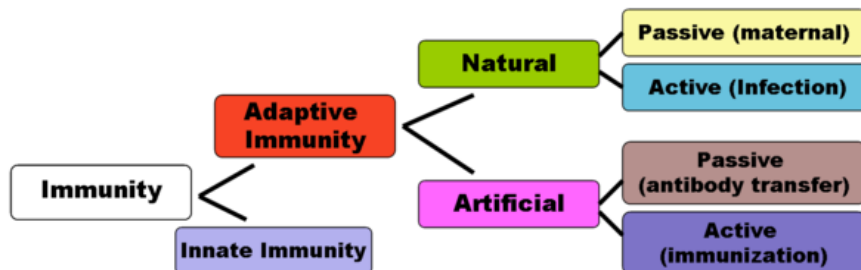
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Immunity

It is defined as the state of protection from infectious disease

It has both a less specific and more specific component.

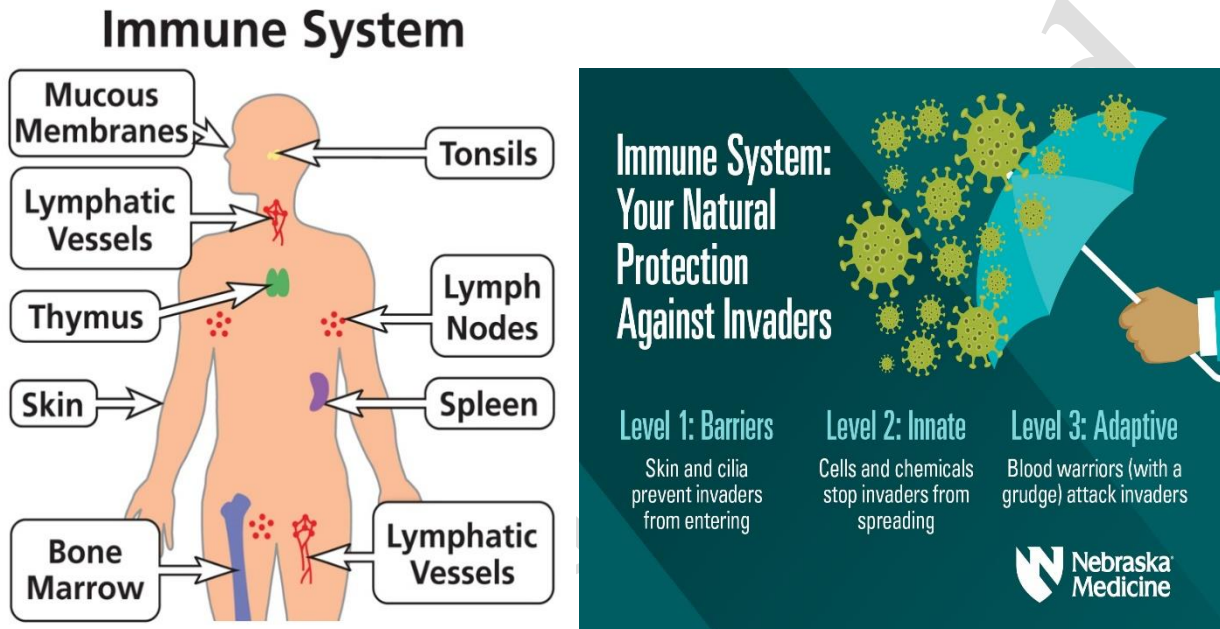
1. The less specific component, innate immunity, provides the first line of defense against infection.
2. The specific component, adaptive immunity, does not come into play until there is an antigenic challenge to the organism.



The Immune system

Immune system is the defense system that has evolved to protect animals from invading pathogenic microorganisms and cancer.

It is able to generate an enormous variety of cells and molecules capable of specifically recognizing and eliminating an apparently limitless variety of foreign invaders.



Immune response

An immune response can be divided into two related activities

1. Recognition
2. Response

Immune recognition is remarkable for its specificity. The immune system is able to recognize subtle chemical differences that distinguish one foreign pathogen from another.

Furthermore, the system is able to discriminate between foreign molecules and the body's own cells and proteins.

Once a foreign organism has been recognized, the immune system recruits a variety of cells and molecules to mount an appropriate response, called an **effector response**, to eliminate or neutralize the organism.

In this way the system is able to convert the initial recognition event into a variety of effector responses, each uniquely suited for eliminating a particular type of pathogen.

Later exposure to the same foreign organism induces a memory response, characterized by a more rapid and heightened immune reaction that serves to eliminate the pathogen and prevent disease

Immunology

The discipline of immunology grew out of the observation that individuals who had recovered from certain infectious diseases were thereafter protected from the disease. The Latin term *immunis*, meaning “exempt,” is the source of the English word immunity, meaning the state of protection from infectious disease.

Reference

Kuby Immunology, 6th Edition, Authored by Kindt, Goldsby and Osborne, Chapter 1, Pages 1, 4

Lesson 2 : Infections vs Non-Infections:

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Infectious disease

Infectious diseases are caused by pathogenic micro-organisms such as bacteria, viruses, fungi or parasites. These micro-organisms are contagious.

Examples of infectious diseases include the common cold, COVID-19, SARS and tuberculosis.

Non-infectious disease

Non-infectious diseases are not caused by pathogens and therefore cannot be spread from one person to another. Instead, non-infectious diseases are caused by factors such as genetics, malnutrition, environment and lifestyle.

Examples of non-infectious diseases include cancer, Alzheimer's disease and epilepsy.

Differences between an infectious and non-infectious disease

1. Infectious diseases are communicable, non-infectious diseases are not

Infectious diseases are contagious, meaning that a person can contract the illness if they are exposed to carriers of the disease. For example, COVID-19

Non-infectious diseases, are non-communicable, meaning they cannot be transmitted through exposure to a carrier. Instead, they appear due to factors such as gene mutations, malnutrition, environmental toxins and lifestyle.

2. Infectious diseases are caused by the transmission of a pathogen

Infectious diseases are transmitted from person-to-person through the transfer of a pathogen such as bacteria, viruses, fungi or parasites.

A non-infectious disease cannot be transmitted through a pathogen.

3. Prevention strategies for infectious disease focuses on stopping the spread

Prevention strategies for infectious disease focus on isolating individuals who have the disease to protect the broader population, and vaccinating against known strains to prevent transmission.

4. Preventative measures for non-infectious diseases are lifestyle and contextually-focused

Preventative measures for non-infectious disease are focused on lifestyle choices and patterns. You can reduce your risk of contracting Type 2 diabetes, for example, by maintaining a healthy weight, exercising regularly and consuming a balanced diet.

Not all non-infectious diseases, however, can be prevented. Some can develop as a result of an individual's genetic history, lifestyle or age.

Reference

<https://www.childfund.org.au/stories/what-is-the-difference-between-an-infectious-and-non-infectious-disease/>

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Lesson 3 : Role of Immune System:

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Sometimes the immune system fails to protect the host adequately or misdirects its activities to cause discomfort, debilitating disease, or even death.

There are several common manifestations of immune dysfunction:

- Allergy and asthma
- Graft rejection and graft-versus-host disease
- Autoimmune disease
- Immunodeficiency

Graft rejection and graft-versus-host disease

When the immune system encounters foreign cells or tissue, it responds strongly to rid the host of the invaders. However, in some cases, the transplantation of cells or an organ from another individual, although viewed by the immune system as a foreign invasion, maybe the only possible treatment for disease. Because the immune system will attack and reject any transplanted organ that it does not recognize as self, it is a serious barrier to this potentially life-saving treatment.

An additional danger in transplantation is that any transplanted cells with immune function may view the new host as nonself and react against it. This reaction, which is termed graft-versus-host disease, can be fatal.

The rejection reaction and graft-versus-host disease can be suppressed by drugs, but this type of treatment suppresses all immune function, so that the host is no longer protected by its immune system and becomes susceptible to infectious diseases. Transplantation studies have played a major role in the development of immunology.

Autoimmune disease

In certain individuals, the immune system malfunctions by losing its sense of self and nonself, which permits an immune attack upon the host. This condition, autoimmunity, can cause a number of chronic debilitating diseases. The symptoms of autoimmunity differ depending on which tissues and organs are under attack.

Immunodeficiency

Since the 1980s, the most common form of immunodeficiency has been acquired immune deficiency syndrome, or AIDS, which results from infection with the retrovirus human immunodeficiency virus, or HIV. In AIDS, T helper cells are infected and destroyed by HIV, causing a collapse of the immune system. It is estimated that 35 million persons worldwide

suffer from this disease, which is usually fatal within 8 to 10 years after infection. Although certain treatments can prolong the life of AIDS patients, there is no known cure for this disease

Reference

Kuby Immunology, 6th Edition, Authored by Kindt, Goldsby and Osborne, Chapter 1, Pages 18, 20,21

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Lesson 4 : Types of Immune system:

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Lesson 4

Innate Immunity

1. Less specific component of Immune system
2. First line of defense against infection
3. Most components of innate immunity are present before the onset of infection and constitute a set of disease-resistance mechanisms that are not specific to a particular pathogen but that include cellular and molecular components that recognize classes of molecules peculiar to frequently encountered pathogens.
4. Phagocytic cells, such as macrophages and neutrophils, barriers such as skin, and a variety of antimicrobial compounds synthesized by the host all play important roles in innate immunity.
5. In contrast to the broad reactivity of the innate immune system, which is uniform in all members of a species,

Adaptive immunity

- the specific component
- does not come into play until there is an antigenic challenge to the organism.

Because adaptive immune responses require some time to marshal, innate immunity provides the first line of defense during the critical period just after the host's exposure to a pathogen. In general, most of the microorganisms encountered by a healthy individual are readily cleared within a few days by defense mechanisms of the innate immune system before they activate the adaptive immune system

Adaptive immunity is capable of recognizing and selectively eliminating specific foreign microorganisms and molecules (i.e., foreign antigens). Unlike innate immune responses, adaptive immune responses are not the same in all members of a species but are reactions to specific antigenic challenges. Adaptive immunity displays four characteristic attributes:

- Antigenic specificity
- Diversity
- Immunologic memory
- Self/nonself recognition

The antigenic specificity of the immune system permits it to distinguish subtle differences among antigens. Antibodies can distinguish between two protein molecules that differ in only a single amino acid.

Reference

Kuby Immunology, 6th Edition, Authored by Kindt, Goldsby and Osborne, Chapter 1, Pages 4,5,9

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Lesson 5 : Components of Innate Immune System:

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Components of Innate Immunity

Innate immunity can be seen to comprise four types of defensive barriers:

- Anatomic
- Physiologic
- Phagocytic
- inflammatory (Table 1-2)

TABLE 1-2 Summary of nonspecific host defenses

Type	Mechanism
<i>Anatomic barriers</i>	
Skin	Mechanical barrier retards entry of microbes. Acidic environment (pH 3–5) retards growth of microbes.
Mucous membranes	Normal flora compete with microbes for attachment sites and nutrients. Mucus entraps foreign microorganisms. Cilia propel microorganisms out of body.
<i>Physiologic barriers</i>	
Temperature	Normal body temperature inhibits growth of some pathogens. Fever response inhibits growth of some pathogens.
Low pH	Acidity of stomach contents kills most ingested microorganisms.
Chemical mediators	Lysozyme cleaves bacterial cell wall. Interferon induces antiviral state in uninfected cells. Complement lyses microorganisms or facilitates phagocytosis. Toll-like receptors recognize microbial molecules, signal cell to secrete immunostimulatory cytokines. Collectins disrupt cell wall of pathogen.
Phagocytic/endocytic barriers	Various cells internalize (endocytose) and break down foreign macromolecules. Specialized cells (blood monocytes, neutrophils, tissue macrophages) internalize (phagocytose), kill, and digest whole microorganisms.
Inflammatory barriers	Tissue damage and infection induce leakage of vascular fluid, containing serum proteins with antibacterial activity, and influx of phagocytic cells into the affected area.

- **The Skin and the Mucosal Surfaces Provide Protective Barriers Against Infection**

Physical and anatomic barriers that tend to prevent the entry of pathogens are an organism's first line of defense against infection

The skin consists of two distinct layers:

- a thinner outer layer—the epidermis
- a thicker layer—the dermis.

The epidermis contains several layers of tightly packed epithelial cells. The outer epidermal layer consists of dead cells and is filled with a waterproofing protein called keratin.

The dermis, which is composed of connective tissue, contains blood vessels, hair follicles, sebaceous glands, and sweat glands.

The sebaceous glands are associated with the hair follicles and produce an oily secretion called sebum. Sebum consists of lactic acid and fatty acids, which maintain the pH of the skin between 3 and 5; this pH inhibits the growth of most microorganisms.

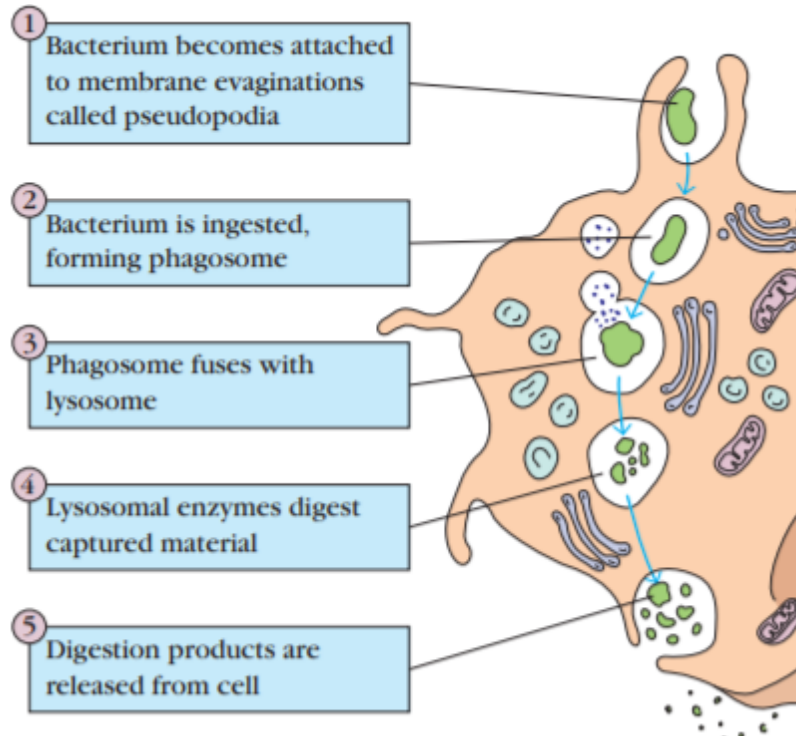
- **Physiologic Barriers to Infection Include General Conditions and Specific**

The physiologic barriers that contribute to innate immunity include temperature, pH, and various soluble and cell associated molecules

A variety of soluble factors contribute to innate immunity, among them the soluble proteins lysozyme, interferon, and complement.

- **Cells That Ingest and Destroy Pathogens Make Up a Phagocytic Barrier to Infection**

Another important innate defense mechanism is the ingestion of extracellular particulate material by phagocytosis. Phagocytosis is one type of endocytosis, the general term for the uptake by a cell of material from its environment. In phagocytosis, a cell's plasma membrane expands around the particulate material, which may include whole pathogenic microorganisms, to form large vesicles called phagosomes



- **Inflammation Represents a Complex Sequence of Events That Stimulates Immune Responses**

Tissue damage caused by a wound or by an invading pathogenic microorganism induces a complex sequence of events collectively known as the inflammatory response.

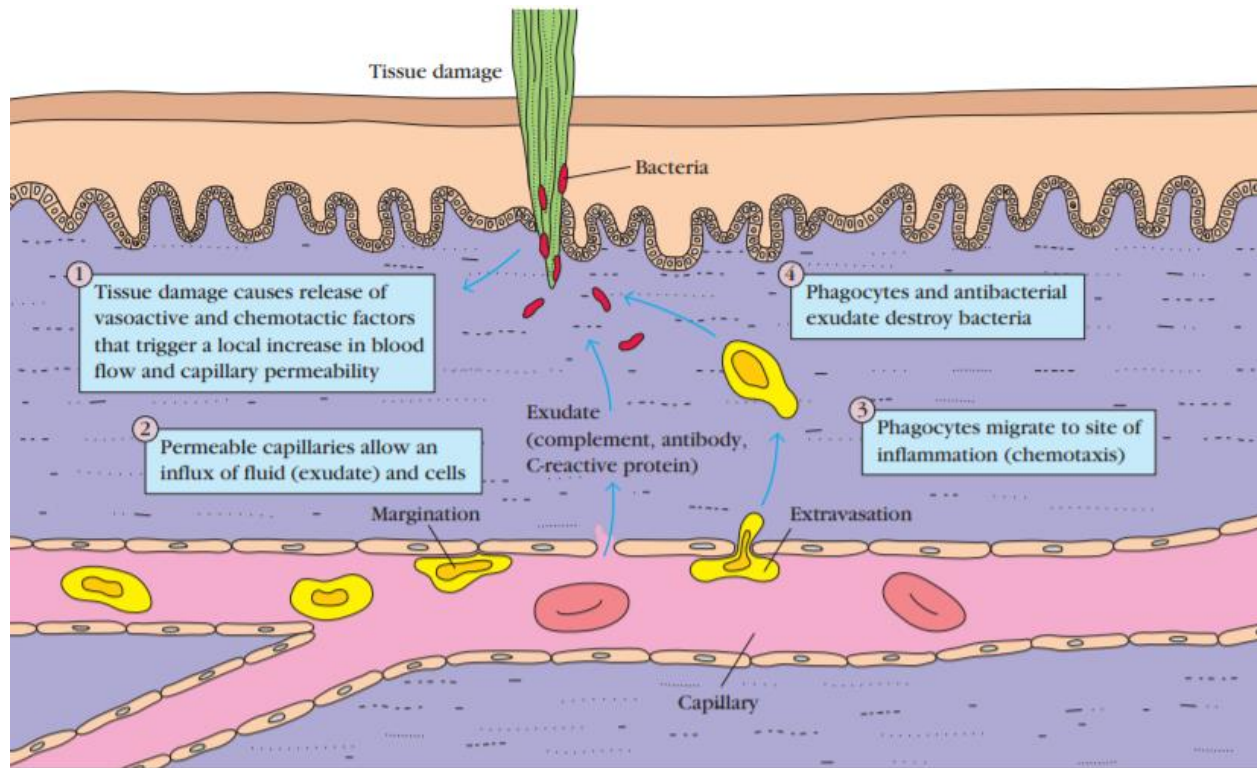


FIGURE 1-4 Major events in the inflammatory response. A bacterial infection causes tissue damage with release of various vasoactive and chemotactic factors. These factors induce increased blood flow to the area, increased capillary permeability, and an influx of white

blood cells, including phagocytes and lymphocytes, from the blood into the tissues. The serum proteins contained in the exudate have antibacterial properties, and the phagocytes begin to engulf the bacteria, as illustrated in Figure 1-3.

Reference

Kuby Immunology, 6th Edition, Authored by Kindt, Goldsby and Osborne, Chapter 1, Pages 5-8

Lesson 6 : Physical barriers of Innate Immune system:

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Anatomical and Physico-chemical barriers of immune system

Anatomical barriers

Skin and mucus membrane are the examples of anatomical barriers that provides immunity.

Skin and mucus membrane:

- The skin consists of two distinct layers; a thin outer layer called epidermis and thick inner layer called dermis.
- Epidermis consists of mostly dead cell filled with keratin. Dermis is composed of connective tissue, hair follicle, sebaceous gland and sweat gland.
- Skin provides first line of defense by preventing entry of microorganisms. However skin may be penetrated by injury or insects.
- Below skin, the mucus membrane prevents the entry of microorganism to the body. And also it secretes mucus that entraps microorganisms.

Anatomical barriers provide immunity by following ways.

- At first skin and mucus membrane prevent entry of microorganism into host body by mechanical separation. For example, Skin surrounds the host body from external and mucus membrane surrounds the body tracts.
- They also have mechanism to kill the pathogen before entry to body. For example; lysozyme, acidic pH, sebum, high salt concentration in sweat are antimicrobial agents found in skin and mucus membrane.
- Skin and mucus membrane provides first line of defense against microorganism as they are first component to encounter with microorganism.

Physico-chemical barriers

Physicochemical barrier includes physiological barrier and chemical barrier.

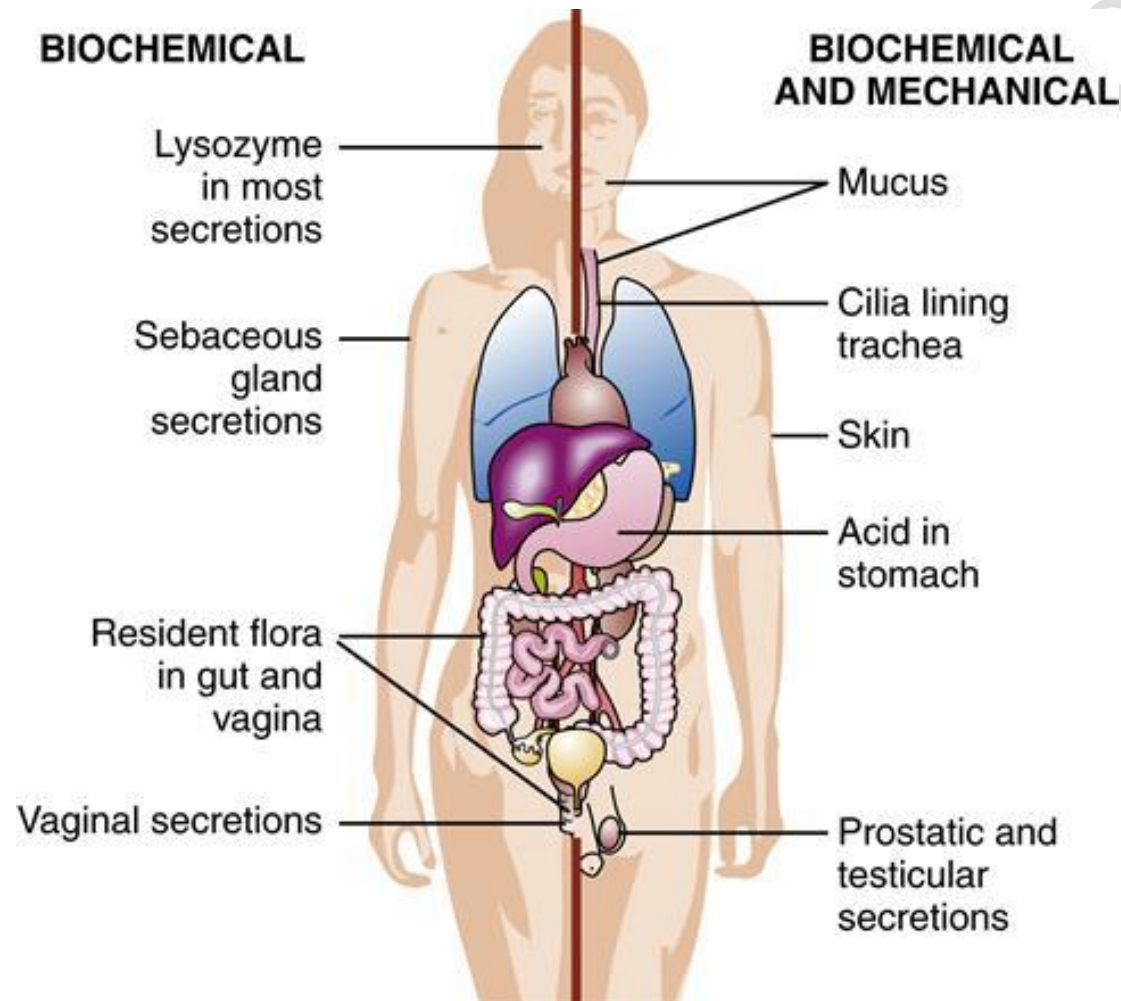
Physiological conditions of body such as normal body temperature, normal body pH etc provides immunity.

- Some species are resistant to certain diseases simply because of their higher body temperature. For example, mammals are susceptible to anthrax but birds are resistant to anthrax. It is because *Bacillus anthracis* are killed by the higher body temperature of birds (39°C).

- Similarly, body pH also provides immunity. For example, acidity of the stomach kills most of the ingested bacteria and provides immunity. In infants, stomach is less acidic. This is the reason why infants suffer more from gastrointestinal disturbance than adults.

Chemical barriers include various antimicrobial chemicals found in body fluids. For example, Lysozyme found in tear and mucus kills many Gram +ve bacteria.

Interferon found in blood and lymph kills viruses. Other antimicrobial chemicals found in body fluids include complement protein, collectins, etc.



Reference

<https://www.onlinebiologynotes.com/anatomical-physico-chemical-barriers-immune-system/>

Lesson 7 : Secretory molecules of Innate Immune System:

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Humoral (Secretory) barriers after breaching the anatomical barriers, infectious agent penetrates the deeper tissue for inflammation which is the tissue response against infection. Humoral barriers in fact mediate that inflammatory process. These barriers consist of different molecules in the form of proteins, enzymes and cytokines which are present in the secretions of body particularly blood plasma including

A variety of soluble factors contribute to innate immunity, among them the soluble proteins

- **Lysozyme**

a hydrolytic enzyme found in mucous secretions and in tears, is able to cleave the peptidoglycan layer of the bacterial cell wall.

- **Interferon**

comprises a group of proteins produced by virus-infected cells. Among the many functions of the interferons is the ability to bind to nearby cells and induce a generalized antiviral state.

- **Complement**

It is a group of serum proteins that circulate in an inactive state. A variety of specific and nonspecific immunologic mechanisms can convert the inactive forms of complement proteins into an active state with the ability to damage the membranes of pathogenic organisms, either destroying the pathogens or facilitating their clearance.

Complement may function as an effector system that is triggered by binding of antibodies to certain cell surfaces, or it may be activated by reactions between complement molecules and certain components of microbial cell walls and trigger activation of cells of the innate or adaptive immune systems.

collectins indicate that these surfactant proteins may kill certain bacteria directly by disrupting their lipid membranes or, alternatively, by aggregating the bacteria to enhance their susceptibility to phagocytosis.

Many of the molecules involved in innate immunity have the property of **pattern recognition**, the ability to recognize a given class of molecules. Because there are certain types of molecules that are unique to microbes and never found in multicellular organisms, the ability to immediately recognize and combat invaders displaying such molecules is a strong feature of innate immunity.

Molecules with pattern recognition ability may be

- soluble, like lysozyme and the complement components described above
- cell-associated receptors

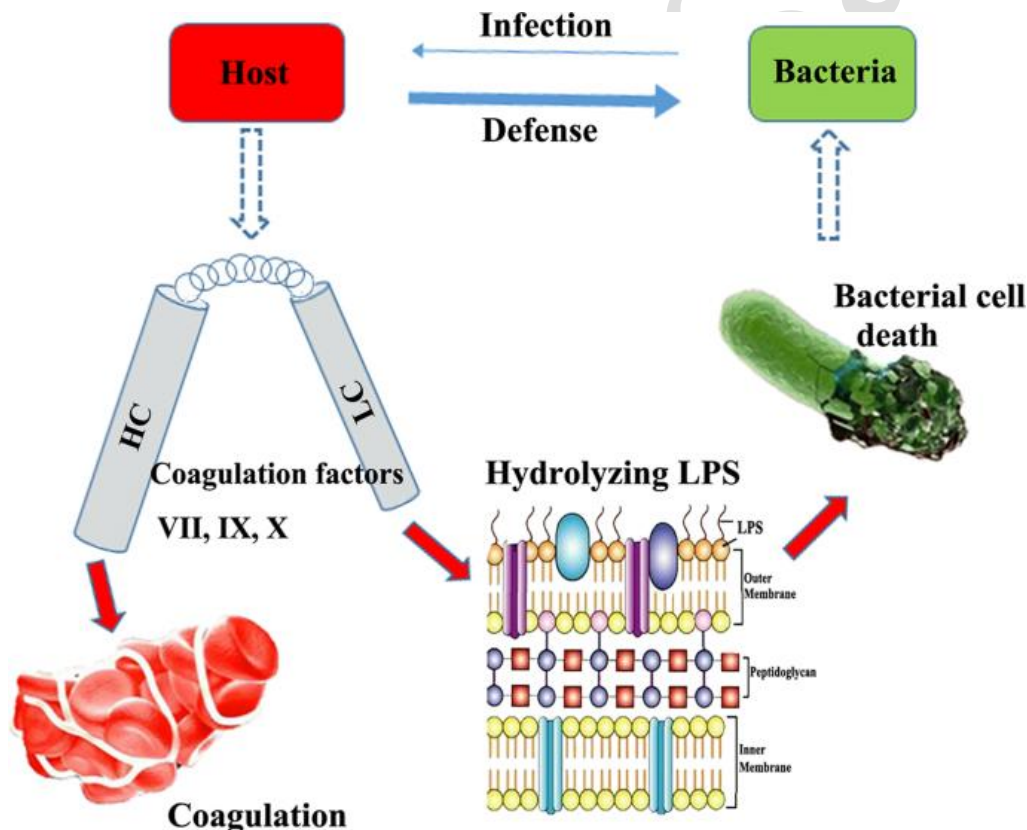
Among the class of receptors designated the toll-like receptors (TLRs), TLR2 recognizes the lipopolysaccharide (LPS) found on Gram-negative bacteria.

Coagulation system

The primary function of the coagulation system is to respond to damage of blood vessels via formation of fibrin, which seals wounds and stops bleeding (hemostasis).

Although clotting is a well-known immune defense mechanism in invertebrates, its role in mammalian host defense is largely unknown. However, multiple lines of evidence now suggest that coagulation is important for the early innate immune response since it supports the inflammatory response, generates antimicrobial peptides and induces local entrapment of bacteria in fibrin clots.

For example, **beta-lysin, a protein produced by platelets during coagulation** can lyse many Gram positive bacteria by acting as a cationic detergent.



Antibacterial mechanism of the host coagulation factors VII, IX, and X against drug-resistant gram-negative bacteria

Lesson 8 : Cells of Innate Immune system:

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The immune system is classified as two types:

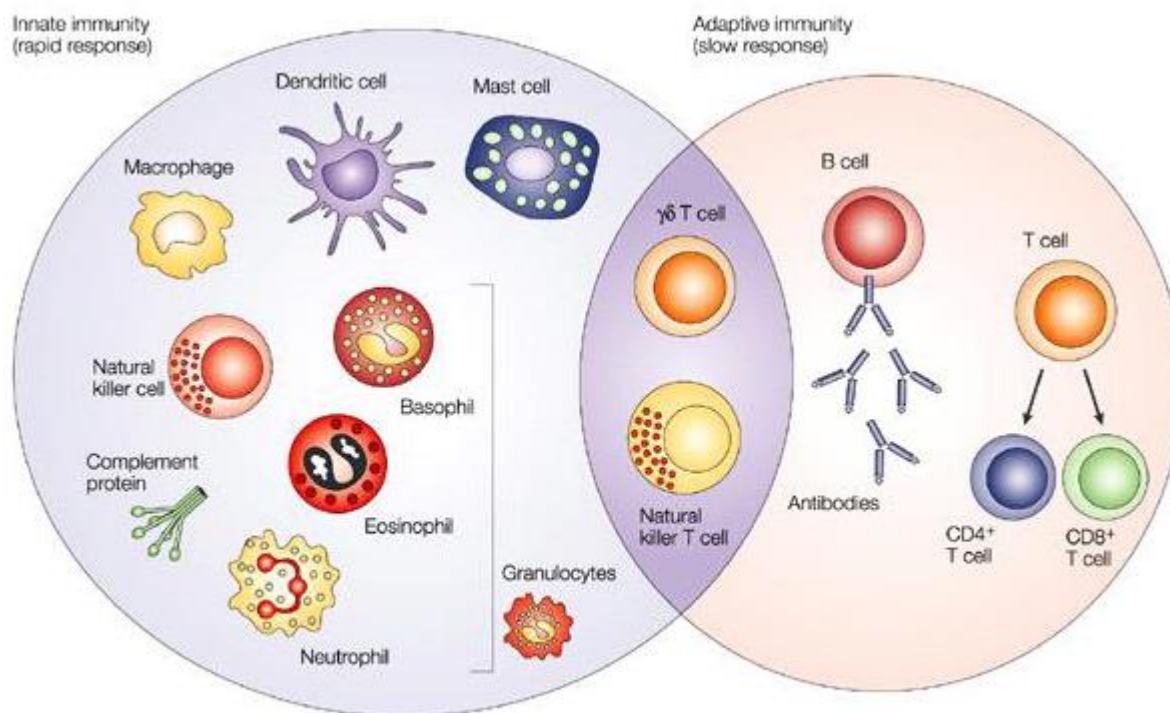
1. Innate immune system
2. Adaptive immune system

All the cells of the immune system develop from bone marrow's Hematopoietic stem cells that later differentiate into Myeloid and Lymphoid progenitors.

Innate immune system:

Innate immune responses are nonspecific and respond to infection quickly and consistently. Though they are general and nonspecific, they play a very important role in protecting the body from external infections. Their activity is based on pattern recognition receptors (PRRs).

The innate immune system comprises mostly the cells from Myeloid Progenitor.



Nature Reviews | Cancer

1. Neutrophils- Neutrophils are the granulocytic leukocytes, that have multi-lobed nuclei. They are the first cells that act at the site of tissue damage to eliminate pathogens

especially bacteria by phagocytosis. A type of cell located in the bloodstream, quickly ingesting and destroying microorganisms.

2. Eosinophils- Eosinophils are also granulocytic leukocytes with a kidney-shaped, lobed nucleus. They release the contents of their granules to kill pathogens. They produce a variety of growth factors and cytokines and helps other immune cells.
3. Basophils- Basophils are the largest granulocytic leukocytes with a bilobed nucleus. They have histamine-rich granules and involve in inflammatory responses. They help in the secretion of cytokines involved in the maturation of T-helper cells.
4. Monocytes- A type of phagocytic cell located in the bloodstream that, when moving to tissues, transforms into a macrophage. Monocytes are the largest white blood cells that produce Macrophages & Dendritic cells.
 - Macrophages engulf and process cell debris, pathogens, and cancer cells. They are involved in wound healing, tissue regeneration, and pro-inflammatory activities.
 - Dendritic cells are present in tissues like skin, lungs, and intestines. They present antigens to B and T cells and also secretes cytokines. Sensitive cells introduce antigen to cells in the immune system.
5. Mast cells- are granulocytes, that secrete heparin, histamine, other factors. They help in wound healing, angiogenesis, and elimination of parasites.
6. Natural killer (NK) cells (May develop from both Myeloid and Lymphoid progenitor). NK cells are produced from the bone marrow and are found in the bloodstream and tissues in reasonably low quantities. They are cytotoxic cells that have small granules with perforins and granzymes and destroys infected cells and cancer cells rapidly.

Lesson 9 : Phagocytosis:

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Phagocytosis and intracellular killing

Phagocytic cells

Neutrophils/Polymorph nuclear cells

PMNs are motile phagocytic cells that have lobed nuclei. They can be identified by their characteristic nucleus or by an antigen present on the cell surface called CD66.

They contain two kinds of granules the contents of which are involved in the antimicrobial properties of these cells.

- The primary or azurophilic granules, which are abundant in young newly formed PMNs, contain cationic proteins and defensins that can kill bacteria, proteolytic enzymes like elastase, and cathepsin G to breakdown proteins, lysozyme to break down bacterial cell walls and myeloperoxidase, which is involved in the generation of bacteriocidal compounds.
- The second type of granule found in more mature PMNs is the secondary or specific granule. These contain lysozyme, NADPH oxidase components, which are involved in the generation of toxic oxygen products, and characteristically lactoferrin, an iron chelating protein and B12-binding protein.

Monocytes/Macrophages

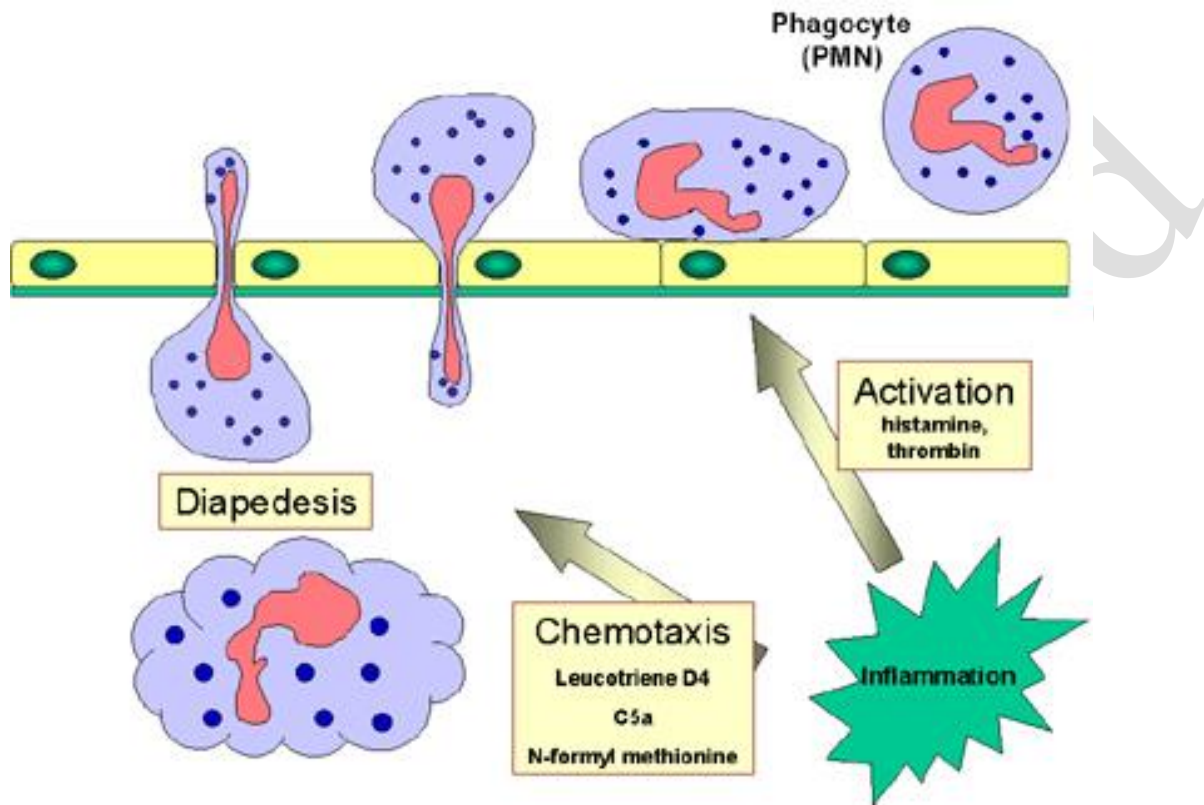
Macrophages are phagocytic cells that have a characteristic kidney-shaped nucleus. They can be identified morphologically or by the presence of the CD14 cell surface marker. Unlike PMNs they do not contain granules but they have numerous lysosomes which have contents similar to the PNM granules.

Response of phagocytes to infection

Circulating PMNs and monocytes respond to danger (SOS) signals generated at the site of an infection.

- SOS signals include N-formyl-methionine containing peptides released by bacteria, clotting system peptides, complement products and cytokines released from tissue macrophages that have encountered bacteria in tissue.
- Some of the SOS signals stimulate endothelial cells near the site of the infection to express cell adhesion molecules such as ICAM-1 and selectins which bind to components on the surface of phagocytic cells and cause the phagocytes to adhere to the endothelium.
- Vasodilators produced at the site of infection cause the junctions between endothelial cells to loosen and the phagocytes then cross the endothelial barrier by “squeezing” between the endothelial cells in a process called diapedesis.

- Once in the tissue spaces some of the SOS signals attract phagocytes to the infection site by chemotaxis (movement toward an increasing chemical gradient). The SOS signals also activate the phagocytes, which results in increased phagocytosis and intracellular killing of the invading organisms.



Chemotactic response to inflammatory stimulus Initiation of Phagocytosis

Phagocytic cells have a variety of receptors on their cell membranes through which infectious agents bind to the cells. These include:

Fc receptors

Bacteria with IgG antibody on their surface have the Fc region exposed and this part of the Ig molecule can bind to the receptor on phagocytes. Binding to the Fc receptor requires prior interaction of the antibody with an antigen. Binding of IgG-coated bacteria to Fc receptors results in enhanced phagocytosis and activation of the metabolic activity of phagocytes.

Complement receptors

Phagocytic cells have a receptor for the 3rd component of complement, C3b. Binding of C3b-coated bacteria to this receptor also results in enhanced phagocytosis and stimulation of the respiratory burst.

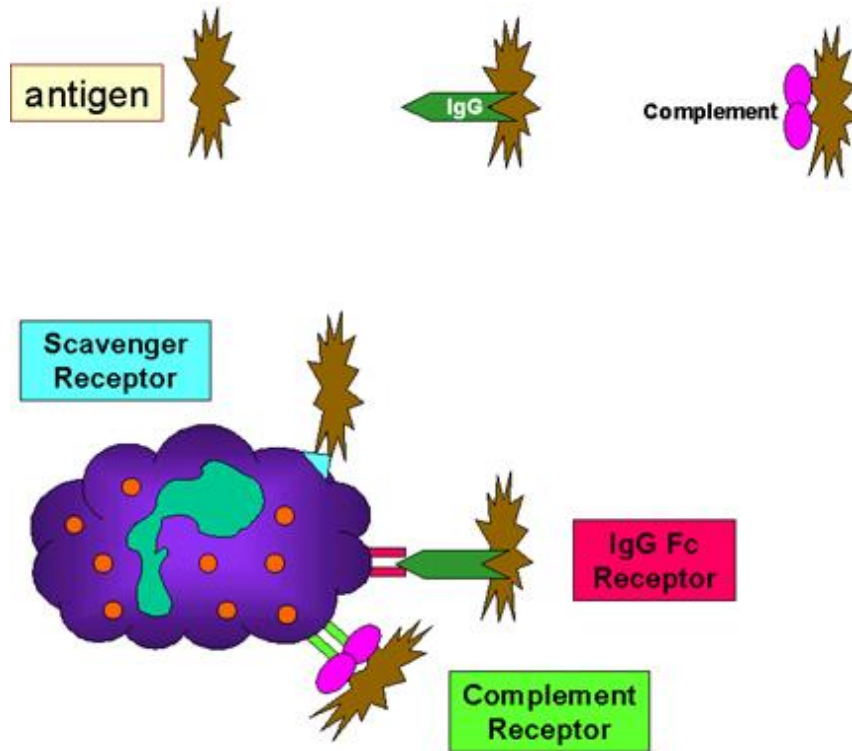
Scavenger receptors

Scavenger receptors bind a wide variety of polyanions on bacterial surfaces resulting in phagocytosis of bacteria.

Toll-like receptors

Phagocytes have a variety of Toll-like receptors (Pattern Recognition Receptors or PRRs) which

recognize broad molecular patterns called PAMPs (pathogen associated molecular patterns) on infectious agents. Binding of infectious agents via Toll-like receptors results in phagocytosis and the release of inflammatory cytokines (IL-1, TNF-alpha and IL-6) by the phagocytes.

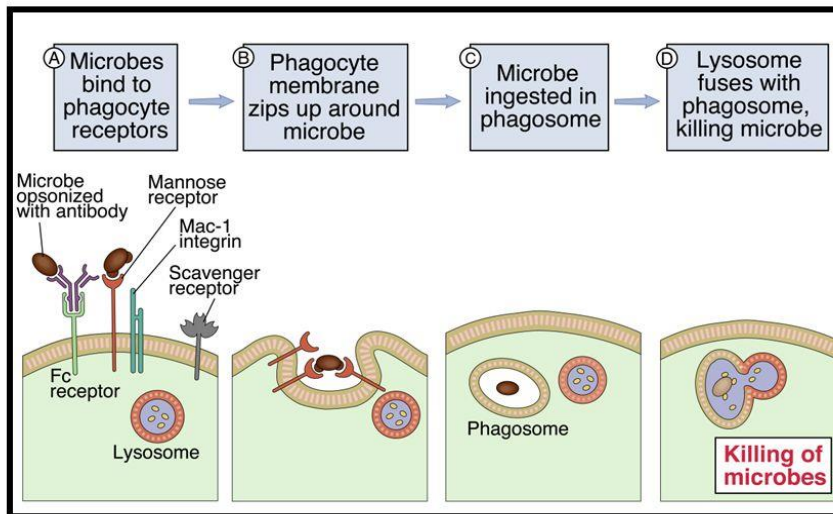


Adherence of bacteria via receptors

Phagocytosis

After attachment of a bacterium, the phagocyte begins to extend pseudopods around the bacterium. The pseudopods eventually surround the bacterium and engulf it, and the bacterium is enclosed in a phagosome. During phagocytosis the granules or lysosomes of the phagocyte fuse with the phagosome and empty their contents. The result is a bacterium engulfed in a phagolysosome which contains the contents of the granules or lysosomes.

Extracellular pathogen phagocytosis and killing



Lesson 10 : Intracellular killing:

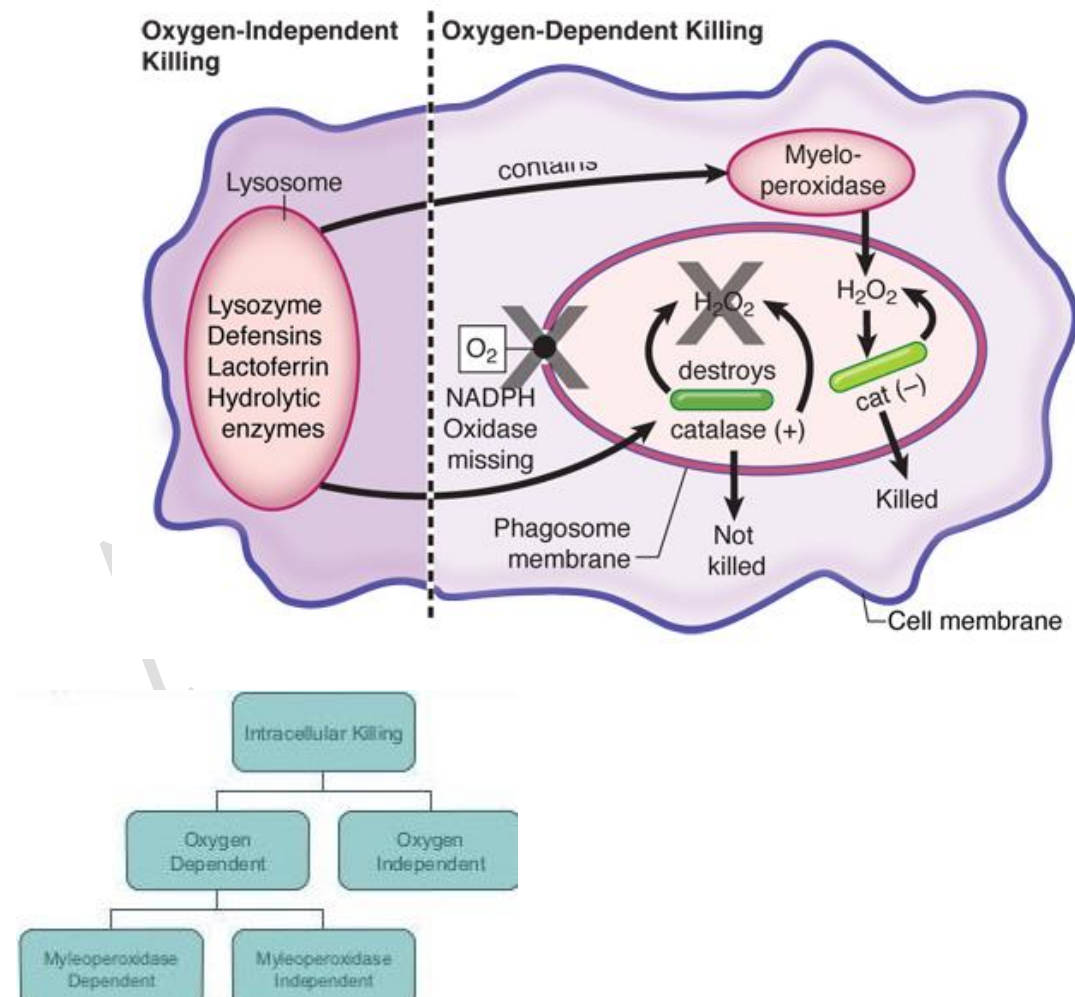
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Intracellular killing of organisms

After phagolysosome formation the first detectable effect on bacterial physiology, occurring within a few minutes after engulfment, is loss of viability (ability to reproduce). The exact mechanism is unknown. Inhibition of macromolecular synthesis occurs later. By 10 to 30 minutes after ingestion many pathogenic and nonpathogenic bacteria are killed followed by lysis and digestion of the bacteria by lysosomal enzymes. The microbicidal activities of phagocytes are complex and multifarious. Metabolic products, as well as lysosomal constituents, are responsible. These activities differ to some extent in neutrophils, monocytes and macrophages.

The microbicidal activities of phagocytes are usually divided into

1. oxygen-dependent
2. Oxygen-independent



Oxygen-independent activity

- Lysosomal granules contain a variety of extremely basic proteins that strongly inhibit bacteria, yeasts and even some viruses. A few molecules of any one of these cationic proteins appear able to inactivate a bacterial cell by damage to their permeability barriers.
- The lysosomal granules of neutrophils contain lactoferrin, an extremely powerful iron-chelating agent, which withholds potential iron needed for bacterial growth. The pH of the phagolysosome may be as low as 4.0 due to accumulation of lactic acid, which is sufficiently acidic to prevent the growth of most pathogens.
- This acidic environment apparently optimizes the activity of many degradative lysosomal enzymes including lysozyme, glycosylases, phospholipases, and nucleases.

Mechanisms of Oxygen Independent Killing

Effector Molecule	Function
Cationic proteins (including cathepsin)	Damage to microbial membranes
Lysozyme	Splits mucopeptide in bacterial cell wall
Lactoferrin	Deprives proliferating bacteria of iron
Proteolytic and hydrolytic enzymes	Digestion of killed organisms

Oxygen-dependent activity

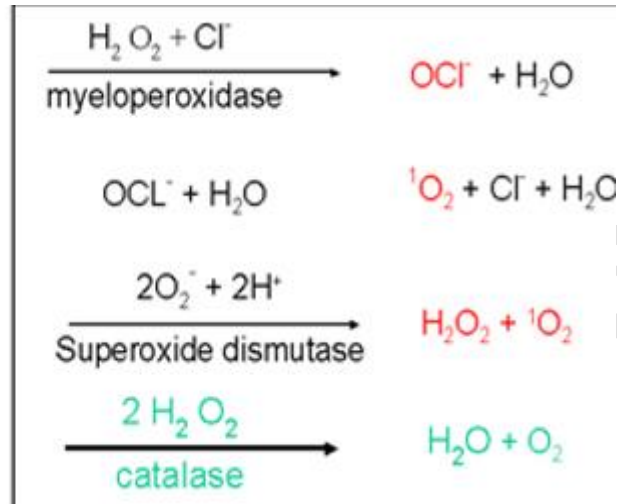
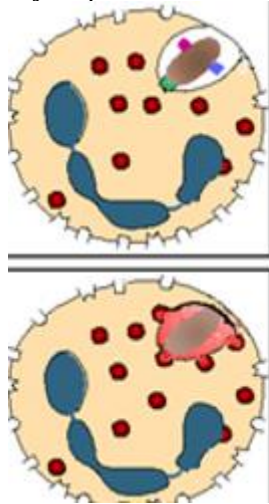
- Liganding of Fc receptors (on neutrophils, monocytes or macrophages) and mannose receptors (on macrophages) increases their O_2 uptake, called the respiratory burst.
- These receptors activate a membrane-bound NADPH oxidase that reduces O_2 to O_2^- (superoxide). Superoxide can be reduced to $OH^\cdot$ (hydroxyl radical) or dismutated to H_2O_2 (hydrogen peroxide) by superoxide dismutase.
- O_2^- , $OH^\cdot$, and H_2O_2 are activated oxygen species that are potent oxidizing agents in biological systems which adversely affect a number of cellular structures including membranes and nucleic acids.

Furthermore, in the case of neutrophils, these reactive oxygen intermediates can act in concert with a lysosomal enzyme called myeloperoxidase to function as the myeloperoxidase system.

- Myeloperoxidase is one of the lysosomal enzymes of neutrophils which is released into the phagocytic vacuole during fusion to form the phagolysosome.

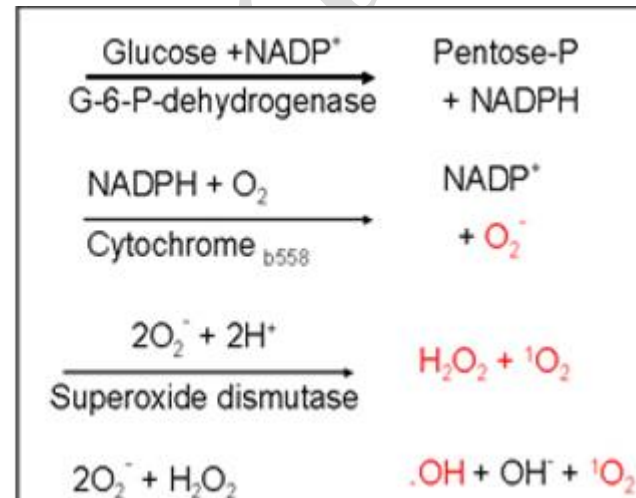
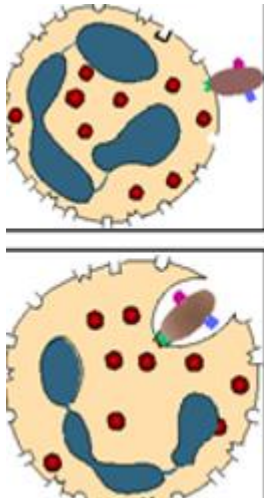
- Myeloperoxidase uses H_2O_2 generated during the respiratory burst to catalyze halogenation (mainly chlorination) of microbes contained within the phagolysosome. Such halogenations are a potent mechanism for killing cells.
- When the NADPH oxidase and myeloperoxidase systems are operating in concert, a series of reactions leading to lethal oxygenation and halogenation of engulfed microbes occurs.

Myeloperoxidase



(MPO)dependent

Myeloperoxidase independent -Involvement of Hexose monophosphate shunt



Detoxification

reactions

PMNs and macrophages have means to protect themselves from the toxic oxygen intermediates. These reactions involve the dismutation of superoxide anion to hydrogen peroxide by superoxide dismutase and the conversion of hydrogen peroxide to water by catalase.

Reaction	Enzyme
$\text{H}_2\text{O}_2 + \text{Cl}^- \rightarrow \text{OCl}^- + \text{H}_2\text{O}$	Myeloperoxidase
$\text{OCl}^- + \text{H}_2\text{O} \rightarrow {}^1\text{O}_2 + \text{Cl}^- + \text{H}_2\text{O}$	
$2\text{O}_2 + 2\text{H}^+ \rightarrow \text{O}_2^- + \text{H}_2\text{O}_2$	Superoxide dismutase
$\text{H}_2\text{O}_2 \rightarrow \text{H}_2\text{O} + \text{O}_2$	Catalase

Intracellular digestion

Dead microbes are rapidly degraded in phagolysosomes to low molecular-weight components. Various hydrolytic enzymes are involved including lysozyme, proteases, lipases, nucleases, and glycosylases. Neutrophils die and lyse after extended phagocytosis, killing, and digestion of bacterial cells. This makes up the characteristic properties of pus.

Macrophages egest digested debris and allow insertion of microbial antigenic components into the plasma membrane for presentation to lymphocytes in the immunological response.

Lesson 11 : Complement System:

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Complement

functions

Historically, the term complement (C) was used to refer to a heat-labile serum component that was able to lyse bacteria (activity is destroyed (inactivated) by heating serum at 56 degrees C for 30 minutes).

However, complement is now known to contribute to host defenses in other ways as well.

- Complement can opsonize bacteria for enhanced phagocytosis; it can recruit and activate various cells including polymorphonuclear cells (PMNs) and macrophages; it can participate in regulation of antibody responses and it can aid in the clearance of immune complexes and apoptotic cells. Complement can also have detrimental effects for the host; it contributes to inflammation and tissue damage and it can trigger anaphylaxis.

Complement comprises over 20 different serum proteins that are produced by a variety of cells including, hepatocytes, macrophages and gut epithelial cells.

- Some complement proteins bind to immunoglobulins or to membrane components of cells.
- Others are proenzymes that, when activated, cleave one or more other complement proteins.

Upon cleavage some of the complement proteins yield fragments that activate cells, increase vascular permeability or opsonize bacteria

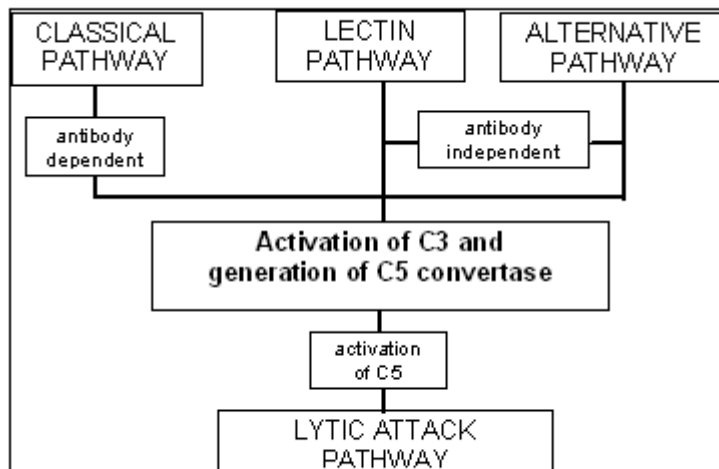


Figure 1. Pathways of complement activation

The Functions of Complement

the realization that interaction of cellular receptors with complement proteins controls B-cell activities gives this system a role in the highly developed acquired immune system. Thus we have a system that straddles innate and acquired immunity, contributing to each in a variety of ways. After initial activation, the various complement components interact, in a highly regulated cascade, to carry out a number of basic functions (Figure 13-1) including:

- Lysis of cells, bacteria, and viruses
- Opsonization, which promotes phagocytosis of particulate antigens
- Binding to specific complement receptors on cells of the immune system, triggering specific cell functions, inflammation, and secretion of immunoregulatory molecules
- Immune clearance, which removes immune complexes from the circulation and deposits them in the spleen and liver

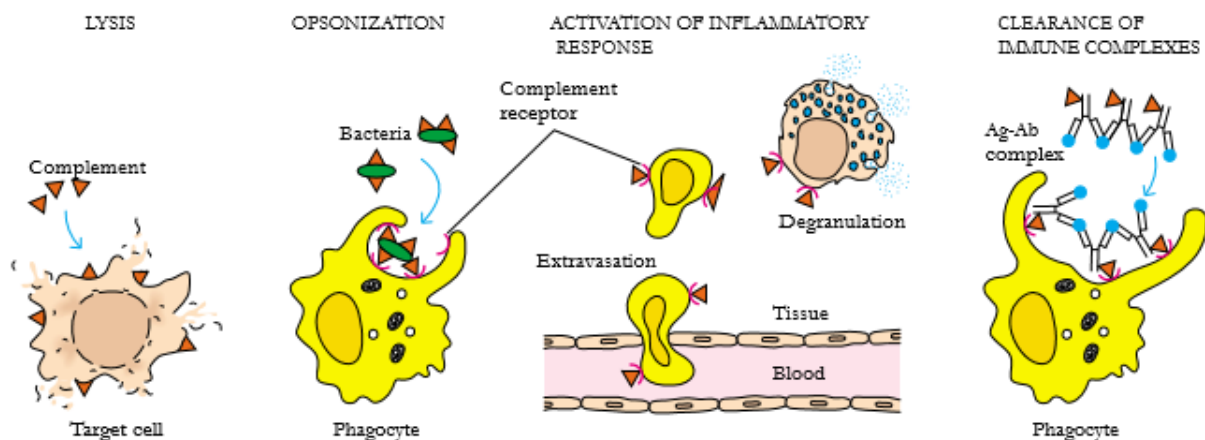


FIGURE 13-1 The multiple activities of the complement system. Serum complement proteins and membrane-bound complement receptors partake in a number of immune activities: lysis of foreign cells by antibody-dependent or antibody-independent pathways; opsonization or uptake of particulate antigens, including bacteria, by

phagocytes; activation of inflammatory responses; and clearance of circulating immune complexes by cells in the liver and spleen. Soluble complement proteins are schematically indicated by a triangle and receptors by a semi-circle; no attempt is made to differentiate among individual components of the complement system here.

Reference

- Kuby Immunology, 6th Edition, Authored by Kindt, Goldsby and Osborne, Chapter 13, Page 299,300
- <http://www.microbiologybook.org/ghaffar/complement.htm>
- <https://www.immunology.org/public-information/bitesized-immunology/systems-and-processes/complement-system>

Lesson 12 : Classical Pathway of Complement Activation:

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Activation of the Classical Pathway

- Complement activation by the classical pathway commonly begins with the formation of soluble antigen-antibody complexes (immune complexes) or with the binding of antibody to antigen on a suitable target, such as a bacterial cell.
- IgM and certain subclasses of IgG (human IgG1, IgG2, and IgG3) can activate the classical complement pathway by recognizing invasive antigens to form the antigen-antibody complexes. C1q, as part of C1 complex (C1q C1r2C1s2), has affinity for the Fc regions of IgG/ IgM. It is the first enzyme in this pathway to trigger a series of enzymatic events. The classical pathway can also be activated by apoptotic/necrotic cells and acute phase proteins.
- The initial stage of activation involves C1, C2, C3, and C4, which are present in plasma in functionally inactive forms

The classical pathway is one of three activation pathways of the complement system, which is a major contributor to the

- defense of infections
- clearance of pathogens
- removal of apoptotic/necrotic cells
- maintenance of homeostasis.

There are at least 21 different serum proteins have been confirmed as components of the classical pathway, in which 11 major protein play the most critical role, including C1q, C1r, C1s, C2, C3, C4, C5, C6, C7, C8, C9.

Activation of the Classical Pathway

The enzyme cascade can be divided into three phases:

- **Initiation**

The recognition of C1q and IgM or hexameric IgG immune complexes, surface proteins of bacteria/viruses, apoptotic cells, is a signal to start the complement classical pathway. Three steps are showed in this stage:

- 1) Zymogen C1 binds directly to the triggers via the globular heads of C1q.
- 2) Ligand binding induces conformational changes in C1q, resulting in subsequent repositioning and autocatalysis of the C1r zymogen dimer to be active form.
- 3) C1r cleaves C1s forming fully activated C1s with catalytic capabilities.

- **Formation of C3 convertase**

Activated C1s binds C4 and enzymatically liberates C4a and C4b, which in turn promotes the cleavage of C2 into C2a and C2b. Surface-attached C4b serves as a platform for the formation of C3 convertase (C4bC2a). C3 convertase can cleave C3 into C3a and C3b, which is essential for the next reaction.

- **Formation of C5 convertase and MAC**

C3b binds to the C4b2a complex to form C5 convertase (C4b2a3b), while C3a plays role in the recruitment of inflammatory cells (anaphylatoxin). C5 convertase then cleaves C5 into C5a and C5b. C5b combines with other terminal components C6, C7, C8, and C9 to form the Membrane Attack Complex (MAC), which leads to lysis of invasive bacteria through insertion into the target cell membranes to create functional pores.

Inhibitors of the Classical Pathway

Because the classical pathway mediates a lot of autoimmune and inflammatory disorders, the regulation is very strict to prevent unwanted complement activation. The complement inhibitors which can antagonize excessive activation of complement have potential clinical application. Scientists have identified a series of inhibitors targeting the proteins in this pathway, including C1q, C1r, C3 etc.

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Teacher: Dr Numan Javed (Phd Holder at SHanghai Institute of Sciences, China)

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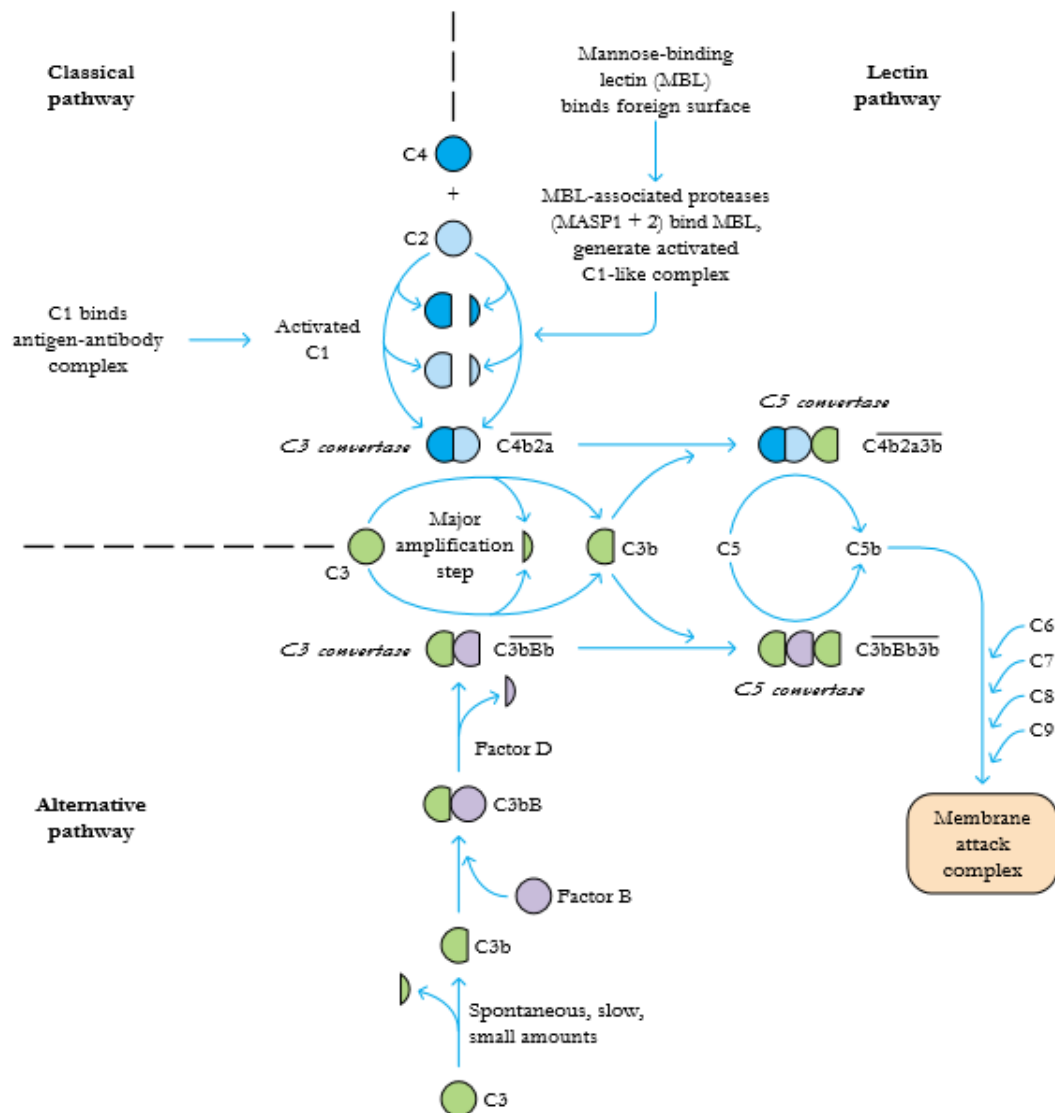


FIGURE 13-2 Overview of the complement activation pathways. The classical pathway is initiated when C1 binds to antigen-antibody complexes. The alternative pathway is initiated by binding of spontaneously generated C3b to activating surfaces such as microbial cell walls. The lectin pathway is initiated by binding of the serum protein MBL to the surface of a pathogen. All three pathways generate C3 and C5 convertases and bound C5b, which is converted into a mem-

brane-attack complex (MAC) by a common sequence of terminal reactions. Hydrolysis of C3 is the major amplification step in all pathways, generating large amounts of C3b, which forms part of C5 convertase. C3b also can diffuse away from the activating surface and bind to immune complexes or foreign cell surfaces, where it functions as an opsonin.

Lesson 13 : Alternate Pathway of Complement Activation:

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The Alternative Pathway Is Antibody-Independent

The alternative pathway generates bound C5b, the same product that the classical pathway generates, but it does so without the need for antigen-antibody complexes for initiation. Because no antibody is required, the alternative pathway is a component of the innate immune system. This major pathway of complement activation involves four serum proteins: C3, factor B, factor D, and properdin. The alternative pathway is initiated in most cases by cell-surface constituents that are foreign to the host

TABLE 13-1

Initiators of the alternative pathway of complement activation

PATHOGENS AND PARTICLES OF MICROBIAL ORIGIN

Many strains of gram-negative bacteria
Lipopolysaccharides from gram-negative bacteria
Many strains of gram-positive bacteria
Teichoic acid from gram-positive cell walls
Fungal and yeast cell walls (zymosan)
Some viruses and virus-infected cells
Some tumor cells (Raji)
Parasites (trypanosomes)

NONPATHOGENS

Human IgG, IgA, and IgE in complexes
Rabbit and guinea pig IgG in complexes
Cobra venom factor
Heterologous erythrocytes (rabbit, mouse, chicken)
Anionic polymers (dextran sulfate)
Pure carbohydrates (agarose, inulin)

SOURCE: Adapted from M. K. Pangburn, 1986, in *Immunobiology of the Complement System*, Academic Press.

Complement Activation Alternative Pathway Mechanisms

alternative pathway is capable of autoactivation because of a process termed "tickover" of C3.

- Tickover occurs spontaneously at a rate of ~1% of total C3 per hour, generating a conformationally altered C3, designated C3(H₂O), that is capable of binding factor B. Once factor B associates with C3(H₂O), factor B itself changes conformation and can then be cleaved by the constitutively active serum protease factor D, generating Ba and Bb. The Bb fragment remains associated with the complex and can then, through its own

serine protease domain, cleave additional C3 molecules, generating a form designated C3b. Once C3b is generated, it associates with factor B to generate more C3-convertase.

- This overall series of successive proteolytic steps is enhanced by the serum protein properdin, which stabilizes protein:protein interactions during the process.
- The alternative pathway can also be initiated as an "amplification loop" when fixed C3b that is generated by classical or lectin pathway activation binds factor B, again resulting in conformational changes in factor B that allow factor D to cleave it similarly to the tickover process.

Complement Activation Alternative Pathway Regulation

Carbohydrate composition and its sialic acid content on the cell surface play an important role in the activation of the alternate pathway.

- Sialic acid blocks activation by favoring the binding of factor H to C3b, which is then inactivated by factor I. Microorganisms lacking sialic acid are killed, whereas human cells covered with glycophorin A, a sialoglycoprotein, are protected.
- C3bBb is relatively labile and undergoes spontaneous decay through dissociation of Bb. Properdin is synthesized by monocytes and T lymphocytes. Properdin binds to C3bBb and stabilizes it, preventing its decay. Factor H competes with factor B for binding to C3b and displaces Bb from C3bBb. It accelerates the inactivation of C3b by factor I. Factor I inactivates C3b to iC3b, a molecule that cannot function enzymatically. Complement receptor 1 (CR1) has factor H-like activity, permitting factor I to cleave C3b. Membrane cofactor protein also has factor H-like activity, mainly for alternative C3 convertase.



① C3 hydrolyzes spontaneously, C3b fragment attaches to foreign surface

② Factor B binds C3a, exposes site acted on by Factor D. Cleavage generates C3bBb, which has C3 convertase activity

③ Binding of properdin stabilizes convertase

④ Convertase generates C3b; some binds to C3 convertase activating C5' convertase. C5b binds to antigenic surface

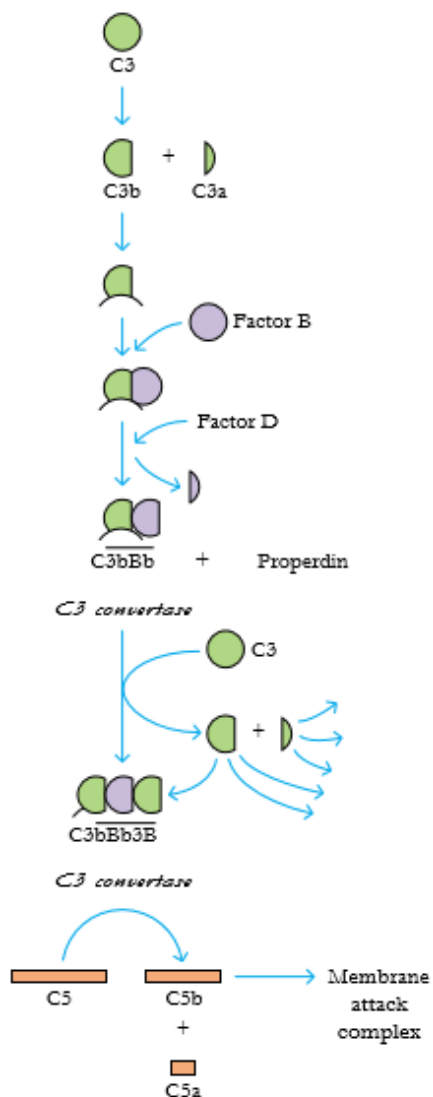


FIGURE 13-7 Schematic diagram of intermediates in the formation of bound C5b by the alternative pathway of complement activation. The C3bBb complex is stabilized by binding of prop-

erlin. Conversion of bound C5b to the membrane-attack complex occurs by the same sequence of reactions as in the classical pathway (see Figure 13-5).

Lesson 14 : Lectin Pathway of Complement Activation:

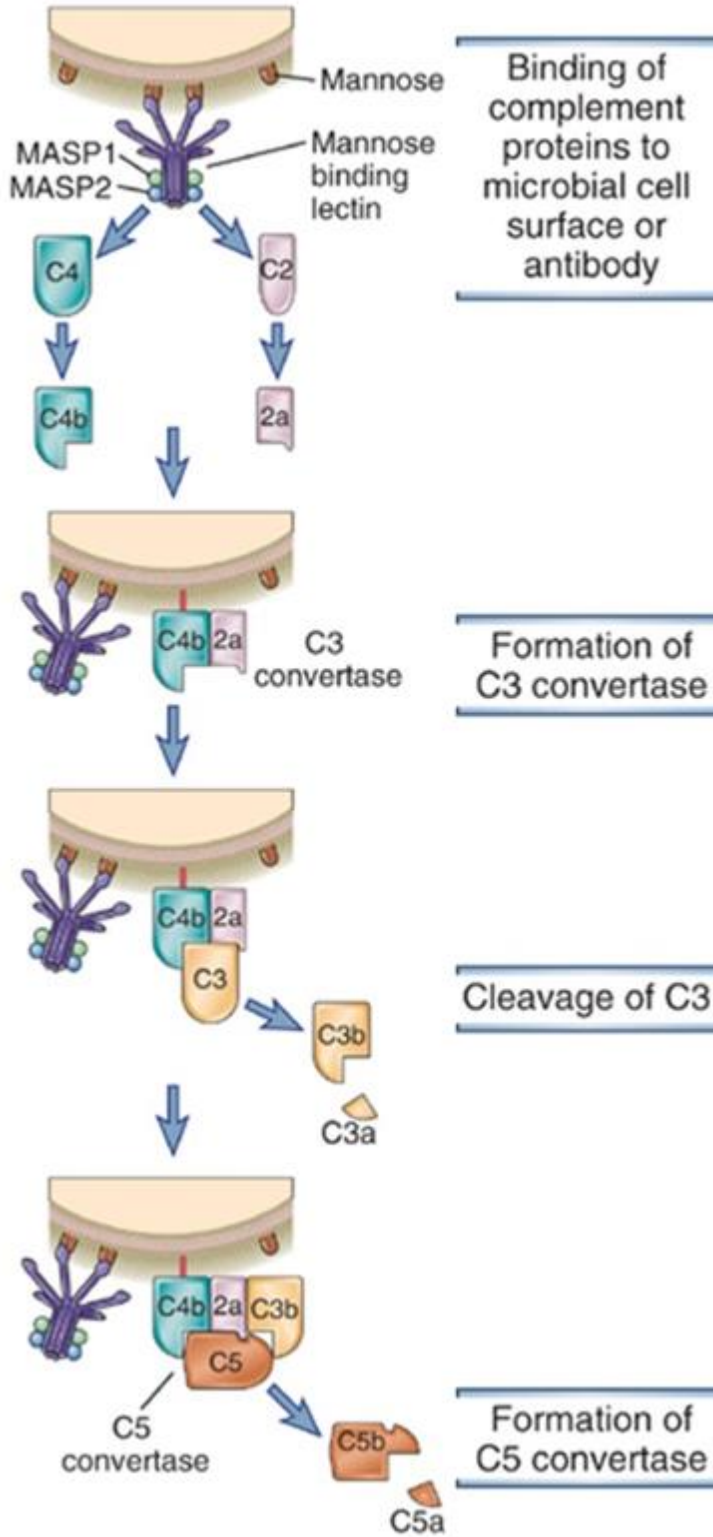
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The Lectin Pathway Originates with Host Proteins Binding Microbial Surfaces

Lectins are proteins that recognize and bind to specific carbohydrate targets. (Because the lectin that activates complement binds to mannose residues.

- The lectin pathway, like the alternative pathway, does not depend on antibody for its activation. However, the mechanism is more like that of the classical pathway, because after initiation, it proceeds, through the action of C4 and C2, to produce a C5 convertase.
- The lectin pathway is activated by the binding of mannose-binding lectin (MBL) to mannose residues on glycoproteins or carbohydrates on the surface of microorganisms.
- MBL is an acute phase protein produced in inflammatory responses. Its function in the complement pathway is similar to that of C1q, which it resembles in structure. After MBL binds to the surface of a cell or pathogen, MBL-associated serine proteases, MASP-1 and MASP-2, bind to MBL. The active complex formed by this association causes cleavage and activation of C4 and C2. The MASP-1 and -2 proteins have structural similarity to C1r and C1s and mimic their activities. This means of activating the C2–C4 components to form a C5 convertase without need for specific antibody binding represents an important innate defense mechanism comparable to the alternative pathway, but utilizing the elements of the classical pathway except for the C1 proteins.

Lectin Pathway



Lesson 15 : Effector Functions of Complement Proteins:

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Effector Functions of Complement Proteins

Complement proteins have following effector functions for providing immunity against infectious agents

- **Opsonization**

(Opsonins) Various complement proteins bind with the surface of pathogens which facilitate the process of phagocytosis by phagocytes. This enhanced process of phagocytosis is called as opsonization, while complement proteins involved in this process are called as opsonins.

- **Chemoattraction**

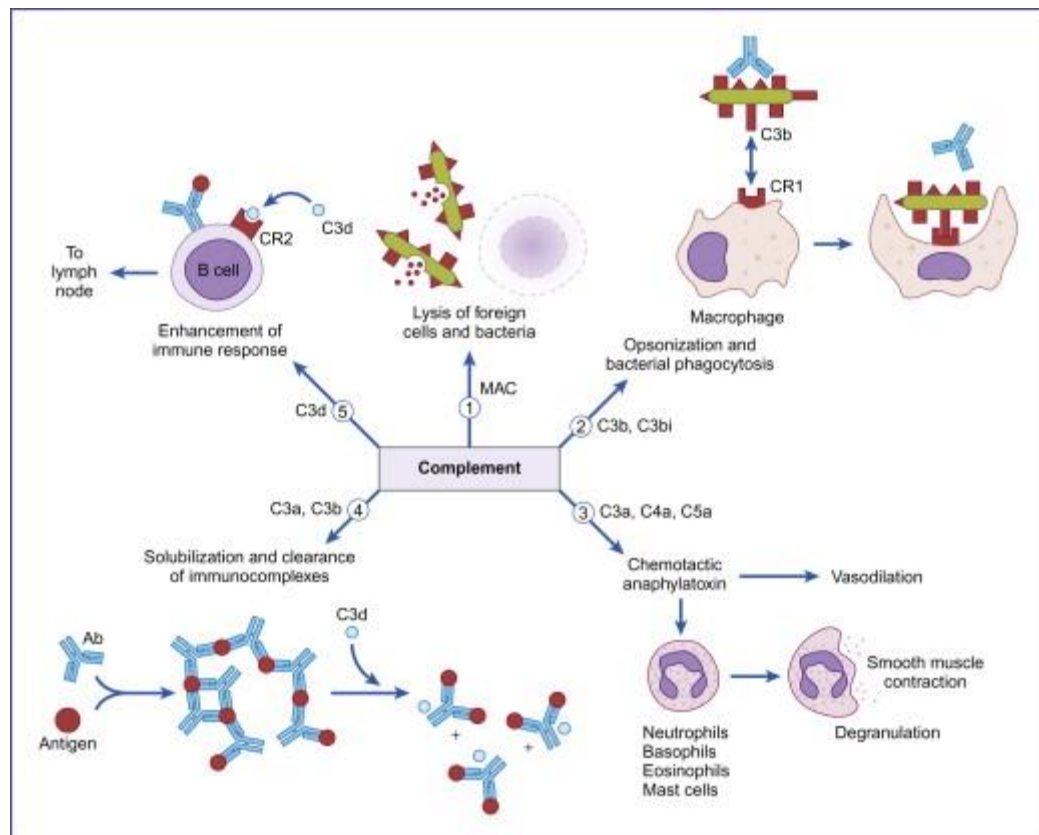
(Chemoattractants) Complement proteins also served as chemoattractants for the movement of phagocytic cells towards the site of infection. The chemoattractant function of complement proteins increased the defense mechanism against invading agents.

- **Anaphylaxis**

(Anaphylatoxins) Complement proteins also involved in hypersensitivity (allergic) reactions which are also called as anaphylactic reactions in the form of exaggerated immune response. Such kind of complement proteins are called as anaphylatoxins.

- **Pro-Inflammatory**

The role of complement proteins in inflammatory process is well established. The tissue response against infectious agent is termed as inflammatory response and the complement proteins contributing towards such kind of response are considered as pro-inflammatory.



Lesson 16 : Pathology related to Complement Proteins:

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Role of Complement in Disease

The complement system plays a critical role in inflammation and defense against some bacterial infections. Complement may also be activated during reactions against incompatible blood transfusions, and during the damaging immune responses that accompany autoimmune disease. Deficiencies of individual complement components or inhibitors of the system can lead to a variety of diseases which gives some indication of their role in protection against disease.

Complement Deficiency	Disease
C3 and Factor B	Severe bacterial infections
C3b-INa, C6 and C8	Severe Neisseria infections
Deficiencies of early C components C1, C4, C2.	Systemic lupus erythematosus (SLE), glomerulonephritis and polymyositis
C1-inhibitor	Hereditary angioedema

Table 5. Complement deficiencies and disease

Pathway/Component	Disease	Mechanism
Classical Pathway		
C1INH	Hereditary angioedema	Overproduction of C2b (prokinin)
C1, C2, C4	Predisposition to SLE	Opsonization of immune complexes help keep them soluble, deficiency results in increased precipitation in tissues and inflammation
Lectin Pathway		
MBL	Susceptibility to bacterial infections in infants or immunosuppressed	Inability to initiate the lectin pathway
Alternative Pathway		
Factors B or D	Susceptibility to pyogenic (pus-forming) bacterial infections	Lack of sufficient opsonization of bacteria
C3	Susceptibility to bacterial infections	Lack of opsonization and inability to utilize the membrane attack pathway
C5, C6, C7 C8, and C9	Susceptibility to Gram-negative infections	Inability to attack the outer membrane of Gram-negative bacteria
Properdin (X-linked)	Susceptibility meningococcal meningitis	Lack of opsonization of bacteria
Factors H or I	C3 deficiency and susceptibility to bacterial infections	Uncontrolled activation of C3 via alternative pathway resulting in depletion of C3

Lesson 17 : Cells & Organs of the Immune System:

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Organs of the Immune System

A number of morphologically and functionally diverse organs and tissues have various functions in the development of immune responses.

These can be distinguished by function as

1. primary lymphoid organs
2. secondary lymphoid organs

primary lymphoid organs

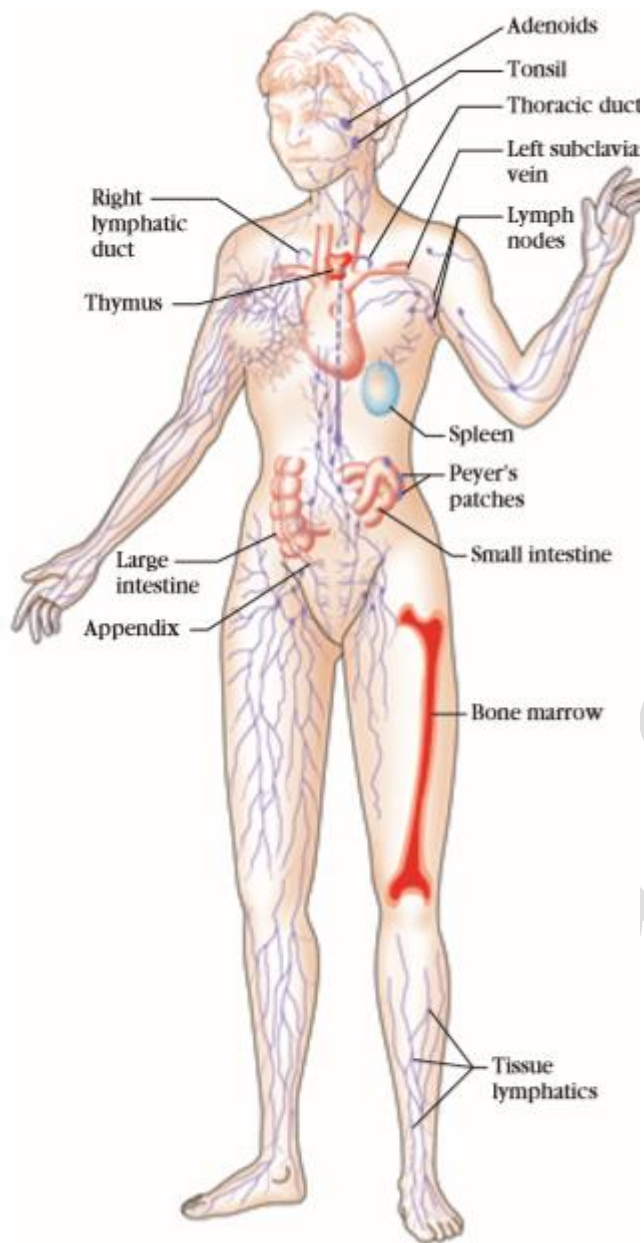
The thymus and bone marrow are the primary (or central) lymphoid organs, where maturation of lymphocytes takes place.

secondary lymphoid organs

The lymph nodes, spleen, and various mucosal-associated lymphoid tissues (MALT) such as gut-associated lymphoid tissue (GALT) are the secondary (or peripheral) lymphoid organs, which trap antigen and provide sites for mature lymphocytes to interact with that antigen.

In addition, tertiary lymphoid tissues, which normally contain fewer lymphoid cells than secondary lymphoid organs, can import lymphoid cells during an inflammatory response. Most prominent of these are cutaneous-associated lymphoid tissues.

Once mature lymphocytes have been generated in the primary lymphoid organs, they circulate in the blood and lymphatic system, a network of vessels that collect fluid that has escaped into the tissues from capillaries of the circulatory system and ultimately return it to the blood.



Lesson 18 : Primary Lymphoid Organs:

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Primary Lymphoid Tissues

Immature lymphocytes generated in hematopoiesis mature and become committed to a particular antigenic specificity within the primary lymphoid organs. Only after a lymphocyte has matured within a primary lymphoid organ is the cell immunocompetent (capable of mounting an immune response).

- T cells arise in the thymus, and in many mammals—humans and mice for example
- B cells originate in bone marrow.

THYMUS

The thymus is the site of T-cell development and maturation.

It is a flat, bilobed organ situated above the heart. Each lobe is surrounded by a capsule and is divided into lobules, which are separated from each other by strands of connective tissue called trabeculae.

Each lobule is organized into two compartments:

- the outer compartment, or cortex, is densely packed with immature T cells, called thymocytes
- the inner compartment, or medulla, is sparsely populated with thymocytes.

The function of the thymus is to generate and select a repertoire of T cells that will protect the body from infection.

As thymocytes develop, an enormous diversity of T-cell receptors is generated by a random process that produces some T cells with receptors capable of recognizing antigen-MHC complexes.

However, most of the T-cell receptors produced by this random process are incapable of recognizing antigen-MHC complexes and a small portion react with combinations of self-antigen-MHC complexes, the thymus induces the death of those T cells that cannot recognize antigen-MHC complexes and those that react with self-antigen– MHC and pose a danger of causing autoimmune disease. More than 95% of all thymocytes die by apoptosis in the thymus without ever reaching maturity.

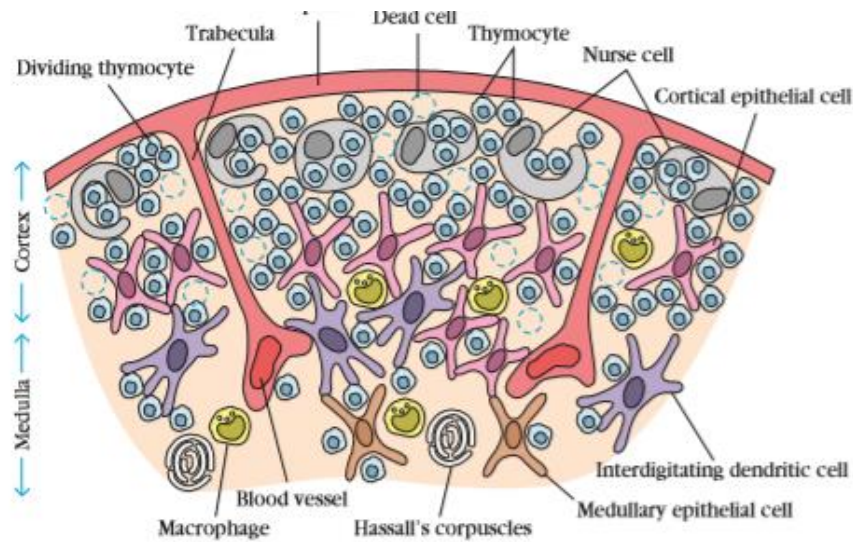


FIGURE 2-14 Diagrammatic cross section of a portion of the thymus, showing several lobules separated by connective tissue strands (trabeculae). The densely populated outer cortex is thought to contain many immature thymocytes (blue), which undergo rapid proliferation coupled with an enormous rate of cell death. Also present in the outer cortex are thymic nurse cells (gray), which are specialized epithelial cells with long membrane extensions that surround as many as 50 thymocytes. The medulla is sparsely populated and is thought to contain thymocytes that are more mature. During their

stay within the thymus, thymocytes interact with various stromal cells, including cortical epithelial cells (light red), medullary epithelial cells (tan), interdigitating dendritic cells (purple), and macrophages (yellow). These cells produce thymic hormones and express high levels of class I and class II MHC molecules. Hassall's corpuscles, found in the medulla, contain concentric layers of degenerating epithelial cells. [Adapted, with permission, from W. van Ewijk, 1991, *Annu. Rev. Immunol.* 9:591, © 1991 by Annual Reviews.]

BONE MARROW

In humans and mice, bone marrow is the site of B-cell origin and development. Arising from lymphoid progenitors, immature B cells proliferate and differentiate within the bone marrow, and stromal cells within the bone marrow interact directly with the B cells and secrete various cytokines that are required for development. Like thymic selection during Tcell maturation, a selection process within the bone marrow eliminates B cells with self-reactive antibody receptors

Lesson 19 : Secondary Lymphoid Organs:

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Secondary Lymphoid Organs

- These are the lymphoid tissues which are involved in providing the specific immune response. These tissues are located distantly from central lymphoid tissues that's why are also called as peripheral lymphoid tissues.
- These tissues provide the specific site for interaction between immune cells like lymphocytes and the antigens.
- Such kind of tissues are located in various anatomical locations for providing continuous surveillance against the foreign agents.
- Secondary lymphoid tissues include lymph nodes in various parts of body like axillary & groin regions. Similarly, tonsils & adenoids are located in throat and larynx area. Spleen is located in the left hypochondrial region. Moreover, appendix and peyer's patches are located in the gastrointestinal tract (GIT) for providing immune response against invading agents

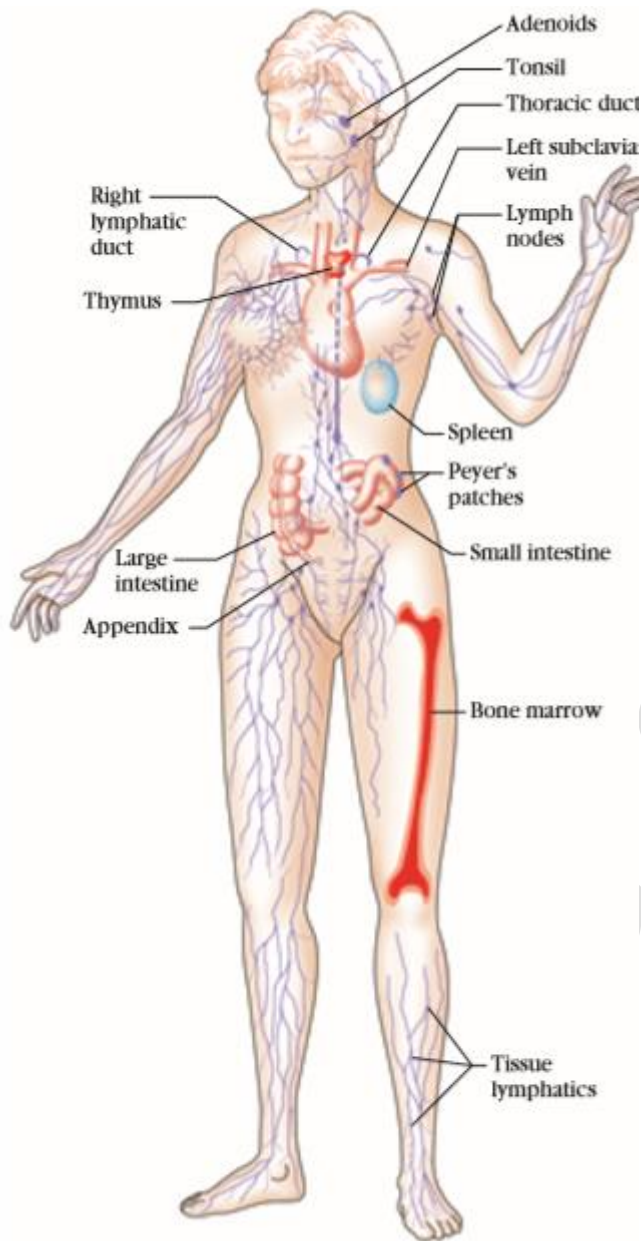
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These organs provide the sites for the interaction of lymphocytes with the antigen, which then proliferate to become effector cells.

These are of following types:

- (a) Spleen,
- (b) Lymph nodes,
- (c) Mucosal associated Lymphoid Tissue (MALT) (i.e., tonsils)

Peyer's patches of small intestine and appendix are also some of the secondary lymphoid organs.

Spleen:

It is a large bean-shaped organ containing lymphocytes and phagocytes. It filters the blood by trapping the pathogens in it.

Lymph Nodes:

These are small solid structures located at different points along the lymphatic system. Their function is to trap the microorganisms or other antigens, that enter the lymph and tissue fluid. Therefore, the antigens trapped in the lymph nodes are responsible for the activation of lymphocytes present there and cause the immune response.

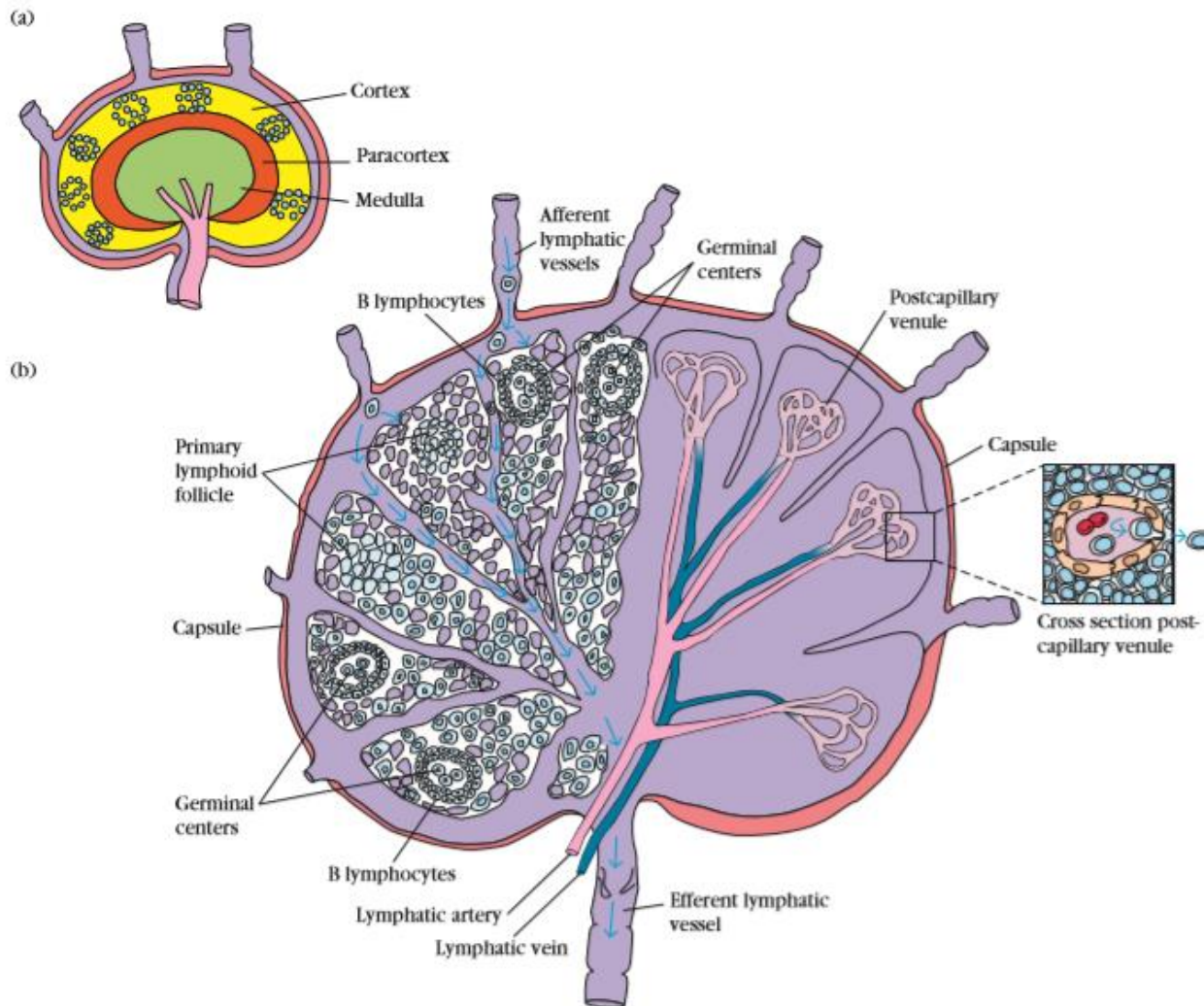


FIGURE 2-18 Structure of a lymph node. (a) The three layers of a lymph node support distinct microenvironments. (b) The left side depicts the arrangement of reticulum and lymphocytes within the various regions of a lymph node. Macrophages and dendritic cells, which trap antigen, are present in the cortex and paracortex. T_H cells are concentrated in the paracortex; B cells are located primarily in the cortex, within follicles and germinal centers. The medulla is popu-

lated largely by antibody-producing plasma cells. Lymphocytes circulating in the lymph are carried into the node by afferent lymphatic vessels; they either enter the reticular matrix of the node or pass through it and leave by the efferent lymphatic vessel. The right side of (b) depicts the lymphatic artery and vein and the postcapillary venules. Lymphocytes in the circulation can pass into the node from the postcapillary venules by a process called extravasation (inset).

(c) Mucosal Associated Lymphoid Tissue (MALT):

This is located within the lining of main tracts in the body like respiratory, digestive, urogenital tracts. MALT constitutes about 50% of the lymphoid tissue in human body.

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Lesson 20 : Cell of Immune Systems:

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Cells of Immune System

These cells are called as immune cells and originate from the hematopoietic stem cells. These stem cells are located in the primary lymphoid tissues like bone marrow. Hematopoietic stem cells differentiate into following two important cellular lineage according to their cellular source

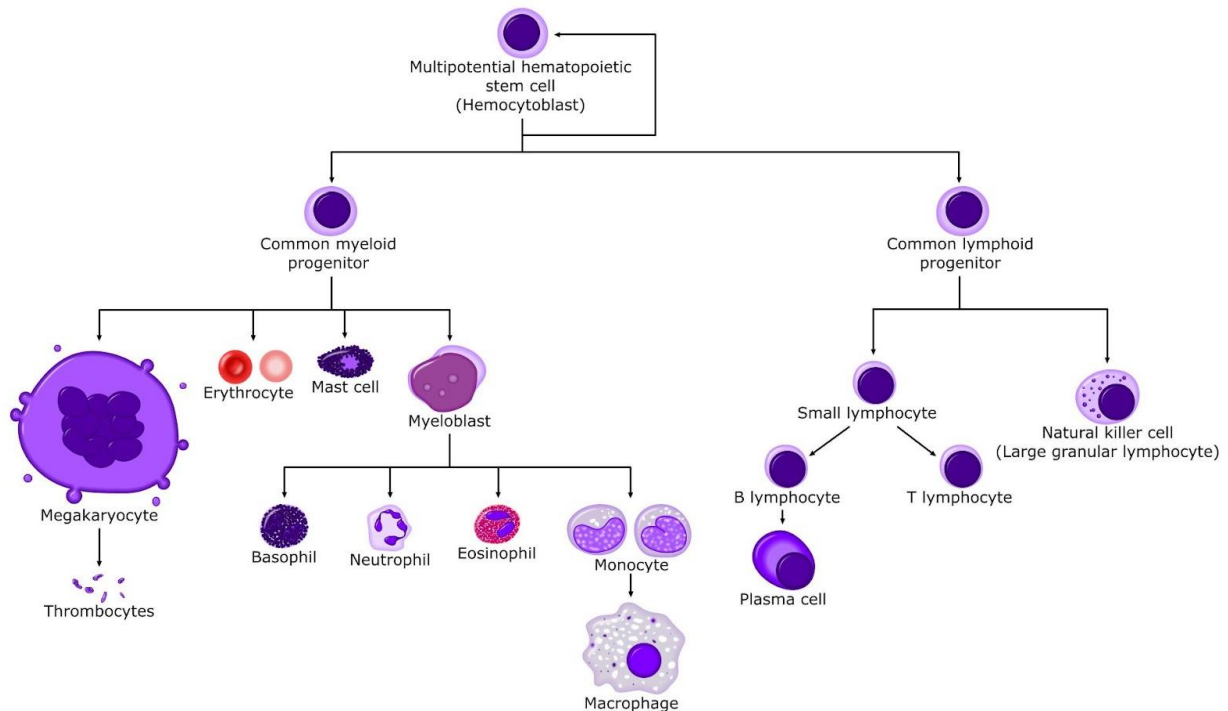
There are two main lineages that derive from the hemopoietic stem cell:

- **The lymphoid lineage**

T lymphocytes	(T cells)
B lymphocytes	(B cells)
Natural killer cells (NK cells)	

- **The myeloid lineage**

Monocytes,		macrophages
Langerhans	cells,	dendritic
Megakaryocytes		cells
Granulocytes (eosinophils, neutrophils, basophils)		



Lesson 21 : Lymphoid Lineage of Immune Cells:

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Lymphoid Lineage of Immune Cells

All blood cells arise from a type of cell called the hematopoietic stem cell (HSC). Stem cells are cells that can differentiate into other cell types; they are self-renewing—they maintain their population level by cell division. Early in hematopoiesis, a multipotent stem cell differentiates along one of two pathways, giving rise to either a common lymphoid progenitor cell or a common myeloid progenitor cell. The types and amounts of growth factors in the microenvironment of a particular stem cell or progenitor cell control its differentiation. During the development of the lymphoid and myeloid lineages, stem cells differentiate into progenitor cells, which have lost the capacity for self-renewal and are committed to a particular cell lineage. Common lymphoid progenitor cells give rise to B, T, and NK (natural killer) cells and some dendritic cells.

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VISUALIZING CONCEPTS

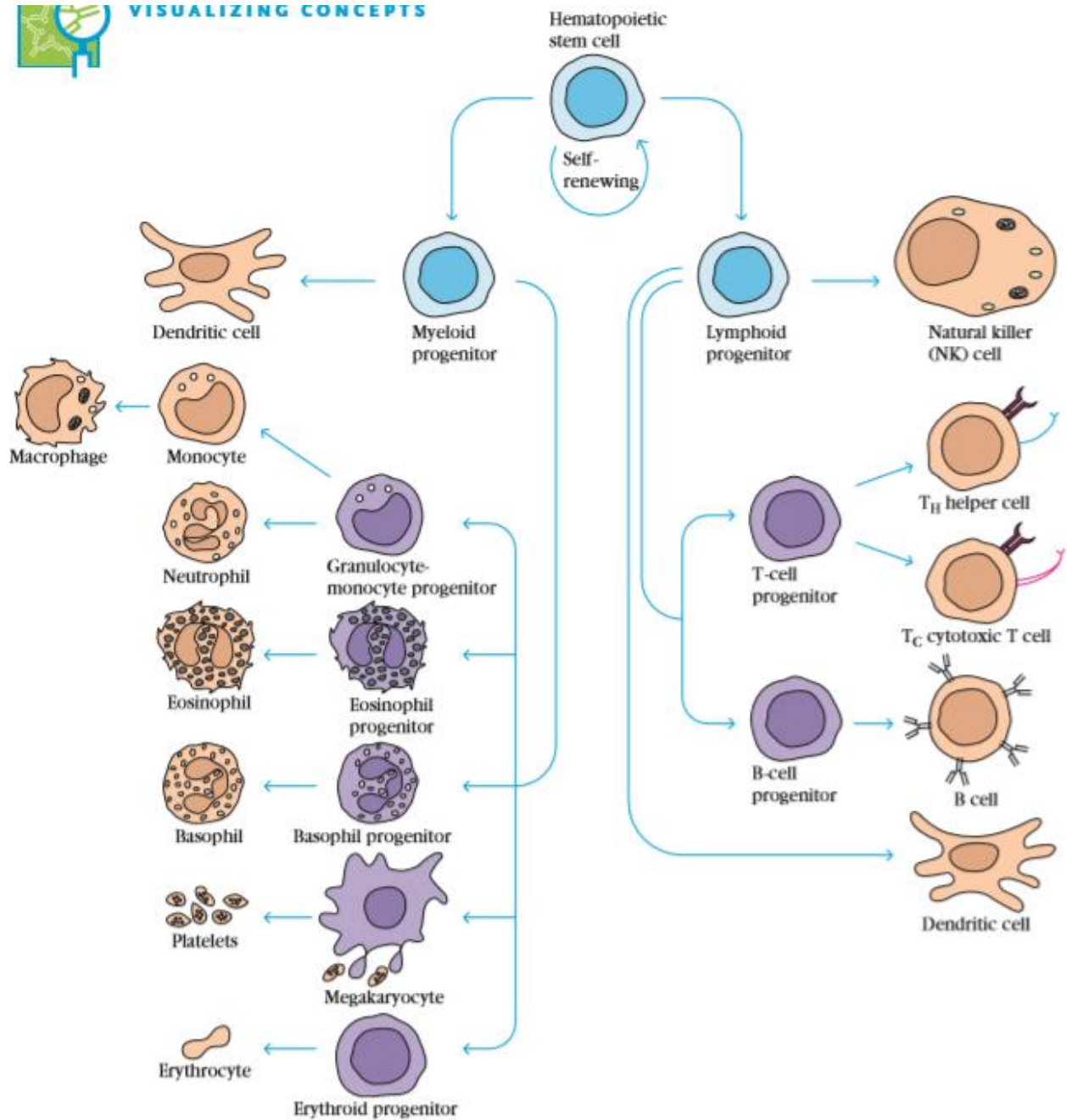


FIGURE 2-1 Hematopoiesis. Self-renewing hematopoietic stem cells give rise to lymphoid and myeloid progenitors. All lymphoid cells descend from lymphoid progenitor cells and all cells

of the myeloid lineage arise from myeloid progenitors. Note that some dendritic cells come from lymphoid progenitors, others from myeloid precursors.

Lesson 22 : Myeloid Lineage of Immune Cells:

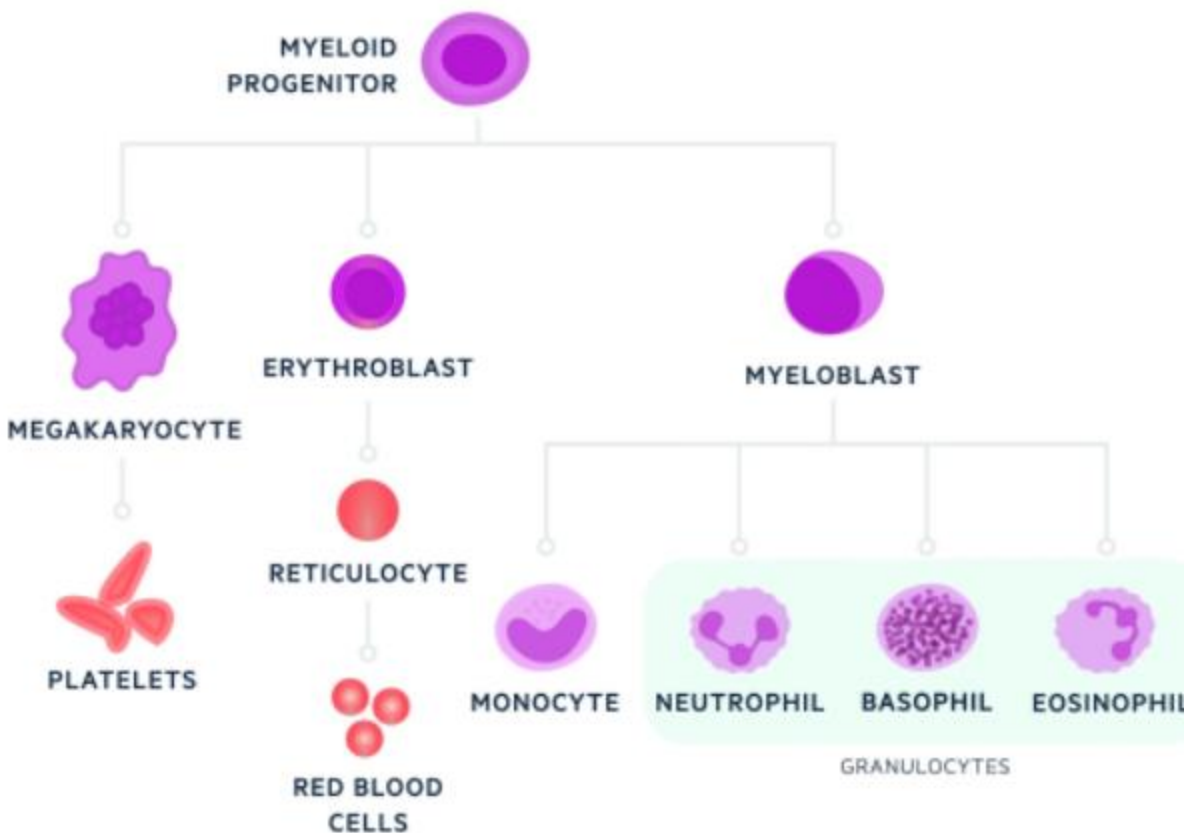
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Myeloid Lineage of Immune Cells

Myeloid stem cells generate progenitors of red blood cells (erythrocytes), many of the various white blood cells (neutrophils, eosinophils, basophils, monocytes, mast cells, dendritic cells), and platelets.

Progenitor commitment depends on the acquisition of responsiveness to particular growth factors and cytokines. When the appropriate factors and cytokines are present, progenitor cells proliferate and differentiate into the corresponding cell type, either a mature erythrocyte, a particular type of leukocyte, or a platelet-generating cell (the megakaryocyte).

Red and white blood cells pass into bone marrow channels; from which they enter the circulation.



Red blood cells

Erythrocytes are **non-nucleated cells**, approximately 7 microns in diameter, which are essential for the transfer of oxygen around the body.

White blood cells

Leucocytes, nucleated cells found in the blood and reticuloendothelial system, are **components of the immune system**.

Platelets

Platelets are essential in the formation of a haemostatic plug at the site of vascular injury.

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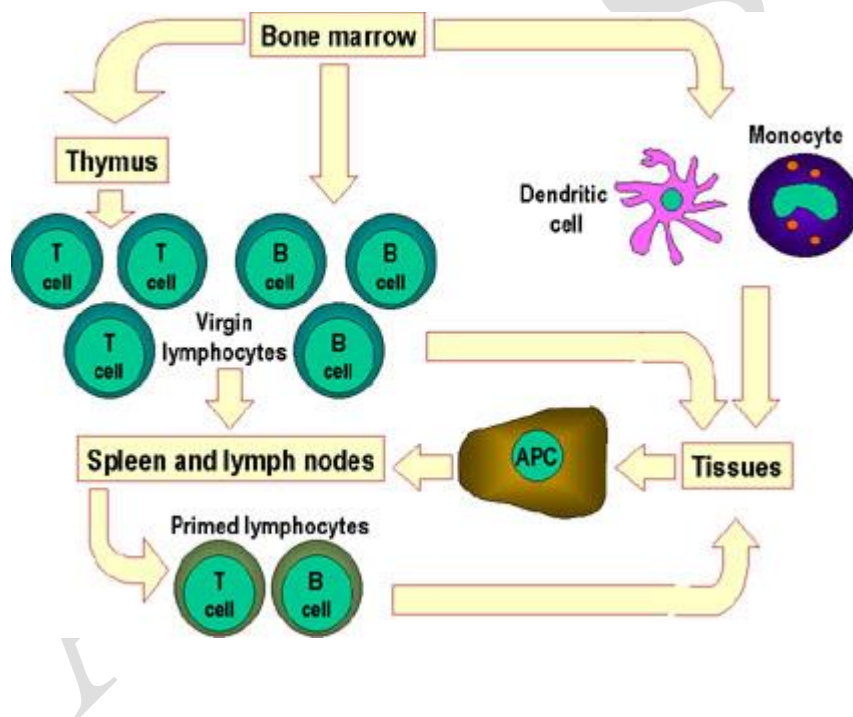
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Lesson 23 : Lymphocyte Recirculation:

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Lymphocyte Recirculation

- Naive (virgin) lymphocytes enter the lymph nodes from the blood via High Endothelial Venules (HEVs)
- Homing receptors on the lymphocytes direct the cells to the HEVs. In the lymph nodes, lymphocytes with the appropriate antigen receptor encounter antigen, which has been transported to the lymph nodes by dendritic cells or macrophages.
- After activation the lymphocytes express new receptors that allow the cells to leave the lymph node and reenter the circulation.
- Receptors on the activated lymphocytes recognize cell adhesion molecules expressed on endothelial cells near the site of an infection and chemokines produced at the infection site help attract the activated cells

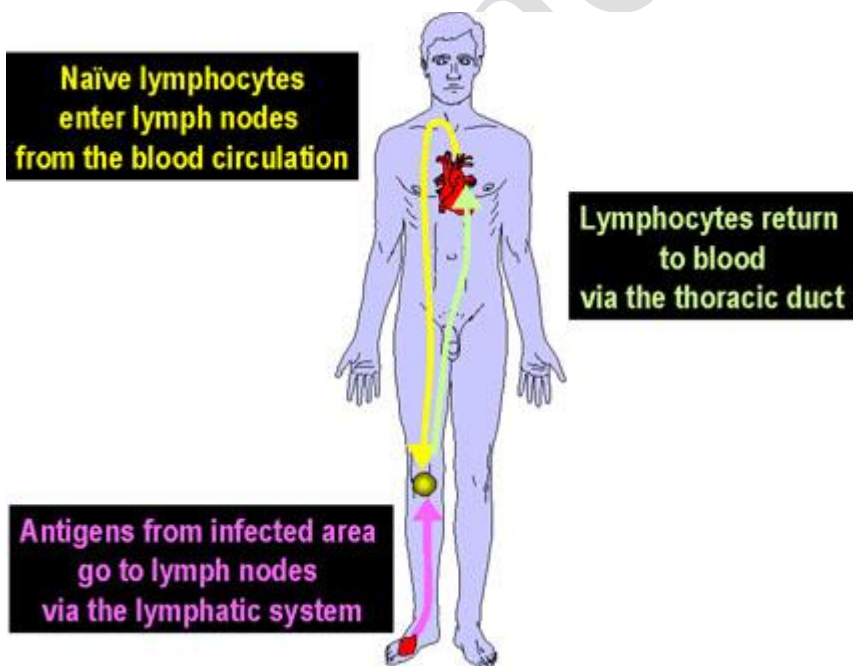


Lesson 24 : Lymphocyte Migration & Localization:

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Lymphocyte Migration & Localization

- Since there are relatively few T or B lymphocytes with a receptor for any particular antigen ($1/10,000 - 1/100,000$), the chances for a successful encounter between an antigen and the appropriate lymphocyte are slim.
- However, the chances for a successful encounter are greatly enhanced by the recirculation of lymphocytes through the secondary lymphoid organs.
- Lymphocytes in the blood enter the lymph nodes and percolate through the lymph nodes. If they do not encounter an antigen in the lymph node, they leave via the lymphatics and return to the blood via the thoracic duct. It is estimated that 1-2% of lymphocytes recirculate every hour.
- If the lymphocytes in the lymph nodes encounter an antigen, which has been transported to the lymph node via the lymphatics, the cells become activated, divide and differentiate to become a plasma cell, Th or Tc cell. After several days the effector cells can leave the lymph nodes via the lymphatics and return to the blood via the thoracic duct and then make their way to the infected tissue site.



Lesson 25 : Clonal Selection of Lymphocyte:

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Antigen Selection of Lymphocytes Causes Clonal Expansion

- A mature immunocompetent animal contains a large number of antigen-reactive clones of T and B lymphocytes; the antigenic specificity of each of these clones is determined by the specificity of the antigen-binding receptor on the membrane of the clone's lymphocytes. As noted above, the specificity of each T and B lymphocyte is determined before its contact with antigen by random gene rearrangements during maturation in the thymus or bone marrow.
- The role of antigen becomes critical when it interacts with and activates mature, antigenically committed T and B lymphocytes, bringing about expansion of the population of cells with a given antigenic specificity.
- In this process of clonal selection, an antigen binds to a particular T or B cell and stimulates it to divide repeatedly into a clone of cells with the same antigenic specificity as the original parent cell.
- Clonal selection provides a framework for understanding the specificity and self/nonself recognition that is characteristic of adaptive immunity.
- Specificity is shown because only lymphocytes whose receptors are specific for a given epitope on an antigen will be clonally expanded and thus mobilized for an immune response. Self/nonself discrimination is accomplished by the elimination, during development of lymphocytes bearing self-reactive receptors or by the functional suppression of these cells in adults. Immunologic memory also is a consequence of clonal selection.
- During clonal selection, the number of lymphocytes specific for a given antigen is greatly amplified. Moreover, many of these lymphocytes, referred to as memory cells, appear to have a longer life span than the naive lymphocytes from which they arise. The initial encounter of a naive immunocompetent lymphocyte with an antigen induces a primary response; a later contact of the host with antigen will induce a more rapid and heightened secondary response. The amplified population of memory cells accounts for the rapidity and intensity that distinguishes a secondary response from the primary response.

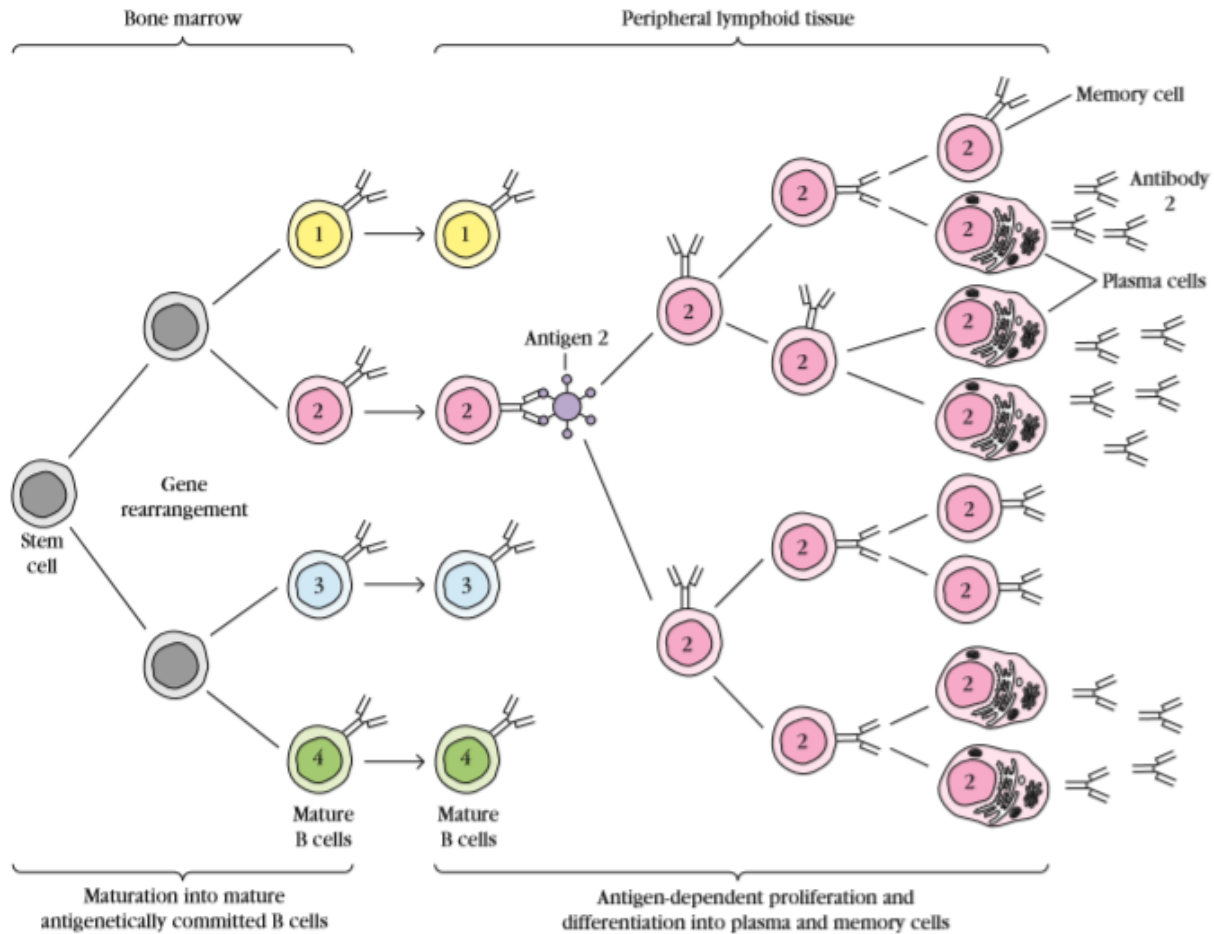


FIGURE 1-10 Maturation and clonal selection of B lymphocytes. Maturation, which occurs in the absence of antigen, produces antigenically committed B cells, each of which expresses antibody with a single antigenic specificity (indicated by 1, 2, 3, and 4). Clonal selection occurs when an antigen binds to a B cell whose membrane-bound antibody molecules are specific for epitopes on that antigen. Clonal expansion of an antigen-activated B cell (number 2 in this ex-

ample) leads to a clone of memory B cells and effector B cells, called plasma cells; all cells in the expanded clone are specific for the original antigen. The plasma cells secrete antibody reactive with the activating antigen. Similar processes take place in the T-lymphocyte population, resulting in clones of memory T cells and effector T cells; the latter include activated T_H cells, which secrete cytokines, and cytotoxic T lymphocytes (CTLs).

Lesson 26 : Control of Extracellular Pathogens:

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Control of Extracellular Pathogens

- Infection by extracellular bacteria induces production of humoral antibodies, which are ordinarily secreted by plasma cells in regional lymph nodes and the submucosa of the respiratory and gastrointestinal tracts.
- The humoral immune response is the main protective response against extracellular bacteria. The antibodies act in several ways to protect the host from the invading organisms, including removal of the bacteria and inactivation of bacterial toxins
- Extracellular bacteria can be pathogenic because they induce a localized inflammatory response or because they produce toxins. The toxins, endotoxin or exotoxin, can be cytotoxic but also may cause pathogenesis in other ways.
- An excellent example of this is the toxin produced by diphtheria, which exerts a toxic effect on the cell by blocking protein synthesis. Endotoxins, such as lipopolysaccharides (LPS), are generally components of bacterial cell walls, while exotoxins, such as diphtheria toxin, are secreted by the bacteria.
- Antibody that binds to accessible antigens on the surface of a bacterium can, together with the C3b component of complement, act as an opsonin that increases phagocytosis and thus clearance of the bacterium.
- In the case of some bacteria—notably, the gram-negative organisms—complement activation can lead directly to lysis of the organism. Antibody-mediated activation of the complement system can also induce localized production of immune effector molecules that help to develop an amplified and more effective inflammatory response. For example, the complement split products C3a, C4a, and C5a act as anaphylatoxins, inducing local mast-cell degranulation and thus vasodilation and the extravasation of lymphocytes and neutrophils from the blood into tissue space. Other complement split products serve as chemotactic factors for neutrophils and macrophages, thereby contributing to the buildup of phagocytic cells at the site of infection. Antibody to a bacteria toxin may bind to the toxin and neutralize it; the antibody-toxin complexes are then cleared by phagocytic cells in the same manner as any other antigenantibody complex.

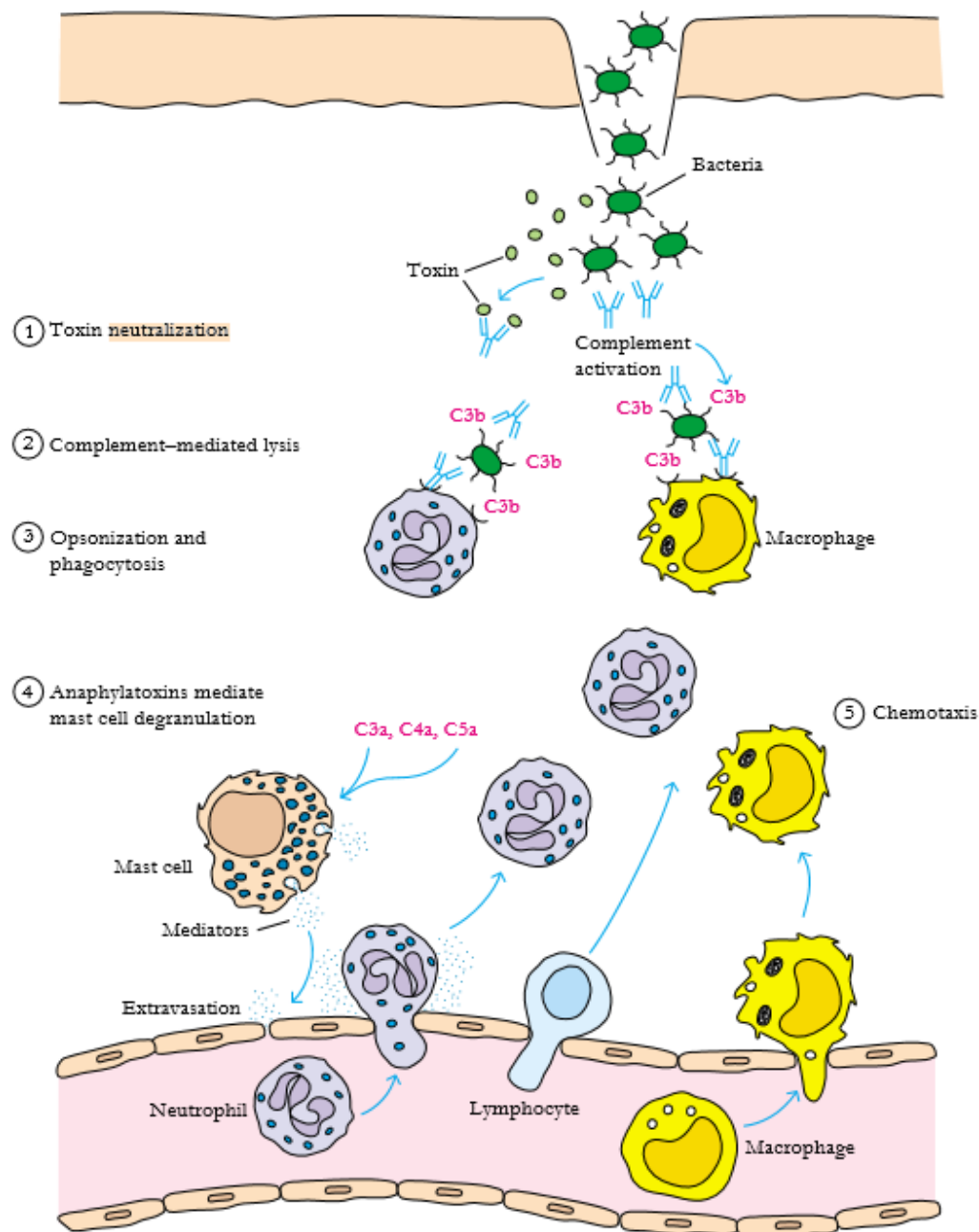
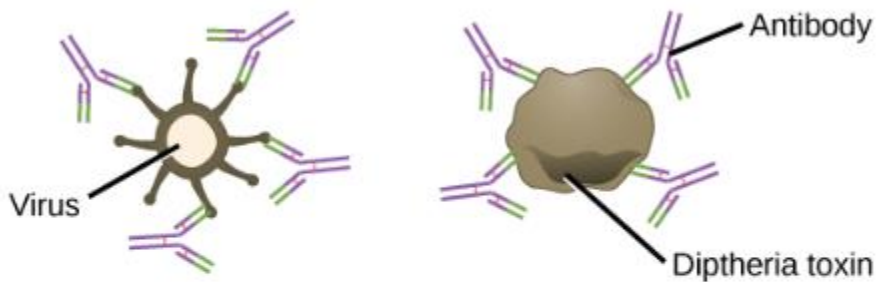


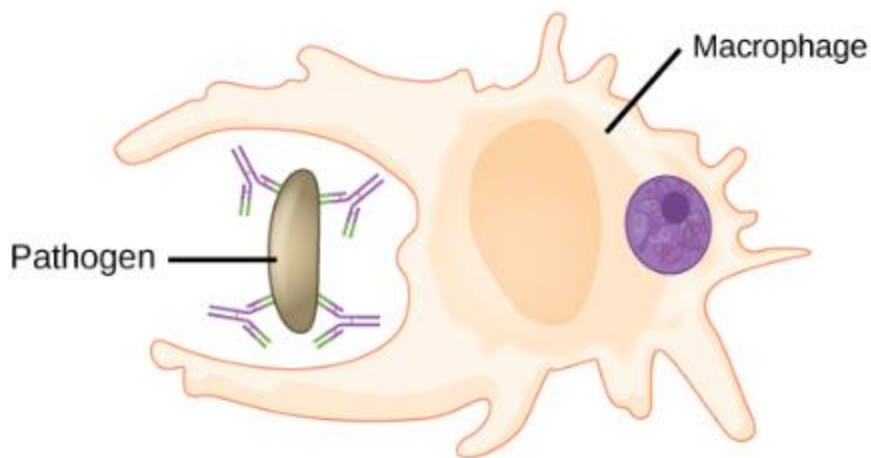
FIGURE 17-8 Antibody-mediated mechanisms for combating infection by extracellular bacteria. (1) Antibody neutralizes bacterial toxins. (2) Complement activation on bacterial surfaces leads to complement-mediated lysis of bacteria. (3) Antibody and the complement split product C_3b bind to bacteria, serving as opsonins to

increase phagocytosis. (4) C_3a , C_4a , and C_5a , generated by antibody-initiated complement activation, induce local mast cell degranulation, releasing substances that mediate vasodilation and extravasation of lymphocytes and neutrophils. (5) Other complement split products are chemotactic for neutrophils and macrophages.

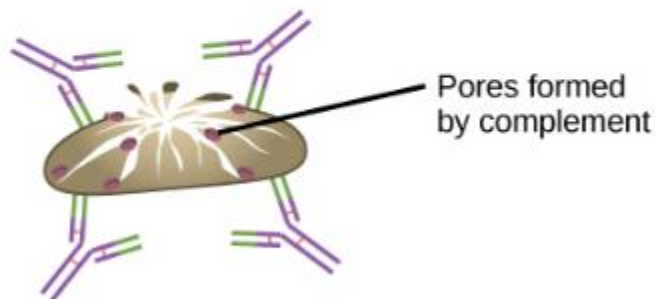
(a) Neutralization Antibodies prevent a virus or toxic protein from binding their target.



(b) Opsonization A pathogen tagged by antibodies is consumed by a macrophage or neutrophil.



(c) Complement activation Antibodies attached to the surface of a pathogen cell activate the complement system.



Mechanisms of antibody action: Antibodies may inhibit infection by (a) preventing the antigen from binding to its target, (b) tagging a pathogen for destruction by macrophages or neutrophils, or (c) activating the complement cascade.

Lesson 27 : Control of Intracellular Pathogens:

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Intracellular pathogens

- Because antibodies do not get into host cells, they are ineffective against intracellular pathogens. The immune system uses a different approach to deal with these kinds of pathogens.
- Cell-mediated responses are the primary defense against intracellular pathogens and the approach is different depending upon where the pathogen resides in the host cell (*i.e.*, in the cytosol or within vesicles). For example, most viruses and some bacteria reside in the cytoplasm of the host cell, however, some bacteria and parasites actually live within endosomes in the infected host cell.

The primary defense against pathogens in the cytosol is the cytotoxic T lymphocyte (Tc or CTL). In contrast, the primary defense against a pathogen within vesicles is a subset of helper T lymphocytes (Th1).

Cytotoxic

T

lymphocytes

CTLs are a subset of T lymphocytes that express a unique antigen on their surface called CD8. These cells recognize antigens from the pathogen that are displayed on the surface of the infected cell and kill the cell thereby preventing the spread of the infection to neighboring cells. CTLs kill by inducing apoptosis in the infected cell.

Th1

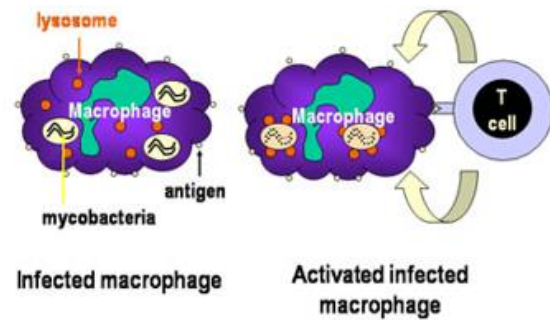
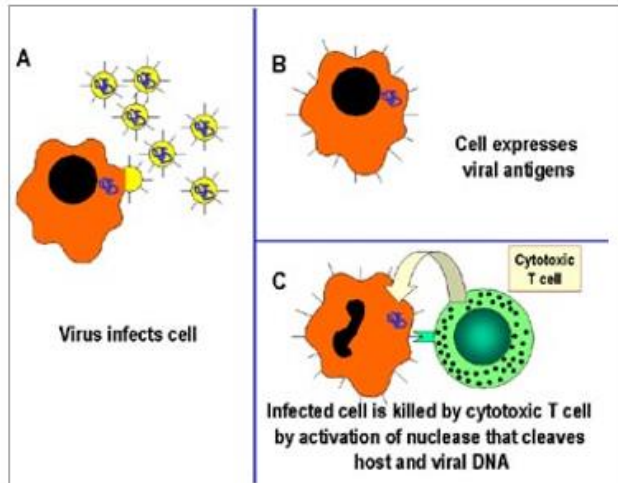
Helper

T

cells

Th cells are a subset of T cells that express a unique antigen on their surface called CD4. A subpopulation of Th cells, Th1 cells, is the primary defense against intracellular pathogens that live within vesicles. Th1 cells recognize antigen from the pathogen that are expressed on the surface of infected cells and release cytokines that activate the infected cell. Once activated, the infected cell can then kill the pathogen.

For example, *Mycobacterium tuberculosis*, the causative agent of tuberculosis, infects macrophages but is not killed because it blocks the fusion of lysosomes with the endosomes in which it resides. Th1 cells that recognize *M. tuberculosis* antigens on the surface of an infected macrophage can secrete cytokines that activate macrophages. Once activated the lysosomes fuse with endosomes and the *M. tuberculosis* bacteria are killed.



Control of Cytosolic Pathogens

Control of Vesicular Pathogens

BT302 - Immunology

Teacher: Dr Numan Javed (Phd Holder at SHanghai Institute of Sciences, China)

1 to 167 Lectures merged

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Lesson 28 : Properties of antibodies and antigens together with their structure, ...:

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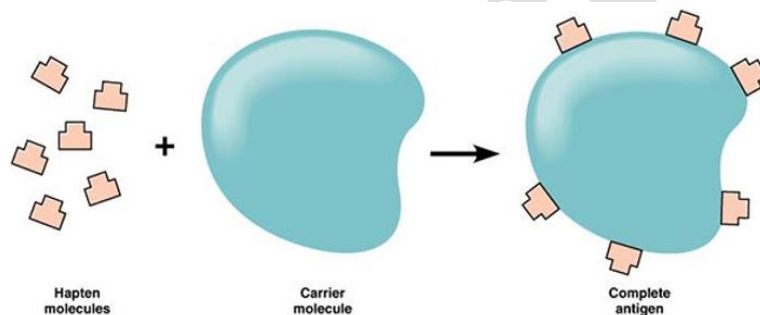
Properties of antibodies and antigens

Immunogen and Antigen

Immunogen is a stimulus that produces a humoral or cell-mediated immune response, whereas antigens are any substance that binds specifically to an antibody or a T-cell receptor. All immunogens are antigens, but all antigens may not be immunogens, some very small molecules called haptens can bind to antibodies or B-cell receptor but they cannot initiate an immune response.

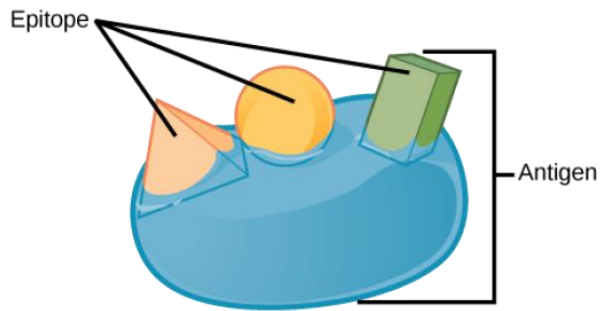
Hapten

Hapten is a molecule that reacts with specific antibody but is not immunogenic by itself, it can be made immunogenic by conjugation to a suitable carrier. Many drugs like penicillins are haptens. A hapten is essentially an incomplete antigen. These small molecules can elicit an immune response only when attached to a large carrier such as a protein; the carrier typically does not elicit an immune response by itself.



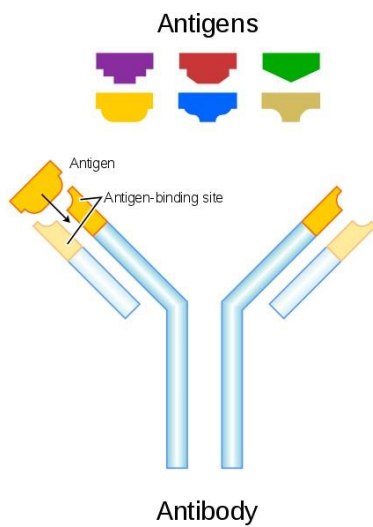
Epitope

An epitope, also known as antigenic determinant, is the part of an antigen that is recognized by an antibody, B-cell receptor or T-cell receptor. Generally, an antigen has several or many different epitopes and reacts with many different antibodies, an epitope is approximately five or six amino acids in length.



Antibody (Ab)

A specific protein which is generated in response to an immunogen and which reacts with an antigen



Lesson 29 : Factors influencing Immunogenicity (Contribution of the Immunogen):

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Factors That Influence Immunogenicity

The Nature of the Immunogen Contributes to Immunogenicity. Immunogenicity is determined, by four properties of the immunogen:

- Foreignness
- Molecular Size
- Chemical Composition
- Complexity
- Ability to Be Processed and Presented with an MHC Molecule On the Surface of an Antigen-Presenting cell or altered self-cell.

FOREIGNNESS

In order to elicit an immune response, a molecule must be recognized as nonself by the biological system. The capacity to recognize nonself is accompanied by tolerance of self, a specific unresponsiveness to self-antigens

MOLECULAR SIZE

There is a correlation between the size of a macromolecule and its immunogenicity. The most active immunogens tend to have a molecular mass of 100,000 daltons (Da). Generally, substances with a molecular mass less than 5000–10,000 Da are poor immunogens,

CHEMICAL COMPOSITION AND HETEROGENEITY

synthetic homopolymers (polymers composed of a single amino acid or sugar) tend to lack immunogenicity regardless of their size. Studies have shown that copolymers composed of different amino acids or sugars are usually more immunogenic than homopolymers of their constituents.

SUSCEPTIBILITY TO ANTIGEN PROCESSING AND PRESENTATION

The development of both humoral and cell-mediated immune responses requires interaction of T cells with antigen that has been processed and presented together with MHC molecules. Large, insoluble macromolecules generally are more immunogenic than small, soluble ones because the larger molecules are more readily phagocytosed and processed. Macromolecules that cannot be degraded and presented with MHC molecules are poor immunogens

Lesson 30 : Factors influencing Immunogenicity (Contribution of the Biological System):

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Contribution of the Biological System

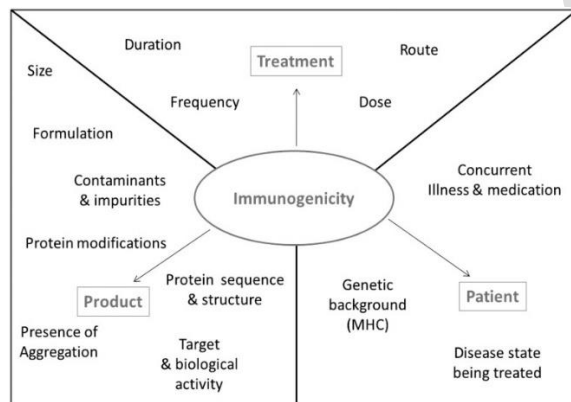
Even if a macromolecule has the properties that contribute to immunogenicity, its ability to induce an immune response will depend on certain properties of the biological system that the antigen encounters.

GENOTYPE OF THE RECIPIENT ANIMAL

The genetic constitution (genotype) of an immunized animal influences the type of immune response the animal manifests, as well as the degree of the response. The response of an animal to an antigen is also influenced by the genes that encode B-cell and T-cell receptors and by genes that encode various proteins involved in immune regulatory mechanisms. Genetic variability in all of these genes affects the immunogenicity of a given macromolecule in different animals

Age

Age is another biological factor which can also influence immunogenicity. Frequently the very young and the very old have a reduced ability to mount an immune response in response to an immunogen.



Lesson 31 : Factors influencing Immunogenicity (Method of Administration):

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IMMUNOGEN DOSAGE AND ROUTE OF ADMINISTRATION

- Each experimental immunogen exhibits a particular dose-response curve, which is determined by measuring the immune response to different doses and different administration routes. An antibody response is measured by determining the level of antibody present in the serum of immunized animals
- An insufficient dose will not stimulate an immune response either because it fails to activate enough lymphocytes or because, in some cases, certain ranges of low doses can induce a state of immunologic unresponsiveness, or tolerance
- The administration route strongly influences which immune organs and cell populations will be involved in the response. Antigen administered intravenously is carried first to the spleen, whereas antigen administered subcutaneously moves first to local lymph nodes. Differences in the lymphoid cells that populate these organs may be reflected in the subsequent immune response.

ADJUVANTS

Adjuvants (from Latin *adjuvare*, to help) are substances that, when mixed with an antigen and injected with it, enhance the immunogenicity of that antigen. Adjuvants are often used to boost the immune response when an antigen has low immunogenicity or when only small amounts of an antigen are available.

they appear to exert one or more of the following effects

- Antigen persistence is prolonged.
- Co-stimulatory signals are enhanced.
- Local inflammation is increased.
- The nonspecific proliferation of lymphocytes is stimulated.

Lesson 32 : Chemical Nature of Immunogen:

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Chemical nature of immunogens

Antigens or immunogens are either proteins or large polysaccharides molecules and rarely, if ever, lipids.

These are cell surfaces or membrane-bound antigens.

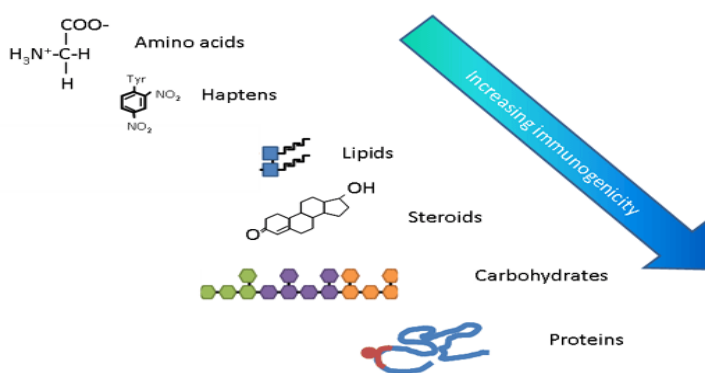
The antigens or immunogens are the large organic molecules which may be:

1. **Proteins:** These are powerful antigens. These are strong antigens because of their high molecular weight and structural complexity.
2. **Lipoprotein:** These are part of the cells.
3. **Lipopolysaccharides:** These are endotoxin, capsular polysaccharides of pneumococci, and other haptens.
4. **Carbohydrates** (polysaccharides) are too small to be antigenic. But in the case of red blood cells, protein or lipid carrier gives rise to an increase in their size, and polysaccharides are present in the form of side chains and give rise to antigenicity.

Red blood cell surface antigens are glycoproteins.

5. **Nucleic acid:** These are non-immunogenic except for single-stranded DNA. These poor antigens because of the simplicity of the structure, molecular flexibility, and rapid degradation.
6. **Polypeptide:** These are hormones, insulin and these are weak antigens.
7. **Nucleoproteins:** These are strong antigens.
8. **Lipids:** These are non-immunogenic except cardiolipin. These are very weak antigens because of simplicity and lack of structural stability. When these are combined with polysaccharides or proteins, then they may act as an antigen.

Immunogenicity



Lesson 33 : Types of Antigens:

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TYPES OF ANTIGENS

T-independent

Antigens

T-independent antigens are antigens which can directly stimulate the B cells to produce antibody without the requirement for T cell help. In general, polysaccharides are T-independent antigens. The responses to these antigens differ from the responses to other antigens.

Properties of T-independent antigens

- Polymeric structure

These antigens are characterized by the same antigenic determinant

Polyclonal activation of B cells
Many of these antigens can activate B cell clones specific for other antigens (polyclonal activation). T-independent antigens can be subdivided into Type 1 and Type 2 based on their ability to polyclonally activate B cells. Type 1 T-independent antigens are polyclonal activators while Type 2 are not.

- Resistance to degradation

T-independent antigens are generally more resistant to degradation and thus they persist for longer periods of time and continue to stimulate the immune system.

T-dependent

Antigens

T-dependent antigens are those that do not directly stimulate the production of antibody without the help of T cells. Proteins are T-dependent antigens. Structurally these antigens are characterized by a few copies of many different antigenic determinants

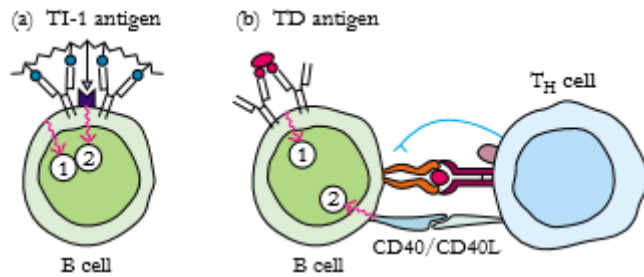


FIGURE 11-6 An effective signal for B-cell activation involves two distinct signals induced by membrane events. Binding of a type 1 thymus-independent (TI-1) antigen to a B cell provides both signals. A thymus-dependent (TD) antigen provides signal 1 by crosslinking mlg, but a separate interaction between CD40 on the B cell and CD40L on an activated T_H cell is required to generate signal 2.

Difference between T dependent Antigen and T independent Antigen

T dependent (TD) Antigen	T independent Antigen
Soluble proteins	Bacterial cell wall components Lipopolysaccharide (LPS), Capsular polysaccharide, flagella etc.
Antigen is processed and displayed on the surface of antigen presenting cells (B Cells) in association with MHC-II.	Antigen processing is not needed
Immunogenic over wide range of dose	Dose dependent immunogenicity
No polyclonal activation i.e. Activate B cells monoclally	Polyclonal activation of B cells occur in high doses of Type-I TI Antigens
Immunologic memory present	No immunologic memory

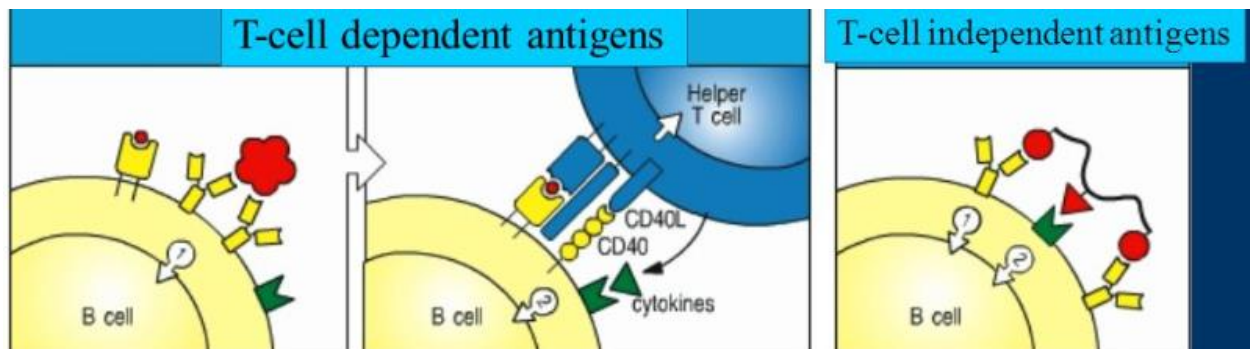
Affinity maturation- Yes	Affinity maturation- No
Isotype switching occurs (<i>i.e. antibodies of all classes are produced</i>)	No isotype switching (Antibody response is restricted to IgM and IgG3)
Activate mature B cells only	Activate both mature and immature B cells

BT302 - Immunology

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Lesson 34 : Superantigens:

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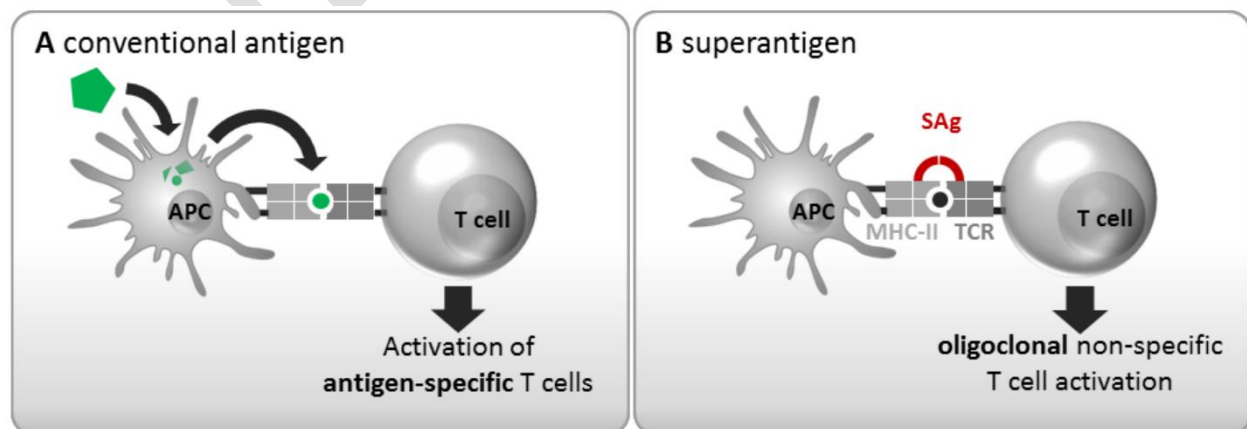
Sperantigens

A variety of microorganisms produce toxins that act as superantigens. superantigens bind simultaneously to a class II MHC molecule and to the V β domain of the T-cell receptor, activating all T cells bearing a particular V β domain. Because of their unique binding ability, superantigens can activate large numbers of T cells irrespective of their antigenic specificity.

Although less than 0.01% of T cells respond to a given conventional antigen, between 5% and 25% of T cells can respond to a given superantigen. The large proportion of T cells responsive to a particular superantigen results from the limited number of TCR V β genes carried in the germ line

A number of bacterial super antigens have been implicated as the causative agent of several diseases such as bacterial toxic shock and food poisoning. Included among these bacterial superantigens are several enterotoxins, exfoliating toxins, and toxic-shock syndrome toxin (TSST1) from *Staphylococcus aureus*; pyrogenic exotoxins from *Streptococcus pyogenes*; and *Mycoplasma arthritidis* supernatant (MAS).

The large number of T cells activated by these superantigens results in excessive production of cytokines. The toxic-shock syndrome toxin, for example, has been shown to induce extremely high levels of TNF and IL-1. As in bacterial septic shock, these elevated concentrations of cytokines can induce systemic reactions that include fever, widespread blood clotting, and shock



Lesson 35 : Hapten-Carrier Conjugate:

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Hapten-Carrier Conjugate

haptens, small organic molecules that are antigenic but not immunogenic. Chemical coupling of a hapten to a large protein, called a carrier, yields an immunogenic hapten-carrier conjugate

Animals immunized with such a conjugate produce antibodies specific for

- the hapten determinant
- unaltered epitopes on the carrier protein
- new epitopes formed by combined parts of both the hapten and carrier.

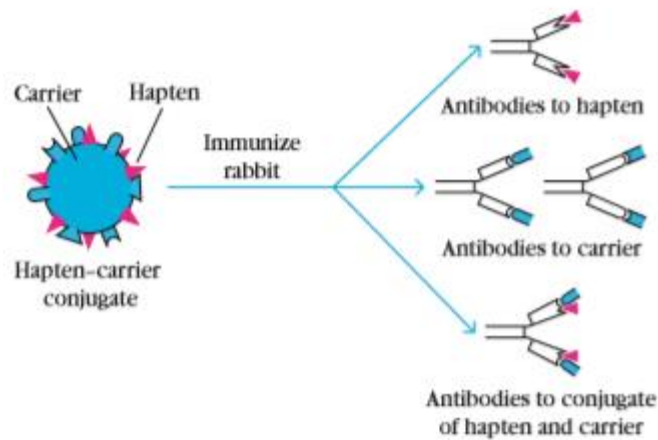
By itself, a hapten cannot function as an immunogenic epitope. But when multiple molecules of a single hapten are coupled to a carrier protein (or nonimmunogenic homopolymer), the hapten becomes accessible to the immune system and can function as an immunogen.

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Injection with:	Antibodies formed:
Hapten (DNP)	None
Protein carrier (BSA)	Anti-BSA
Hapten-carrier conjugate (DNP-BSA)	Anti-DNP (major) Anti-BSA (minor) Anti-DNP/BSA (minor)

FIGURE 3-10 A hapten-carrier conjugate contains multiple copies of the hapten—a small nonimmunogenic organic compound such as dinitrophenol (DNP)—chemically linked to a large protein carrier such as bovine serum albumin (BSA). Immunization with DNP alone elicits no anti-DNP antibodies, but immunization with DNP-BSA elicits three types of antibodies. Of these, anti-DNP antibody is predominant, indicating that in this case the hapten is the immunodominant epitope in a hapten-carrier conjugate, as it often is in such conjugates.

Lesson 36 : Antigenic determinants recognized by Innate Immune System:

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The antigenic determinants recognized by innate immune system is purely non-specific in nature due to non-specific attribute of innate immune system. These determinants are present on the surface of the pathogens in a form of various molecular configurations called as pathogen associated molecular patterns (PAMPS).

PAMPs include cell membrane proteins, lipopolysaccharide and flagellular proteins. For these antigenic determinants immune cells carry pattern recognition receptors (PRRs) which bind non-specifically with corresponding PAMPs in order to generate immune response

Pathogen-Associated Molecular Patterns (PAMPs)

Pathogens, especially bacteria, have molecular structures that

- are not shared with their host;
- are shared by many related pathogens;
- are relatively invariant; that is, do not evolve rapidly (in contrast, for example, to such pathogen molecules as the hemagglutinin and neuraminidase of influenza viruses).

Examples:

- the flagellin of bacterial flagella;
- the peptidoglycan of Gram-positive bacteria;
- the **lipopolysaccharide (LPS)**, also called **endotoxin** of Gram-negative bacteria;
- double-stranded RNA. (Some viruses of both plants and animals have a genome of dsRNA. And many other viruses of both plants and animals have an RNA genome that in the host cell is briefly converted into dsRNA.
- unmethylated DNA (eukaryotes have many times more cytosines, in the dinucleotide CpG, with methyl groups attached.

Pattern Recognition Receptors (PRRs)

There are three groups:

1. secreted molecules that circulate in blood and lymph;
2. surface receptors on phagocytic cells like macrophages that bind the pathogen for engulfment;
3. cell-surface receptors that bind the pathogen initiating a signal leading to the release of effector molecules (cytokines).

1. Secreted PRRs

Example: Circulating proteins (e.g., C-reactive protein) that bind to PAMPs on the surface of many pathogens. This interaction triggers the complement cascade leading to the opsonization of the pathogen and its speedy phagocytosis.

2. Phagocytosis Receptors

Macrophages have cell-surface receptors that recognize certain PAMPs, e.g., those containing mannose. When a pathogen covered with polysaccharide with mannose at its tips binds to these, it is engulfed into a phagosome.

3. Toll-Like Receptors (TLRs)

Macrophages, dendritic cells, and epithelial cells have a set of transmembrane receptors that recognize different types of PAMPs. These are called Toll-like receptors (TLRs) because of their homology to receptors first discovered and named in *Drosophila*.

Mammals have 12 different TLRs each of which specializes — often with the aid of accessory molecules — in a subset of PAMPs. In this way, the TLRs identify the nature of the pathogen and turn on an effector response appropriate for dealing with it. These signaling cascades lead to the expression of various **cytokine genes**.

Examples:

- **TLR-1**

Forms a heterodimer with TLR-2 at the cell surface which binds to the peptidoglycan of Gram-positive bacteria like Streptococci and Staphylococci.

- **TLR-2**

With TLR-1, binds cell-wall components of Gram-positive bacteria.

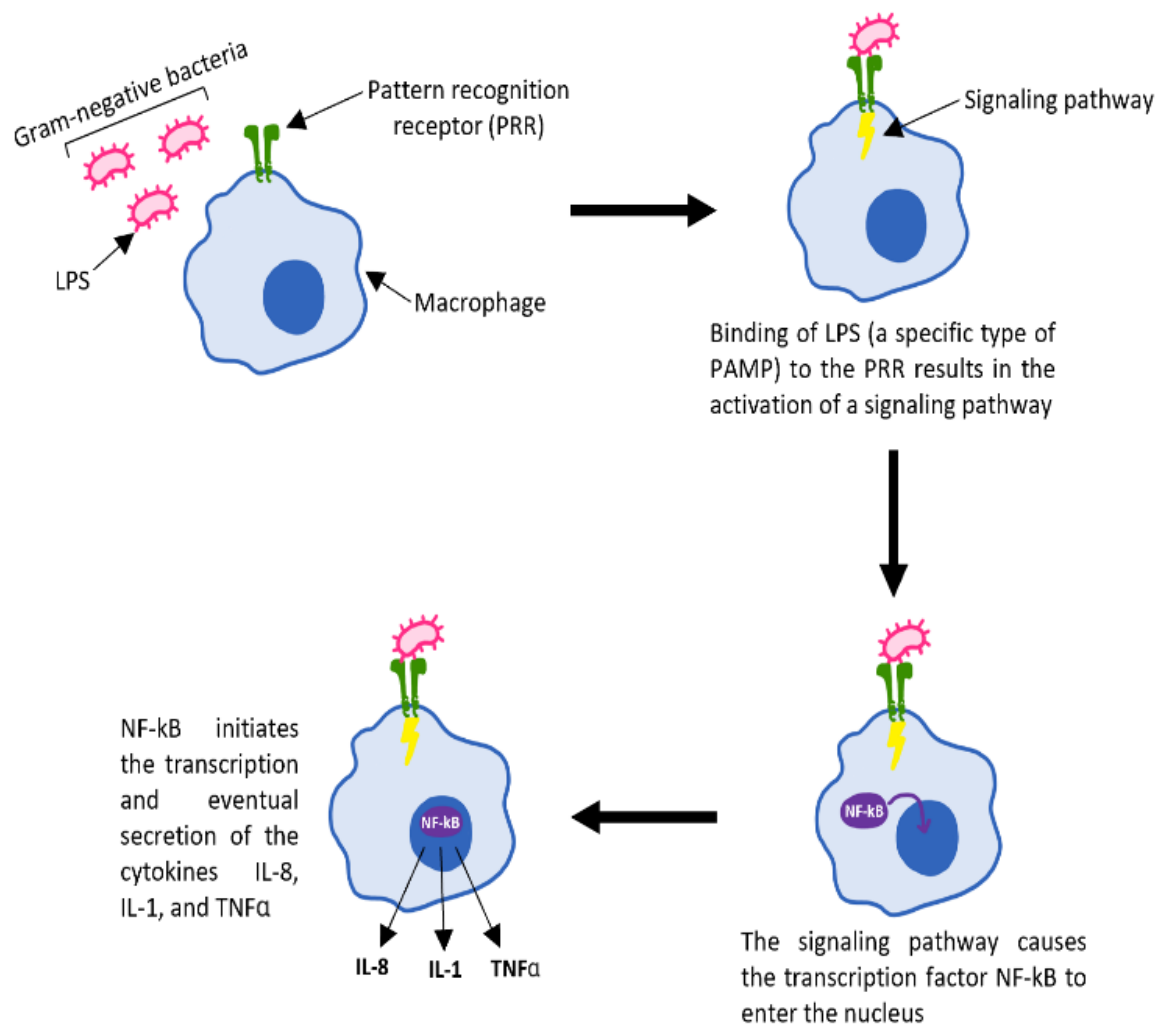


Table 1 Examples of pathogen associated molecular patterns and their receptors

PAMP	PRR	Biological Consequence of Interaction
Microbial cell wall components	Complement	Opsonization, Complement activation
Mannose-containing carbohydrates	Mannose-binding protein	Opsonization Complement activation
Polyanions	Scavenger receptors	Phagocytosis
Lipoproteins of Gram + bacteria Yeast cell wall components	TLR-2 (Toll-like receptor 2)	Macrophage activation, secretion of inflammatory cytokines
Double stranded RNA	TLR-3	Production of interferon (antiviral)
LPS (lipopolysaccharide of Gram negative bacteria)	TLR-4	Macrophage activation, secretion of inflammatory cytokines
Flagellin (bacterial flagella)	TLR-5	Macrophage activation, secretion of inflammatory cytokines
U-rich Single stranded viral RNA	TLR-7	Production of interferon (antiviral)
CpG containing DNA	TLR-9	Macrophage activation, secretion of inflammatory cytokines

Lesson 37 : Characteristics of Immunoglobulins:

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Characteristics of Immunoglobulins (Antibodies)

- Antibodies are antigen binding proteins present on the B-cell membrane and secreted by plasma cells.
- Membrane-bound antibody confers antigenic specificity on B cells; antigen-specific proliferation of B-cell clones is elicited by the interaction of membrane antibody with antigen.
- Secreted antibodies circulate in the blood, where they serve as the effectors of humoral immunity by searching out and neutralizing antigens or marking them for elimination.
- All antibodies share structural features, bind to antigen, and participate in a limited number of effector functions.
- The antibodies produced in response to a particular antigen are heterogeneous.
- Most antigens are complex and contain many different antigenic determinants, and the immune system usually responds by producing antibodies to several epitopes on the antigen. This response requires the recruitment of several clones of B cells. Their outputs are monoclonal antibodies, each of which specifically binds a single antigenic determinant. Together, these monoclonal antibodies make up the polyclonal and heterogeneous serum antibody response to an immunizing antigen.

The immunoglobulins derive their name from the finding that they migrate with globular proteins when antibody-containing serum is placed in an electrical field

GENERAL FUNCTIONS OF IMMUNOGLOBULINS

Antigen

binding

Immunoglobulins bind specifically to one or a few closely related antigens. Each immunoglobulin actually binds to a specific antigenic determinant. Antigen binding by antibodies is the primary function of antibodies and can result in protection of the host. The valency of antibody refers to the number of antigenic determinants that an individual antibody molecule can bind. The valency of all antibodies is at least two and in some instances more.

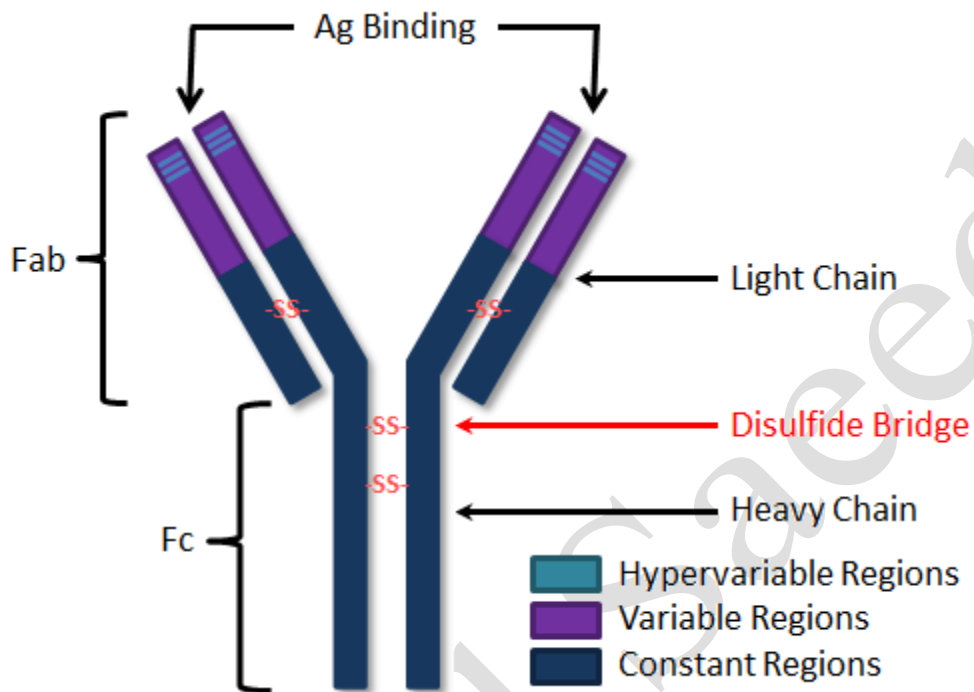
Effector

Functions

Frequently the binding of an antibody to an antigen has no direct biological effect. Rather, the significant biological effects are a consequence of secondary "effector functions" of antibodies. The immunoglobulins mediate a variety of these effector functions. Usually the ability to carry out a particular effector function requires that the antibody bind to its antigen. Not every immunoglobulin will mediate all effector functions. Such effector functions include:

- Fixation of complement - This results in lysis of cells and release of biologically active molecules
- Binding to various cell types - Phagocytic cells, lymphocytes, platelets, mast cells, and basophils have receptors that bind immunoglobulins. This binding can activate the cells

to perform some function. Some immunoglobulins also bind to receptors on placental trophoblasts, which results in transfer of the immunoglobulin across the placenta. As a result, the transferred maternal antibodies provide immunity to the fetus and newborn



Lesson 38 : Basic Structure of Antibodies:

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Basic Structure of Antibodies

Antibodies Are Heterodimers

Antibody molecules have a common structure of four peptide chains. This structure consists of

- two identical light (L) chains, polypeptides of about 25,000 molecular weight
- two identical heavy (H) chains, larger polypeptides of molecular weight 50,000 or more.

Like the antibody molecules they constitute, H and L chains are also called **immunoglobulins**.

Each light chain is bound to a heavy chain by a disulfide bond, and by such noncovalent interactions as salt linkages, hydrogen bonds, and hydrophobic bonds, to form a heterodimer (H-L). Similar noncovalent interactions and disulfide bridges link the two identical heavy and light (H-L) chain combinations to each other to form the basic four-chain (H-L)₂ antibody structure, a dimer of dimers.

The first 110 or so amino acids of the amino-terminal region of a light or heavy chain varies greatly among antibodies of different specificity. These segments of highly **variable sequence** are called **V regions**:

- VL in light chains
- VH in heavy chains.

All of the differences in specificity displayed by different antibodies can be traced to differences in the amino acid sequences of V regions. In fact, most of the differences among antibodies fall within areas of the V regions called **complementarity-determining regions** (CDRs), and it is these CDRs, on both light and heavy chains, that constitute the antigen binding site of the antibody molecule.

By contrast, within the same antibody class, far fewer differences are seen when one compares sequences throughout the rest of the molecule. The regions of relatively **constant sequence** beyond the variable regions have been dubbed C regions, CL on the light chain and

CH on the heavy chain.

Antibodies are glycoproteins; with few exceptions, the sites of attachment for carbohydrates are restricted to the constant region.

Hinge region

This region makes antibodies flexible to change its shape while performing function after binding with its corresponding antigens.

These are folded regions which contain an intra-chain disulfide bond. Usually, heavy chains contain four domains while light chains composed on two domains.

These are carbohydrates which are attached on CH2 domain in most of immunoglobulins



FIGURE 4-2 Schematic diagram of structure of immunoglobulins derived from amino acid sequencing studies. Each heavy and light chain in an immunoglobulin molecule contains an amino-terminal variable (V) region (aqua and tan, respectively) that consists of 100–110 amino acids and differs from one antibody to the next. The remainder of each chain in the molecule—the constant (C) regions (purple and red)—exhibits limited variation that defines the two light-chain subtypes and the five heavy-chain subclasses. Some heavy chains (γ , δ , and α) also contain a proline-rich hinge region (black). The amino-terminal portions, corresponding to the V regions, bind to antigen; effector functions are mediated by the other domains. The μ and ϵ heavy chains, which lack a hinge region, contain an additional domain in the middle of the molecule.

Lesson 39 : Structure of the Variable Region:

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Structure of the Variable Region

- Diversity in the Variable-Region Domain Is Concentrated in CDRs
- Detailed comparisons of the amino acid sequences of a large number of VL and VH domains revealed that the sequence variation is concentrated in a few discrete regions of these domains. The pattern of this variation is best summarized by a quantitative plot of the variability at each position of the polypeptide chain.

The variability is defined as:

Variability = # of different amino acids at a given position / Frequency of the most common amino acid at given position

Thus if a comparison of the sequences of 100 heavy chains revealed that a serine was found in position 7 in 51 of the sequences (frequency 0.51), it would be the most common amino acid. If examination of the other 49 sequences showed that position 7 was occupied by either glutamine, histidine, proline, or tryptophan, the variability at that position would be 9.8 (5/0.51).

Variability plots of VL and VH domains of human antibodies show that maximum variation is seen in those portions of the sequence that correspond to the loops that join the strands. These regions were originally called **hypervariable regions in recognition of their high variability**. Hypervariable regions form the antigen binding site of the antibody molecule. Because the antigen binding site is complementary to the structure of the epitope, these areas are now more widely called **complementarity determining regions (CDRs)**.

The three heavy-chain and three light-chain CDR regions are located on the loops that connect the strands of the VH and VL domains. The remainder of the VL and VH domains exhibit far less variation; these stretches are called the **framework regions (FRs)**. The wide range of specificities exhibited by antibodies is due to variations in the length and amino acid sequence of the six CDRs in each Fab fragment. The framework region acts as a scaffold that supports these six loops. The three-dimensional structure of the framework regions of virtually all antibodies analyzed to date can be superimposed on one another; in contrast, the hypervariable loops (i.e., the CDRs) have different orientations in different antibodies.

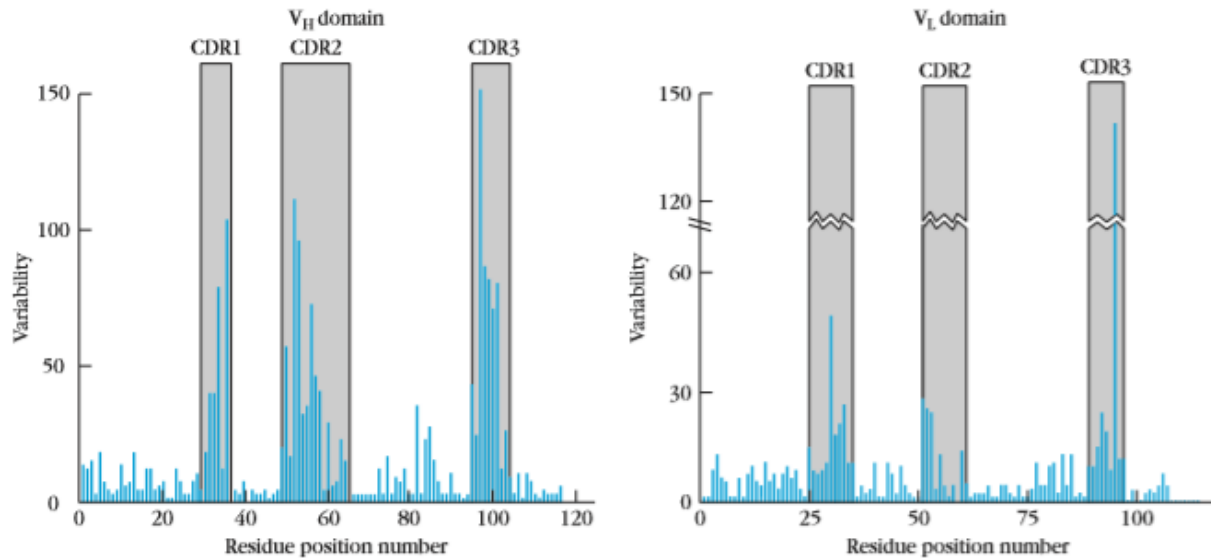


FIGURE 4-9 Variability of amino acid residues in the V_L and V_H domains of human antibodies with different specificities. Three hyper-variable (HV) regions, also called complementarity-determining regions (CDRs), are present in both heavy- and light-chain V domains. As shown in Figure 4-7 (right), the three HV regions in the

light-chain V domain are brought into proximity in the folded structure. The same is true of the heavy-chain V domain. [Based on E. A. Kabat *et al.*, 1977, *Sequence of Immunoglobulin Chains*, U.S. Dept. of Health, Education, and Welfare.]

BT302 - Immunology

Teacher: Dr Numan Javed (Phd Holder at SHanghai Institute of Sciences, China)

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Lesson 40 : Antibodies Fragments Structure Function:

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Antibodies Fragments Structure/Function

When the -globulin fraction of serum is separated into high- and low-molecular weight fractions, antibodies of around 150,000-MW, designated as immunoglobulin G (IgG) are found in the low molecular-weight fraction.

In a key experiment, brief digestion of IgG with the enzyme papain produced three fragments, two of which were identical fragments and a third that was quite different.

- The two identical fragments (each with a MW of 45,000), had antigen-binding activity and were called **Fab fragments (“fragment, antigen binding”)**.
- The other fragment (MW of 50,000) had no antigen binding activity at all. Because it was found to crystallize during cold storage, it was called the **Fc fragment (“fragment, crystallizable”)**.

Digestion with pepsin, a different proteolytic enzyme, also demonstrated that the antigen-binding properties of an antibody can be separated from the rest of the molecule

This crosslinking of the FcR by binding to an array of antibody Fc regions initiates a signal-transduction pathway that results in the phagocytosis of the antigen-antibody complex.

Inside the phagocyte, the pathogen becomes the target of various destructive processes that include enzymatic digestion, oxidative damage, and the membrane-disrupting effects of antibacterial peptides.

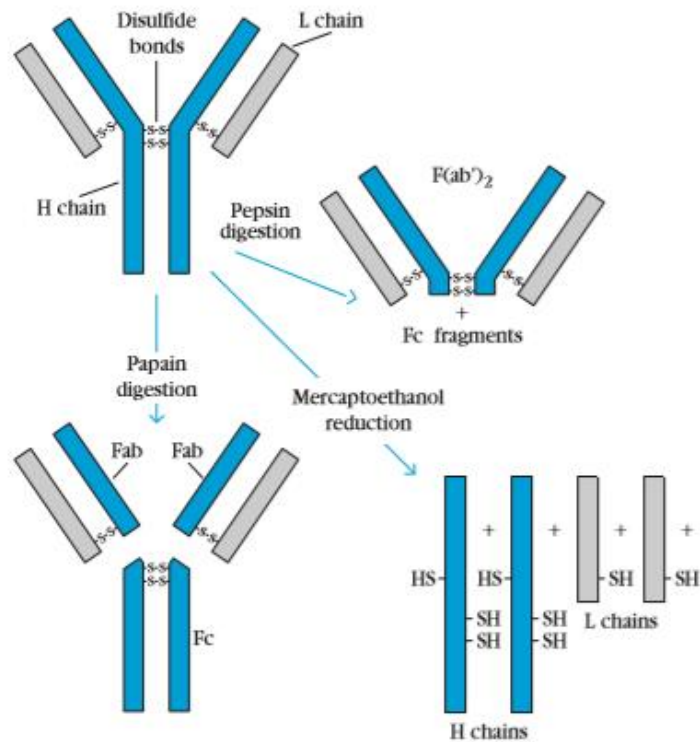


FIGURE 4-3 Prototype structure of IgG, showing chain structure and interchain disulfide bonds. The fragments produced by various

treatments are also indicated. Light (L) chains are in gray and heavy (H) chains in blue.

Lesson 41 : Classes of Antibodies:

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Immunoglobulin

classes

The immunoglobulins can be divided into five different classes, based on differences in the amino acid sequences in the constant region of the heavy chains. All immunoglobulins within a given class will have very similar heavy chain constant regions. These differences can be detected by sequence studies or more commonly by serological means (*i.e.* by the use of antibodies directed to these differences).

- IgG - Gamma heavy chains
- IgM - Mu heavy chains
- IgA - Alpha heavy chains
- IgD - Delta heavy chains
- IgE - Epsilon heavy chains

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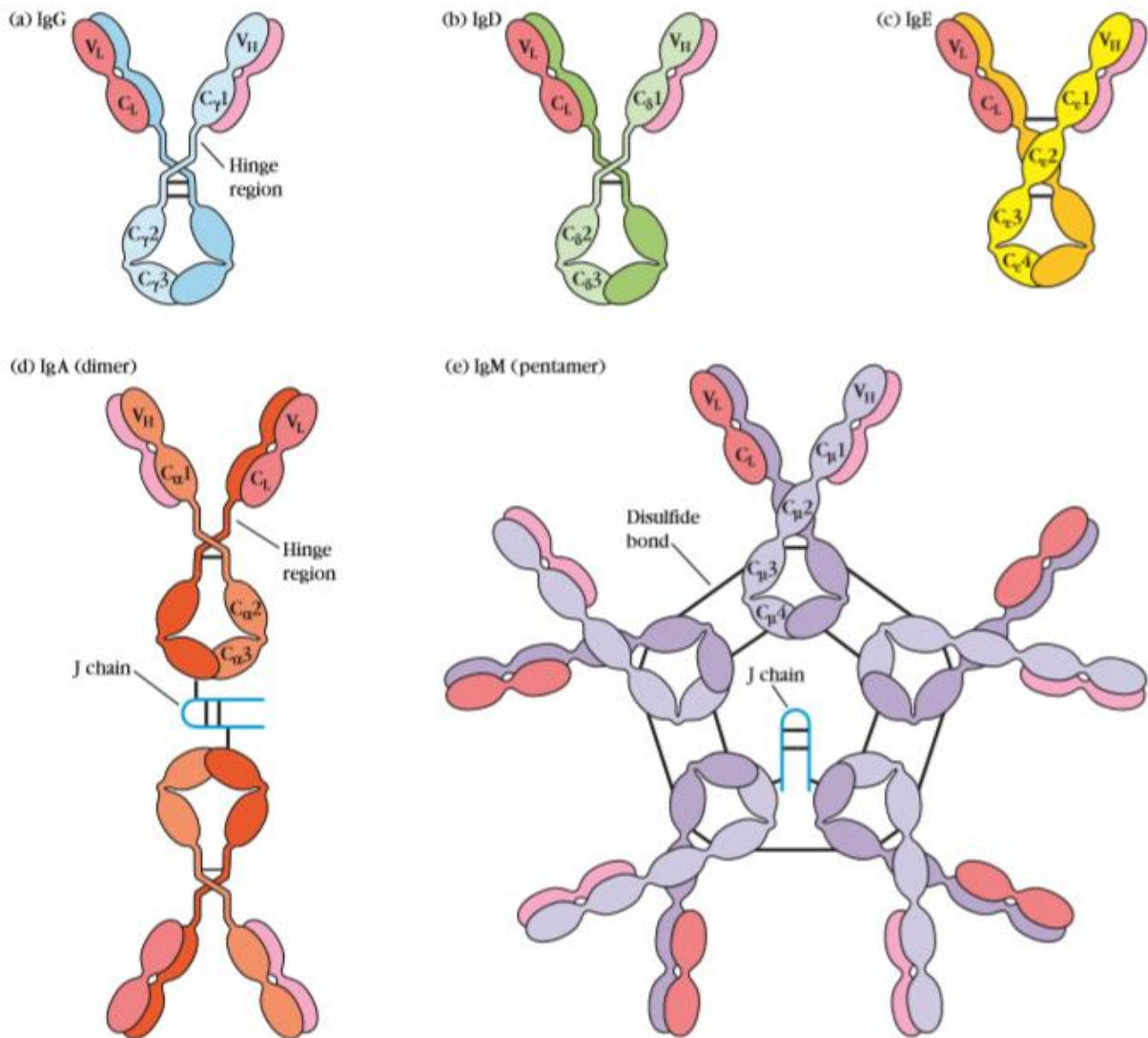


FIGURE 4-13 General structures of the five major classes of secreted antibody. Light chains are shown in shades of pink, disulfide bonds are indicated by thick black lines. Note that the IgG, IgA, and IgD heavy chains (blue, orange, and green, respectively) contain four domains and a hinge region, whereas the IgM and IgE heavy chains (purple and yellow, respectively) contain five domains but no hinge region. The polymeric forms of IgM and IgA contain a polypeptide,

called the J chain, that is linked by two disulfide bonds to the Fc region in two different monomers. Serum IgM is always a pentamer; most serum IgA exists as a monomer, although dimers, trimers, and even tetramers are sometimes present. Not shown in these figures are intrachain disulfide bonds and disulfide bonds linking light and heavy chains (see Figure 4-2).

Lesson 42 : IgM IgG Properties Functions:

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IgG

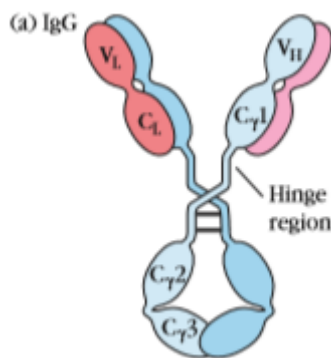
Structure

All IgG's are monomers (7S immunoglobulin). The subclasses differ in the number of disulfide bonds and length of the hinge region.

Properties

IgG is the most versatile immunoglobulin because it is capable of carrying out all of the functions of immunoglobulin molecules.

- IgG is the major Ig in serum - 75% of serum Ig is IgG
- IgG is the major Ig in extra vascular spaces
- Placental transfer - IgG is the only class of Ig that crosses the placenta. Transfer is mediated by a receptor on placental cells for the Fc region of IgG. Not all subclasses cross equally well; IgG2 does not cross well.
- Fixes complement - Not all subclasses fix equally well; IgG4 does not fix complement
- Binding to cells - Macrophages, monocytes, PMNs and some lymphocytes have Fc receptors for the Fc region of IgG. Not all subclasses bind equally well; IgG2 and IgG4 do not bind to Fc receptors. A consequence of binding to the Fc receptors on PMNs, monocytes and macrophages is that the cell can now internalize the antigen better. The antibody has prepared the antigen for eating by the phagocytic cells. The term opsonin is used to describe substances that enhance phagocytosis. IgG is a good opsonin. Binding of IgG to Fc receptors on other types of cells results in the activation of other functions.



IgM

Structure

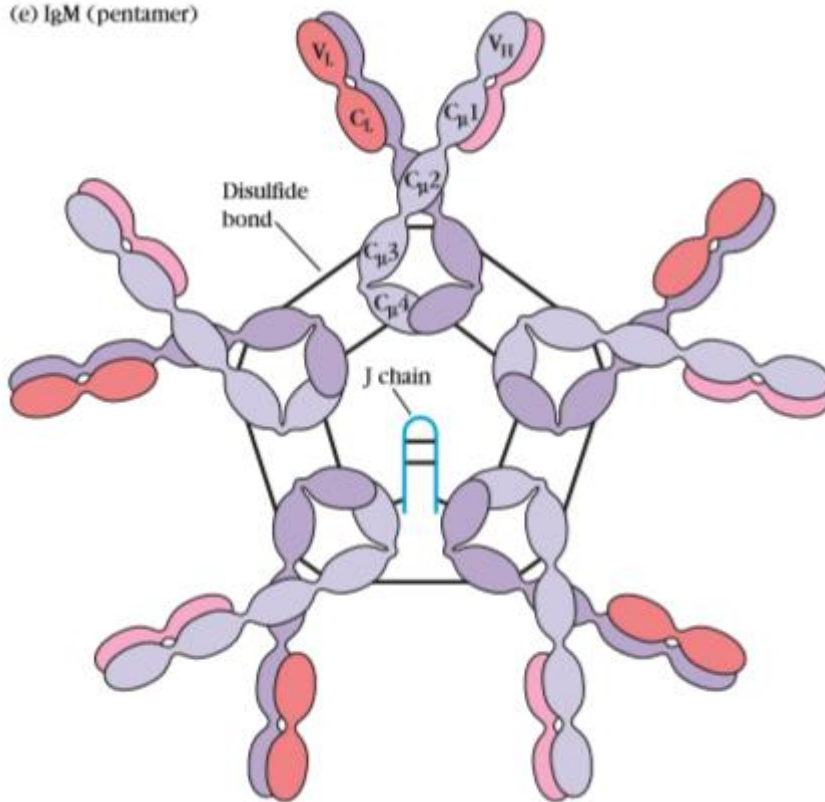
IgM normally exists as a pentamer (19S immunoglobulin) but it can also exist as a monomer. In the pentameric form all heavy chains are identical and all light chains are identical. Thus, the valence is theoretically 10. IgM has an extra domain on the mu chain (C_{H4}) and it has another

protein covalently bound via a S-S bond called the J chain. This chain functions in polymerization of the molecule into a pentamer.

Properties

- **IgM is the third most common serum Ig.**
- IgM is the first Ig to be made by the fetus and the first Ig to be made by a virgin B cells when it is stimulated by antigen.
- As a consequence of its pentameric structure, IgM is a good complement fixing Ig. Thus, IgM antibodies are very efficient in leading to the lysis of microorganisms.
- As a consequence of its structure, IgM is also a good agglutinating Ig. Thus, IgM antibodies are very good in clumping microorganisms for eventual elimination from the body.
- IgM binds to some cells via Fc receptors.
- Surface IgM exists as a monomer and lacks J chain but it has an extra 20 amino acids at the C-terminus to anchor it into the membrane. Cell surface IgM functions as a receptor for antigen on B cells.

(e) IgM (pentamer)



Lesson 43 : IgA IgE IgD Properties Functions:

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IgA

Structure

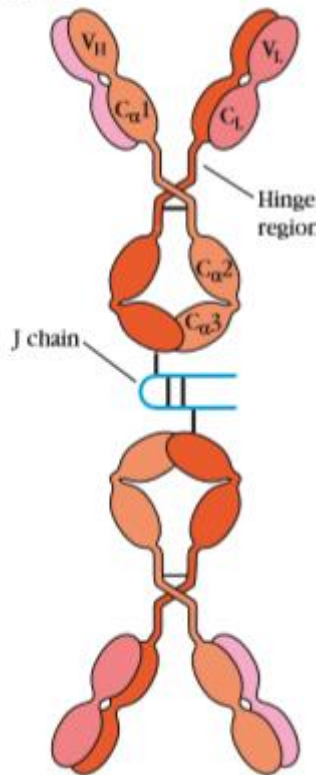
Serum IgA is a monomer but IgA found in secretions is a dimer. When IgA exists as a dimer, a J chain is associated with it.

When IgA is found in secretions is also has another protein associated with it called the secretory piece or T piece; sIgA is sometimes referred to as 11S immunoglobulin. Unlike the remainder of the IgA which is made in the plasma cell, the secretory piece is made in epithelial cells and is added to the IgA as it passes into the secretions. The secretory piece helps IgA to be transported across mucosa and also protects it from degradation in the secretions.

Properties

- IgA is the 2nd most common serum Ig.
- IgA is the major class of Ig in secretions - tears, saliva, colostrum, mucus. Since it is found in secretions secretory IgA is important in local (mucosal) immunity.
- Normally IgA does not fix complement, unless aggregated.
- IgA can bind to some cells - PMN's and some lymphocytes.

(d) IgA (dimer)



IgD

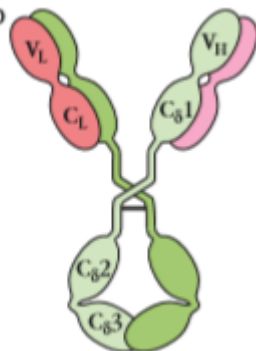
Structure

IgD exists only as a monomer.

Properties

- IgD is found in low levels in serum; its role in serum uncertain.
- IgD is primarily found on B cell surfaces where it functions as a receptor for antigen. IgD on the surface of B cells has extra amino acids at C-terminal end for anchoring to the membrane. It also associates with the Ig-alpha and Ig-beta chains.
- IgD does not bind complement.

(b) IgD



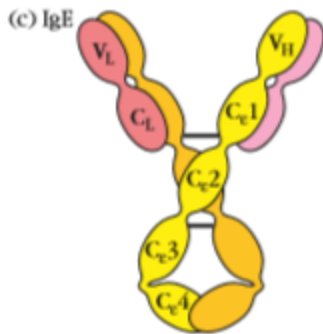
IgE

Structure

IgE exists as a monomer and has an extra domain in the constant region.

Properties

- IgE is the least common serum Ig since it binds very tightly to Fc receptors on basophils and mast cells even before interacting with antigen.
- Involved in allergic reactions - As a consequence of its binding to basophils and mast cells, IgE is involved in allergic reactions. Binding of the allergen to the IgE on the cells results in the release of various pharmacological mediators that result in allergic symptoms.
- IgE also plays a role in parasitic helminth diseases. Since serum IgE levels rise in parasitic diseases, measuring IgE levels is helpful in diagnosing parasitic infections. Eosinophils have Fc receptors for IgE and binding of eosinophils to IgE-coated helminths results in killing of the parasite.
- IgE does not fix complement.



Lesson 44 : Types vs Sub-types of Antibodies:

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Immunoglobulin

Types

Immunoglobulins can also be classified by the type of light chain that they have. Light chain types are based on differences in the amino acid sequence in the constant region of the light chain. These differences are detected by serological means.

- Kappa light chains
- Lambda light chains

Immunoglobulin

Subtypes

The light chains can also be divided into subtypes based on differences in the amino acid sequences in the constant region of the light chain.

Lambda subtypes

- Lambda 1
- Lambda 2
- Lambda 3
- Lambda 4

Kappa chains have following two subtypes e.g K1 & K2

For example, the subtypes of IgG include IgG Kappa 1 & IgG Lambda 4. Similarly, the subtypes of IgM can be IgM Kappa2 & IgM lambda 3 respectively

Nomenclature

Immunoglobulins are named based on the class, or subclass of the heavy chain and type or subtype of light chain. Unless it is stated precisely, you should assume that all subclass, types and subtypes are present. IgG means that all subclasses and types are present.

Heterogeneity

Immunoglobulins considered as a population of molecules are normally very heterogeneous because they are composed of different classes and subclasses each of which has different types and subtypes of light chains. In addition, different immunoglobulin molecules can have different antigen binding properties because of different V_H and V_L regions.

Lesson 45 : Antigen & Antibody Reactions:

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Antigen & Antibody Reactions

NATURE OF ANTIGEN-ANTIBODY BONDS

The combining site of an antibody is located in the F(ab) portion of the antibody molecule and is assembled from the hypervariable regions of the heavy and light chains. The binding between this site and the antigen takes place with the following characteristics and processes:

- The bonds that hold the antigen to the combining site of any antibody are noncovalent, and, hence, they are reversible in nature.
- These bonds may be hydrogen bonds, electrostatic bonds, or Van der Waals forces.
- Usually there are multiple bond formations observed, ensuring relatively tight binding between antibody and antigen.
- The specific binding between the antigenic determinant on the cell (known as epitope) and the antigen combining site (paratope) on the antibody involves very small portions of the molecules, usually comprising only a few amino acids.
- These sites are critical in antigen-antibody reactions as specific binding has to overcome repulsion between the two molecules.
- When the epitope comes in contact with paratope they are first attracted to each other by ionic and hydrophobic forces.
- These forces help them overcome their hydration energies and allow for the expulsion of water molecules as epitope and paratope approach each other.
- This attraction becomes even stronger when Van der Waals forces are employed later on to bring epitope and paratope even closer.

Lesson 46 : Factors Effecting on Antigen & Antibody Tests:

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Antibody Affinity Is a Quantitative Measure of Binding Strength

The combined strength of the noncovalent interactions between a single antigen-binding site on an antibody and a single epitope is the affinity of the antibody for that epitope. Low-affinity antibodies bind antigen weakly and tend to dissociate readily, whereas high-affinity antibodies bind antigen more tightly and remain bound longer

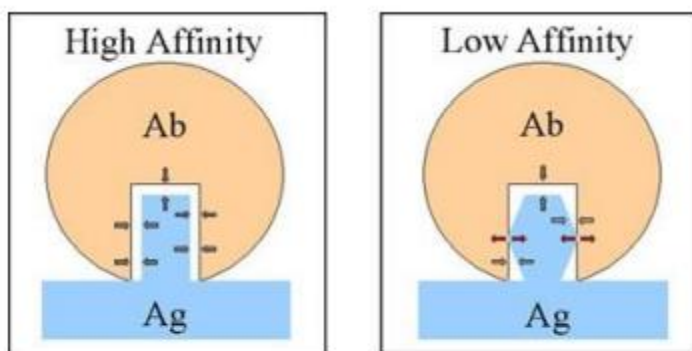


Fig 3.14: The binding interactions between antigen (Ag) and antibody (Ab).

Antibody Avidity Incorporates Affinity of Multiple Binding Sites

- The affinity at one binding site does not always reflect the true strength of the antibody-antigen interaction.
- When complex antigens containing multiple, repeating antigenic determinants are mixed with antibodies containing multiple binding sites, the interaction of an antibody molecule with an antigen molecule at one site will increase the probability of reaction between those two molecules at a second site.
- The strength of such multiple interactions between a multivalent antibody and antigen is called the avidity.
- The avidity of an antibody is a better measure of its binding capacity within biological systems (e.g., the reaction of an antibody with antigenic determinants on a virus or bacterial cell) than the affinity of its individual binding sites. High avidity can compensate for low affinity. For example, secreted pentameric IgM often has a lower affinity than IgG, but the high avidity of IgM, resulting from its higher valence, enables it to bind antigen effectively

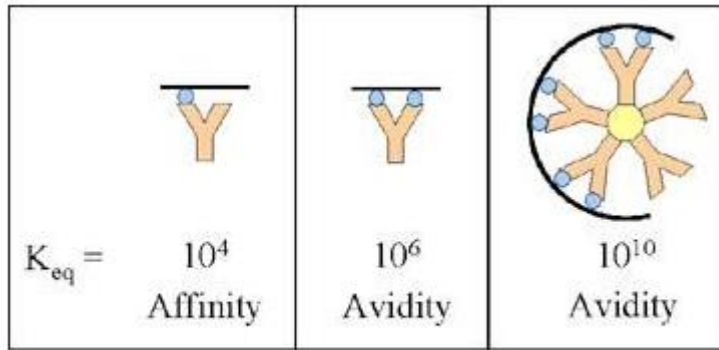


Fig 3.15: Effect of avidity on Ag (Blue colored) & antibody (orange colored) reactions

Antigen to antibody ratio

The ratio between the antigen and antibody influences the detection of antigen-antibody complexes because the size of the complexes formed is related to the concentration of the antigen and antibody.

Physical form of the antigen

The physical form of the antigen influences how one detects its reaction with an antibody. If the antigen is a particulate, one generally looks for agglutination of the antigen by the antibody. If the antigen is soluble one generally looks for the precipitation of the antigen after the production of large insoluble antigen-antibody complexes.

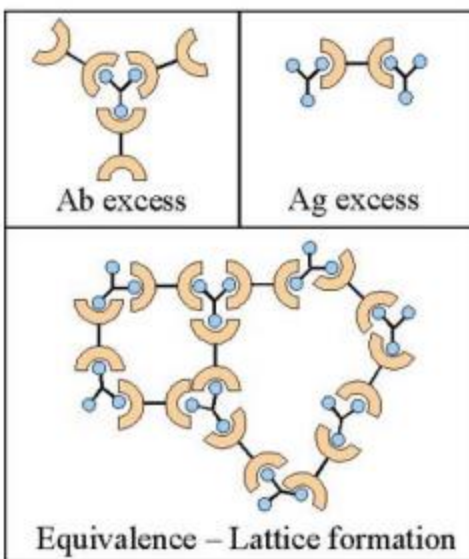


Fig 3.16: Ag/Ab ratio effects on Ag/Ab reactions

Lesson 47 : Agglutination Reactions:

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Agglutination Reactions

These are antigen and antibody reactions in which the particulate nature of antigen causes the reaction with its specific antibody to form visible clumps. Such visible clumps are called as **agglutinates**. The antibodies which render such kind of reactions are called as “agglutinins”. As the antigens are particulate in nature and the nature of particle determines the kind of agglutination reactions. There are following three important kinds of agglutination reactions based upon the particle

1. Hemagglutination

In these reactions, the antigens are present on the surface of red blood cells (RBCs). That's why these are called as hemagglutination reactions.

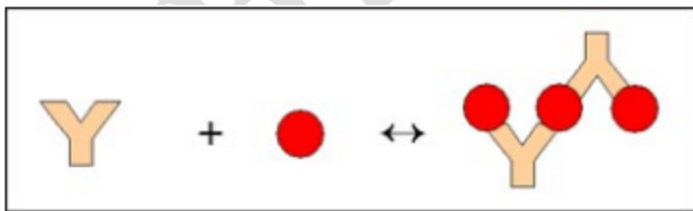
2. Bacterial agglutination

The bacterial antigens present on the surface of bacteria react with their corresponding specific antibodies and form visible agglutinates. Such kind of agglutination reactions are called as bacterial agglutinations.

3. Latex Agglutination

In this kind of agglutination reactions, the antigens are present on the surface of latex particles which form visible agglutinates after reacting with specific antibodies.

Hemagglutination These are the reactions in which antigens are present on the surface of red blood cells (RBCs). That's why these are called as hemagglutination reactions



Hemagglutination reactions: Antibodies (Orange color) react with RBC (red color) carrying antigens

Hemagglutination reactions can be of two types based on the nature of reaction

A. Qualitative Hemagglutination:

These hemagglutination reactions determine the presence of antigen or antibody in the serum. For example, the Blood group typing (ABO & Rh).

B. Quantitative Hemagglutination

This kind of heagglutination is used for determining the quantity of antibodies or antigens in the serum of the patient. In this kind of agglutination, the titer of antibodies is determined which is the lowest concentration of Antibody which causes agglutination or the maximum dilution of serum which causes agglutination

Application of agglutination reactions

Agglutination reactions are used for following ways

- Blood group determination: ABO & Rh for both either on the basis of antigens of RBC or antibodies of serum
- Assessing bacterial infections e.g Widal test for diagnosis of Typhoid fever

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Lesson 48 : Precipitation Reactions:

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Precipitation Reactions

These are the kind of antigen/antibody reactions in which soluble nature of antigen causes the reaction of antibody to form precipitate in a medium. These reactions are called as immunoprecipitation due to involvement of antigen & antibody. The antibodies which cause such kind of reactions are called as precipitins. These reactions occur in a medium which is used for visualization of Ag & Ab reaction e.g Agar. Both Ag & Ab are soluble form in serum.

Types of Precipitation reactions

On the basis of diffusion of each molecule i.e Ag & Ab, following two types of precipitation reactions are

- 1) Single immunodiffusion or precipitation
- 2) Double immunodiffusion or precipitation

Single Immunodiffusion

These kind of precipitation reactions are also called as Radial Immunodiffusion. In such kind of reactions, antibody is already incorporated in agar. While the patient's serum containing antigen is added in the agar well. Antigen diffuses in agar & react with corresponding antibody to form visible precipitate in the form of ring. The size of the ring determines the concentration of antigen in the patient's sera as there is a direct relationship between the diameter of precipitation ring and the antigen concentration (Fig) These kind of immunoassays are used for determining the level of serum immunoglobulins e.g IgA, IgG & IgM.

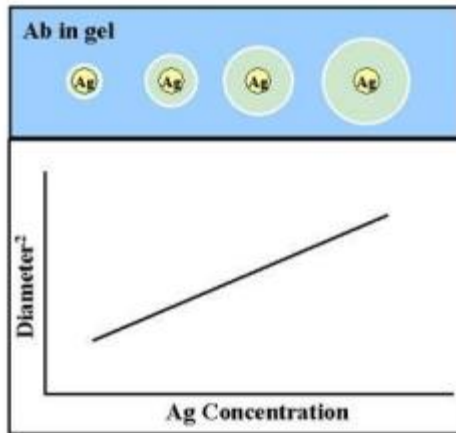


Figure: Appearance of precipitation ring in Single immunodiffusion

Double Immunodiffusion

These are the precipitation reactions in which both antigen & antibody is added in the medium separately in contrast to single immunodiffusion where antibody is already incorporated in the agar. Antigen & antibody are allowed to react together and reached at equivalence to form a visible precipitate. Such kind of reactions are used for qualitative analysis of complex mixture of antigens in a sample can be determined. Moreover, these reactions are also used for checking the purity of isolated serum proteins.

Lesson 49 : Complement Fixation Tests:

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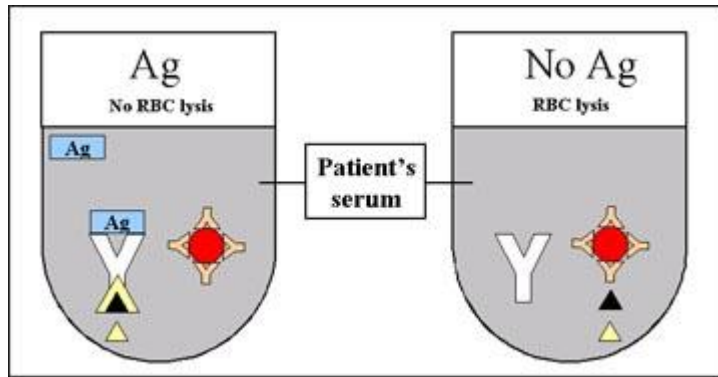
Complement Fixation

Antigen/antibody complexes can also be measured by their ability to fix complement because an antigen/antibody complex will "consume" complement if it is present, whereas free antigens or antibodies do not.

Tests for antigen/antibody complexes that rely on the consumption of complement are termed complement fixation tests and are used to quantitate antigen/antibody reactions. This test will only work with complement fixing antibodies (IgG and IgM are best).

The principle of the complement fixation test

- Antigen is mixed with the test serum to be assayed for antibody and antigen/antibody complexes are allowed to form.
- A control tube in which no antigen is added is also prepared.
- If no antigen/antibody complexes are present in the tube, none of the complement will be fixed.
- However, if antigen/antibody complexes are present, they will fix complement and thereby reduce the amount of complement in the tube.
- After allowing complement fixation by any antigen/antibody complexes, a standard amount of red blood cells, which have been pre-coated with anti-erythrocyte antibodies is added. The amount of antibody-coated red blood cells is predetermined to be just enough to completely use up all the complement initially added, if it were still there. If all the complement was still present (i.e. no antigen/antibody complexes formed between the antigen and antibody in question), all the red cells will be lysed.
- By simply measuring the amount of red cell lysis by measuring the release of hemoglobin into the medium, one can indirectly quantitate antigen/antibody complexes in the tube.
- Complement fixation tests are most commonly used to assay for antibody in a test sample but they can be modified to measure antigen.



Interpretation of CFT

- Lysis of RBC: No Complement fixation by Ag & Ab complex: No Ag: Negative CFT
- No Lysis: Complement fixation by AG & Ab complex: Ag is present: Positive CFT

Applications of Complement Fixation Test

Complement Fixation tests are used for determining Gonococcal Ab in serum

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Lesson 50 : Enzyme Linked Immunosorbent Assay (ELISA):

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Enzyme Linked Immunosorbent Assay (ELISA)

Enzyme Linked Immunosorbent Assays (ELISA) are those that are based on the measurement of an enzymatic reaction associated with immune complexes. In any particular assay, the enzyme may be linked to either the antigen or the antibody.

As are a type of immunoassay that are commonly used to quantify levels of a specific target within a sample. Samples routinely used in ELISAs include serum, plasma, cell culture supernates, cell lysates, saliva, tissue lysates, and urine.

ELISA Types

The four main types of ELISAs are direct, indirect, sandwich, and competitive. Each type of ELISA has its own advantages and disadvantages.

Direct ELISA

In a direct ELISA, an antigen or sample is immobilized directly on the plate and a conjugated detection antibody binds to the target protein. Substrate is then added, producing a signal that is proportional to the amount of analyte in the sample. Since only one antibody is used in a direct ELISA, they are less specific than a sandwich ELISA.

When to Use: Assessing antibody affinity and specificity. Investigating blocking/inhibitory interactions.

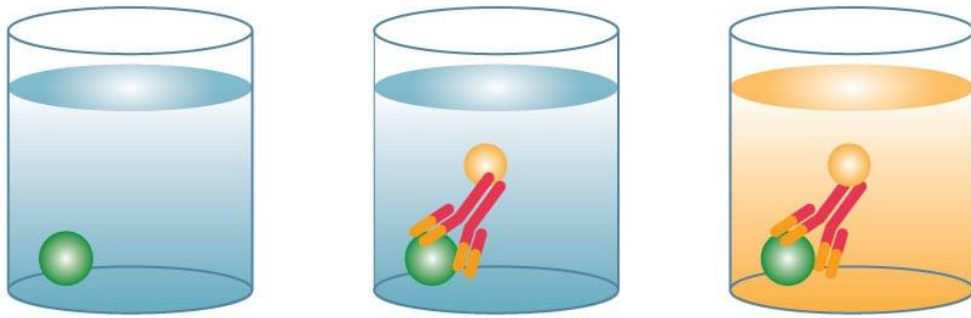
Advantages:

Fast and simple protocol

Disadvantages:

Less specific since you are only using 1 antibody.

Potential for high background if all proteins from a sample are immobilized in well.



Indirect ELISA

An indirect ELISA is similar to a direct ELISA in that an antigen is immobilized on a plate, but it includes an additional amplification detection step. First, an unconjugated primary detection antibody is added and binds to the specific antigen. A conjugated secondary antibody directed against the host species of the primary antibody is then added. Substrate then produces a signal proportional to the amount of antigen bound in the well.

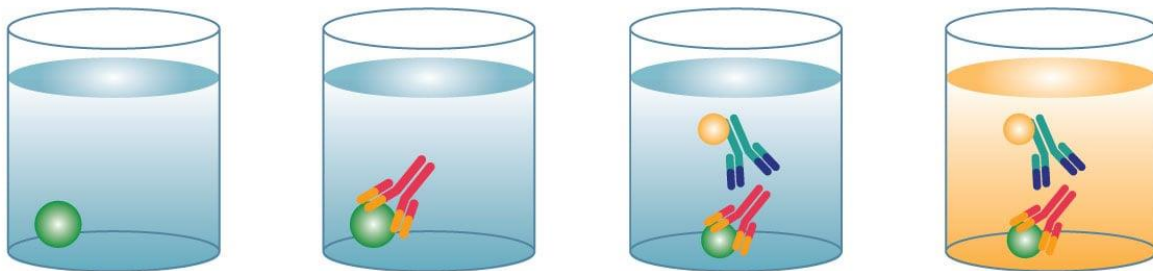
When to Use: Measuring endogenous antibodies.

Advantages:

Amplification using a secondary antibody

Disadvantages:

Potential for cross-reactivity caused by secondary antibody



Sandwich ELISA

Sandwich ELISAs are the most common type of ELISA. Two specific antibodies are used to sandwich the antigen, commonly referred to as matched antibody pairs. Capture antibody is coated on a microplate, sample is added, and the protein of interest binds and is immobilized on the plate. A conjugated-detection antibody is then added and binds to an additional epitope on the target protein. Substrate is added and produces a signal that is proportional to the amount of analyte present in the sample. Sandwich ELISAs are highly specific, since two antibodies are required to bind to the protein of interest.

When to Use: Determining analyte concentration in a biological sample.

Advantages:

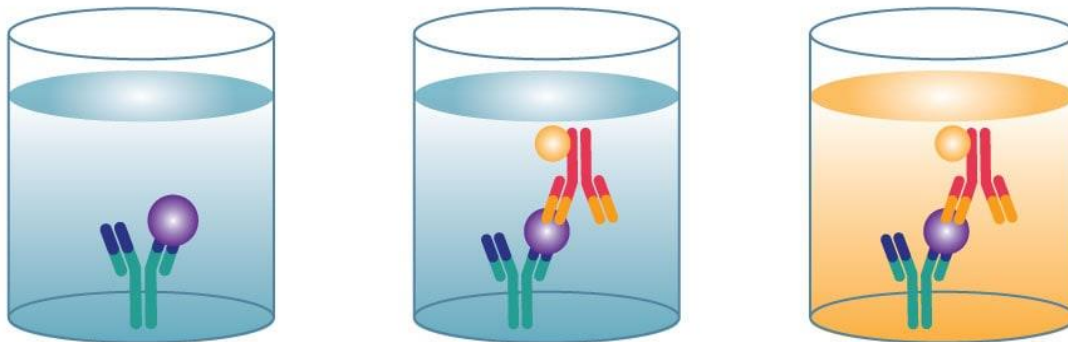
Highest specificity and sensitivity

Compatible with complex sample matrices

Disadvantages:

Longer protocol

Challenging to develop



Competitive ELISA

Competitive ELISAs are commonly used for small molecules, when the protein of interest is too small to efficiently sandwich with two antibodies. Similar to a sandwich ELISA, a capture antibody is coated on a microplate. Instead of using a conjugated detection antibody, a conjugated antigen is used to compete for binding with the antigen present in the sample. The more antigen present in the sample, the less conjugated antigen will bind to the capture antibody. Substrate is added and the signal produced is inversely proportional to the amount of protein present in the sample.

When to Use: Determining concentrations of a small molecules and hormones.

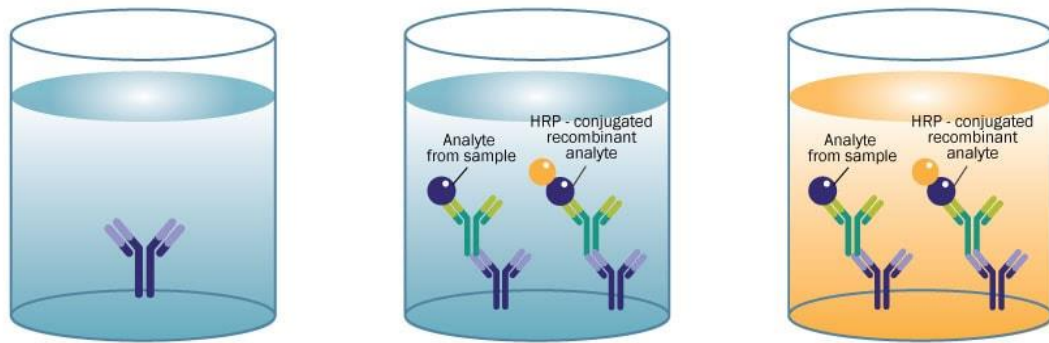
Advantages:

Ability to quantitate small molecules

Disadvantages:

Less specific since you are only using 1 antibody

Requires a conjugated antigen



BT302 - Immunology

**Teacher: Dr Numan Javed (Phd Holder at SHanghai
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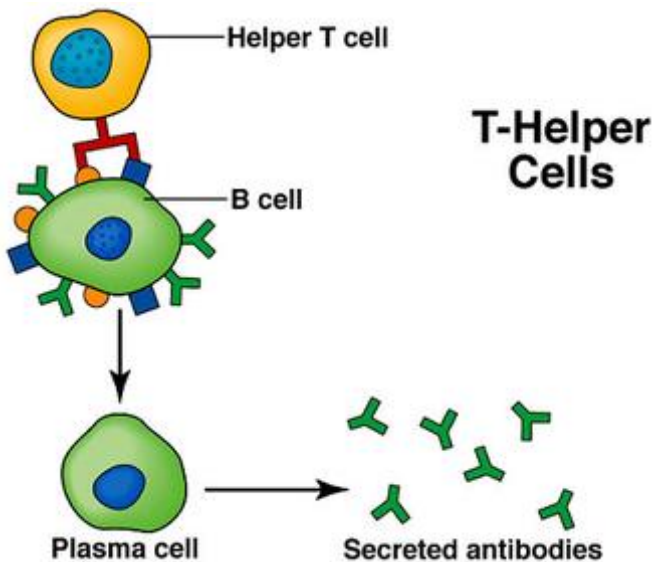
1 to 167 Lectures merged

Effort by AHMAD SAEED

Lesson 51 : Genetics of Antibody Structure & Diversity:

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The immune system is the body's defence against disease and foreign bodies, under the form of antibody production, tissue rejection and phagocytosis.



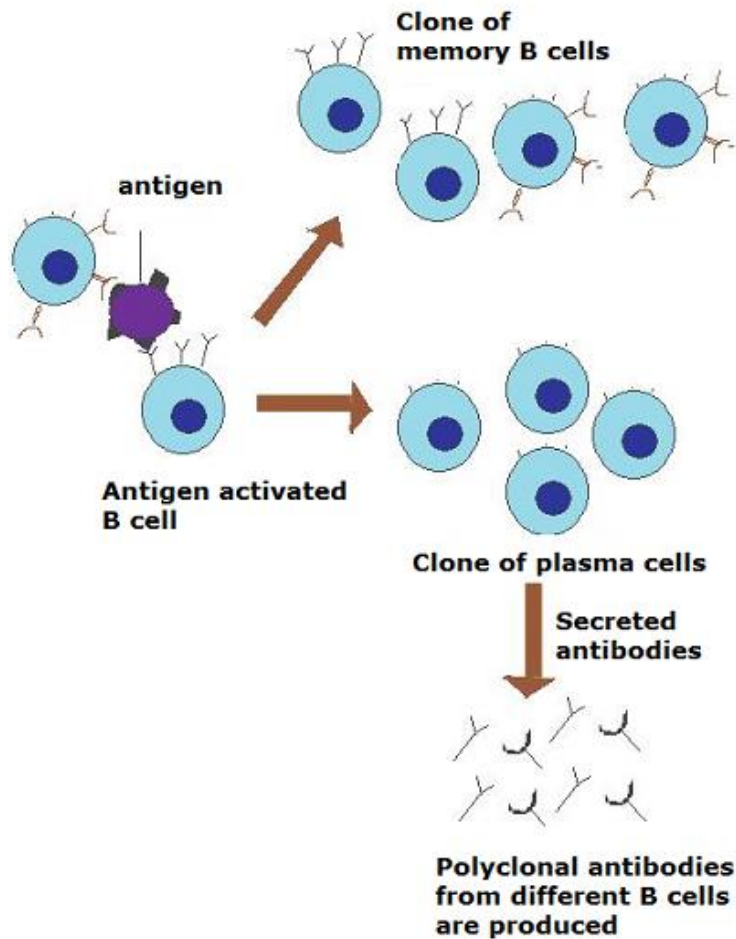
Antibody

production

Antibodies are produced by lymphocytes, which are formed in lymph nodes. Lymphocytes produced antibodies in response to the presence of pathogens such as bacteria. This is because alien cells have chemicals called antigen on their surface. A different antibody is produced for each antigen.

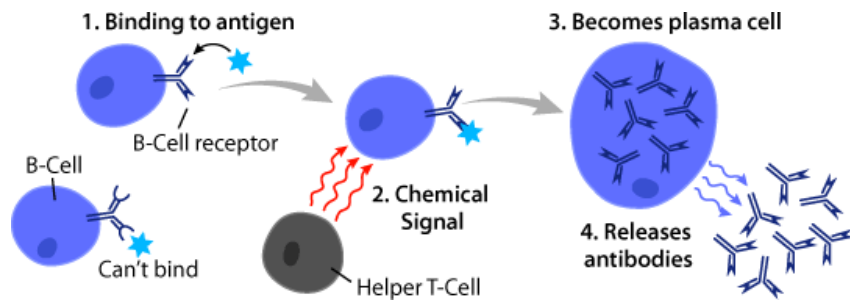
The antibodies make bacteria clump together in preparation for action by phagocytes, or neutralize toxins produced by the bacteria. Once antibodies have been made, they remain in the blood to provide long-term protection.

Some lymphocytes memorize the antigens the body has been exposed to. They can rapidly reproduce and produce antibodies to respond to further infections by the same pathogen (disease-causing organism).



The T and B lymphocytes (T and B Cells) are involved in the acquired or antigen-specific immune response given that they are the only cells in the organism able to recognize and respond specifically to each antigenic epitope. The B Cells have the ability to transform into plasmocytes and are responsible for producing antibodies (Abs). Thus, humoral immunity depends on the B Cells while cell immunity depends on the T Cells.

Antibody Formation process can be T cell-independent or dependent based on the dependency of T-lymphocytes in this process. During the process of antibody formation, B-cells recognize the antigen with its specific receptor which is also called as B-cell receptor (BCR). After binding with specific antigen with BCR on B-cell, there is transformation of B-cells into a specialized variety of cells which are called as plasma cells. Plasma cells are also called as antibody secreting cells as these secrete antibodies



Events of antibody formation

Lesson 52 : General Characteristics of Antibody Response:

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GENERAL CHARACTERISTICS OF THE ANTIBODY RESPONSE

Self / non-self discrimination

One characteristic feature of the specific immune system is that it normally distinguishes between self and non-self and only reacts against non-self.

Memory

A second feature of the specific immune response is that it demonstrates memory. The immune system "remembers" if it has seen an antigen before and it reacts to secondary exposures to an antigen in a manner different than after a primary exposure. Generally only an exposure to the same antigen will illicit this memory response.

Specificity

A third characteristic feature of the specific immune system is that there is a high degree of specificity in its reactions. A response to a particular antigen is specific for that antigen or a few closely related antigens.

Lesson 53 : Fate of Immunogen:

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Fate of the immunogen

Immunogen enter in the body first time which is called as primary injection of immunogen while if there is second time exposure of body with the same immunogen is called as secondary injection of immunogen.

The fate of immunogen is different in both primary and secondary cases

Clearance after primary injection
Immunogen enters in the body first time (Primary Injection).

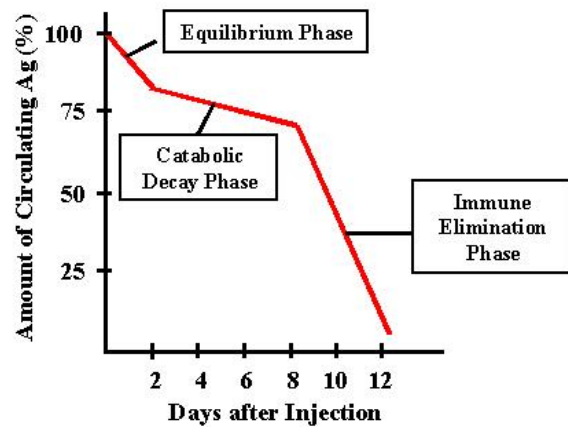
The kinetics of antigen clearance from the body after a primary administration is depicted in Figure

Equilibrium phase
The first phase is called the equilibrium or equilibration phase. During this time the antigen equilibrates between the vascular and extravascular compartments by diffusion. This is normally a rapid process. Since particulate antigens don't diffuse, they do not show this phase.

Catabolic decay phase
In this phase the host's cells and enzymes metabolize the antigen. Most of the antigen is taken up by macrophages and other phagocytic cells. The duration will depend upon the immunogen and the host.

Immune elimination phase
In this phase, newly synthesized antibody combines with the antigen producing antigen/antibody complexes which are phagocytosed and degraded. Antibody appears in the serum only after the immune elimination phase is over.

Clearance after secondary injection
If there is circulating antibody in the serum, injection of the antigen for a second time results in a rapid immune elimination. If there is no circulating antibody then injection of the antigen for a second time results in all three phases but the onset of the immune elimination phase is accelerated due to memory response.



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Lesson 54 : Primary & Secondary Antibody Response:

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Specificity of primary and secondary responses

Antibody elicited in response to an antigen is specific for that antigen, although it may also cross react with other antigens which are structurally similar to the eliciting antigen. In general secondary responses are only elicited by the same antigen used in the primary response. However, in some instances a closely related antigen may produce a secondary response, but this is a rare exception.

Qualitative changes in antibody during primary and secondary responses

Immunoglobulin

class

variation

In the primary response, the major class of antibody produced is IgM, Antibodies produced during this response are with less affinity and less persistent whereas in the secondary response it is IgG (or IgA or IgE). The antibodies that persist in the secondary response are the IgG antibodies. Antibodies produced in this response have high affinity. The affinity of antibodies depend upon the dosage of antigen, the low dosage of antigen causes increase in the affinity of antibodies which is called as affinity maturation

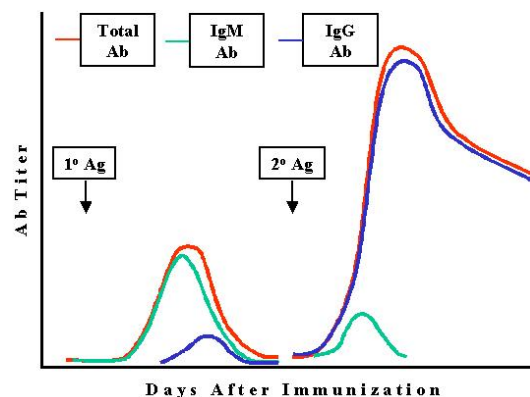


Figure: Antibody formation during primary & secondary response

Affinity

The affinity of the IgG antibody produced increases progressively during the response, particularly after low doses of antigen. This is referred to as affinity maturation. Affinity maturation is most pronounced after secondary challenge with antigen.

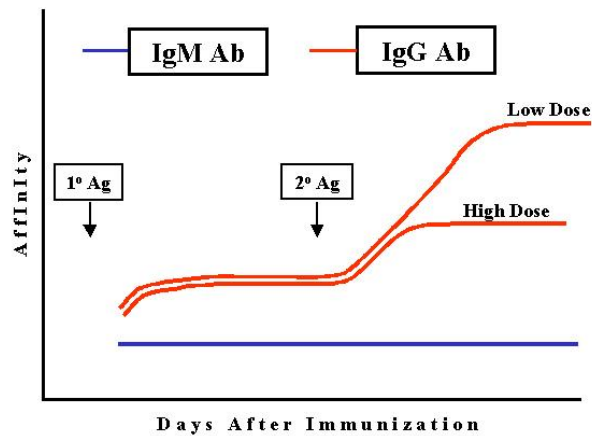


Figure : Affinity of antibodies during primary & secondary response

One explanation for affinity maturation is clonal selection as illustrated in above Figure. A second explanation for affinity maturation is that, after a class switch has occurred in the immune response, somatic mutations occur which fine tune the antibodies to be of higher affinity. There is experimental evidence for this mechanism, although it is not known how the somatic mutation mechanism is activated after exposure to antigen.

Lesson 55 : Kinetics of antibody response against T-dependent antigens:

[Please click here to download ppts](#)

Kinetics of antibody responses to T-dependent Antigen

Primary (1°) Antibody response
The kinetics of a primary antibody response to an antigen is illustrated in Figure 2.

Inductive, latent or lag phase
In this phase, the antigen is recognized as foreign and the cells begin to proliferate and differentiate in response to the antigen. The duration of this phase will vary depending on the antigen but it is usually 5 to 7 days.

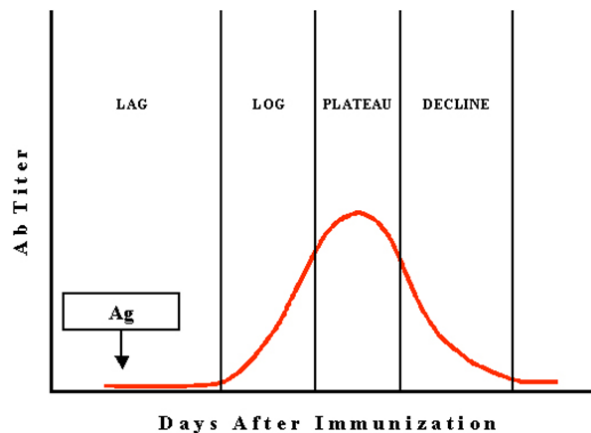


Figure: Kinetics of Primary response for T-dependent antigens

Log or Exponential Phase
In this phase, the antibody concentration increases exponentially as the B cells that were stimulated by the antigen differentiate into plasma cells which secrete antibody.

Plateau or steady-state phase
In this phase, antibody synthesis is balanced by antibody decay so that there is no net increase in antibody concentration.

Decline or decay phase
In this phase, the rate of antibody degradation exceeds that of antibody synthesis and the level of antibody falls. Eventually the level of antibody may reach base line levels.

Secondary (2°), memory or anamnestic response

Lag phase
In a secondary response, there is a lag phase by it is normally shorter than that observed in a primary response.

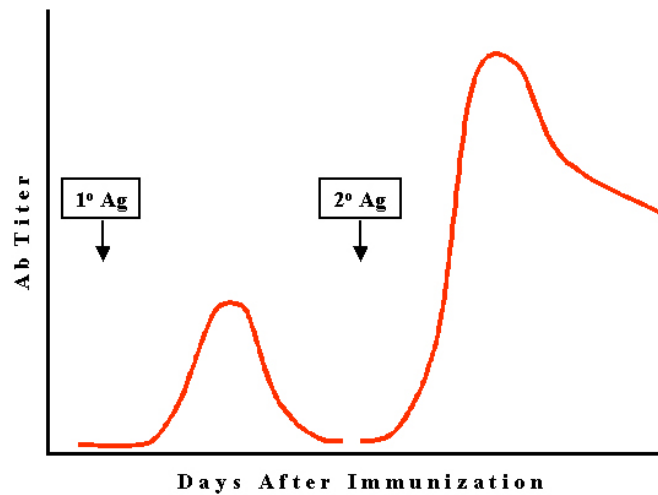


Figure: Kinetics of secondary response for T- dependent antigens

Log

The log phase in a secondary response is more rapid and higher antibody levels are achieved.

phase

Steady state phase

Decline

The decline phase is not as rapid and antibody may persist for months, years or even a lifetime.

phase

Lesson 56 : Kinetics of antibody response against T-independent antigens:

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Kinetics of antibody response against T-independent antigens

In case of T-independent antigens, mainly IgM class of antibodies are produced in contrast with T-dependent antigens. Likewise T-dependent antigens, there are all those four phases in primary response. However, secondary response also has same phases in second time exposure to antigens. Moreover, there is no memory response in case of T-independent antigens

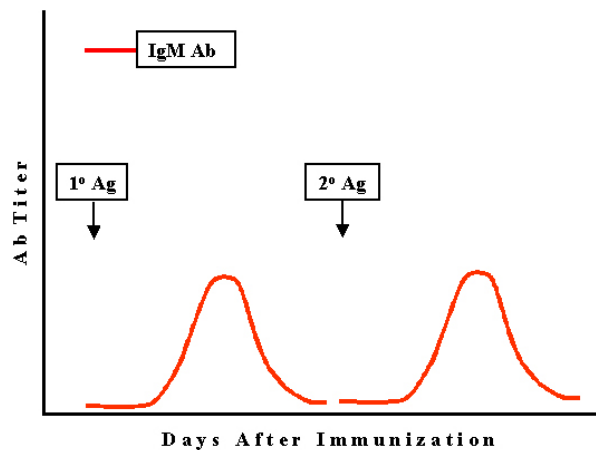


Figure : Kinetics of antibody formation against T-independent antigens

Lesson 57 : Cellular Events during Antibody Response

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Cellular events during primary and secondary responses to T-dependent antigen

Primary response

Lag **phase**
Clones of T and B cells with the appropriate antigen receptors bind antigen, become activated and begin to proliferate. The expanded clones of B cells differentiate into plasma cells which begin to secrete antibody.

Log

phase

The plasma cells initially secrete IgM antibody since the C_{μ} heavy chain gene is closest to the rearranged VDJ gene. Eventually some B cells switch from making IgM to IgG, IgA or IgE. As more B cells proliferate and differentiate into antibody secreting cells the antibody concentration increases exponentially.

Stationary

phase

As antigen is depleted, T and B cells are no longer activated. In addition, mechanisms which down regulate the immune response come into play. Furthermore, plasma cells begin to die. When the rate of antibody synthesis equals the rate of antibody decay the stationary phase is reached.

Decline

phase

When no new antibody is produced because the antigen is no longer present to activate T and B cells and the residual antibody slowly is degraded, the decay phase is reached.

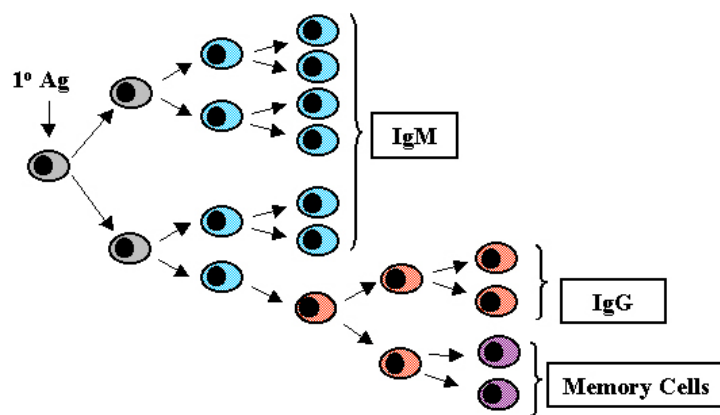


Figure: Cellular events during Primary response of antibody formation

Secondary response

Not all of the T and B cells that are stimulated by antigen during primary challenge with antigen die. Some of them are long lived cells and constitute what is referred to as the memory cell pool. Both memory T cells and memory B cells are produced and memory T cells survive longer than memory B cells. Upon secondary challenge with antigen not only are virgin T and B cells activated, the memory cells are also activated and thus there is a shorter lag time in the secondary response. Since there is an expanded clone of cells being stimulated the rate of antibody production is also increased during the log phase of antibody production and higher levels are achieved. Also, since many if not all of the memory B cells will have switched to IgG (IgA or IgE) production, IgG is produced earlier in a secondary response. Furthermore since there is an expanded clone of memory T cells which can help B cells to switch to IgG (IgA or IgE) production, the predominant class of Ig produced after secondary challenge is IgG (IgA or IgE).

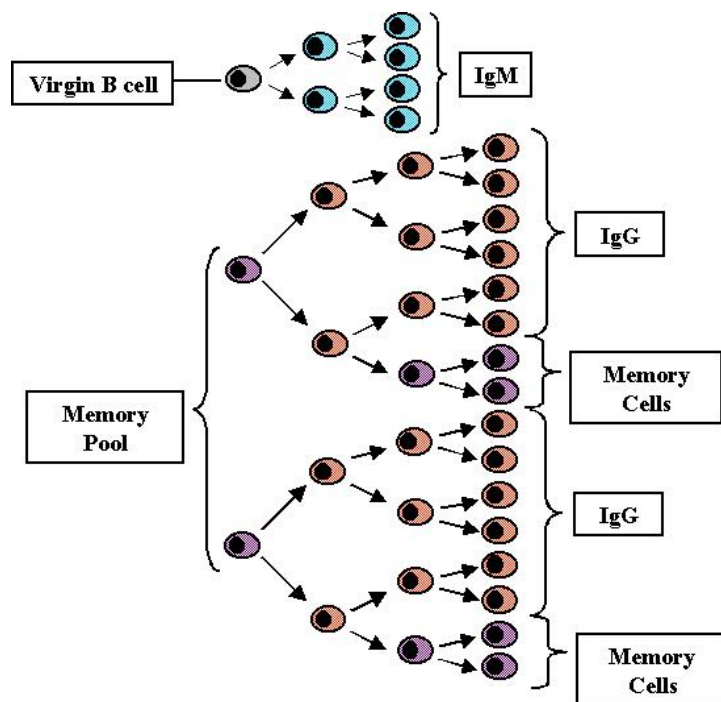


Figure: Cellular events during Primary response of antibody formation

Lesson 58 : Membrane & Secreted Antibodies:

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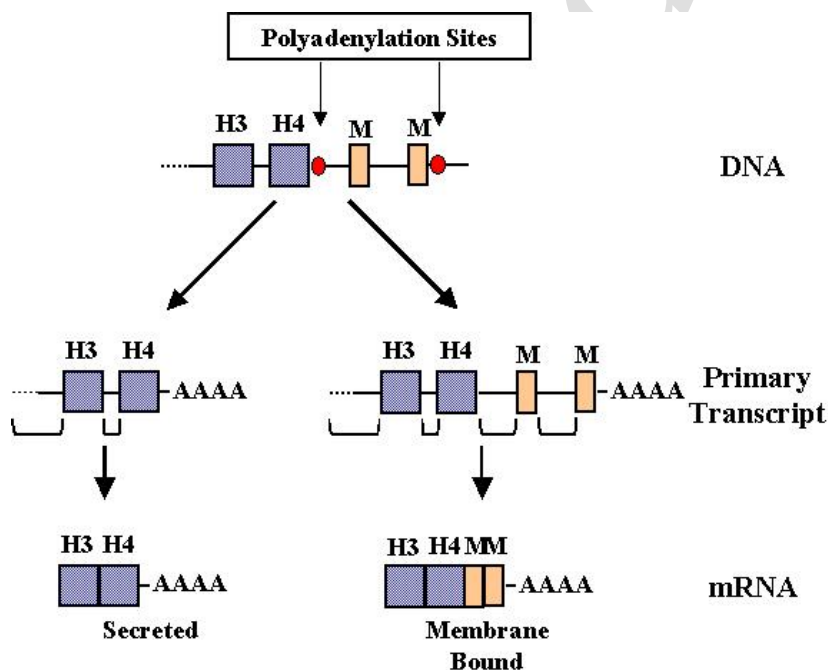
Membrane and secreted immunoglobulin

The specificity of membrane immunoglobulin on a B cell and the Ig secreted by the plasma cell progeny of a B cell is the same. An understanding of how the specificity of membrane and secreted Ig from an individual B cell can be the same comes from an understanding of immunoglobulin genes (Figure).

There are two potential polyA sites in the immunoglobulin gene.

- One after the exon for the last heavy chain domain
- Other after the exons that code for the trans- membrane domains.
- If the first polyA site is used, the pre-mRNA is processed to produce a secreted protein.
- If the second polyA site is used, the pre-mRNA is processed to produce a membrane form of the immunoglobulin.

However, in all cases the same VDJ region is used and thus the specificity of the antibody remains the same. All C regions genes have these additional membrane pieces associated with them and thus after class switching other classes of immunoglobulins can be secreted or expressed on the surface of B cells.



Lesson 59 : Expression of Immunoglobulin Genes & V(D)J Recombination:

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Structure of Human Antibody Gene (Loci)

Antibody Gene

As one Gene encode for one Protein, similarly the immunoglobulin which is glycoproteins are encoded by a single gene. The Immunoglobulin repertoire is encoded by multiple germ line gene segments. The immunoglobulin heavy and light chains are coded for by three separate gene families each one on a separate chromosome - one for the heavy chain and one for each of the light chain types. Each of these gene families has several V region genes and one or more C region genes. The V and C regions genes are not however immediately adjacent to each other.

These segments undergo somatic recombination during development of B-Cells. The basic component of gene is inherited but alteration occur during lifetime for diversification of antibodies. Immunoglobulin gene is divided into following two families based upon the structure of antibody

1. Light chain gene family
2. Heavy chain gene family

Light Chain Gene Family

This family encode for following light chains of immunoglobulin

1) Lambda Light Chains

This gene is located on chromosome 22. Lambda light chain gene is composed of 4 constant (C) region genes. Among these everyone encodes for each subtype of Lambda. It also contain 30 Variable (V) genes. Each of V region gene composed of two exons, one L that encodes for Ladder exons and the other V that encodes for variable region. Upstream of each of the C genes there is and additional exon called J (joining). The L, V, J and C exons are separated by [introns](#) (intervening non-coding sequences) (Fig 1).

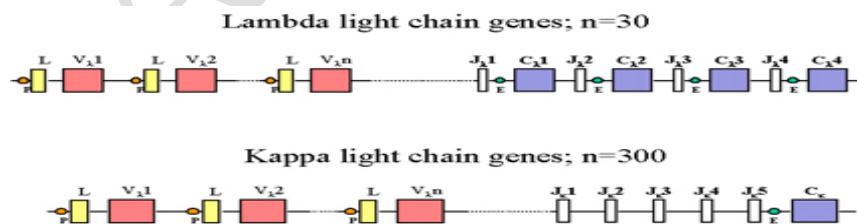


Fig 1: Structure of Lambda Light Chain gene

2. Kappa Light Chains

The kappa light chain gene family contains only one C region gene, since there is only one type of kappa light chain. There are many V region genes (approximately 250) each of which has a leader exon and a V exon. In the κ gene family there are several J exons located between the V and C genes. All of the exons are separated by introns.

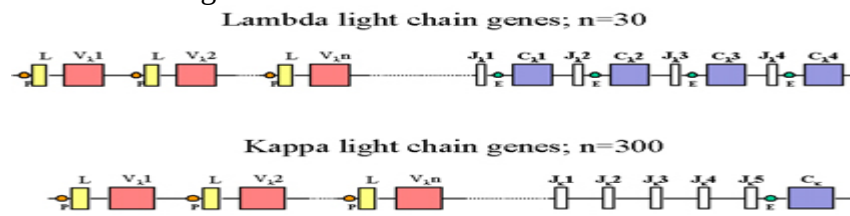
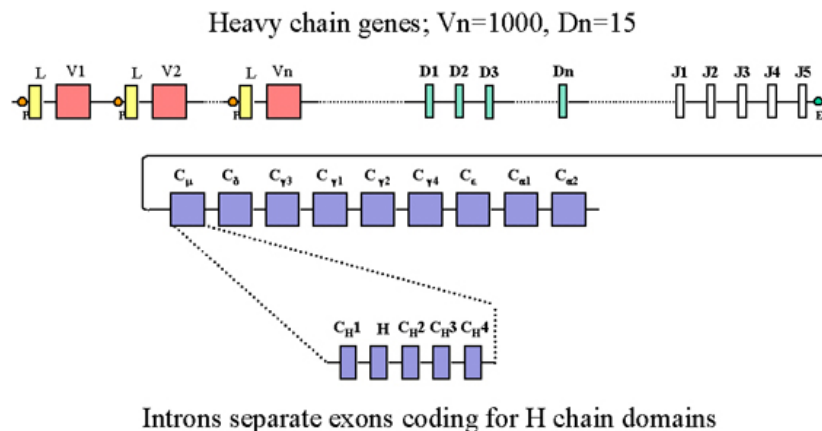


Figure 2: Structure of Kappa light chain gene

Heavy Chain Gene Family

Heavy chain gene is located at chromosome 14. It is composed of many constant C region gene for a each class & sub class. Each of C contain exons for hinge region & domains. There are approx.1000 variable genes. Each of V gene composed of L (Ladder) exons. Also contain V (variable) exons. In contrast to light chain gene family, it also carry additional D (diversity) exons which determines the diversity of immunoglobulin. Moreover, J (Joining) exons also exist. There are also introns in between exons (Fig 3)



Introns separate exons coding for H chain domains

Fig 3: Structure of heavy chain gene

Lesson 60 : Somatic Recombination:

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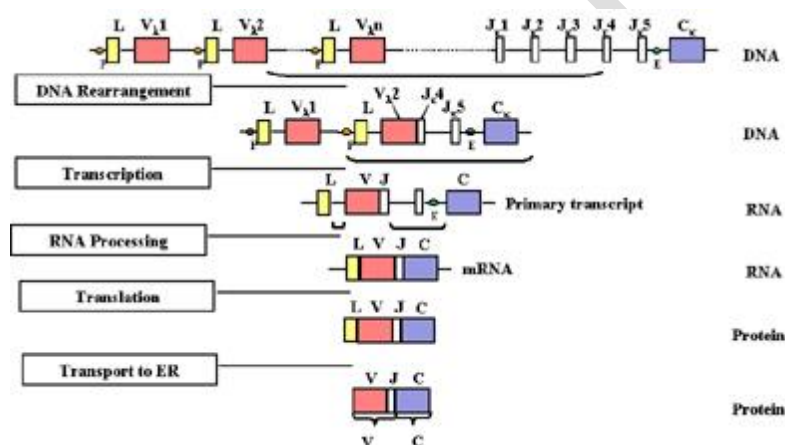
Somatic Recombination

- Recombination involves rearrangement of DNA in somatic cells.
- The newly recombined genes are not inherited in contrast to germ cells.
- Primary Ig repertoire differ slightly from one individual to next one.
- Also differ in individual's lifetime by their exposure to different antigens.

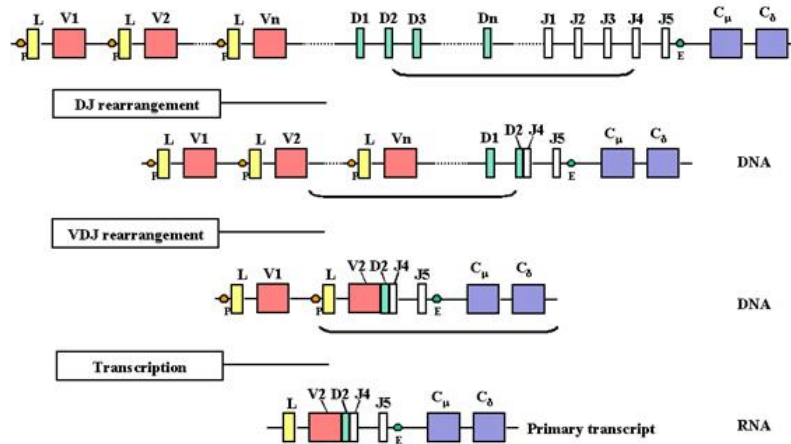
As B-cell differentiate into mature one, it make light chain (Kappa). There is rearrangement of genes (exons) as introns are removed and genes start to express. In this recombination, V genes become close to J genes by removing introns.

Recombined DNA processed into mature RNA by splicing. As Bcell differentiate into mature one, it make heavy chains. There is rearrangement of genes (exons) as introns are removed and genes start to express. In this recombination, D genes become close to J & then V recombines with DJ. Finally recombined DNA processed into mature RNA by splicing

Fig: The process of somatic recombination



As a cell differentiates into a mature B cell that will make a light chain, there is rearrangement of the various genes (exons)



As transcription initiates from the promoter a pre-mRNA is made which contains sequences from the L, V, D, J C μ and C δ regions as well as sequences for the introns between L and V, between J and C μ , and between C μ and C δ

BT302 - Immunology

Teacher: Dr Numan Javed (Phd Holder at SHanghai Institute of Sciences, China)

1 to 167 Lectures merged

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Lesson 61 : Somatic Hypermutation:

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Somatic Hyper mutation

The phenomenon of enhanced rate of point mutation in the immunoglobulin V region genes is called as Somatic hyper mutation. It occurs particularly in V gene which codes for 2nd hypervariable region. As a result, there is an increase in immunoglobulin diversity after various antigenic stimulation. Moreover, it also causes increase the affinity of immunoglobulin in order to compete for limited amount of available antigens.

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Lesson 62 : Role of Somatic Hypermutation in diversity of antibodies:

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Role of Somatic Hyper mutation in diversity of antibodies

Somatic hyper mutation has very significant role in the diversity of antibody. As a result, there is generation of sum of all the possible antibody specificities that an organism can produce. Normally, humans can make 10^7 - 10^8 different antibody molecules. Immunoglobulin diversity increase after various antigenic stimulation. So increase in antibody specificities due to this phenomenon of somatic mutation. It occurs at high rate approximately at 10^6 times higher. The exact mechanism by which mutation occurs in V region gene without effecting the C region is still under research. However, there is some role of activation induced cytidine deaminase (AID) which is considered essential in DNA deamination for somatic hyper mutation.

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Lesson 63 : V(D)J combinational Diversity:

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V (D) J combinational Diversity

The component of antibody diversity that is generated by joining of various antibody gene segments. As in case of light chain genes, the V & J region genes combination occurs, while for heavy chain, all V, D & J region genes combination occurs. The D region combines first with J region and then V to form a complete heavy chain.

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Lesson 64 : Regulation of V(D)J Recombination:

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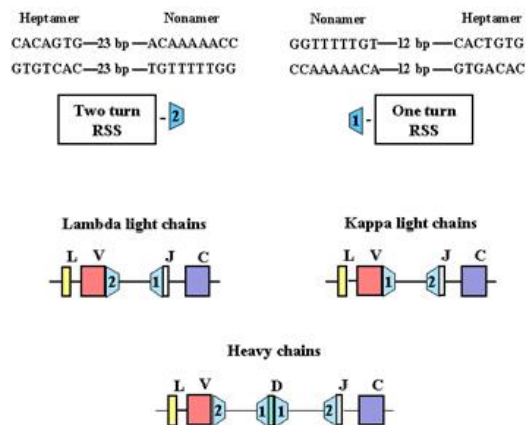
Flanking the V, J and D exons, there are unique sequences referred to as recombination signal sequences (RSS), which function in recombination. Each RSS consists of a conserved nonamer and a conserved heptamer that are separated by either 12 or 23 base pairs (bp) as illustrated in Figure. The 12bp and 23 bp spaces correspond to one or two turns of the DNA helix.

Recombination only occurs between a 1 turn and a 2 turn signal. In the case of the λ light chains there is a 1 turn signal upstream of the J exon and a 2 turn signal downstream of V_{λ} . In the case of the κ light chains there is a 1 turn signal downstream of the V_{κ} gene and a 2 turn signal upstream of the J exon. In the case of the heavy chains there are 1 turn signals on each side of the D exon and a 2 turn signal downstream of the V gene and a 2 turn signal upstream of the J exon. Thus, this ensures that the correct recombination events will occur.

The recombination event results in the removal of the introns between V and J in the case of the light chains or between the V, D, and J in the case of the heavy chains. The recombination event is catalyzed by two proteins, Rag-1 and Rag-2. Mutations in the genes for these proteins results in a severe combined immunodeficiency disease (both T and B cells are deficient), since these proteins and the RSS are involved in generating both the B and T cell receptors for antigen.

Mechanism of DNA Rearrangements

- Recombination signal sequences (RSS)
 - Nonmer
 - Heptamer
 - 1 or 2 turn signals
- Rag-1 and Rag-2



Flanking the V, J and D exons there are unique sequences referred to as recombination signal sequences (RSS), which function in recombination. Each RSS consists of a conserved nonamer and a conserved heptamer that are separated by either 12 or 23 base pairs

Lesson 65 : Affinity Maturation:

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Affinity Maturation

- Increase in average affinity of Immunoglobulins
- Following Ag activation, V regions of heavy & light chains are diversified by somatic hypermutation
- This results in increase in binding affinity of BCR for its cognate ligand
- B-cells with higher affinity Ig can compete better for limited amount of Ag
- Increase in binding strength of Ab produced by immune response with Ag
- Affinity maturation occurs during secondary Ab response

BT302 - Immunology

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Lesson 66 : Class Switching or Class Switch Recombination:

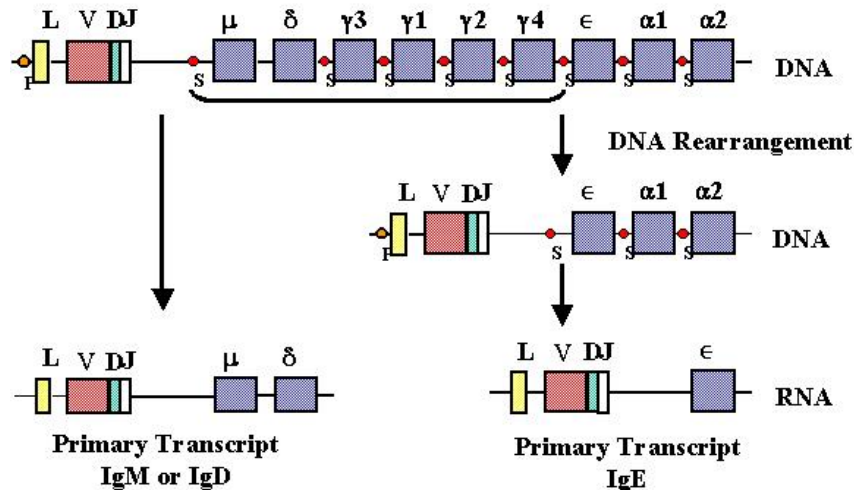
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Class Switching

During an antibody response to a T-dependent antigen a switch occurs in the class of Ig produced from IgM to some other class (except IgD). Our understanding of the structure of the immunoglobulin genes, helps explain how class switching occurs (Figure).

During class switching another DNA rearrangement occurs between a switch site (S_{μ}) in the intron between the rearranged VDJ regions and the C_{μ} gene and another switch site before one of the other heavy chain constant region genes. The result of this recombination event is to bring the VDJ region close to one of the other constant region genes, thereby allowing expression of a new class of heavy chain. Since the same VDJ gene is brought near to a different C gene and since the antibody specificity is determined by the hypervariable regions within the V region, the antibody produced after the switch occurs will have the same specificity as before.

Cytokines secreted by T helper cells can cause the switch to certain isotypes.



Lesson 67 : Antigen Processing & Presentation:

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Review of B & T-Cell Receptors for Antigens

B-Cell Receptors (BCR)

B cells and T cells recognize different substances as antigens and in a different form. The B cell uses cell surface-bound immunoglobulin as a receptor and the specificity of that receptor is the same as the immunoglobulin that it is able to secrete after activation. B cells recognize the following antigens in soluble form:

- Proteins (both conformational determinants and determinants exposed by denaturation or proteolysis)
- Nucleic acids
- Polysaccharides
- Some lipids
- Small chemicals (haptens)

T-Cell Receptors (TCR)

In contrast, the overwhelming majority of antigens for T cells are proteins, and these must be fragmented and recognized in association with MHC products expressed on the surface of nucleated cells, not in a soluble form. T cells are grouped functionally according to the class of MHC molecules that associate with the peptide fragments of the protein: helper T cells recognize only those peptides associated with class II MHC molecules, and cytotoxic T cells recognize only those peptides associated with class I MHC molecules

Lesson 68 : Introduction to Antigen Processing & Presentation:

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ANTIGEN PROCESSING AND PRESENTATION

Antigen processing and presentation are processes that occur within a cell that result in fragmentation (proteolysis) of proteins, association of the fragments with MHC molecules, and expression of the peptide- Major Histocompatibility Complex (MHC) molecules at the cell surface of antigen presenting Cells (APC) e.g Macrophages. where they can be recognized by the T cell receptor on a T cell. However, the path leading to the association of protein fragments with MHC molecules differs for class I and class II MHC.

MHC class I molecules present degradation products derived from intracellular (endogenous) proteins in the cytosol.

MHC class II molecules present fragments derived from extracellular (exogenous) proteins that are located in an intracellular compartment.

The presentation of processed antigen with MHC molecule to T-cell determine the function of T-cell too which is as follows

- Cytotoxic T-Cells: MHC I
- Helper T-Cells- MHC II

Lesson 69 : Antigen Processing & Presentation of Endogenous antigens:

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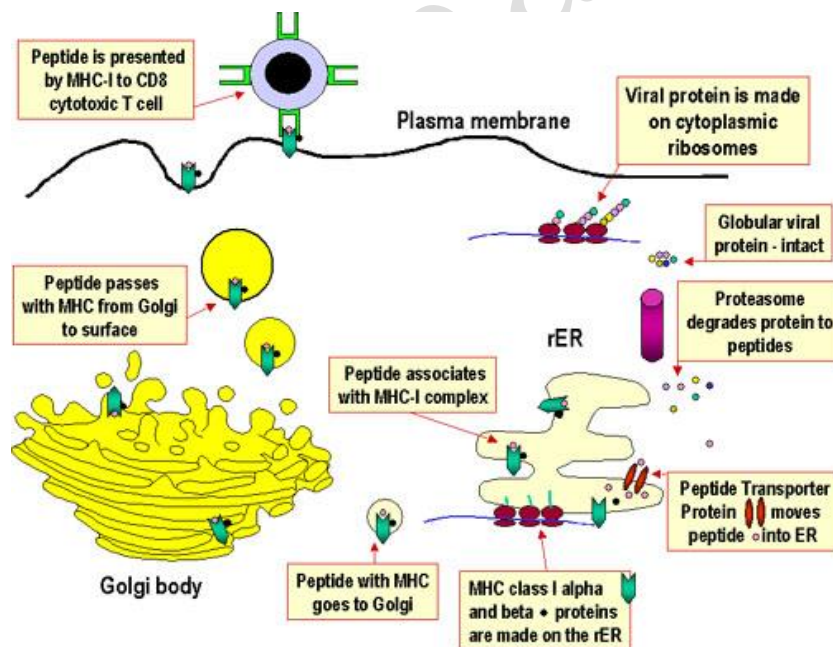
Antigen processing and presentation in cells expressing class I MHC

- The process in which the antigen is being processed is called as antigen processing. This process varies according to the source of antigen. The intracellular source of antigen which is also called as Endogenous antigen reside in the cytosol of antigen presenting cell like Viruses.
- For presenting such kind of endogenous antigens MHC I is required which is present in antigen presenting cells.

All nucleated cells express class I MHC. As shown in Figure, proteins are fragmented in the cytosol by proteasomes (a complex of proteins having proteolytic activity) or by other proteases. The fragments are then transported across the membrane of the endoplasmic reticulum by transporter proteins. (The transporter proteins and some components of the proteasome have their genes in the MHC complex).

Synthesis and assembly of class I heavy chain and beta2 microglobulin occurs in the endoplasmic reticulum.

Within the endoplasmic reticulum, the MHC class I heavy chain, beta2microglobulin and peptide form a stable complex that is transported to the cell surface.



Pathway of class I MHC restricted presentation of an endogenously synthesized antigen. An example of such an antigen would be a viral protein made in the cell as a result of infection

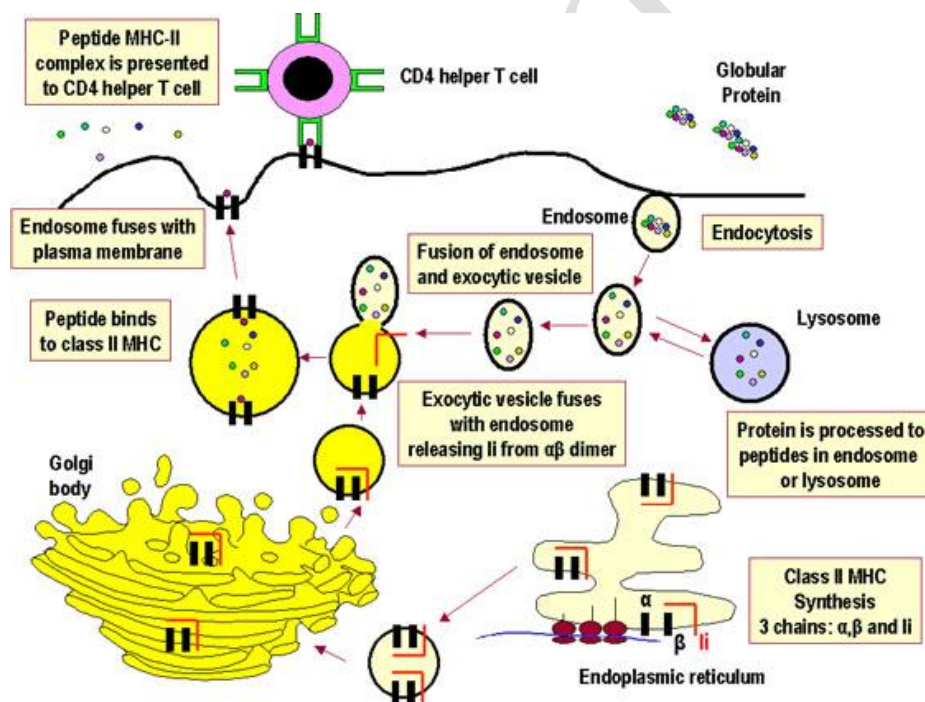
Lesson 70 : Antigen Processing & Presentation of Exogenous antigens:

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Antigen processing and presentation in cells expressing class II MHC

Whereas all nucleated cells express class I MHC, only a limited group of cells express class II MHC, which includes the antigen presenting cells (APC). The principal APC are macrophages, dendritic cells (Langerhans cells), and B cells, and the expression of class II MHC molecules is either constitutive or inducible, especially by interferon-gamma in the case of macrophages.

As shown in Figure, exogenous proteins taken in by endocytosis are fragmented by proteases in an endosome. The alpha and beta chains of MHC class II, along with an invariant chain, are synthesized, assembled in the endoplasmic reticulum, and transported through the Golgi and trans-Golgi apparatus to reach the endosome, where the invariant chain is digested, and the peptide fragments from the exogenous protein are able to associate with the class II MHC molecules, which are finally transported to the cell surface.



Pathway of class II MHC-restricted presentation of an exogenous antigen

Lesson 71 : Major Histocompatibility Complex (MHC):

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Major Histocompatibility Complex (MHC)

Introduction to MHC

Cell-cell interactions of the adaptive immune response are critically important in protection from pathogens. These interactions are orchestrated by the immunological synapse whose primary components are

1. T cell antigen receptor (TCR)
2. Major histocompatibility complex (MHC) molecule

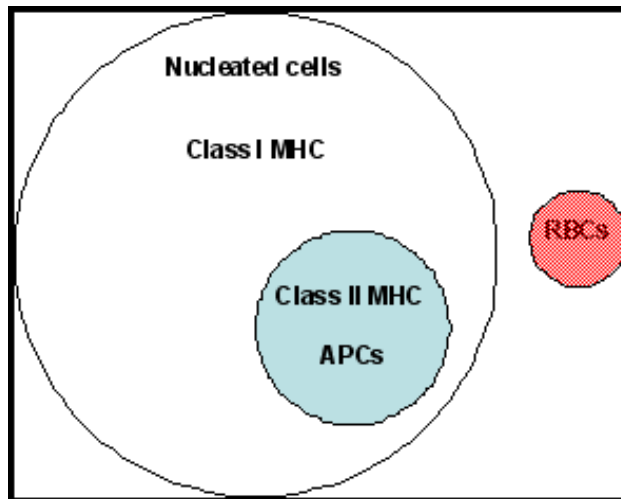
The major function of the TCR is to recognize antigen in the correct context of MHC and to transmit an excitatory signal to the interior of the cell. Since binding of peptide within the MHC is not covalent, there are many factors which help stabilize the immunological synapse.

Gene products encoded in the MHC were first identified as being important in rejection of transplanted tissues. Furthermore, genes in the MHC were found to be highly polymorphic (i.e. in the population there were many different allelic forms of the genes).

Studies with inbred strains of mice showed that genes in the MHC were also involved in controlling both humoral and cell-mediated immune responses. For example, some strains of mice could respond to a particular antigen but other strains could not and these strains differed only in one or more of the genes in the MHC.

Subsequent studies showed that there were two kinds of molecules encoded by the MHC – Class I molecules and class II molecules which are recognized by different classes of T cells.

1. Class I molecules were found on all nucleated cells (not red blood cells)
2. class II molecules were found only on antigen presenting cells, (APCs) which included dendritic cells, macrophages, B cells and a few other types (Figure).



Distribution of class I and class II MHC molecules on human cells

It was not until the discovery of how the TCR recognizes antigen that the role of MHC genes in immune responses was understood. The TCR was shown to recognize antigenic peptides in association with MHC molecules. T cells recognize portions of protein antigens that are bound non-covalently to MHC gene products.

- Cytotoxic T cells (T_c) recognize peptides bound to class I MHC molecules
- Helper T cells (T_h) recognize peptides bound to class II MHC molecules.

Lesson 72 : Structure of Class I MHC molecules:

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STRUCTURE OF CLASS I MHC MOLECULES

The molecule

Class I MHC molecules are composed of two polypeptide chains,

- a long α chain and
- a short β chain called β 2-microglobulin

The α chain has four regions.

- A cytoplasmic region, containing sites for phosphorylation and binding to cytoskeletal elements.
- A transmembrane region containing hydrophobic amino acids by which the molecule is anchored in the cell membrane.
- A highly conserved α 3 immunoglobulin-like domain to which CD8 binds.
- A highly polymorphic peptide binding region formed from the α 1 and α 2 domains. The β 2- microglobulin associates with the α chain and helps maintain the proper conformation of the molecule.

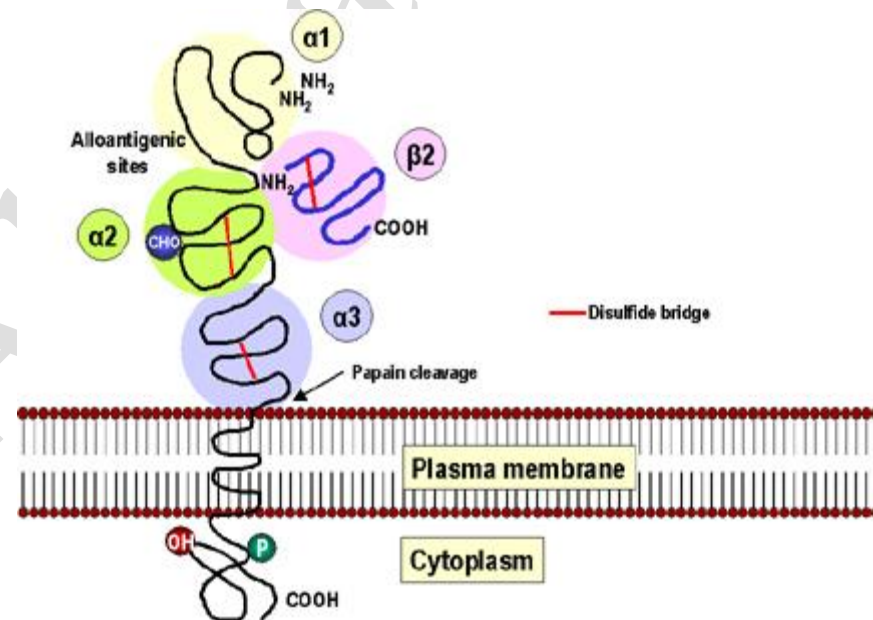


Figure: The MHC class 1 molecule has three globular domains alpha 1 (yellow), alpha 2 (green) and alpha 3 (blue). The alpha 3 domain is closely associated with the non-MHC -encoded beta 2

microglobulin (pink). The latter is stabilized by a disulfide bridge (red) and is similar to an immunoglobulin domain in three-dimensional structure. The alloantigenic sites which carry determinants specific to each individual are found in the alpha 1 and 2 domains. The latter also has a carbohydrate chain (blue, CHO). There is a phosphate in the cytoplasmic domain. Papain cleaves near the outer surface of the plasma membrane

The antigen-binding groove

The structure of the peptide binding groove, revealed by X-ray crystallography, shows that the groove is composed of two α helices forming a wall on each side and eight β -pleated sheets forming a floor.

The peptide is bound in the groove and the residues that line the groove make contact with the peptide (Figure). These are the residues that are the most polymorphic. The groove will accommodate peptides of approximately 8-10 amino acids long.

Whether a particular peptide will bind to the groove will depend on the amino acids that line the groove.

Because class I molecules are polymorphic, different class I molecules will bind different peptides. Each class I molecule will bind only certain peptides and will have a set of criteria that a peptide must have in order to bind to the groove.

For every class I molecule, there are certain amino acids that must be at a particular location in the peptide before it will bind to the MHC molecule. These sites in the peptide are referred to as the “**anchor sites**”. The ends of the peptide are buried within the closed ends of the class I binding groove while the center bulges out for presentation to the TCR.

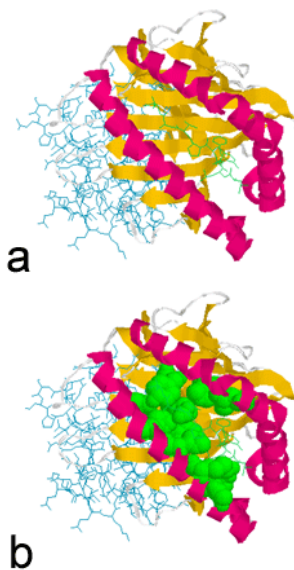


Figure a. Peptide binding groove of class I MHC molecules.
b. Groove with highlighted highly variable residues. The variable residues are clustered around the peptide-binding pocket

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Lesson 73 : Structure of Class II MHC molecules:

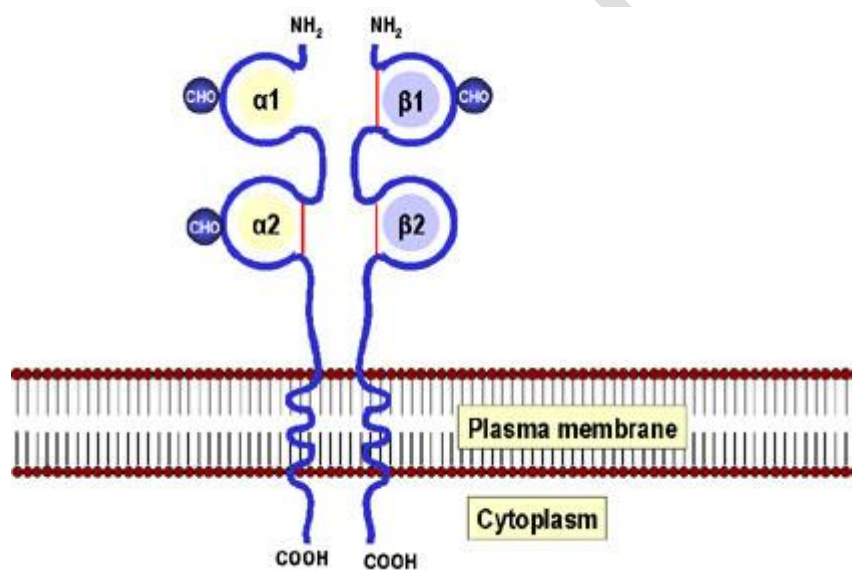
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STRUCTURE OF CLASS II MHC MOLECULES

The molecule

Class II MHC molecules are composed of two polypeptide chains an α and a β chain of approximately equal length (Figure). Both chains have four regions:

- A cytoplasmic region containing sites for phosphorylation and binding to cytoskeletal elements
- A transmembrane region containing hydrophobic amino acids by which the molecule is anchored in the cell membrane
- A highly conserved $\alpha 2$ domain and a highly conserved $\beta 2$ domain to which CD4 binds
- A highly polymorphic peptide binding region formed from the $\alpha 1$ and $\beta 1$ domains



The antigen-binding groove

The structure of the peptide binding groove, revealed by X-ray crystallography, shows that, the groove is composed of two α helices forming a wall on each side and eight β -pleated sheets forming a floor.

Both the $\alpha 1$ and $\beta 1$ chain contribute to the peptide binding groove. The peptide is bound in the groove and the residues that line the groove make contact with the peptide. These are the residues that are the most polymorphic.

The groove of Class II molecules is open at one end so that the groove can accommodate longer peptides of approximately 13-25 amino acids long with some of the amino acids located outside of the groove. Whether a particular peptide will bind to the groove will depend on the amino acids that line the groove. Because class II molecules are polymorphic, different class II molecules will bind different peptides. Like class I molecules, each class II molecule will bind only certain peptides and will have a set of criteria that a peptide must have in order to bind to the groove (i.e. “anchor sites”).

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Lesson 74 : Important Aspects of MHC:

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IMPORTANT ASPECTS OF MHC

- Each MHC molecule has only one binding site. The different peptides a given MHC molecule can bind all bind to the same site, but only one at a time.
- Because each MHC molecule can bind many different peptides, binding is termed degenerate.
- MHC polymorphism is determined only in the germline. There are no recombinational mechanisms for generating diversity.
- MHC molecules are membrane-bound; recognition by T cells requires cell-cell contact.
- Alleles for MHC genes are co-dominant. Each MHC gene product is expressed on the cell surface of an individual nucleated cell.
- A peptide must associate with a given MHC of that individual, otherwise no immune response can occur. That is one level of control.
- Mature T cells must have a T cell receptor that recognizes the peptide associated with MHC. This is the second level of control.
- Cytokines (especially interferon- γ) increase level of expression of MHC.
- Peptides from the cytosol associate with class I MHC and are recognized by Tc cells. Peptides from within vesicles associate with class II MHC and are recognized by Th cells.
- Polymorphism in MHC is important for survival of the species.

Lesson 75 : Role of MHC in Tissue Matching:

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Role of MHC in Tissue Matching

MHC has very important role in tissue transplantation from one individual to other. The individual who donates the tissue is called as donor while the individual who receive the tissue is called as recipient. The prospective donor & recipient are tested for compatibility prior to transplantation. That's why MHC is also called as Human leucocyte antigen (HLA). These molecules are the primary target of immune responses against allogeneic transplants Moreover, HLA molecules are highly diverse in human population. HLA typing or tissue matching detects & classifies this diversity. There is extensive polymorphism in HLA that's why there is less chances of HLA matched donors for transplants. Approximately 25% of siblings inherit same HLA type

Lesson 76 : Self MHC Restriction:

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SELF MHC RESTRICTION

In order for a T cell to recognize and respond to a foreign protein antigen, it must recognize the MHC on the presenting cell as self MHC. This is termed self MHC restriction.

- Helper T cells recognize antigen in context of class II self MHC.
- Cytolytic T cells recognize antigen in context of class I self MHC.

The process whereby T cells become restricted to recognizing self MHC molecules occurs in the thymus.

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Lesson 77 : Differences Between Monoclonal & Polyclonal Antibodies:

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Monoclonal & Polyclonal Antibodies

Differences B/W Monoclonal & Polyclonal Antibodies

Monoclonal Antibodies

The antibodies which are produced from a single antibody producing B-cell are called as monoclonal antibodies. These antibodies also able to bind with single and unique antigen binding site (epitope). Monoclonal antibodies mainly consist of single subtype of IgG for example IgG1, IgG2 & IgG3.

Polyclonal Antibodies

The collection of antibodies from a different antibody producing B lymphocytes are called as polyclonal antibodies. These antibodies also able to bind with multiple epitopes on a same antigen. These antibodies are obtained from serum which has antibodies having different affinities. Polyclonal antibodies mainly belong to IgG class.

Lesson 78 : Generation of Polyclonal Antibodies:

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Generation of Polyclonal Antibodies

The general procedure for generation of polyclonal antibodies include following phases

1. Antigen preparation/Antigen & Adjuvant conjugate

In this phase, the peptides of antigens are synthesized. The peptides of various length are designed for effective immune response generation. Moreover, the peptides are also conjugated with other molecules like carbohydrate & lipids in order to produce more diverse immune response.

2. Antigen Immunization

After peptide synthesis, these are administered in animal like rabbit for generating immune response. The antigen immunization can last for 6-8 rounds depending upon the levels of generated antibodies which are detected through ELISA.

3. Antibody Purification from animal's serum

As immune response is generated in the form of antibodies after challenging with antigens, those antibodies are purified from animal's serum through various methods. These methods usually based on antigen affinity or specific affinity. This phase lasts from 1-2 weeks.

4. Antibody Validation

In this phase, antibodies are validated for their specificity and function. Moreover, the support and services for the purpose of commercialization are being done.

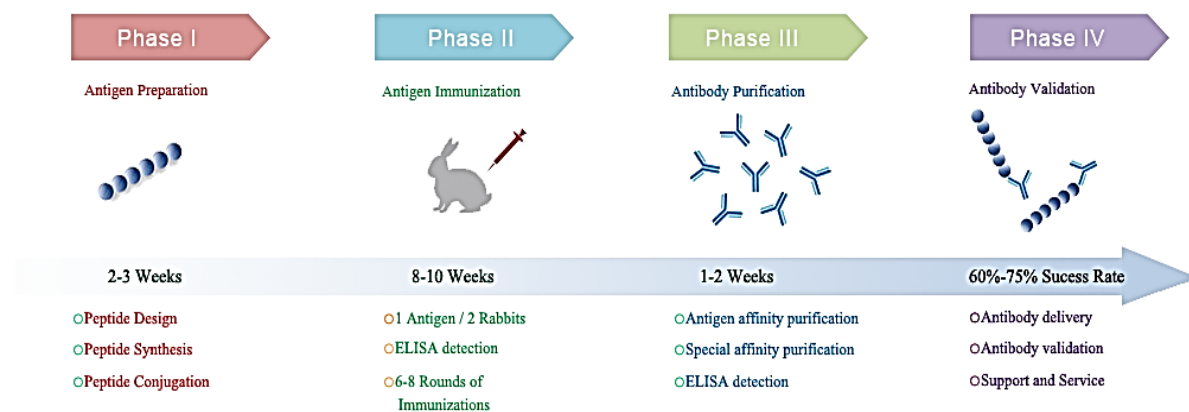


Figure: Phases of generation of polyclonal antibodies

Lesson 79 : Role of Adjuvants in Polyclonal Antibodies Production:

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Role of Adjuvants in Polyclonal Antibodies Production

Adjuvants are the substances which potentiates immune response against antigens. These have the ability to modulate towards desired immune response. However, with adjuvants there are mostly undesirable effects like cellular toxicity. That's why the selection of adjuvant is being made in order to get maximum immunostimulation. For the production of polyclonal antibodies following adjuvants are used

1. Freund's Complete Adjuvants
2. Freund's Incomplete Adjuvants

Freund's Complete Adjuvants (FCA)

Freund's complete adjuvant (FCA) are water-in-oil emulsion containing antigens with heat killed Mycobacterium tuberculosis. These adjuvants have the ability to stimulate both humoral & cell mediated immune response for antibody production.

Freund's Incomplete Adjuvants

These adjuvants are also water-in-oil emulsion containing antigens without heat killed Mycobacterium tuberculosis. These are used as booster antigen dose. These are used by mixing equal parts of the antigens.

Lesson 80 : Production of Monoclonal Antibodies:

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The traditional monoclonal antibody (mAb) production process usually starts with generation of mAb-producing cells (i.e. hybridomas) by fusing myeloma cells with desired antibody-producing splenocytes (e.g. B cells). These B cells are typically sourced from animals, usually mice. After cell fusion, large numbers of clones are screened and selected on the basis of antigen specificity and immunoglobulin class.

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Monoclonal Antibody Production

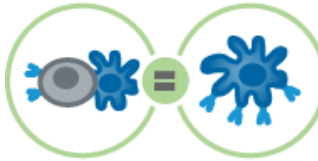
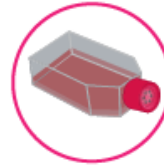


Immunization of mice & isolation of splenocytes

Mice are immunized with an antigen and later their blood is screened for antibody production. The antibody-producing splenocytes are then isolated for *in vitro* hybridoma production.

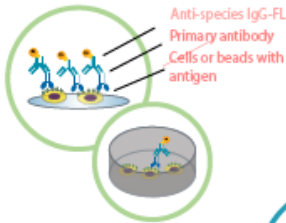
Preparation of myeloma cells

Myeloma cells are immortalized cells that, once fused with spleen cells, can result in a hybridoma capable of unlimited growth. Myeloma cells are prepared for fusion.



Fusion

Myeloma cells and isolated splenocytes are fused together to form hybridomas in the presence of polyethylene glycol (PEG), which causes cell membranes to fuse.



Clone screening and picking

Clones are screened and selected on the basis of antigen specificity and immunoglobulin class.

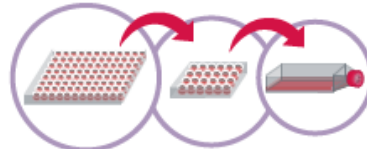


Functional characterization

Confirm, validate, and characterize (e.g. ELISA) each potentially high-producing colony.

Scale up and wean

Scale up clones producing desired antibodies and wean off selection agent(s).



Expansion

Expand clones producing desired antibodies (e.g. bioreactors or large flasks).



Lesson 81 : Hybridoma Formation:

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Hybridoma Formation

The fusion of lymphoid tumor cell with normal B-Lymphocyte with single specificity hybrid cell line is called as hybrid cell line. This hybrid cell line has dual function.

1. Immortality of tumor cell
2. The production of antibody with single specificity

In hybridoma formation, following principle involves

Animal (mouse) is being injected with a specific antigen for a specific time point. After that, Blymphocytes are isolated and sorted from the spleen of mouse. In the next phase, the Blymphocytes are fused with lymphocyte immortal cell which are myeloma cell. These cells provide the immortality to fused B-cells. The fusion between these two cells leads towards the generation of fused cell called as hybridoma. After fusion, the selection Hybridoma cell is usually done by using HAT medium (Hypoxanthine, aminopterin, thymidine) which allows the growth of hybridoma cells. While unfused cells undergo programmed cell death (apoptosis).

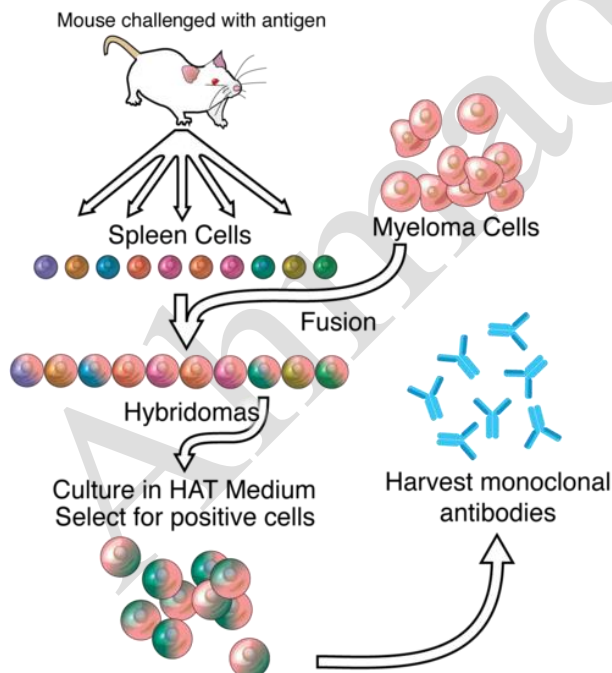


Figure: The process of hybridoma generation

The selected hybridoma cells are harvested for production of monoclonal antibodies. For large amount of monoclonal antibodies, hybridoma cells are propagated.

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Lesson 82 : Usage of Monoclonal Antibodies:

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Usage of Monoclonal Antibodies

Monoclonal antibodies have very significant applications in clinical and basic medical sciences which are as follows

1. Usage in prevention, diagnosis & treatment of disease
2. Immunophenotyping of Immune cells, Monoclonal antibodies are used against cell surface molecules like Cluster of Differentiation (CD)
3. Typing of various tumors e.g Leukemia
4. Immunohistochemistry of solid tumors
5. Use in Immunotherapy: Against various tumor marker e.g CD20 B-Cell lymphoma
6. Use in Fluorescence Activated cell sorting (FACS)

Lesson 83 : T-Lymphocyte Receptors, Maturation, Activation & Differentiation:

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STRUCTURE OF THE T CELL RECEPTOR (TCR)

The surface receptor which is present on T-lymphocytes for binding with antigen bounded with major histocompatibility complex (MHC) is called as T-cell receptor (TCR). This receptor is just like immunoglobulin in its structure, all those structures which resemble with immunoglobulin structures are called as immunoglobulin superfamily. That's why TCR is the part of immunoglobulin superfamily. TCR is a heterodimer surface receptor

- The TCR is a heterodimer composed of one α and one β chain of approximately equal length (Figure 8).
- Each chain has a short cytoplasmic tail but it is too small to be able to transduce an activation signal to the cell.
- Both chains have a transmembrane region comprised of hydrophobic amino acids by which the molecule is anchored in the cell membrane.
- Both chains have a constant region and a variable region similar to the immunoglobulin chains.
- The variable region of both chains contains hypervariable regions that determine the specificity for antigen. Each T cell bears a TCR of only one specificity (*i.e.* there is allelic exclusion).

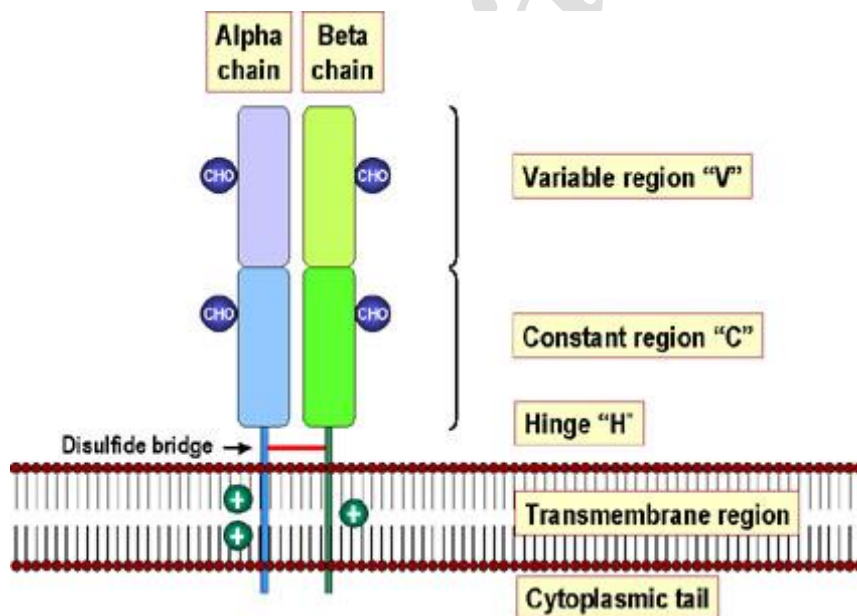


Figure: The T cell receptor heterodimer comprises two transmembrane glycoproteins, the alpha and beta chains. There are two domains in the external part of each chain and these resemble

immunoglobulin variable and constant regions. There are sugar chains on each domain. There is a short sequence similar to the immunoglobulin hinge region that connects the immunoglobulin-like domains to the transmembrane sequence. This contains cysteines that form a disulfide bridge. The hydrophobic transmembrane helical structures are unusual in that they contain positively charged amino acids (basic amino acids). The alpha chain has two positively charged residues while the beta chain has one.

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Lesson 84 : Diversity of T-Cell Receptor (TCR):

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Diversity of T-Cell Receptor (TCR)

The diverse array of TCR is based on the gene which encodes for TCR. Structurally, the TCR gene is composed of various gene segments which are joined together. T-cell maturation occurs in an organ which is called as thymus where the expression of the gene for TCR occurs. The germ line gene for α chain contains only two segments which are called as V (Variable) & J (Joining) segments. However, the germ line gene for $\alpha\beta$ chain is composed of three segments called as V (variable), D (diversity) & J (Joining) segments. The diversity for specificity of TCR is based on combinatorial diversity which in fact is based on various combinations of V, D & J segments for TCR. In majority, the specificity of TCR is based on the combination of α & β chains which constitutes the major T-cell population. However, the small population T-cells also exist which carry the TCR with γ (Gamma) & δ (Delta) chains. This subset of T-cells is called as $\gamma\delta$ T-cells. Usually the T-cells predominate in mucosal epithelium for elimination of certain bacterial & viral antigens in MHC dependent antigen processing. This predominant subset of T-cells belongs to $\alpha\beta$ variety.

$\gamma\delta$ T-cells also contain the repertoire which is also based on combinatorial diversity of various gene segments. The Gamma (γ) chain gene contains V & J segment, while the Delta (δ) chain germline gene contains V, D & J segments. The $\gamma\delta$ T-cells recognize antigens independent of MHC association in contrast with $\alpha\beta$ variety.

Lesson 85 : CD3 complex Structure & Functions:

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CD3 complex Structure & Function

The CD3 stands for Cluster of differentiation 3. This structure is present adjacent to TCR. This is basically the transducing element of TCR. It exists as a complex form which is composed of following proteins (as shown in fig), that's why it is also called as CD3 complex

1. One Gamma(γ)
2. One delta(δ)
3. Two Epsilon(ϵ)
4. Two Zeta(ζ)

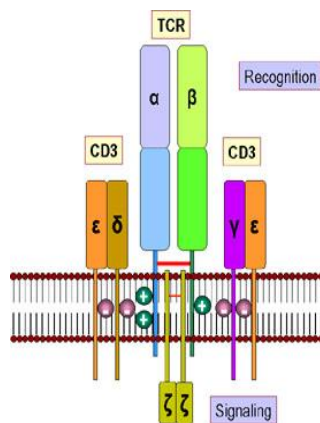


Figure: The structure of CD3 complex: The receptor for antigens on the T cell surface comprises eight proteins.

(a) Two disulfide-bonded chains of the T cell receptor which form a heterodimer. These recognize peptides associated with MHC molecules.
(b) Four chains, collectively termed CD3, that associate with the T cell receptor dimer and participate in its transport to the surface of the cell. The CD3 complex together with the zeta chains, which form a homodimer, transduce the signal after antigen has bound

The CD3 complex is considered as invariant proteins means with conserved regions without variability. It has no role in determining the specificity of T-Cell. However, it is necessary for surface expression of TCR during development of T-cell. This has very important role in transducing signals from T-cell surface receptor to interior of cell which is called as Signal transduction. This transduction signals arise after engaging of antigen with TCR in association with MHC complex.

Lesson 86 : Cell Surface Molecules involved in T-Cell & Other Interaction:

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Cell Surface Molecules involved in T-Cell & Other Interaction

The interface between antigen presenting cell (Target Cell) and T-lymphocyte occur for mounting a strong immune response against antigen. This interaction between TCR & MHC molecules are not so strong and called as immunological synapse. In this immunological synapse, the cell surface molecules on T-cells & their interacting molecule on antigen presenting cells are involved. T-cells also express co-receptor for MHC molecule (e.g CD8 & CD4). Moreover, the expression of supporting molecules are increased by the action of cytokines released from either T-cells or antigen presenting cells. The increased expression of supporting molecules for making immunological synapse strong are called as modulators of immune system. In addition to these molecules, some molecules are also required for T-cell activation. For the activation of T-cells following two signals are required

1. Signal 1 is generated after engagement of TCR with Antigen/MHC complex
2. Signal 2 is generated after engagement of supporting molecules with their corresponding ligands

For signal 2, the co-stimulatory molecules are also required for activation of T-cells e.g C28 is a co-stimulatory molecule on T-cell surface & its ligand exists on APC which is known as B7-1. If there is lack of co-stimulation, then there is no activation of T-cells occur and this process is called as cellular anergy. For the full activation of T-cells, the engagement of Immunological synapse is mandatory. As a result the antigen presenting cells (APC) must possess & present peptides to T-cells.

Lesson 87 : Accessory Molecules:

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Accessory Molecules

The interaction between TCR on T-cells & antigen/MHC complex on APC is considered as weak for mounting a strong immune response. In order to stabilize this interaction, there is the need of additional molecules in addition to TCR & antigen/MHC which are called as accessory molecules. These molecules are invariant molecules and have no role in determining the specificity of T-cells. However, these accessory molecules can modulate the immune system in either positive or negative manner. These molecules are critical to success or failure of controlling the immune response to foreign antigen like infectious agents. Moreover, these are also involved in aberrant response to self-antigens like in autoimmune diseases and response to tumors or cancerous cells. These molecules mainly promote or suppress the immune response. The nomenclature of these molecules are according to their function.

Followings are the important accessory molecules (Fig 1 & 2)

1. CD4

This molecule of T-cell is for binding to class II MHC which ensures the binding of T-helper (Th) cells to antigen presenting cells

2. CD8

This accessory molecule is for binding to Class I MHC which ensures the binding of cytotoxic T (Tc) cells to target cell

3. CD2

This accessory molecule on the surface of T-cell has a ligand known as Leucocyte Function Antigen 3 (LFA-3) on antigen presenting cells (APC)

4. Leucocyte Function Antigen 1 (LFA-1)

This molecule has ligand known as Intracellular Adhesion Molecule-1 (ICAM-1)

5. CD28

CD28 is the accessory molecule on T-cell and has ligand called as B7-1/B7-2

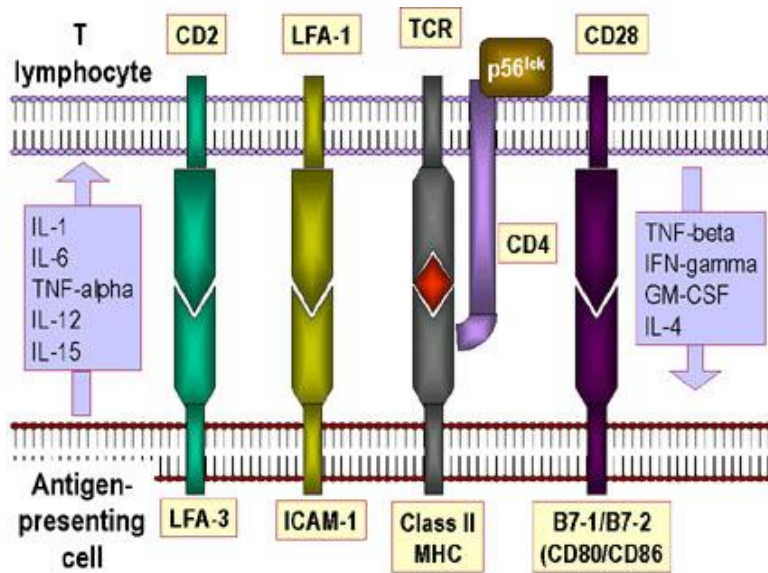


Figure 1: Accessory molecules of Helper T-cells & their specific ligands on APC: Some cytokines produced by each cell type are shown

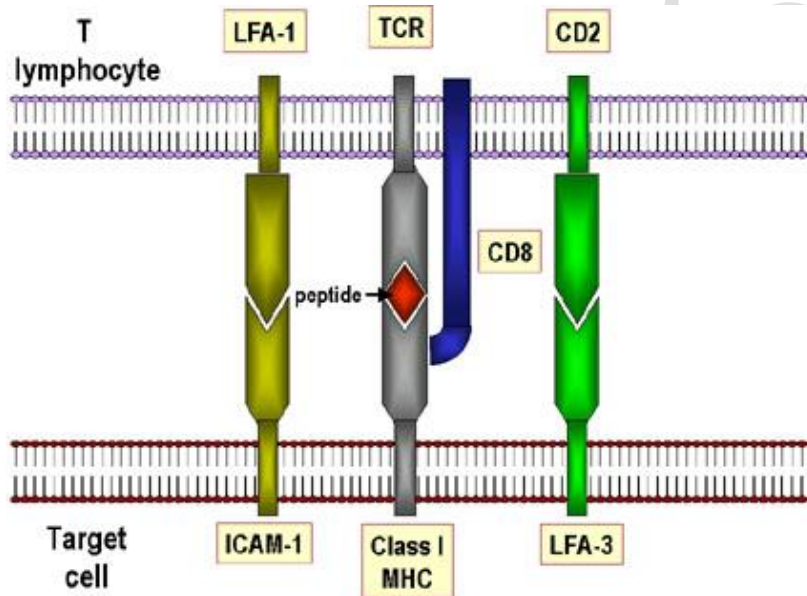


Figure 2: Accessory molecules of Cytotoxic T-cells & their specific ligands on APC

Lesson 88 : Co-Stimulatory Molecules for Activation & Maturation of T-Cells:

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T-cell activation

The engagement of TCR with MHC containing antigen on antigen presenting cell. Furthermore, the engagement of co-stimulatory & accessory molecules leads towards the secretion of cytokines after T-cell activation. These cytokines causes the maturation & differentiation of T-Cells (as shown in fig 1)

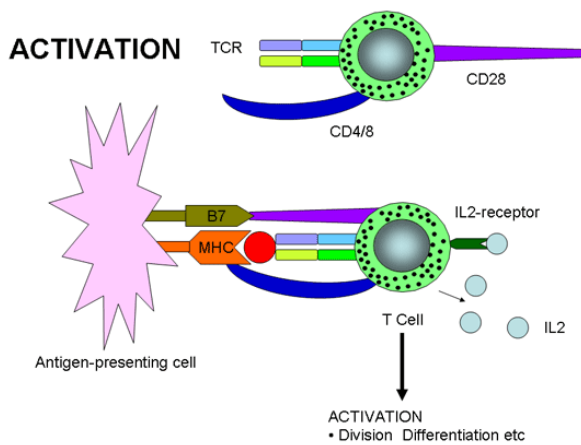


Figure 1: The process of T-cell activation

The lack of co-stimulation even after engagement of TCR & antigen/MHC molecule leads the T-cell towards a state called as anergy. This is non-responsiveness of T-cells towards antigen as a result no T-Cell activation occur (Fig 2)

ANERGY: Engagement of TCR and Ag/MHC in absence of co-stimulation can lead to anergy

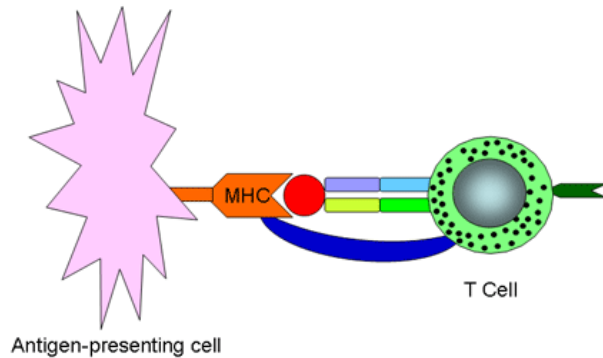


Figure 2: Anergy of T-Cells

The lack of co-stimulation also leads to down regulation of T-Cell activation which is due to absence of CD28. There is expression of CTLA-4 which engages CD28 molecule. This cause no responsiveness of T-cell to antigen. Finally, there is no T-Cell activation (Fig 3)

DOWN REGULATION CTLA4 interacts with B7

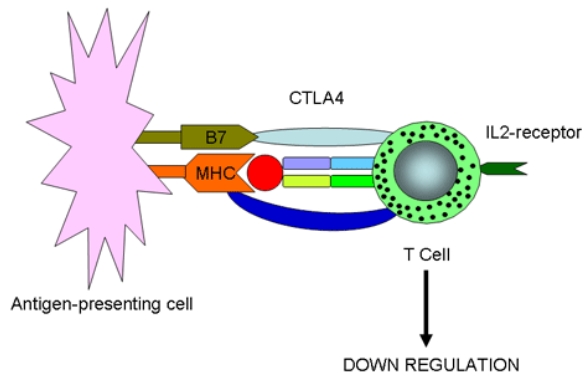


Figure 3: Inactivation of T-cell due to down regulation of accessory molecule CD28

No Engagement of TCR, MHC with antigen, interaction of CD28 with B7. As a result, there is no signal for activation of T-cells (Fig.4)

ENGAGEMENT OF CO-STIMULATORY MOLECULES IN THE ABSENCE OF TCR: NO RESPONSE

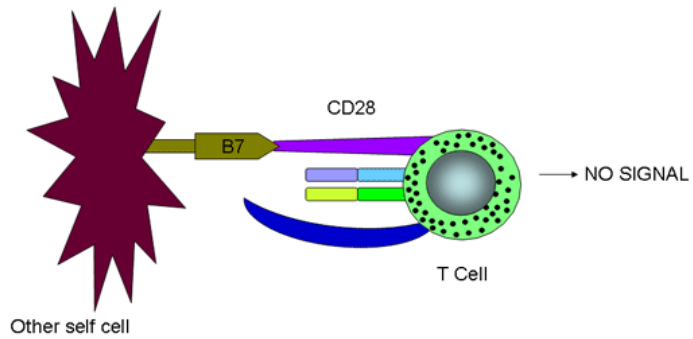


Figure 4: Inactivation of T-cells due to no engagement of MHC

Lesson 89 : B-Lymphocytes Receptors, Maturation, Activation & Differentiation:

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B-Lymphocytes Receptors, Maturation, Activation & Differentiation

Antigens Responding to B-Cells

B-cells are stimulated only by antigens which are called as T-Independent Antigens because these antigens have the ability to stimulate the B-cells directly without the help of T-lymphocytes. Chemically, these antigens are polysaccharides in nature for example the antigens of *Streptococcus Pneumococci*. These antigens exist in a polymeric structure which mean that antigens are characterized by the same antigenic determinants (epitopes). These antigenic determinates are repeated many times on a single antigen (Fig.1)

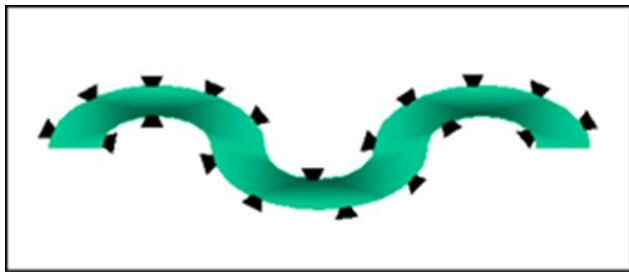


Figure 1: Structure of T-independent antigen with antigenic determinants (black in color)

T-independent antigens have the ability to activate the various clones of B-Cells at a same time. This kind of antigenic activation is called as polyclonal activation. Most of these antigens can activate B-cell clones specific for other antigens. T-Independent antigens can be classified broadly in to two main following types based on their ability to activate B-cells

1. Type 1 T-independent antigens

These antigens are polyclonal activator as these have the ability to activate a large number of B-cell clones

2. Type 2 T-independent antigens

These antigens are not polyclonal activator of B-cells in contrast to type-1 antigens. These can activate usually few clones of B-lymphocytes.

T-independent antigens are more resistant to degradation by phagocytic cells. That's why such antigens can persist for longer time in the body for providing effective immune response for longer time points. Moreover, these antigens are also involved in consistent stimulation of

immune system. These antigens also have the attribute of mitogenicity for various immune cells in order to increase the number of cells accordingly.

Like T-independent antigens, T-dependent antigens can activate the B-cells not directly but indirectly with the help of T-cells for B-cell activation. T-dependent antigens didn't fulfil the molecular requirements for direct stimulation of B-cells that's why help of T-cells is required. These T-dependent antigens readily degraded by phagocytes afterward these are presented to T-cells with class II MHC for B-cells activation. These antigens are protein in nature unlike T-independent antigens. Moreover, these antigens contain variety of epitopes with few copies on the surface of antigen (Fig.2).

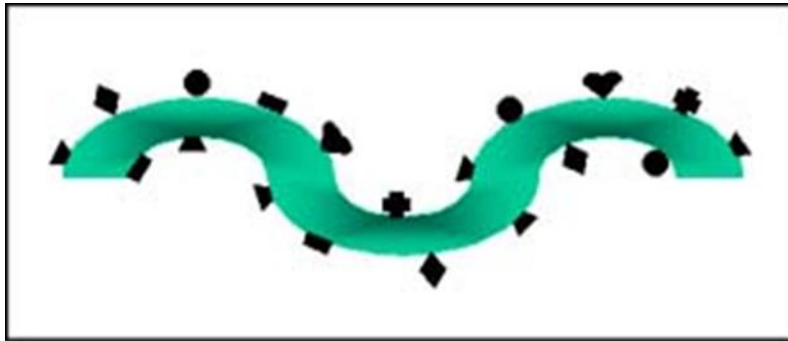


Figure 2: Structure of T-dependent antigen with antigenic determinants (in black color)

Lesson 90 : Antigens Processing by B-Cells:

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Antigens Processing by B-Cells

For T-independent antigens

These antigens have the ability to stimulate the B-cells directly. These also have the ability to activate a substantial proportion of the B-cell pool. This kind of B-cell activation is polyclonal activation that is without reference to antigen specificity of the surface receptor hypervariable regions for example the type I T-independent antigens. B-cells are activated when there is high concentration of antigen for example in case of bacteria the lipopolysacchrides antigens exist in greater quantity. B-cell activation occur after binding of antigen to a surface molecule which is known as B-cell receptor. Afterward there is bypassing of the early part of the biochemical pathway mediated by specific antigen receptor. Cross linking of immunoglobulin receptor occur due to repeating determinants on antigenic structure. As a result the transformation of B-cells occur into plasma cell for immunoglobulin secretion (Fig.1)

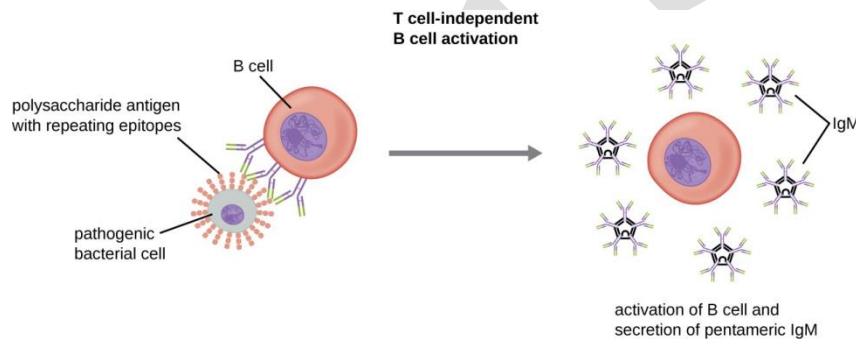


Figure 1: Antigen processing by B-cells for T-independent antigens

T-dependent antigens

In case of T-dependent antigens, there is no direct stimulation of B-cells without the help of T-cells. The T-dependent antigens are recognized by surface immunoglobulin receptor known as BCR. These antigens are univalent with respect to specificity of each determinant on the surface of antigen. The antigens are internalized within endosomes of B-cells where those are processed by fusion with lysosomes into simple peptides. Afterward, the processed peptides are expressed on the surface of B-cells with Class II MHC molecule to Helper T-cells with co-stimulatory molecules. As a result, the clonal expansion & differentiation of B-cells into plasma cells occur (Fig.2).

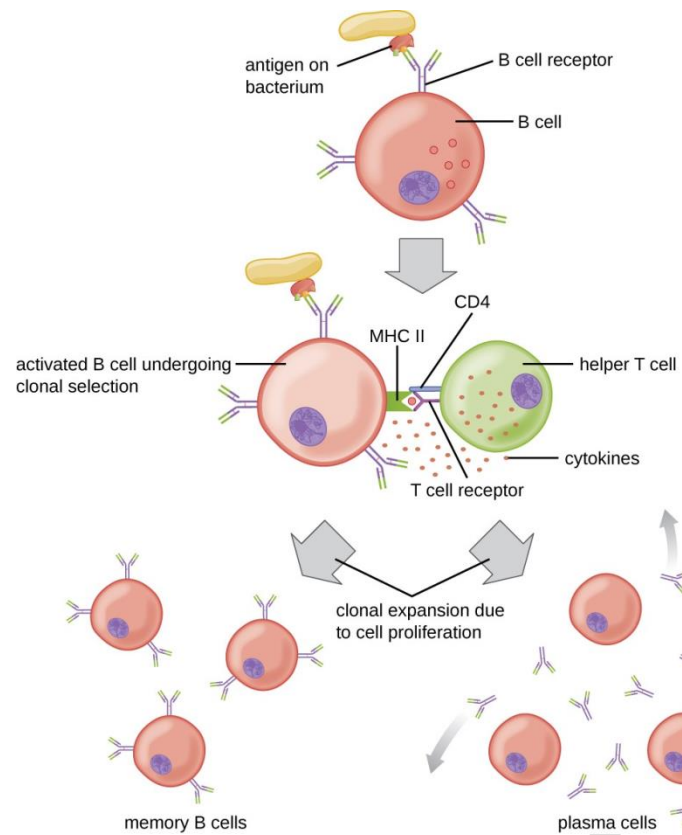


Figure 2: Antigen processing by B-cells for T-dependent antigens

Lesson 91 : Nature of B-Cell Activation:

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Nature of B-Cell Activation

The naïve or resting B-cells are non-dividing as like to T-cells. The B-cells Activation occur through BCR which is a surface molecule of B-cells. Like T-cell, BCR (surface immunoglobulin) doesn't process any intrinsic enzymatic activity which means there is need of certain other molecules for full activation of B-cells. These molecules are called as accessory molecules associated with antigen receptor. These molecules propagate activation signals into B-cells for differentiation into plasma cells for antibody secretion

Structure of B-Cell Receptor (BCR)

BCR exist as a complex structure. It is composed of membrane anchored immunoglobulin (IgM) in the cell surface of B-cell. Additionally, $Ig\alpha$ & $Ig\beta$ heterodimer exist in the chain forms which are associated with disulfide linkage between these two chains. The cytoplasmic tail of $Ig\alpha$ & $Ig\beta$ contain a single Immune receptor tyrosine based activation motif (ITAM). The cross linking of BCR through antigen results in the initiation of phosphotyrosine kinase (PTK) driven signals which are seeded by ITAM (Fig.1)

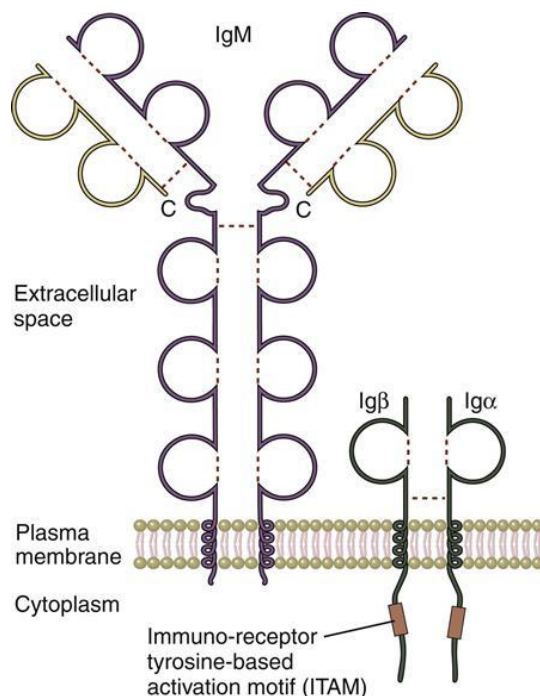


Figure 1: The structure of BCR

Like T-cells activation, the B-Cells also require co-stimulation to mount efficient effector responses. As like C28 in case of T-cells, a complex of co-stimulatory molecules perform this function. This complex of co-stimulatory molecules composed of following molecules (Fig.2)

1. CD81
2. CD21
3. CD19

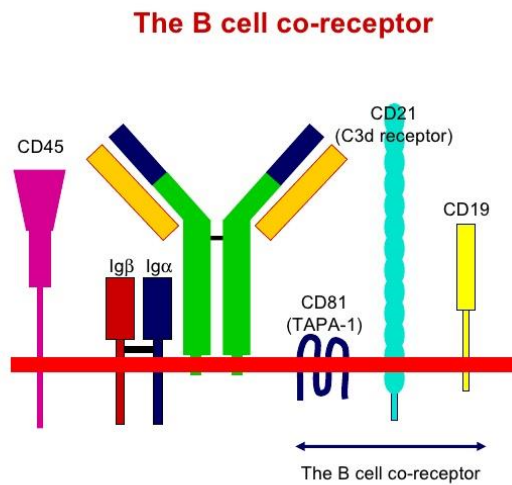


Figure 2: The structure of BCR with co-stimulatory complex

Lesson 92 : B-Cell Receptor & Co-stimulation For Maturation:

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B-Cell Receptor & Co-stimulation for Maturation

The process of activation of B-cells occur after cross linking of BCR through antigen. As the cross linking initiates, there is activation of a molecule called as Src for cellular signaling. The Src phosphorylates tyrosine on ITAM of $I\alpha$ & β . Then the phosphorylated ITAM act as docking site for Syk protein. After that, the downstream signaling for antibody synthesis occur. There are following two biochemical processes are activated during antibody synthesis

1. The Activation of Phospholipase C ($PLC\gamma$)
2. The Phosphorylation of adapter protein called as SLP which in turn cause the activation of Ras & Rac proteins

In turn, the mitogen activation proteins (MAP) for example JNK are activated. Afterward, the AP-1 moves to nucleus for B-cell proliferation & differentiation into plasma cells (Fig.1)

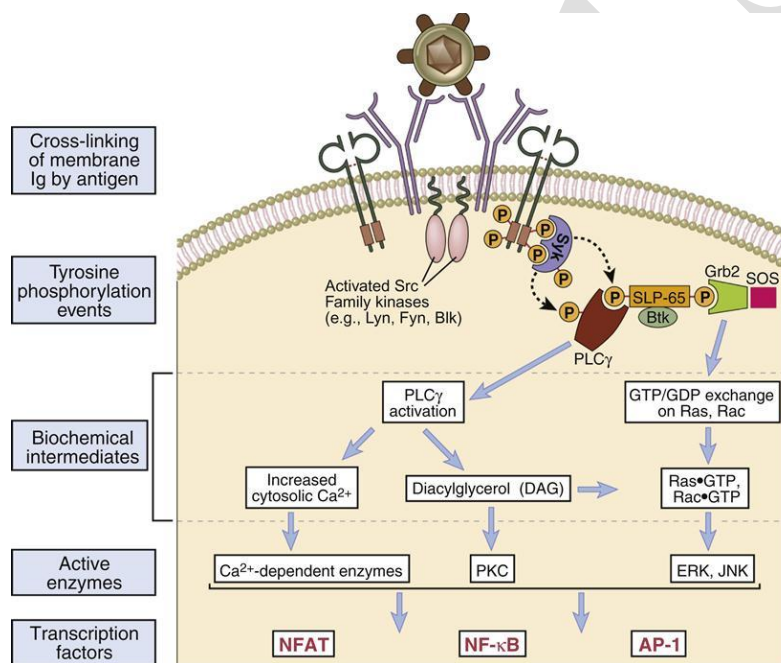


Figure 1: The signaling process of B-cell after antigenic stimulation

A second signal is also required after BCR engagement during which the accessory molecules are involved for amplification of B-cell signals. In this case, the antigen molecule is bounded with complement protein like C3d. The amplification process occur when multivalent antigen with C3d molecule engage the complement receptor-2 (CR2) with BCR. The Signals from

immunoglobulin & CR2 cause the phosphorylation of Syk protein. Moreover, another accessory molecule called as CD19 also reorients for PI-3 kinase for B-cell activation & differentiation (Fig.2).

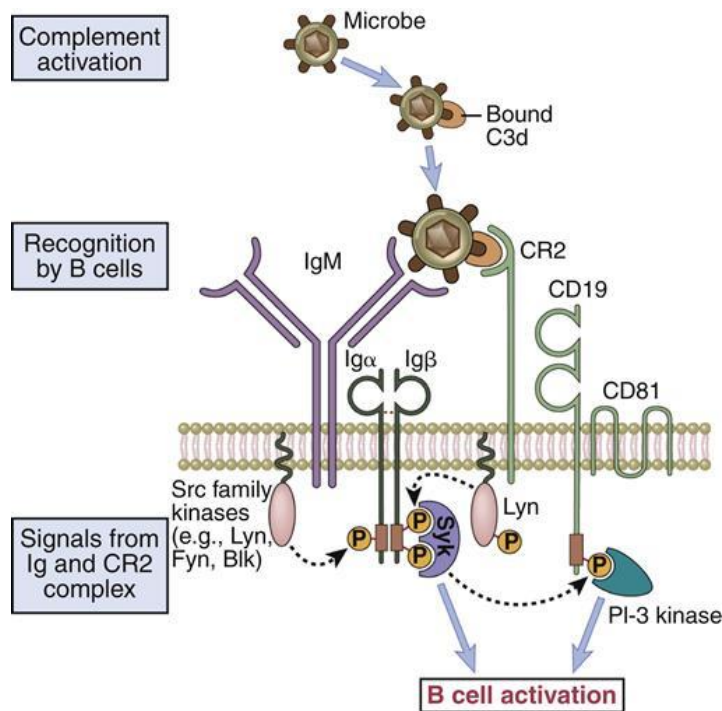


Figure 2: Activation of B-cell through accessory molecules like complement

Lesson 93 : Role of CD40 in B-Cell Activation:

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Role of CD40 in B-Cell Activation

The T-dependent antigens are univalent in nature with single type of antigenic determinant on it. For such antigens, the B-cells process & present such antigens. There is the process of up regulation of co-stimulatory molecule on B-cell which are known as CD40. The engagement of CD 40 occur with its ligand on T-cells which is called as CD40 ligand. For fully activation of B-cells there is also ensured co-stimulatory signaling in B-cells. In this case of T-dependent antigens, the co-stimulatory signal is the CD40 & CD40 ligand interaction. After that the cytokines are released from T-cell like IL-2/4/5 for acting on B-cell for activation (fig.1). These cytokines are involved in B-cell proliferation & differentiation into plasma cells. Moreover, the process of class switching and affinity maturation also occur for an effective humoral response. Clinically, the absence of CD40L due to defect in the gene of CD40 ligand on T-cells as a result there is absence of class switching process. This defect manifest the hyper IgM globunemia with no IgG due to absence of class switching process. Consequently, the patients having this defect present with recurrent infections and immunodeficiency.

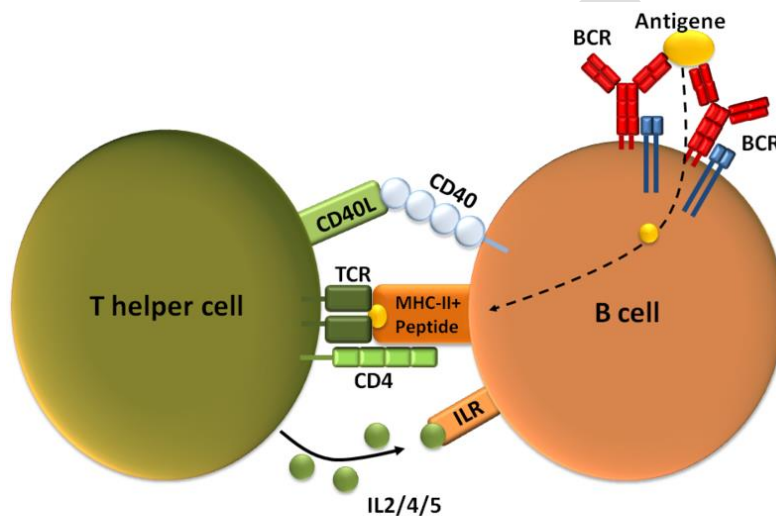


Figure 1: Activation of B-Cells for T-dependent antigen with CD40/CD40 ligands

Lesson 94 : Complement System Introduction:

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Introduction to Complement System

Complement system is comprised of heat labile serum proteins which are able to destroy or lyse pathogens. This form of host defense is termed as one way of host defense against potential pathogens. Complement proteins become inactive by heating serum at 56°C for 30 minutes. This system is composed of more than twenty (20) proteins. These proteins are produced by variety of body's cells including hepatocytes (Liver cells), macrophages & gut epithelial cells. Among complement proteins, some proteins has the ability to bind with immunoglobulin molecules while some are involved in binding on the membrane component of various immune cells. Most of complement proteins exist in inactive form which are also called as Proenzymes. Such proteins need activation into active form for proper functioning. These proteins when activated can cleave one or more other complement proteins. Such kind of activation process is called as cascading for complement activation.

BT302 - Immunology

**Teacher: Dr Numan Javed (Phd Holder at SHanghai
Institute of Sciences, China)**

1 to 167 Lectures merged

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Lesson 95 : Functions of Complement Proteins:

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Functions of Complement Proteins

Complement system provides specific as well as non-specific resistance against infections. This kind of resistance is exhibited with following two important functions

- Primary function

This function of complement proteins is considered primary as these activated proteins used to kill or lyse infectious agents like bacteria. For this complement proteins get deposited on the surface of infectious agents which make hole in those cells for their lysis or killing.

- Secondary or Effector functions

These functions of complement proteins effect on other functions of immune system. These functions of complement proteins are as follows

1. Opsonization

It is the process by which phagocytosis of microbes by immune cells get enhanced. Those proteins which undertake this process are called as opsonins. For example, some complement proteins like C3b, iC3b and C4b are good opsonins. These proteins attach to microorganism first making a complex. This complex of complement protein & microbe bind with complement receptor on the surface of phagocytic cell as a result phagocytosis of microbe occur by phagocytes (Fig.1)

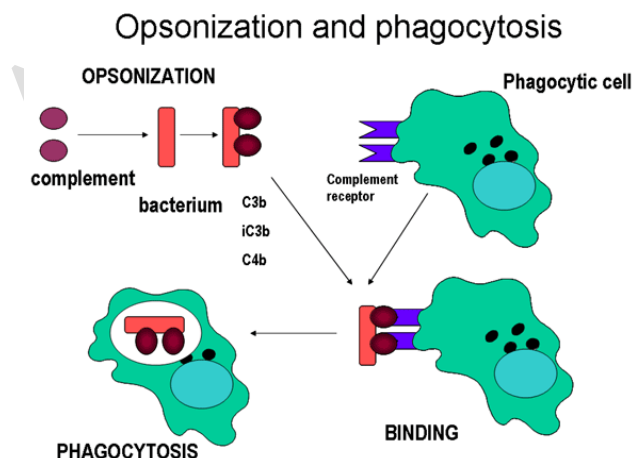


Figure 1: The opsonization function of complement proteins

2. Chemotaxis

The phenomenon by which phagocytic cells move towards the site of infection is called as chemotaxis, and the molecules involved in this process are called as chemotactic factors. The complement proteins are considered as good chemotactic factors like C5a. These act as potent activator of neutrophils, basophils & macrophages. These cause the induction of adhesion molecules on surface of blood vessel's endothelial cells, which bind the phagocytic cells on it (Fig.2). One of the important complex of complement proteins called as membrane attack complex (MAC) composed of C5b, C6 and C7 also act as good chemotactic factor.

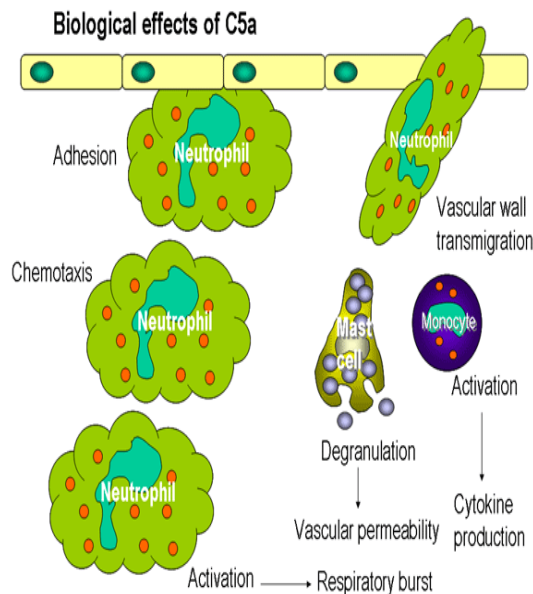


Figure 2: The chemotactic function of complement proteins

3. Anaphylaxis

Anaphylaxis is the form of immune response in which exaggerated response is generated against allergen. The molecules which produced such response are called as anaphylatoxins. The complement proteins like C4a, C3a & C5a act as potent anaphylatoxins. These molecules cause basophils and mast cells degranulation as a result various mediators or cytokines are generated (Fig 3). Those mediators involved in smooth muscles contraction, vasodilatation and bronchoconstriction.

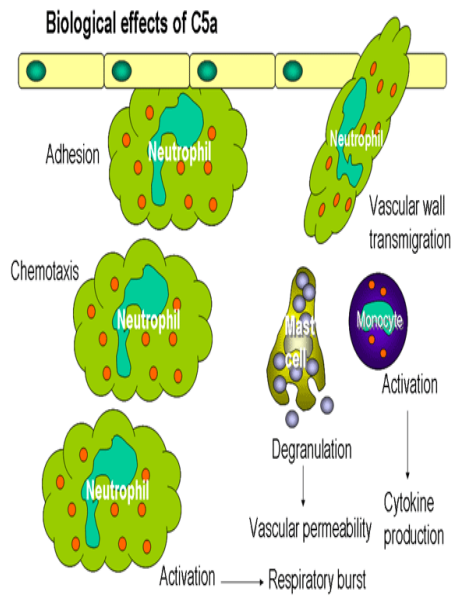


Figure 3: The anphylatoxis function of complement proteins

Lesson 96 : Classical Pathway of Complement Activation In Detail:

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Complement Activation

Complement proteins are activated through a series of cascading reactions which are called as complement activation. Followings are the important pathways for complement activation

1. Classical Pathway
2. Alternate Pathway
3. Mannose Binding Lectin (MBL) Pathway

These pathways generate the activated form of complement like C5 which leads towards a pathway called as common pathway as a result the membrane attack complex is generated for the lysis of microorganism (Fig 1)

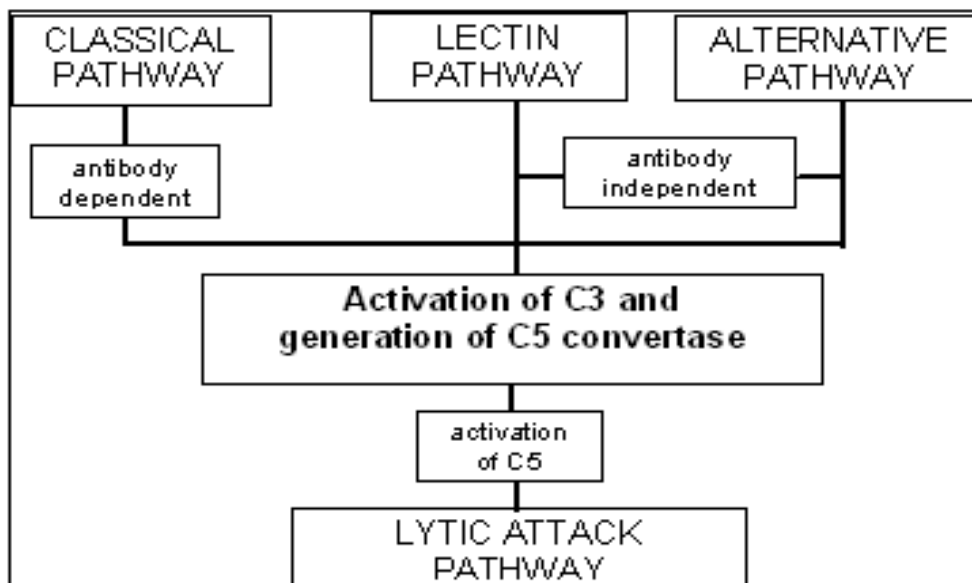


Figure 1. Pathways of complement activation

Figure 1: The pathways of complement activation

Classical Pathway

This pathway of complement activation is also called as antibody dependent complement activation. As in this pathway, antibody bind to microbe which links as the first molecule of

classical pathway. This pathway begins with C1 activation which is a multi subunit protein containing three sub proteins

- i.C1q,
- ii.C1r
- iii.C1s

C1 protein binds with antibody on the surface of microorganism

Followings are important events of this pathway

1. Activation of C1

In this event, there is binding of C1q to Fc portion of Immunoglobulin (IgM & IgG) which have bounded with antigens on bacterial surface. Binding of C1q in turn activates another subunit C1r and ultimately C1s (Fig 2). The activated C1rs act as enzyme for C4 to cleave it in C4a & C4b

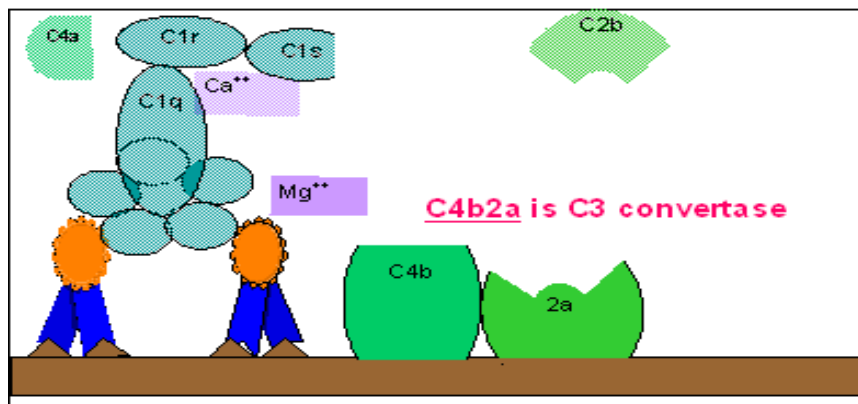


Figure 2. Generation of C3 convertase in the classical pathway

Figure 2: Initiation of classical pathway with C3 convertase generation

2. Generation of C3 convertase

As C1rs is activated, this complex also act on C2. As a result, C2 cleaves into two subunits C2a & C2b. C2a binds on bacterial surface with C4b which is also a subunit of C4 generated after its cleavage. This complex of C4b & C2a on bacterial surface is called as C3 convertase which further acts on C3 to c3a and c3b (as shown in Fig. 2).

3. Generation of C5 convertase

In this stage, the activated C3 convertase (C4b&C2a) act on C3 to convert into C3a & C3b. C3a moves into microenvironment for other immunological functions while C3b binds with C4b & C2a. As a result, a complex of C4bC2aC3b is formed. This complex is called as C5 convertase for the conversion of C5 into C5a & C5b.

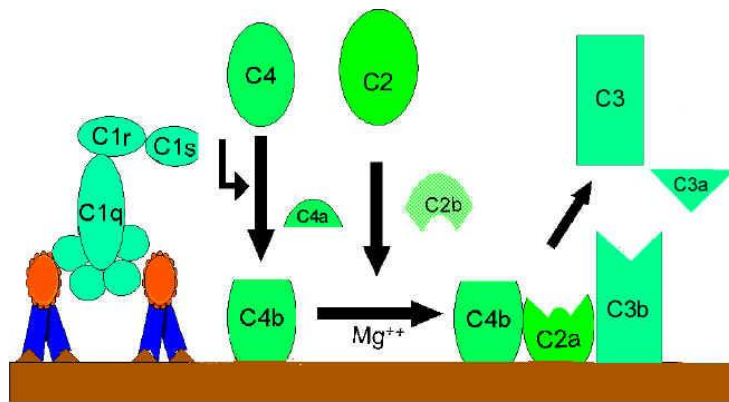


Figure 3: An overview of events during classical pathway of complement activation

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Lesson 97 : Alternative Pathway of Complement Activation In Detail:

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Alternative Pathway of Complement Activation

This pathway of complement activation is antibody independent complement activation as there is no need of antibodies for this. This is initiated by direct conversion of C3 into C3a & C3b. Moreover, there is need of various serum proteins & factors like Factor B, D & Mg^{++} ions. Generally, in serum there is low level spontaneous hydrolysis of C3 to produce C3i. In serum, then factor B binds with C3i to form a complex of C3iB. This complex becomes susceptible to another serum factor called as factor D which cleaves B into Bb. As a result, a complex C3iBb is formed which acts as C3 convertase for conversion of C3 into C3a & C3b. The resulting C3b reacts again with factor B and become susceptible to factor D, There is continuous formation of C3bBb which acts as C3 convertase for cleavage of C3 (as shown in fig.1).

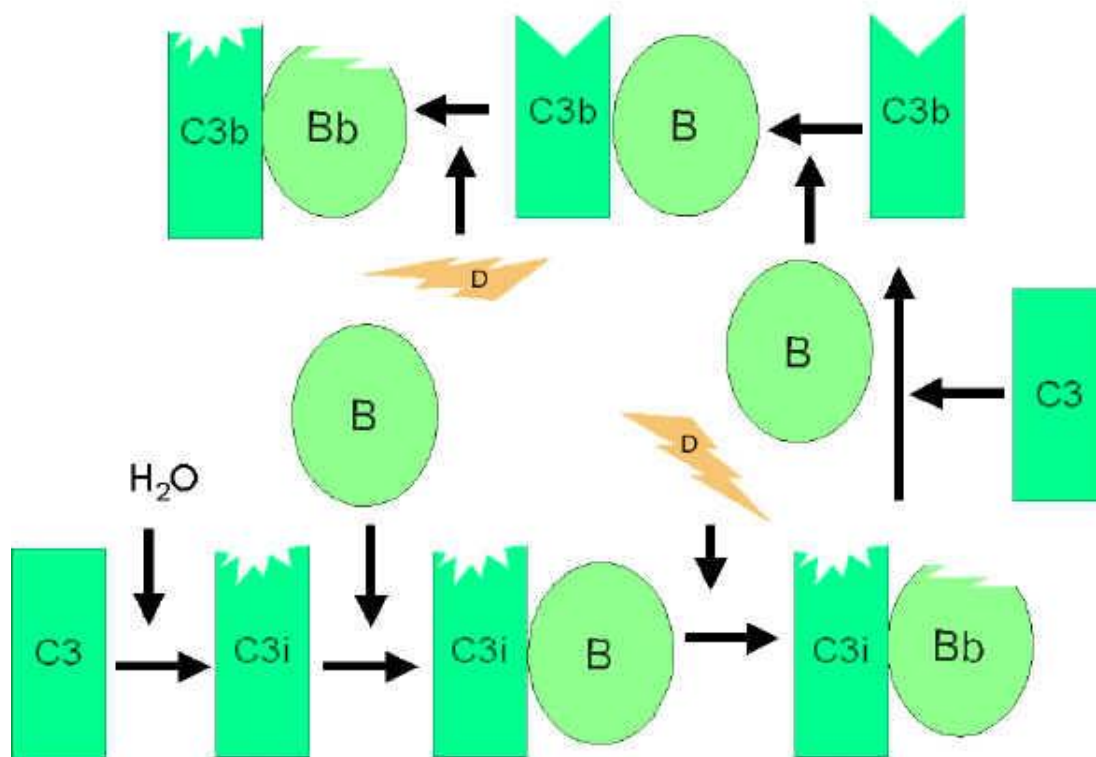


Figure 1: Conversion of C3 into C3b during alternative pathway

As a result of this C3b amplification loop, there is more production of C3b. This amplification loop of C3b is triggered by various molecules of microorganisms like lipopolysaccharides (LPS) of Gram negative bacteria, cell wall of bacteria & yeasts. These are also called as activator of

alternative pathway. Physiologically, the autonomous activation of C3 is controlled by following mechanisms

1. By Decay accelerating factor (DAF)
2. Control of C3b amplification is done by a molecule called as Decay accelerating factor (DAF), as a result the formation of C3 convertase is blocked (Figure 2)

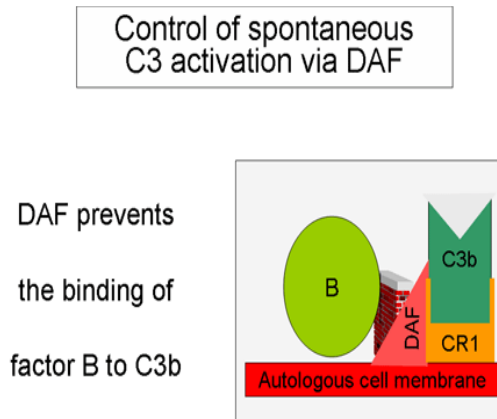


Figure 2: Control of spontaneous activation of C3 via DAF

2. By dissociating C3 convertase after cleavage of Bb from C3b

In this control mechanism, the spontaneous amplification of C3b loop is controlled by enzymatic degradation of C3b by serum factor i.e factor H & I (Figure 3)

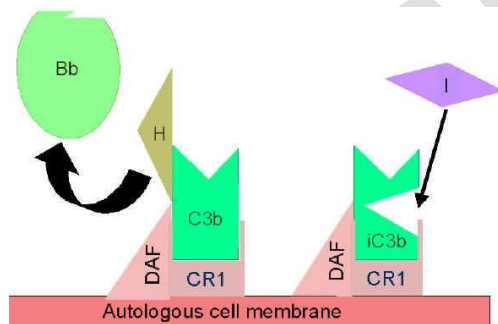


Figure 3: Dissociation of C3 convertase after cleavage of Bb from C3b

Clinically, the deficiency of Factor H & I in serum leads to increased susceptibility of patients to various infections due to non-availability of active C3.

In alternative pathway, the activated C3b is stabilized on the surface of infectious agent which is also called as activation surface by a protein called as protector (P) protein. This protector protein is absent on homologous cells of body (Fig. 4)

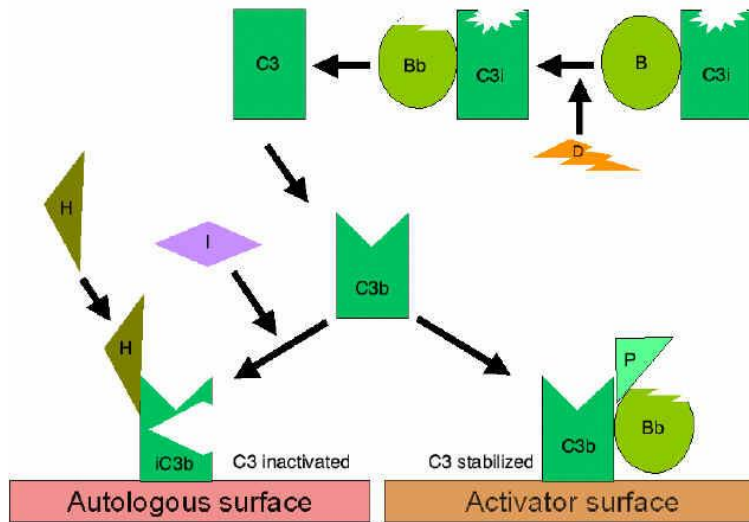


Figure 4: Stabilization of C3 on Activator surface by Protector Protein

As C3b is stabilized on activator surface, two molecules of C3b, Bb and protector protein (P) form a complex called as C5 convertase which is the fate of alternative pathway (Fig.5).

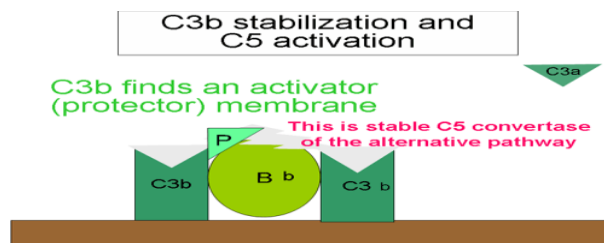


Figure 5: Structure of C5 convertase of alternative pathway

Lesson 98 : Lectin Pathway of Complement Activation In Detail:

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Lectin Pathway of Complement Activation

This pathway is also called as antibody independent complement activation like alternative pathway. However, it is initiated by mannose binding Lectins (MBL) on bacterial surface with mannose containing polysaccharides (Mannans). After binding of MBL on bacterial surface, there is association of two serine proteases called mannose-associated serine proteases (MASP). There are following two types of MASPs

1. MASP-1
2. MASP-2

In this pathway, MASP-1 corresponds like C1r & MASP-2 like C1s of classical pathway respectively. The MBL acts like C1q of classical pathway. The activation of MASPs happen after the formation of MBL/MASP-1 & MASP-2 tri molecular complex. This complex cleaves C4 & C2 into C4b & C2a respectively for the formation of C3 convertase (Fig.1)

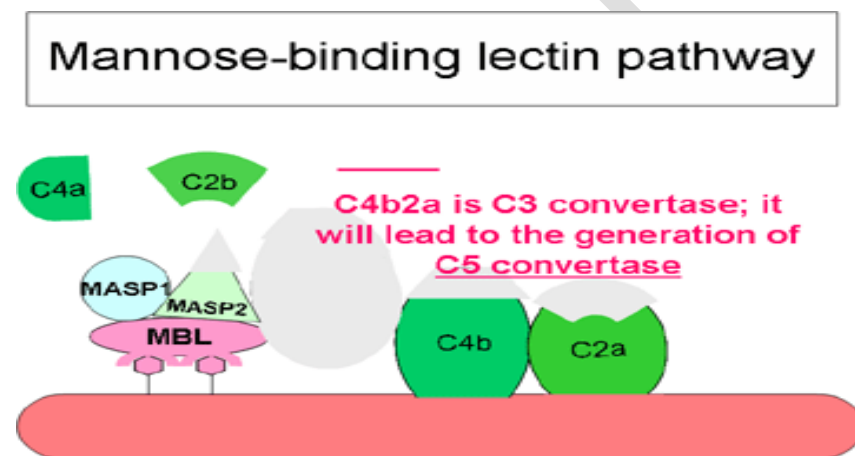


Figure 1: Mannose binding Lectin (MBL) pathway for complement activation

This C3 convertase cleaves C3 into C3a & C3b. The C3b binds with C4b & C2a for the generation of C5 convertase (C4b, C2a & C3b complex). While C3a move to microenvironment for other immunological functions. The biological activities of C4a, C2b & C3a & regulatory proteins of MBL pathway are same like classical pathway.

Lytic (Common) Pathway

This pathway is also called as Membrane attack complex (MAC) pathway as it is involved in the lysis of infectious agents. C5 convertase is generated from all three following pathways

- Classical: C4b2a3b
- Alternative: C3bBb3b
- Lectin: C4b2a3b

This C5 convertase converts C5 into C5b & C5a. This C5b rapidly associates with other complement proteins like C6 & C7 and insert into membrane. Subsequently, another protein called C8 binds with these membrane inserted proteins followed by several molecules of C9. These C9 proteins make pore in the membrane which leads towards leakage of cellular contents for cytolysis. The C5bC6C7C8C9 is called as membrane attack complex (MAC) (Fig 2)

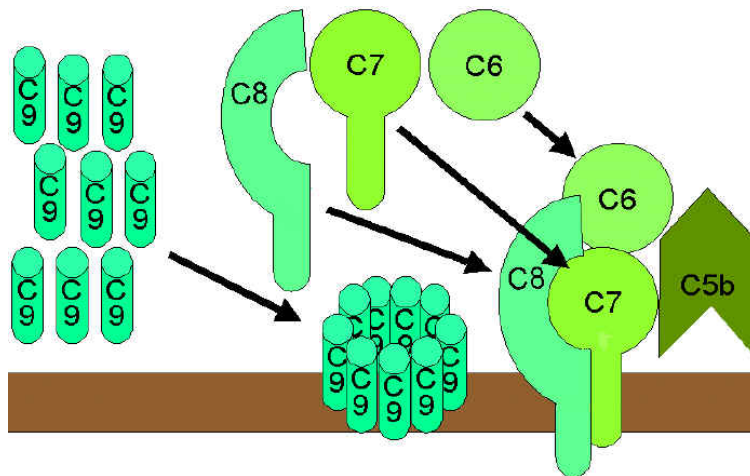


Figure 2: The lytic (common) pathway of complement activation

Lesson 99 : Biological Active Products of Complement:

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Biological Active Products of Complement

The complement system provides specific & non-specific resistance against infections. As a result of complement activation, there is production of various biological active molecules which are involved in resistance, anaphylaxis & inflammation. Followings are the important biological active products of complement

Kinin production

C2b which is produced in classical pathway is called as Pro-Kinin. This Pro-Kinin is activated by serum factor called as Plasmin into Kinin which is a potent biological active compound. The excess C2b production causes undesirable effects like smooth muscles contraction and vasodilatation

Anaphylatoxins

Anaphylaxis is the form of immune response in which exaggerated response is generated against allergen. The molecules which produce such response are called as anaphylatoxins. The complement proteins like C4a, C3a & C5a act as potent anaphylatoxins. These molecules cause basophils and mast cells degranulation as a result various mediators or cytokines are generated. Those mediators involved in smooth muscles contraction, vasodilatation and bronchoconstriction.

Chemotactic factors

The phenomenon by which phagocytic cells move towards the site of infection is called as chemotaxis, and the molecules involved in this process are called as chemotactic factors. The complement proteins are considered as good chemotactic factors like C5a. These act as potent activator of neutrophils, basophils & macrophages. These cause the induction of adhesion molecules on surface of blood vessel's endothelial cells, which bind the phagocytic cells on it (Fig.11.2). One of the important complex of complement proteins called as membrane attack complex (MAC) composed of C5b, C6 and C7 also act as good chemotactic factor.

Opsonins

It is the process by which phagocytosis of microbes by immune cells gets enhanced. Those proteins which undertake this process are called as opsonins. For example, some complement proteins like C3b, iC3b and C4b are good opsonins. These proteins attach to microorganism first making a complex. This complex of complement protein & microbe binds with complement

receptor on the surface of phagocytic cell as a result phagocytosis of microbe occur by phagocytes.

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Lesson 100 : Hypersensitivity

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Introduction to Hypersensitivity

- Hypersensitivity refers to excessive, undesirable (damaging, discomfort-producing and sometimes fatal) reactions produced by the normal immune system. It is the state of undesirable reaction by normal immune system, in which immune system overreacts against various stimuli. These are also known as hypersensitivity reaction or intolerance.
- This state of body is considered as damaging, uncomfortable and fatal. This clinical state requires pre-sensitize state of host, in which the host is already exposed to stimulus in the form of allergens.
- In hypersensitivity, the protective role of immune system become harmful for its host. This condition is also termed as Allergy which is an abnormal response against the otherwise harmless environment stimulus e.g food, pollen & animal dander.
- The hypersensitive state of immune response also exhibits in the form of autoimmune disorders which are the abnormal immune response against host's self- tissues.
- Hypersensitivity reactions can be divided into four types: type I, type II, type III and type IV, based on the mechanisms involved and time taken for the reaction.
- Frequently, a particular clinical condition (disease) may involve more than one type of reaction.

Lesson 101 : Classification of Hypersensitivity:

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Classification of Hypersensitivity

Hypersensitivity reactions are classified on the basis of following two factors

- Nature of immune response involved
- The time required for immune response

Followings are the important types of hypersensitivity reactions

- i.Type I
- ii.Type II
- iii.Type III
- iv.Type IV

Lesson 102 : Type I Hypersensitivity

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Type I Hypersensitivity Reactions

These reactions are also known as Immediate or Anaphylactic reactions which involve various tissues of the body with following clinical conditions

- Skin: Urticaria & Eczema
- Eyes: Conjunctivitis
- Nose: Rhinorrhea, Rhinitis
- Lungs: Asthma
- Gastro-Intestinal Tract (GIT): Gastroenteritis

These reactions manifest various undesirable symptoms from minor inconveniences to fetal death. In these reactions, usually reaction is quick & immediate ranging 15 to 30 minutes, however, sometimes can be 10-12 hours.

- These are mediated by IgE antibodies.
- The primary cells involved in these reactions include basophils and mast cells.
- The reaction is amplified by involvement of other cells like platelets, neutrophils & eosinophils.

The allergens of type I reactions have preferential involvement of IgE antibodies which are produced as a result of class switching phenomenon. During these reactions, there is sensitization of mast cells with IgE via a specific FcR receptor. The fixed IgE is cross linked with allergen (Fig. 1). As a result, degranulation of mast cells occur which release various mediators of significant clinical features of type I reactions as shown in Table 1

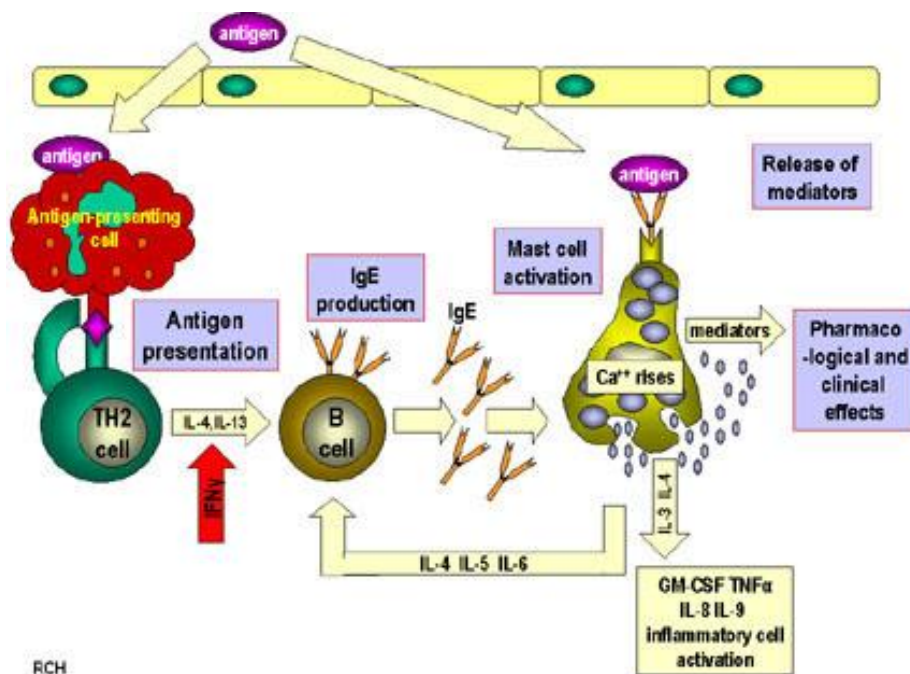


Figure 1: Mechanism of Type I hypersensitivity reactions

Mediators in the granules of Mast Cells Attract eosinophil and neutrophils

Table 1: Various mediators of Mast cell granules with their functions

MEDIATOR

Preformed mediators in granules

Histamine	Bronchoconstriction, mucus secretion, vasodilatation, vascular permeability
Trypsin	Proteolysis
Kininogenase	Kinins and vasodilatation, vascular permeability, edema
ECF-A(tetrapeptides)	Attract eosinophil and neutrophils

Newly formed mediators

Leukotriene B₄ Basophil attractant

Leukotriene C₄, D₄ Same as histamine but 1000x more potent

Prostaglandins D₂ Edema and pain

PAF Platelet aggregation and heparin release: microthrombi

Diagnosis of Type I Hypersensitivity

Type I hypersensitivity reactions are diagnosed in the following important ways

- **By Skin (Prick & Intradermal) test**

In this test, the allergen is administered through skin and the reaction is observed in the form of redness, itching and swelling of the skin

- **Measurement of total IgE levels**

In patient's serum, the measurement of total IgE also help in the diagnosis of type I hypersensitivity reactions.

- **Allergen specific IgE levels**

In addition to total IgE levels, the allergen specific IgE levels also help in the diagnosis of immediate hypersensitivity reactions.

Lesson 103 : Type II Hypersensitivity:

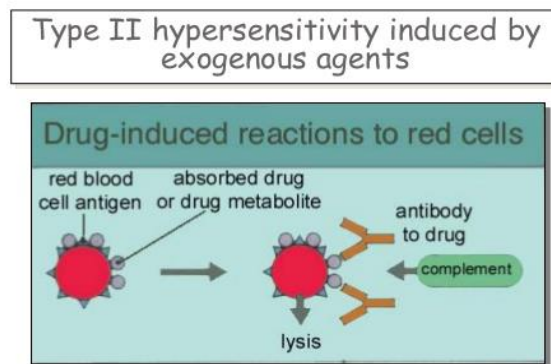
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Type II Hypersensitivity

- This type of hypersensitivity is also known as cytotoxic hypersensitivity.
- This condition affects various organs & tissues of body which determines the tissue specific nature of hypersensitivity.
- For such reactions there is the involvement of various antigens. These antigens are usually either endogenous (from inside of body) like antigens of body's self-tissues or exogenous (from outside of body) like various chemicals in the form of haptens.

These **exogenous antigens** bind to the surface of cell membrane. There is generation of antibodies against these antigens which are IgM or IgG in nature. Antigen and antibody reaction occur on the surface of the target cells. The complement proteins get fixed on immune complex on the surface of target cell like RBC surface. As a result, lysis of target cell occurs (Fig.1). These kind of reactions have reaction time from minutes to hour.

The examples of exogenous antigens include Drug induced hemolytic anemia, granulocytopenia & thrombocytopenia.



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Figure 1: Drug (exogenous antigen) induced hemolytic anemia

Similarly, endogenous antigens also induce immune response in the form of generation of antibodies. The example of endogenous antigens induced cytotoxic hypersensitivity include erythroblastosis fetalis which is also called as hemolytic disease of newborns (HDN). This condition occurs due to Rh incompatibility between mother & fetus. As a result, there is the development of anti-Rh antibodies in maternal serum (IgG) which has the ability of crossing placenta in subsequent pregnancy (Fig. 2)

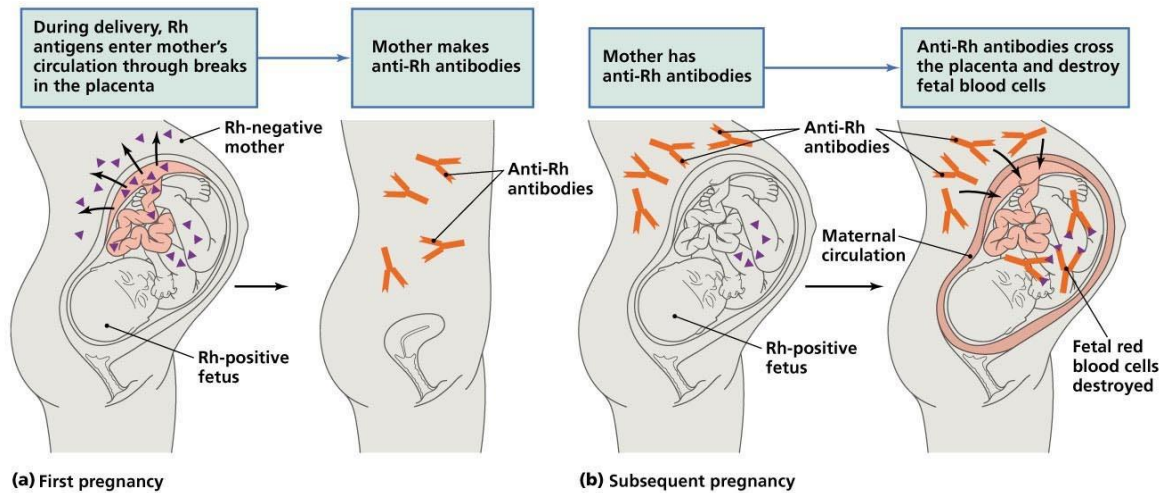


Figure 2: Events of Endogenous antigen induced cytotoxicity in erythroblastosis fetalis or hemolytic disease of newborns (HDN)

Likewise, other examples of **endogenous antigen** induced cytotoxic hypersensitivity include following autoimmune diseases like

- Goodpasture's Syndrome in which there is generation of Autoantibodies against glomerulus basement membrane
- Hashimoto's thyroiditis in which Autoantibodies are generated against thyroid

Antibody dependent Cellular Cytotoxicity (ADCC) is also another form of Type II hypersensitivity. This type hypersensitivity is independent of complement system. However, the effector immune cell lyse the target cell coated with antibody. These effector cells can be the primary effector cells like natural killer (NK) cells while neutrophils, macrophages and eosinophils are also included in such kind of hypersensitivity reactions.

Examples of such kind of reactions include the cytotoxicity of tumorous cells by NK cells (Fig. 3)

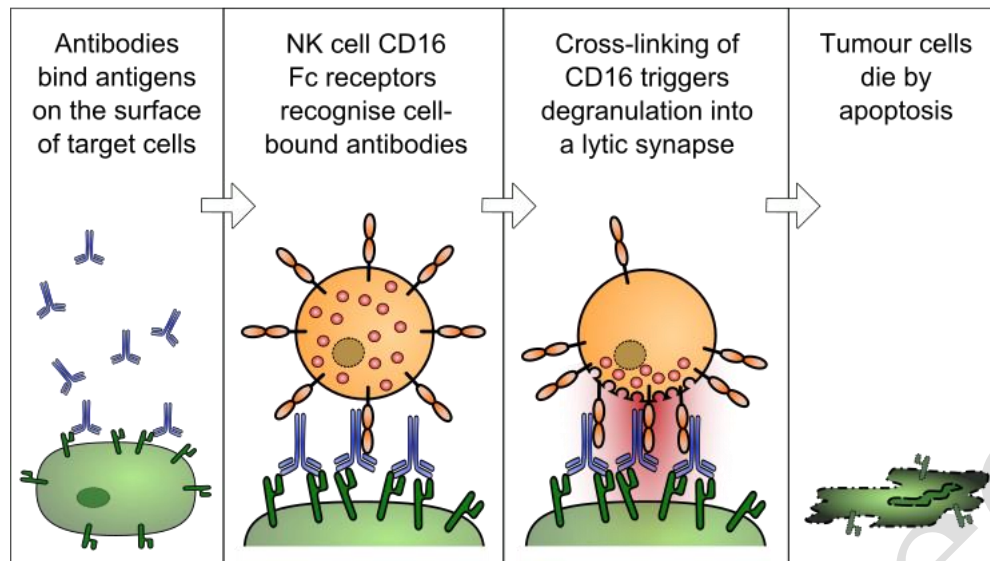


Figure 3: Cytotoxicity of tumor cell through cytotoxic hypersensitivity

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Lesson 104 : Type III Hypersensitivity:

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Type III Hypersensitivity

- These reactions are also known as immune complex mediated hypersensitivity.
- The reaction time required for such reactions is 3-10 hours after exposure to antigen.

There is generation of antibodies which bind with corresponding antigens as a result soluble immune complex formation occur. These immune complex effect generally the host and that condition is called as serum sickness. Otherwise, these reactions are specific to various organs and cause the following conditions

- Skin: Systemic Lupus erythematosus (SLE), Arthus reaction
- Kidney: Lupus nephritis
- Lung: Aspergillosis
- Blood vessels: Polyarteritis
- Joints: Rheumatoid Arthritis

For the pathogenesis of these reactions there can be the involvement of microorganisms



Figure 1: Clinical picture of SLE & Rheumatoid arthritis patients

The antigens involved in these reactions are soluble in its state which can be of two types

1. Exogenous

These include various microbial infections like chronic bacterial, viral & parasitic infections

2. Endogenous

Non-organ specific autoimmunity e.g SLE

Antibodies involved in these reactions are mostly IgG but IgM are also involved. However, the primary components of these reactions are soluble immune complexes with complement proteins like (C3a & C5a). Moreover, platelets and neutrophils are also involved in this damage. The lesion of these reactions contain primarily neutrophils with immune complex and complement. Furthermore, infiltrating macrophages also contribute later in healing process

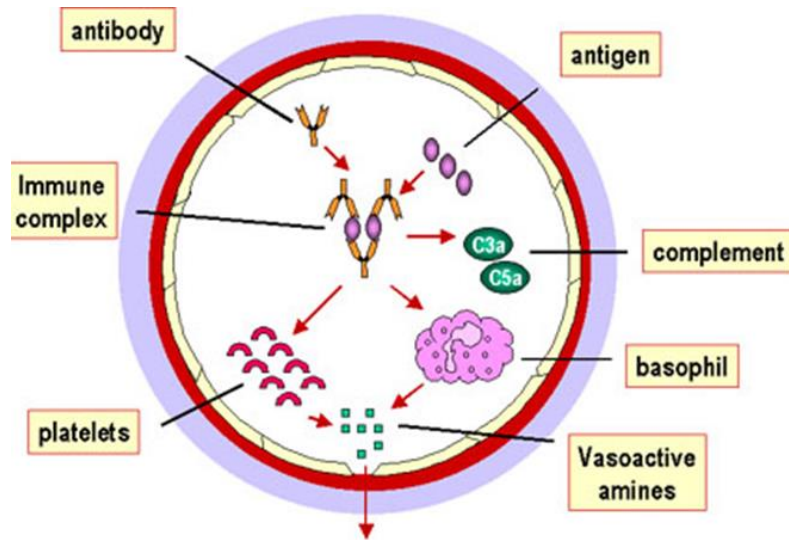
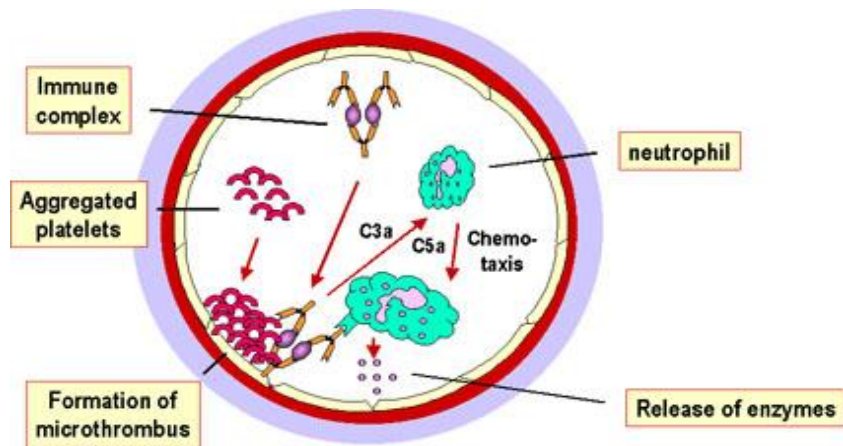


Figure 2: Components involved in Type III hypersensitivity reactions

The severity of these reactions also based on affinity of antibodies, the Size of immune complex and the type of tissue involved. In the lesion of these reactions, aggregated platelets form micro thrombus which increase the vascular permeability (Fig. 3). The diagnosis of these reactions is done by presence of immune complex with complement in tissue biopsies through immunofluorescence.



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Figure 3: Mechanisms of tissue destruction in type III hypersensitivity reactions

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Lesson 105 : Type IV Hypersensitivity:

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Type IV Hypersensitivity

- These reactions are also termed as cell mediated hypersensitivity.
- Due to more time requirement like 48-72 hours, these reactions are also called as delayed type of hypersensitivity.

Immunologically, T-cells, Macrophages and monocytes are involved in these reactions. In these reactions, the response is against intracellular pathogens that's why these reactions are antibody independent in nature. Lymphocytes like CD8 (Cytotoxic) T-cells cause direct damage to target cells. Likewise, CD4 (Helper) T-cells for example Th1 cells recognize foreign antigen with MHC class II. Moreover, the secretion of Cytokines, activation of macrophages and Inflammation increase the magnitude of tissue injury & damage.

Type IV Hypersensitivity

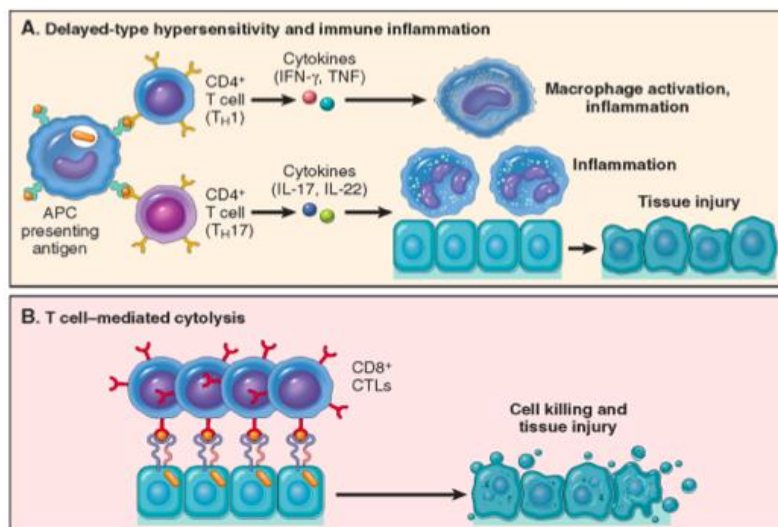


Figure 1: Involvement of immune mechanisms in Type IV hypersensitivity reactions

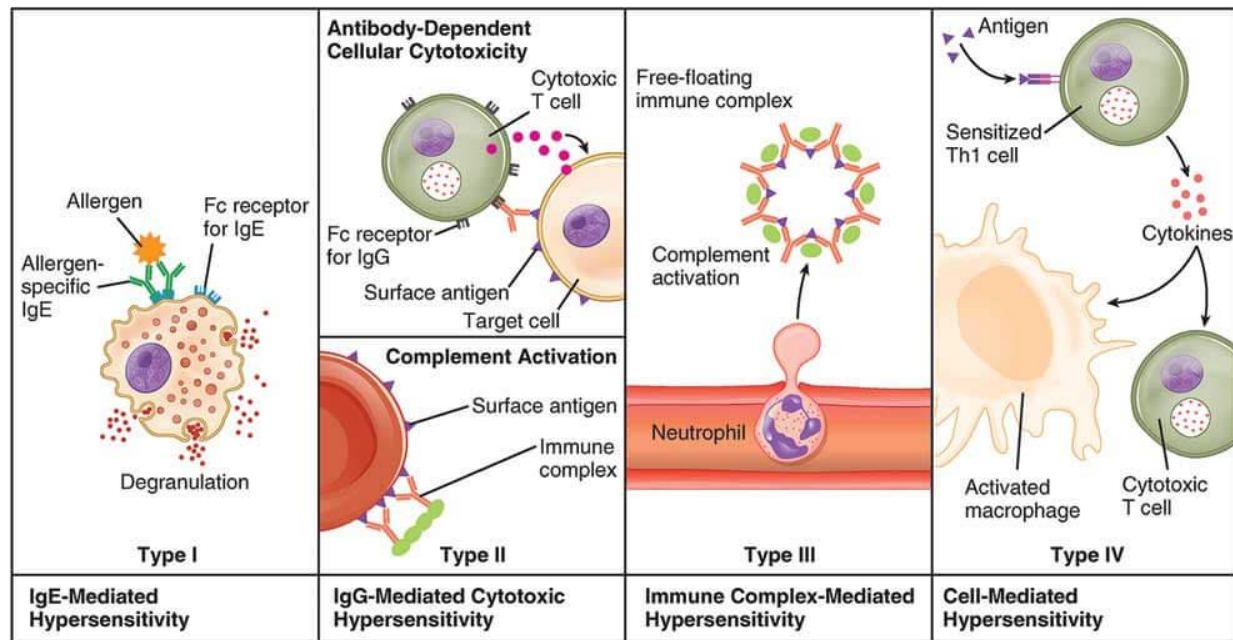
Type	Reaction type	Clinical Appearance	Histology	Antigen & Site
Contact	48-72 hrs	Eczema	Lymphocytes, followed by macrophages; edema of epidermis	Epidermal (organic chemicals, poison ivy, heavy metals, <i>etc.</i>)
Tuberculin	48-72 hrs	Local induration	Lymphocytes, monocytes, macrophages	Intradermal (tuberculin, lepromin, <i>etc.</i>)
Granuloma	21-28 days	Hardening	Macrophages, epitheloid and giant cells, fibrosis	Persistent antigen or foreign body presence (tuberculosis, leprosy, <i>etc.</i>)

Lesson 106 : Comparison of different Type Hypersensitivity:

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Table - Comparison of Different Types of hypersensitivity

Characteristics	Type-I (Anaphylactic)	Type-II (Cytotoxic)	Type-III (Immune complex)	Type-IV (Delayed type)
Antibody		IgG, IgM	IgG, IgM	None
Antigen	Exogenous	Cell surface	Soluble	Tissues and organs
Response time	30 minutes	Minutes-hours	Hours	72 hours
Appearance	Wheal and flare	Lesions and necrosis	Thrombocytopenia and edema, vasculitis and necrosis	Thrombocytopenia and induration
Pathology	Eosinophils and lymphocytes	Antibody complement	Antibody complement and eosinophils	Antibody complement and lymphocytes
Transferred	Antibody	Antibody	Antibody	Cells
Examples	Allergic asthma, hay fever	Thrombocytopenic purpura, hemolytic disease of newborn, Goodpasture's syndrome	Serum sickness, farmer's lung, rheumatoid arthritis	Delayed hypersensitivity test, poison ivy, contact dermatitis, tumor rejection



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Lesson 107 : Cytokines:

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Cytokines

Overview about Cytokines

Cytokines belong to diverse group of non-antibody proteins which act as mediators between cells.

- Initially, these proteins were considered as products of immune cells. More importantly, these were also considered to act as mediators for immune cells.
- Later on, it was also found that these are also produced from non-immune cells and served as mediators too.

Cytokines are used as biological response modifier for treatment of various diseases. In general, cytokine is a general term but it can be more specific term which is used according to their cell of origin as follows

- **Monokines**

These are produced by mononuclear phagocytic cells.

- **Lymphokines**

These are produced by activated lymphocytes.

- **Interleukins**

These are the mediators between leucocytes

- **Chemokines**

These are responsible for leucocytes migration and inflammatory process

- Cytokines usually perform their functions as cascade signaling in the form of a network which is termed as cytokine network.
- Their functions are for enhancing their effects of other cytokines in an additive or synergistic manner.
- Moreover, these act for suppressing the effects of other cytokines in an opposing or antagonistic way.
- Cytokines are not as preformed proteins like antibodies. These are produced by gene transcription as needed because mRNA for cytokines are short lived in nature.

- Furthermore, the individual cytokine can act on many cells of the host which is termed as pleotropic response of cytokines.

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Lesson 108 : Mechanism of acting of Cytokines:

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Mechanism of acting of Cytokines

- Cytokines act through specific receptors present on various cells. These are considered as redundant in nature. The redundancy of cytokines is due to the nature of cytokine receptors.
- The complex interaction between host's cells and cytokines occur through cytokine network. Different cells can respond to same cytokine due to structural similarities among cytokine receptors (Fig.1).
- The cytokine signaling is flexible which can have both protective and damaging effects. Similarly, these can also influence the synthesis of other cytokines.

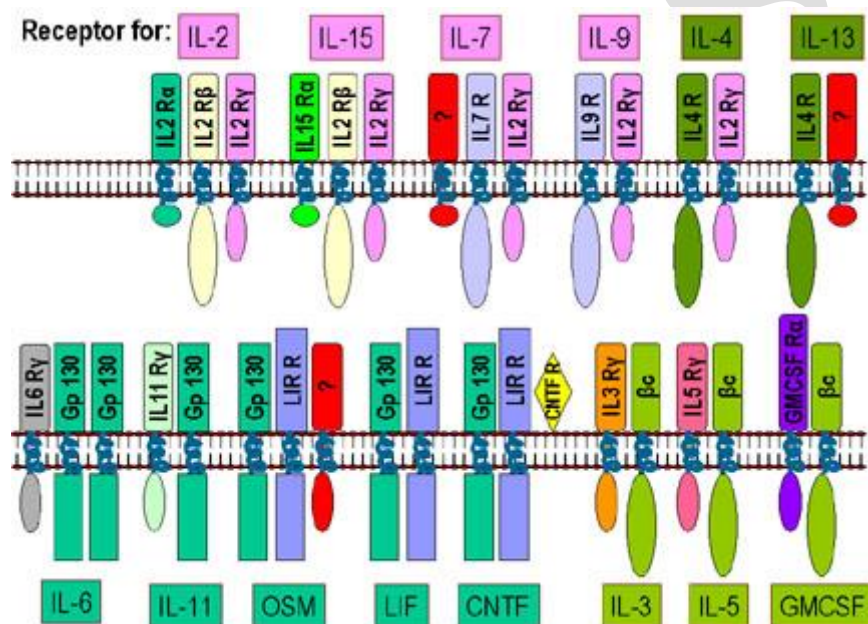


Figure 1: The structure of various cytokine receptors

Cytokines bind to specific receptors with high affinity and respond in following three ways

1. Autocrine

These effect on same cells which secrete the cytokine

2. Paracrine

In this way, the cytokines effect on nearby cells of cytokine secreted cells.

3. Endocrine

Through this way, the cytokines effect on distant cells through circulation

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Lesson 109 : Categories of Cytokines:

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Categories of Cytokines

Cytokines are categorized on following two basis

1. Their source of origin
2. Their functions

I. Categories based on source cells

Followings are the categories of cytokines on the basis of their source of origin

i) Monokines

As the name indicates, these are produced by mononuclear phagocytic cells for example Interferon gamma and Interleukin (IL-1) from macrophages & Monocytes

ii) Lymphokines

These are produced by activated lymphocytes for example IL-3, IL-4 & IL-5.

iii) Interleukins

These are mediators between leucocytes for example IL-1, IL-10 & IL-18.

iv) Chemokines

These are responsible for leucocytes migration for example IL-8.

II. Based on functions

On the basis of functions, cytokines are categorized into following two categoryies

1. Cytokines which act for Innate (non-Specific) immune system **e.g TNF- α , IL-1, IL-10 etc**
2. Cytokines which act for Adaptive (Specific) immune system **e.g IL-2, IL-4 & IL-5**

Lesson 110 : Cytokines of Innate Immune System:

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Cytokines of Innate Immune System

These cytokines which act primarily for innate (non-Specific) immune system. These are as follows

1) Tumor Necrosis Factor-Alpha (TNF- α)

It is produced by activated macrophages in response to microbes or their products for example lipopolysaccharides (LPS) of gram negative bacteria. It acts a mediator of acute inflammation. It recruits neutrophils and macrophages at the site of infection either by stimulating endothelial cells to secrete adhesion proteins or by secreting chemokines. Moreover, TNF- α induces fever and acute phase proteins in host.

2) IL-1

This is also an inflammatory cytokine as like TNF- α . It is produced by activated macrophages. Furthermore, it also helps to activate T-cells.

3) IL-10

It is an inhibitory cytokine which inhibits the cytokine production of cytokines from macrophages.

4) IL-12

This cytokine is produced by dendritic cells and macrophages. It enhances the cytolytic activity of cytotoxic T-cells

5) Type I interferon

This class of interferon include Interferon α & β . These inhibit viral replication in cells by increasing the expression of class I MHC which in turn make them susceptible to cytotoxic Tlymphocytes (CTL).

6) IFN- γ

It is produced primarily by Th1 cells. It enhances the cytolytic activity of NK cells by induction of class I & II MHC molecules.

7) Chemokines

These are chemotactic cytokines produced by many leucocytes. These recruits inflammatory cells at site of infection for an inflammatory response.

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Lesson 111 : Cytokines of Adaptive Immune System:

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Cytokines of Adaptive Immune System

Followings are the cytokines which act primarily for Adaptive (Specific) immune system

1) IL-2

It is produced by T-helper (Th) cells but Cytotoxic T-cells (Tc) also produced it in lesser extent. It stimulates B-cell division. It can also activate NK cells and monocytes. Moreover, IL-2 acts on T cells in an autocrine manner.

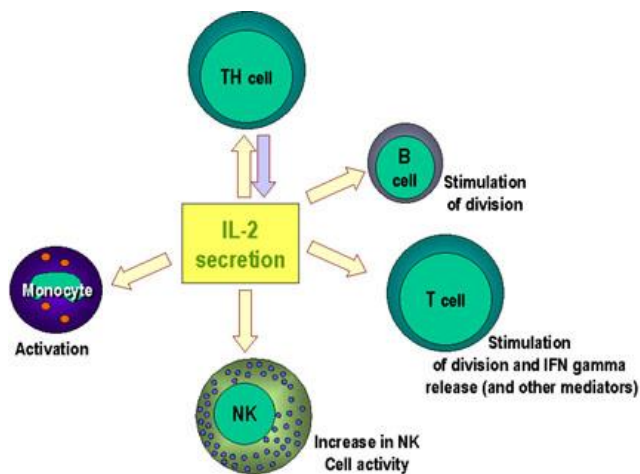


Figure 1: Functions of IL-2 on various other immune cells

Activation of T-cells results in expression of IL-2R receptors and also production of IL-2. IL-2 binds to IL-2R and promotes cell division. However, IL-2 receptor decays if T-cells are not activated and eventually the proliferative phase ends (Figure 2).

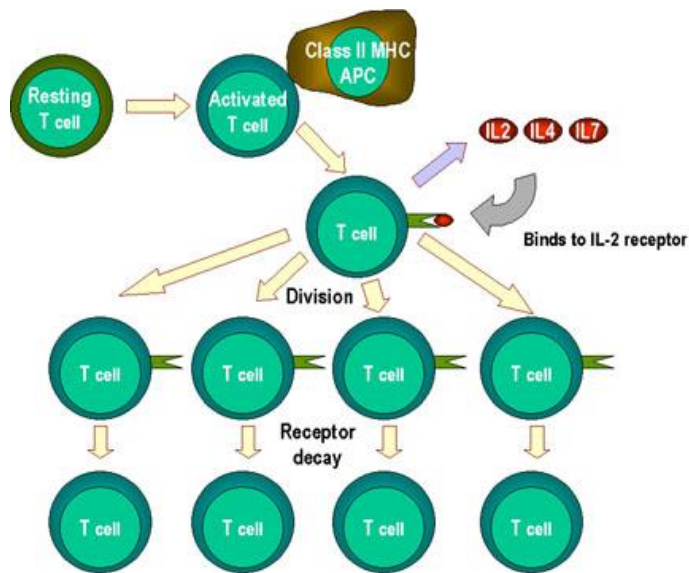


Figure 2: Activation of T-cells by IL-2

2) IL-4

IL-4 is produced by macrophages and Th2 cells. It promotes the development of Th2 from naïve T-helper (Th) cells. The Differentiated Th2 cells result in the production of antibodies. IL-4 also involves in the class switching of antibody into IgE

3) IL-5

IL-5 is produced by Th2 cells. It promotes the development of B-cells and eosinophils. It also activates mature eosinophils.

4) Tumor Growth Factor (TGF)- β

TGF- β is produced by T cells and many other cells types. It acts as inhibitory cytokine which inhibits the proliferation of T-cells and activation of macrophages. It also acts on neutrophils and endothelial cells to block the effects of pro-inflammatory cytokines.

Lesson 112 : Cytokine Network:

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Cytokine Network

A complex series of overlapping and inter-related connections among cytokines. As cytokines are secreted from one kind of cells and have effect on other type of cells & organs. Within this network, some cytokines have synergistic effects on other cytokines while some have antagonistic effects. Through this network, lymphocytes, macrophages and other tissues are regulated for their functions (Fig. 1)

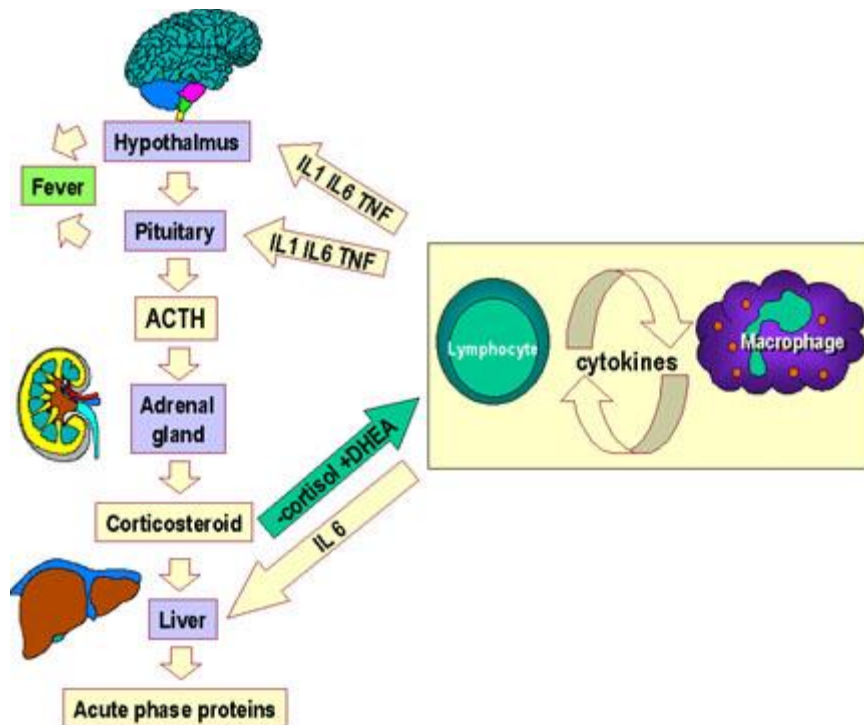


Figure 1: Communication B/W Lymphocytes, Macrophages & Other Tissues

Similarly, monocytes, lymphocytes and other components are also communicated through this network (Fig. 2)

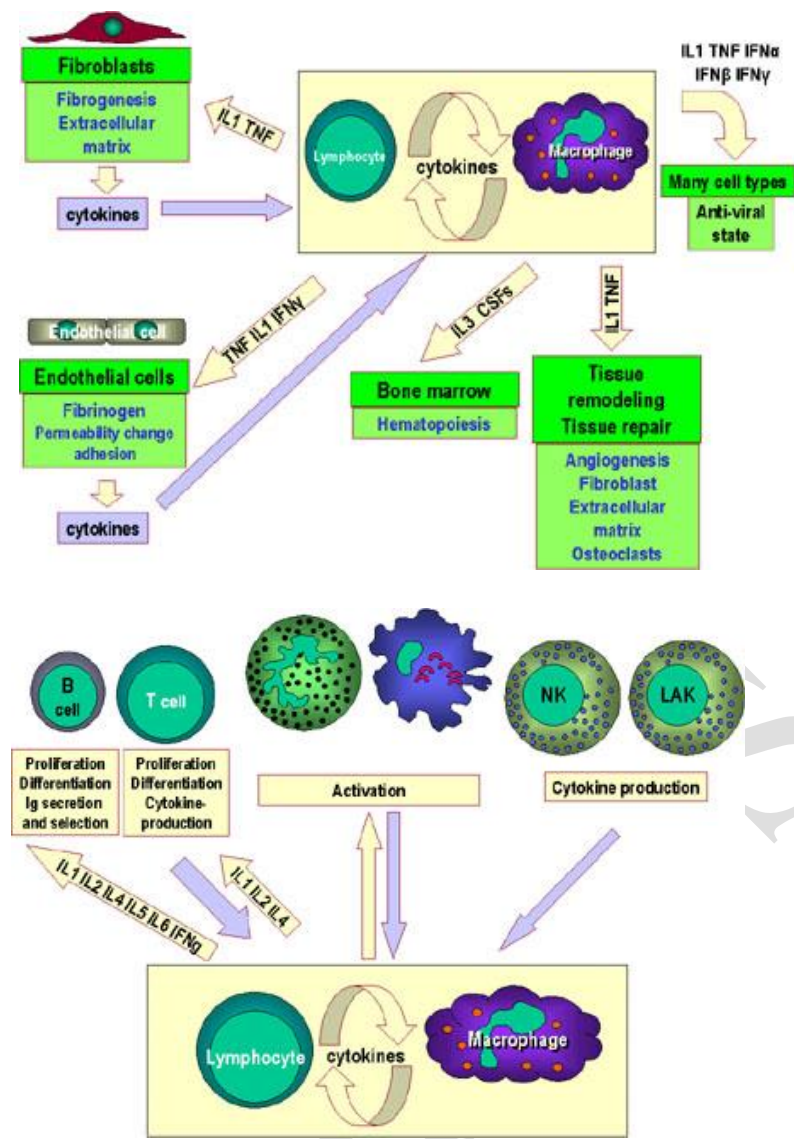


Figure 2: Communication B/W Lymphocytes, Macrophages & Other Components of Immune System

Lesson 113 : Immuno-Regulation by Cytokine:

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Immuno-Regulation by Cytokine

The control of immune response between lymphocytes and macrophages. The balance is required between antigen driven activation of lymphocytes and regulatory influences which is called as immune-regulation. Immuno-regulation occurs at following three phases of immune responses

1. Recognition phase
2. Activation Phase
3. Effector Phase

The cytokines are considered as positive or negative regulator of immune response. The cytokines can act on many stages of immune responses. Their activity is dependent on presence of other cytokines in the tissue's microenvironment. It also depends upon receptor expression on effector cells. This kind of regulation regulates the nature and extent of generated immune response.

Lesson 114 : Resistance & Immune Response to Infectious diseases:

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Resistance and Immune Response to Infectious diseases

What is Immune Evasion?

All the strategies used by pathogenic organisms and tumor cells to evade host immune's system are called as immune evasion mechanisms. Immune system has one of the important characteristics of immune surveillance for protection against pathogenic organisms and transformed cells. However, pathogens maximize their ability to flourish and develop infection. Likewise, tumor cells also have the potential to evade the host's immune surveillance and sustain inside the body. Immune-evasion occurs either due to weak host immune response or by strategies devised by pathogens. In immune evasion, immune system can be bypassed by evading humoral immunity. Moreover, cell mediated immunity can also be evaded by pathogenic microbes and transformed cells.

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Lesson 115 : Mechanisms of Immune Evasion:

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Mechanisms of Immune Evasion

The strategy to escape from immune system is caused by different microbes differently as follows

a) Extracellular organisms

These organisms usually inactivate the humoral components of immune system

b) Intracellular organisms

These organisms usually bypass the host immune system by inactivating intracellular killing and other cell mediated immune mechanisms which are as follows

Antigenic Variation

Microorganisms mutate their antigenic surface molecules for immune-evasion which is called as antigenic variation. As a result, there is not available longer protection by antibodies produced under previous exposure of antigenic forms. There are following two forms of antigenic variations

• Antigenic shift

This is the major form of change in antigenic structure of antigen as a result the new strains of microbes are generated (Fig.1)

• Antigenic drift

This is the minor form of antigenic variation within the same strain for example influenza (Flu) virus exhibits both antigenic shift and drift for pandemic and seasonal flu respectively. Similarly, bacteria like E.coli, Neisseria gonorrhoeae and Salmonella typhimurium also exhibit antigenic drift (Fig. 1).

Differences Between Antigenic shift & Antigenic drift

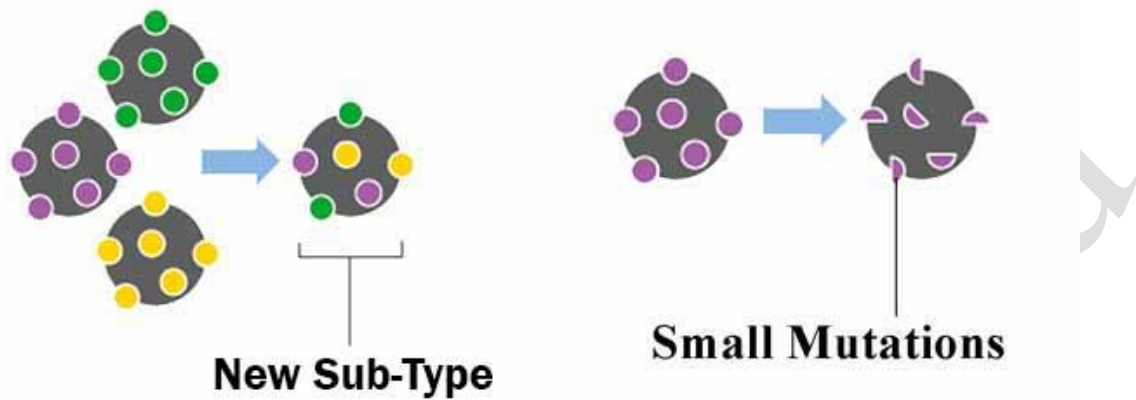


Figure 1: Comparison between antigenic shift and drift

Inhibition of Complement activation

The degradation of complement proteins through complement deviation which occurs by deviation of the complement activation site on bacterial cell. Moreover, the resistance to insertion of

Membrane Attack Complex (MAC) in bacterial due to thick bacterial cell wall and capsule also responsible for inhibition of complement activation.

Resistance to Phagocytosis

The inhibition of phagocytosis process of bacteria due to cell surface molecules of bacteria for example capsule of bacteria like pneumococcus. Similarly, the trapping of bacteria by bacterial enzymes for example coagulase of *S. aureus*. Moreover, the killing of phagocytes due to various secreted toxins from different bacteria for example Leucocidin & Lysins from *S. aureus* and Streptococci respectively. The different immune-evasion mechanisms of *S. aureus* (Fig.2)

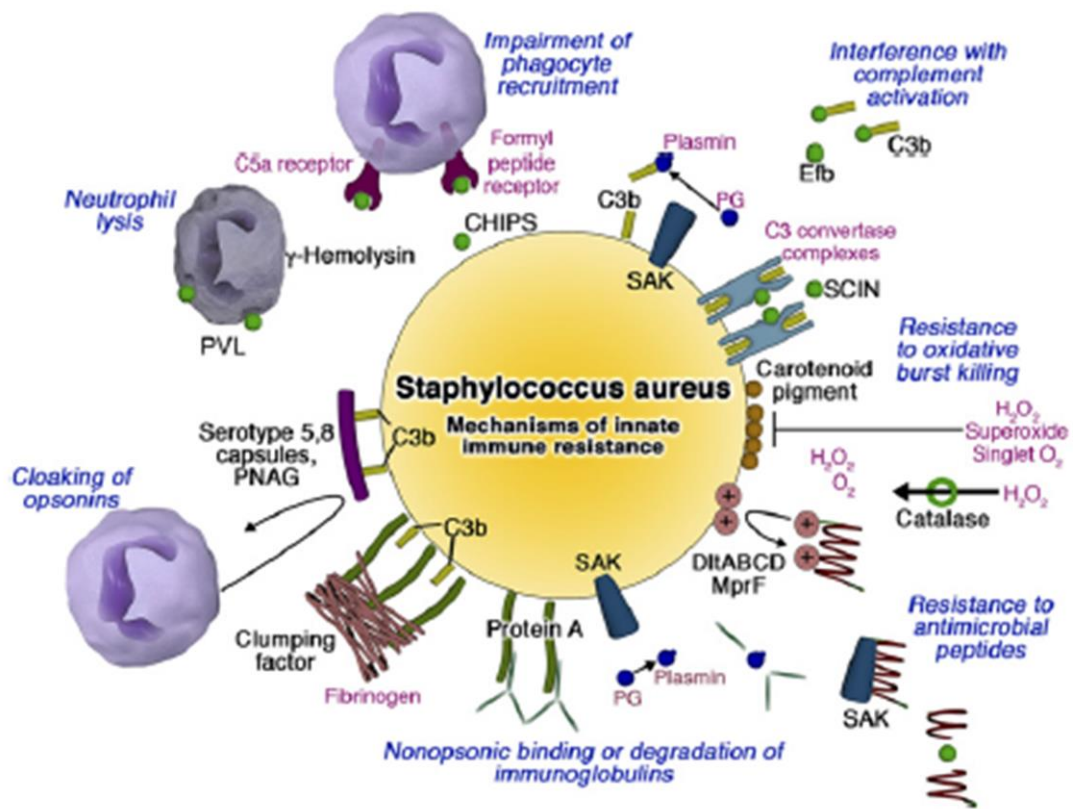


Figure 2: Strategies of Immune evasion by *S. aureus*

Lesson 116 : Immune Response against Extracellular Pathogens:

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The immune system has developed to protect the host from pathogens and other foreign substances. Self/non-self discrimination is one of the hallmarks of the immune system.

There are two main sites where pathogens may reside:

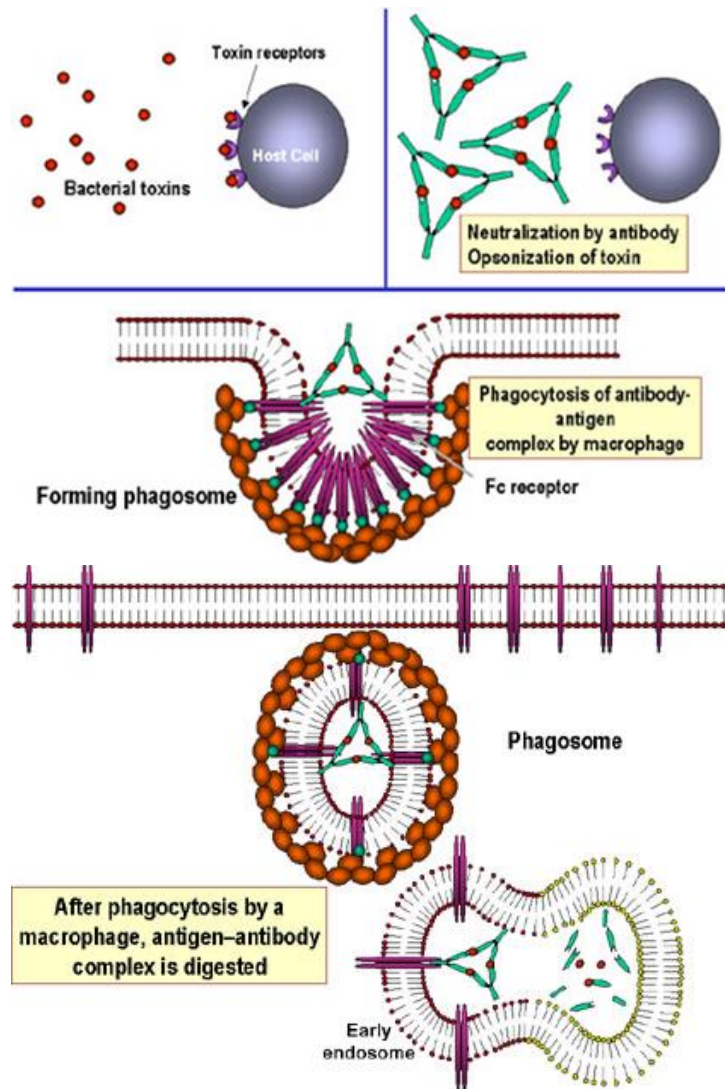
1. Extracellularly in tissue spaces
2. Intracellularly within a host cell

the immune system has different ways of dealing with pathogens at these sites. Although immune responses are tailored to the pathogen and to where the pathogen resides, most pathogens can elicit both an antibody and a cell-mediated response, both of which may contribute to ridding the host of the pathogen. However, for any particular pathogen an antibody or a cell-mediated response may be more important for defense against the pathogen.

Extracellular pathogens

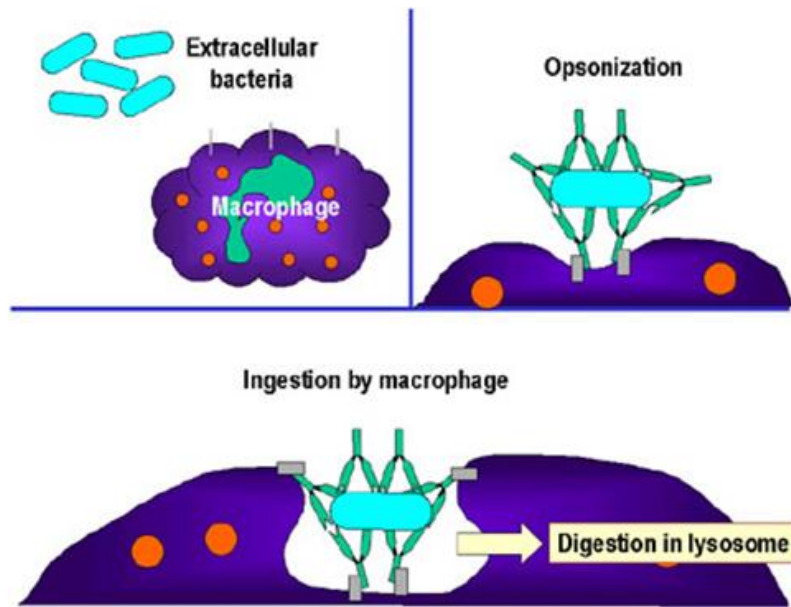
Antibodies are the primary defense against extracellular pathogens and they function in three major ways:

- **Neutralization**
By binding to the pathogen or foreign substance antibodies, can block the association of the pathogen with their targets. For example, antibodies to bacterial toxins can prevent the binding of the toxin to host cells thereby rendering the toxin ineffective. Similarly, antibody binding to a virus or bacterial pathogen can block the attachment of the pathogen to its target cell thereby preventing infection or colonization.



- **Opsonization**

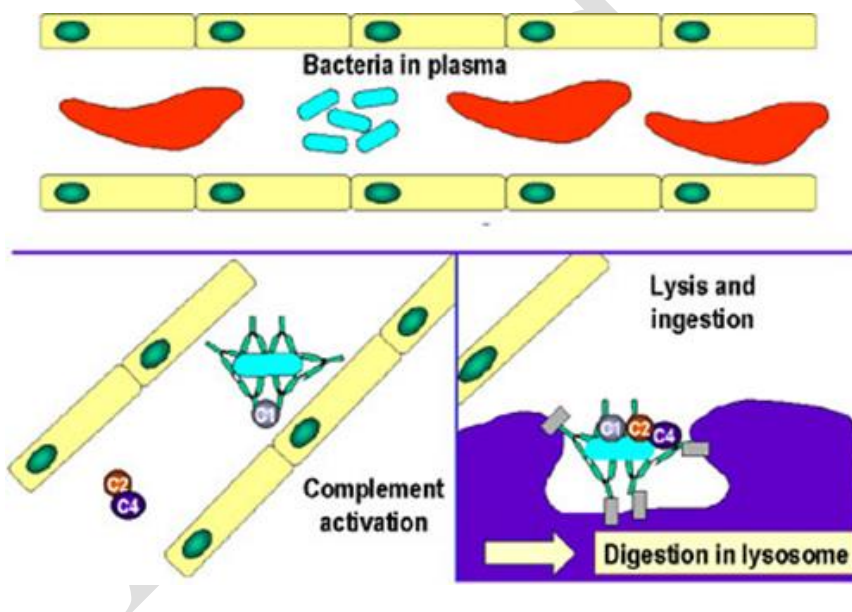
Antibody binding to a pathogen or foreign substance can opsonize the material and facilitate its uptake and destruction by phagocytic cells. The Fc region of the antibody interacts with Fc receptors on phagocytic cells rendering the pathogen more readily phagocytosed.



- **Complement**

Activation of the complement cascade by antibody can result in lysis of certain bacteria and viruses. In addition, some components of the complement cascade (e.g. C3b) opsonize pathogens and facilitate their uptake via complement receptors on phagocytic cells.

activation



Lesson 117 : Immune Response against Intracellular Pathogens:

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Intracellular pathogens

- Because antibodies do not get into host cells, they are ineffective against intracellular pathogens. The immune system uses a different approach to deal with these kinds of pathogens.
- Cell-mediated responses are the primary defense against intracellular pathogens and the approach is different depending upon where the pathogen resides in the host cell (*i.e.*, in the cytosol or within vesicles).
- For example, most viruses and some bacteria reside in the cytoplasm of the host cell, however, some bacteria and parasites actually live within endosomes in the infected host cell.
- The primary defense against pathogens in the cytosol is the cytotoxic T lymphocyte (Tc or CTL).
- In contrast, the primary defense against a pathogen within vesicles is a subset of helper T lymphocytes (Th1).
- Cytotoxic T lymphocytes (Figure 2)
CTLs are a subset of T lymphocytes that express a unique antigen on their surface called CD8. These cells recognize antigens from the pathogen that are displayed on the surface of the infected cell and kill the cell thereby preventing the spread of the infection to neighboring cells. CTLs kill by inducing apoptosis in the infected cell.

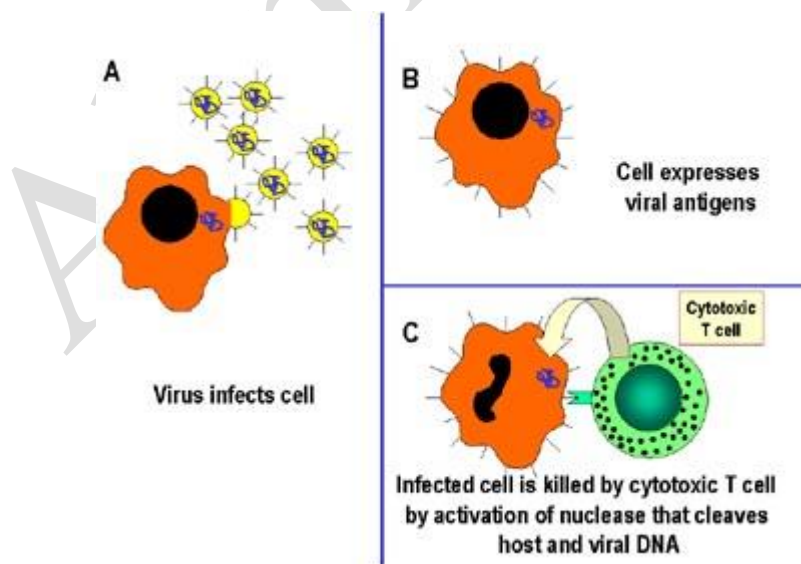


Figure 1: Mechanism of host defense against intracellular infection by viruses. Cells infected by viruses are recognized by specialized T cells called cytotoxic T lymphocytes (CTLs), which kill the infected cells directly. The killing mechanism involves the activation of nucleases in the infected cell, which cleave host and viral DNA.

- Th1 Helper T cells (Figure 3)
Th cells are a subset of T cells that express a unique antigen on their surface called CD4. A subpopulation of Th cells, Th1 cells, is the primary defense against intracellular pathogens that live within vesicles. Th1 cells recognize antigen from the pathogen that are expressed on the surface of infected cells and release cytokines that activate the infected cell. Once activated, the infected cell can then kill the pathogen. For example, *Mycobacterium tuberculosis*, the causative agent of tuberculosis, infects macrophages but is not killed because it blocks the fusion of lysosomes with the endosomes in which it resides. Th1 cells that recognize *M. tuberculosis* antigens on the surface of an infected macrophage can secrete cytokines that activate macrophages. Once activated the lysosomes fuse with endosomes and the *M. tuberculosis* bacteria are killed.

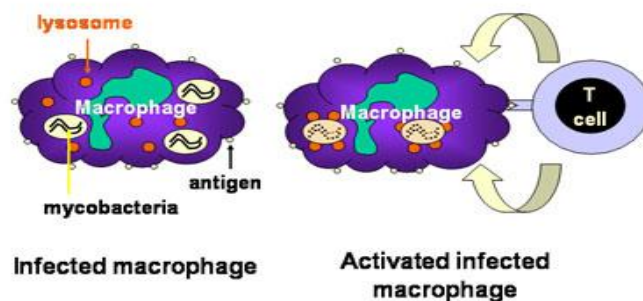


Figure 2: Mechanism of host defense against intracellular infection by mycobacteria. Mycobacteria infecting macrophages live in cytoplasmic vesicles that resist fusion with lysosomes and consequent destruction of the bacteria by macrophage bacteriocidal activity. However, when the appropriate T cell recognizes an infected macrophage it releases macrophage-activating molecules that induce lysosomal fusion and the activation of macrophage bacteriocidal activities

Although immune responses are tailored to the pathogen and to where the pathogen resides, most pathogens can elicit both an antibody and a cell-mediated response, both of which may contribute to ridding the host of the pathogen. However, for any particular pathogen an antibody or a cell-mediated response may be more important for defense against

the

pathogen.

Mechanisms of Immune Evasion by intercellular Bacteria

Mechanism of Immune Evasion	Examples
Inhibition of phagolysosome formation	<i>Mycobacterium tuberculosis</i> , <i>Legionella pneumophila</i>
Inactivation of reactive oxygen and nitrogen species	<i>Mycobacterium leprae</i> (phenolic glycolipid)
Disruption of phagosome membrane, escape into cytoplasm	<i>Listeria monocytogenes</i> (hemolysin protein)

9

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Lesson 118 : Cell Mediated Effector Response:

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Cell Mediated Effector Response

Cell Mediated Immunity

The immune response which is independent of antibodies and based on cell to cell interaction is called as cell mediated immune response or immunity. In this kind of immunity, there is activation of phagocytes and antigen specific cytotoxic T-lymphocytes which further release various cytokines from activated cells in response to various antigen. Moreover, naïve & mature T-cells are differentiated into effector T-cells after interacting with antigens via antigen presenting cells (APC). On contrary, the humoral immunity is based on cell free body fluid like serum which contain various protective molecules like antibodies. However, in cellular immunity is associated with cells for protective functions of immunization. Both innate and adaptive immune systems have humoral & cellular immunity

Cellular mediated immunity provides protection by following ways

- T-cell immunity

In this form of protection, cytotoxic T-cells (CTL) are activated for the killing of infected and transformed cells

- By activating macrophages

By this way, macrophages are activated for destroying various pathogens

- By activating natural killer (NK) cells

In this way of protection, NK cells are activated for killing of transformed and viral infected cells

Lesson 119 : Role of Cytotoxic T-Lymphocytes in Cell Mediated Immunity:

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Lesson 119

Role of Cytotoxic T-Lymphocytes in Cell Mediated Immunity

The cytotoxic T-Lymphocytes are also called as CTLs and CD8+ve. These cells are not mature after exiting thymus where these have functional TCR which can recognize antigens but cannot kill the target cells. There is need of differentiation for full activation of CTLs. CTL differentiates from Pre-CTLs in response to following two signals

1. The specific antigen in context of Class I MHC
2. The cytokines produced from T-helper 1 (Th1) cells especially IL-2 & INF- γ (Fig. 1)

CTLs cause the killing of target cells. This of killing is antigen specific where target cell must bear same antigen with class I MHC molecule

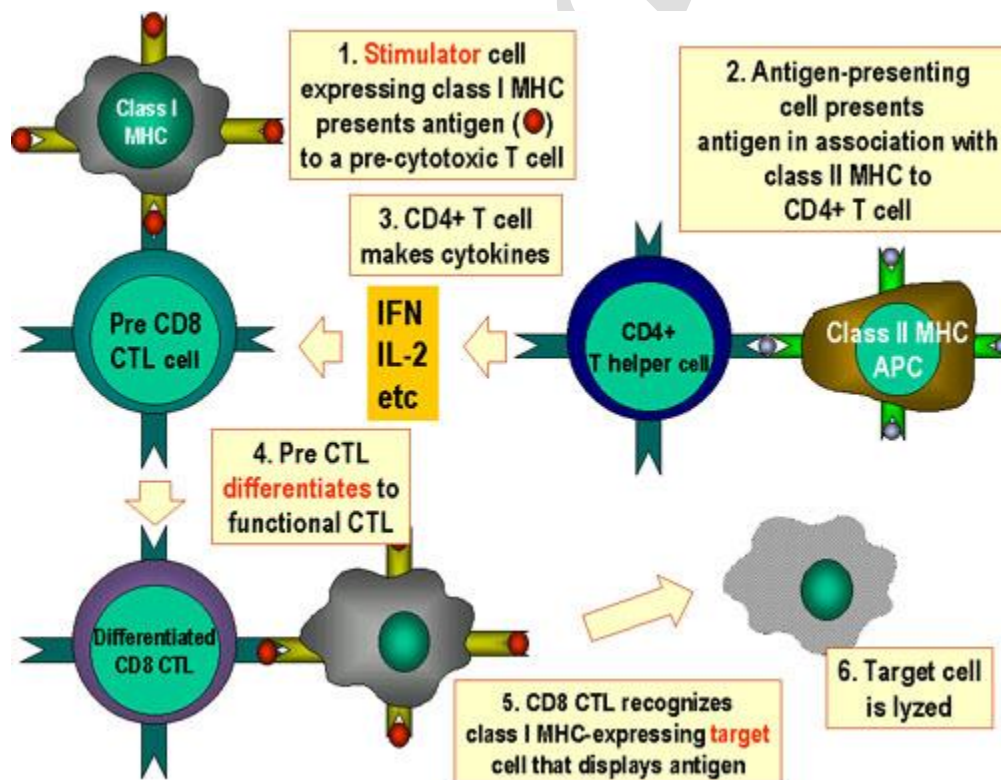


Figure 1: Maturation & Differentiation of cytotoxic T-cells (CTLs)

Moreover, CTL killing also requires cell to cell contact in which target cell should contain cell surface MHC molecule. However, CTLs are not injured while targeting the target cell. One CTL can kill sequentially numerous target cells.

Mechanisms of CTL killing

Mechanistically, CTLs rely on granule-mediated killing. In this form of killing, fully differentiated CTL have numerous granules which contain perforin & granzymes in its cytoplasm. Upon CTL degranulation, perforin monomers are polymerized to produce polyperforin molecules. These polyperforin molecules are assembled in the form of channels in the presence of calcium ions (Fig.2)

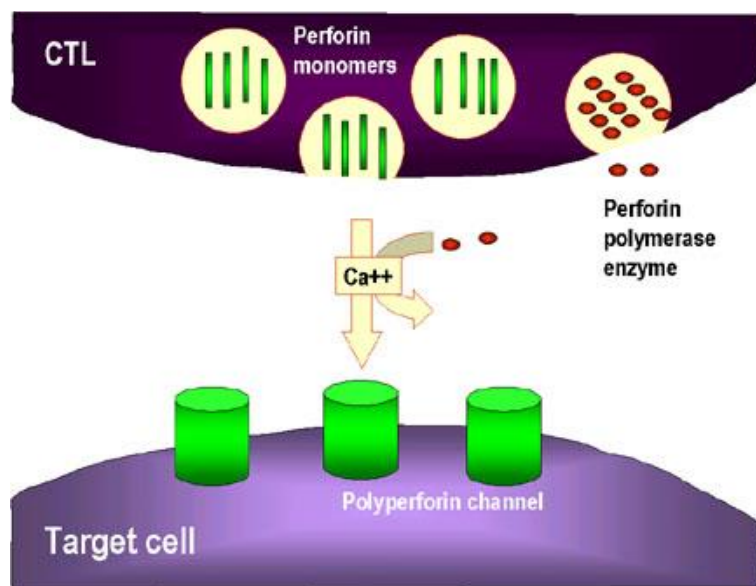


Figure 2: Release of perforin molecules from CTL and formation of polyperforin channel in the membrane of target cell

After developing polyperforin channels in the membrane of target cell, the CTL release degradative enzyme or toxins which travel through perforin channels towards target cells (Fig. 3). The killing of target cell occur with the help of degradative enzymes. Furthermore, the cytokines like TNF- α & INF- γ from CTL bind to target cells and induce programmed cell death or apoptosis of target cell

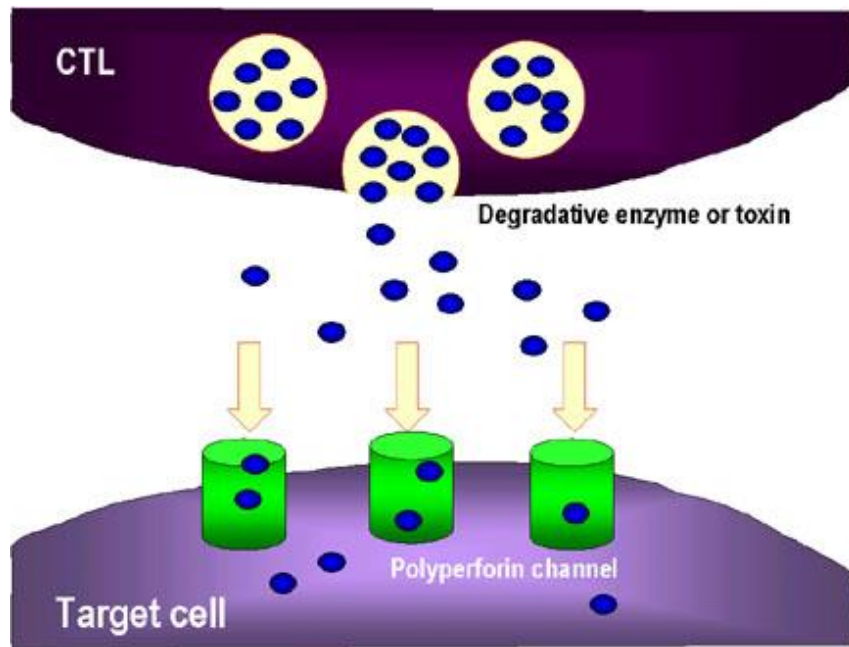


Figure 3: Killing of target cell after releasing of degradative enzyme or toxin

BT302 - Immunology

**Teacher: Dr Numan Javed (Phd Holder at SHanghai
Institute of Sciences, China)**

1 to 167 Lectures merged

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Lesson 120 : Role of Helper T-Lymphocytes in Cell-Mediated Immunity:

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Role of Helper T-Lymphocytes in Cell-Mediated Immunity

Helper T (Th)-Lymphocytes which are also called as CD4+ve cells. These cells are at the center of cell mediated immunity. Helper T-lymphocytes recognize specific antigens presented with class II MHC. Moreover, these cells are also involved in B-lymphocytes activation. Helper T-cells also release various cytokines which further activate other immune cells (Fig.1)

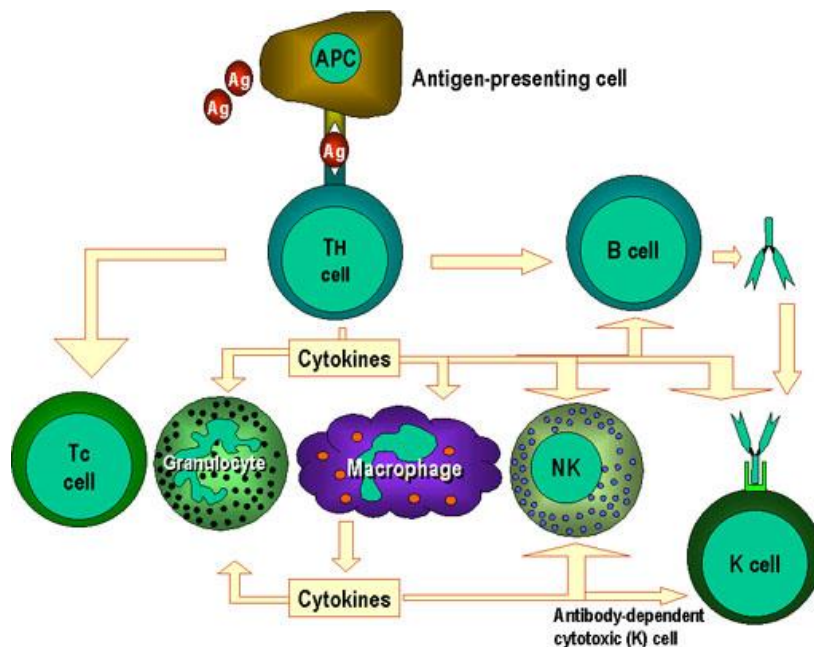


Figure 1: Functions of Helper T-lymphocytes

There are two major subtypes of Th cells

- 1) Th1 cells
- 2) Th2 cells

• Th1 cells

These cells release cytokine like IFN γ upon activation which then activate macrophages. In turn, macrophages participate in the generation & differentiation of CTL for cell mediated immunity. However, IFN γ acts as inhibitory cytokine for Th2 (Fig.2)

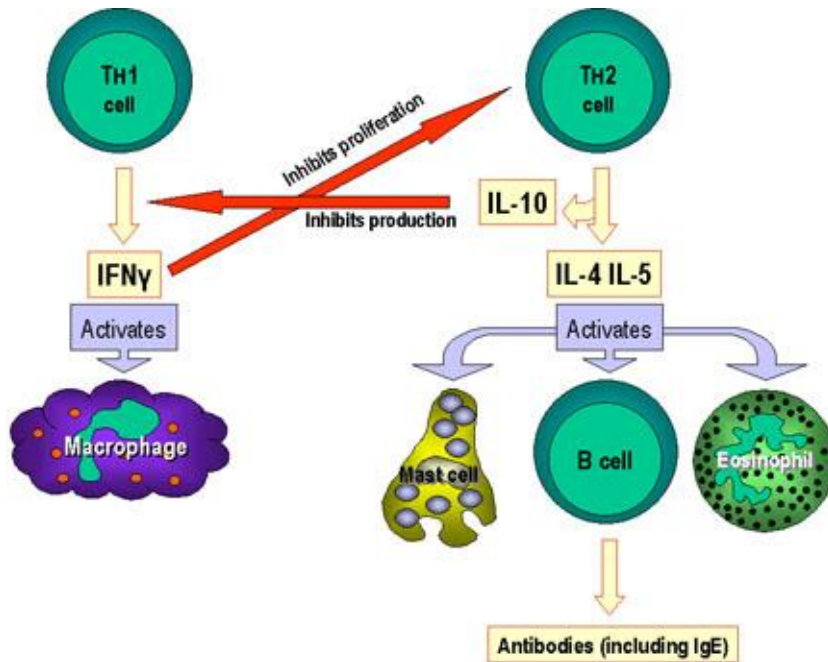


Figure 2: Function of Th1 & Th2 cells

• Th2 cells

Upon activation, Th2 cells secrete cytokines like IL-4 & IL-5 which further activate B-cells for humoral immunity. This kind of humoral response is also involved in the generation of IgE antibodies through the process of class switching. Moreover, Th2 cells are also involved in the activation of mast cells and eosinophils for providing immunity against allergen and parasitic infections. Th2 cell also secrete IL-10 which is inhibitory for IFN γ production from Th1 cells (Fig.2).

Lesson 121 : Role of NK cells in Cell Mediated Immunity:

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Role of Natural Killer (NK) cells in Cell Mediated Immunity

NK cells are also called as Large Granular Lymphocytes (LGL) as these resemble with lymphocytes except that they are larger and have numerous granules. These have surface molecules like C16 and CD56 but lack in CD3 molecule. NK cells have the ability of killing virus infected and malignant cells after activation. These are activated upon exposure to cytokines like IL-2 & IFN γ and then recognize the infected and malignant cells (Fig.1). The recognition is based on the absence of class I MHC molecule on the surface of infected cells in contrast to normal cells. Afterward, NK cells kill the target cell like CTL using perforin & graenzymes

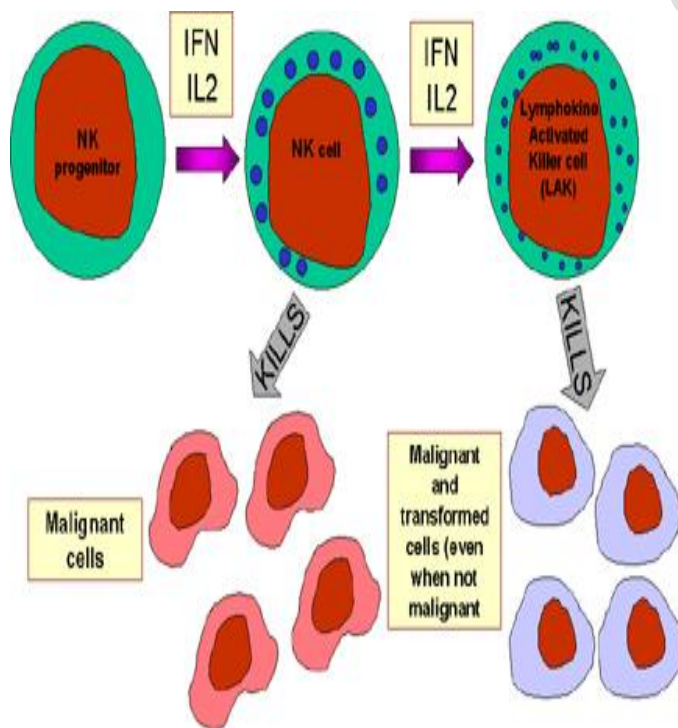


Figure 1: Activation & killing of malignant cells by NK cells

NK cells need cell to cell interaction for its effector function to kill malignant cells. NK cell cannot kill normal cells which are self-cells due to normal expression of Class I MHC. However, the viral infected and malignant cells are recognized due to absence of Class I MHC (Fig.2). In this way viral infected & tumor cells are rejected

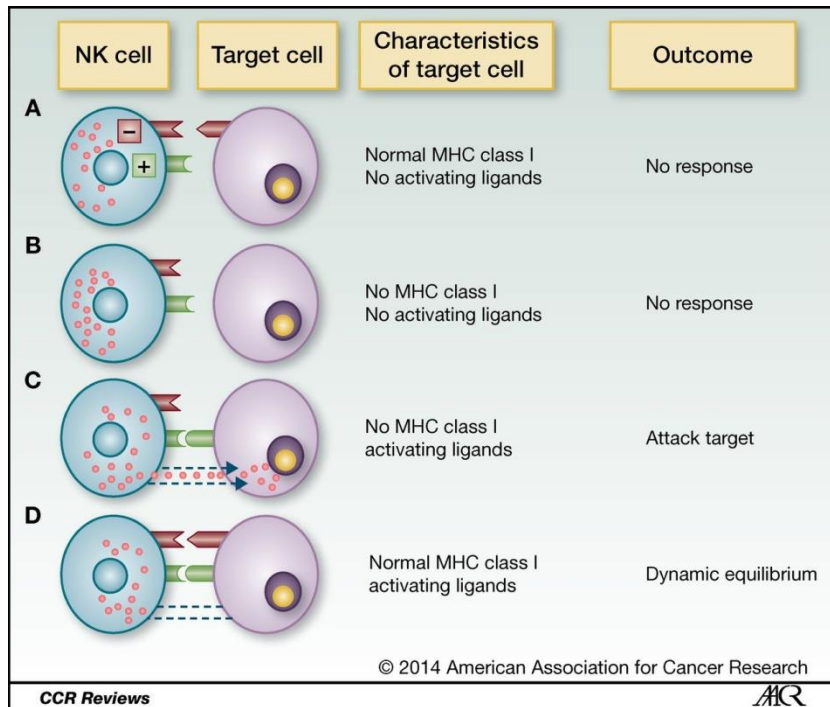


Figure 2: Functioning of NK cells based on characteristics of target cells

Lesson 122 : Antibody dependent Cellular Toxicity by Killer Cells:

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Antibody dependent Cellular Toxicity by Killer Cells

Antibody dependent Cellular Cytotoxicity or (ADCC) is a kind of cellular killing which is based on antibody and killer (K) cells. K cells are not morphologically distinct type of cells. These are any cell which can mediate ADCC like NK cell, macrophages & PMNLs. Antibody act as to bring K cell close to target cell leading towards cell to cell interaction. K cells have Fc receptors (FcR) on their surface. For such kind of killing firstly antibodies like IgG which are specific to antigens of target cell bind to target cell. The coated target cell binds to Fc receptor for IgG present on NK, LAK cells and macrophages. Moreover, the killing of the target cell is being done through perforin/granzyme mediated mechanisms (Fig.1).

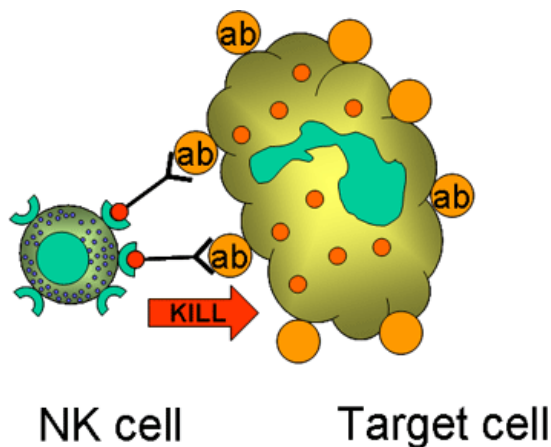


Figure 1: The process of Antibody dependent cellular cytotoxicity (ADCC)

Furthermore, ADCC can also be mediated through IgE. For this the mast cells and eosinophils are involved which have Fc receptors for IgE. Firstly, IgE coat the parasite surface which then bring coated parasitic cell to mast cells. Mast cells or eosinophils get degranulated for killing of parasite (Fig.2)

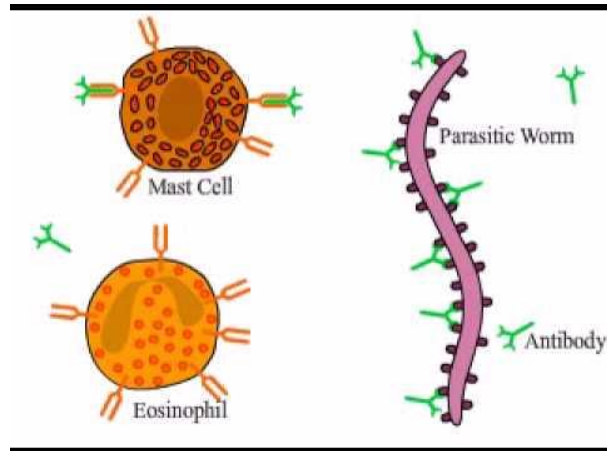


Figure 2: IgE mediated killing of parasite via ADCC

BT302 - Immunology

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Lesson 123 : Lymphokine Activated Killer Cells in Cell Mediated Immunity:

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Lymphokine Activated Killer Cells in Cell Mediated Immunity

Lymphokine activated killer (LAK) cells are NK cells which are differentiated by continuous exposure to IL-2 and interferon (Fig.1). Upon activation, LAK cells kill the transformed and malignant cells. This killing process is selective for sparing normal cells but needs cell-to-cell interaction.

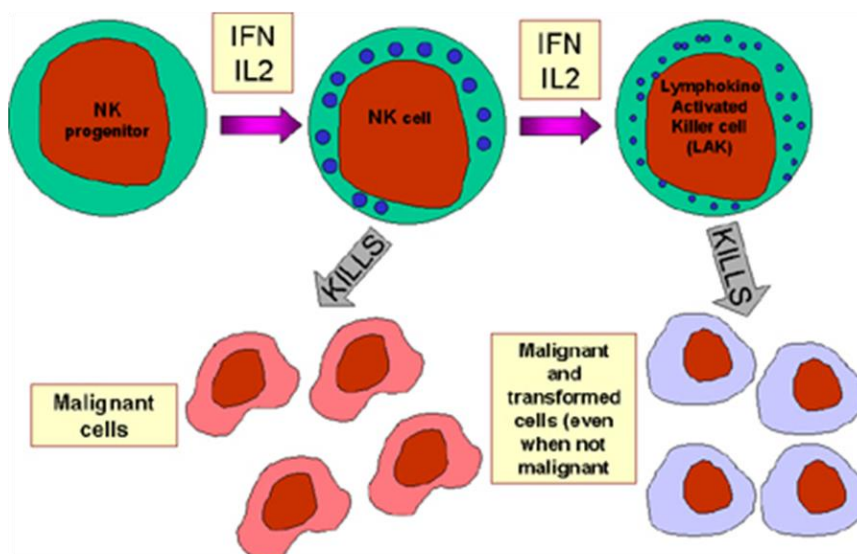


Figure 1: Activation of LAK cells

The killing is executed as tumor cells lack class I MHC, LAK cells are specific for tumor cells. LAK cells respond to those tumor cells which are resistant to NK cells killing. LAK cell therapy is used for treating various tumors.

Lesson 124 : Leucocytes Migration & Inflammation:

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Leucocytes Migration and Inflammation

Inflammation vs Infection

Inflammation

It is the body's attempt in order to protect from harmful stimulus. It is one of the complex tissue response which is in fact the part of host immune system. Inflammation is exhibited in a series of events which is also termed as inflammatory response. This is for wound healing as well as for clearance of infections. Inflammation can be beneficial for host in order to remold or restore the damaged tissue. Inflammatory response is exhibited in the form of undesirable effects which are in the form of following signs of inflammation (Fig.16.1)

1. Dolor (Pain)
2. Calor (Heat)
3. Rubor (Redness)
4. Tumor (Swelling) 5. Functio laesa (Loss of function)



Figure 16.1: Signs of Inflammation

Inflammation is indicated with a suffix “itis” for example appendicitis is the inflammation of appendix.

Outcomes of Inflammation

In response of inflammatory process, following outcomes occur

- There is restoration of normal tissue
- Large amount of tissue is destroyed which is called as fibrosis
- There is also pus formation which is also termed as abscess formation
- Inflammation can be prolonged for longer time points which is also called as chronic inflammation

Infection

Infections is not synonym to inflammation. It is the invasion of host tissue by pathogen which is the disease causing microbes for example bacteria, viruses, parasites, and fungi etc. It is also termed as infectious disease which is transmittable or communicable. Infection should comply the following Koch’s postulates

I. Infectious agent is present in patients suffering from infection however, absent in healthy individuals II. Infectious agent can grow as pure culture

Infection can be of the following nature based on its outbreak

- Epidemic: which is the sudden outbreak of infection
- Endemic: which is the constant occurrence of a particular infection in a community or population
- Pandemic: it is the global occurrence of an infection

Infections also describe the action of body after invasion in the form of inflammatory response. Infectious diseases can be various kinds according to anatomical locations like

- Respiratory tract infections
- Gastrointestinal (GIT) infections
- Skin infections
- Genital tract infections

Lesson 125 : Role of Phagocytes in Inflammation:

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Part of the inflammatory response is the recruitment of polymorphonuclear eosinophiles and macrophages to sites of infection. These cells are the main line of defense in the non-specific immune system.

Neutrophils

- Polymorphonuclear cells are recruited to the site of infection where they phagocytose invading organisms and kill them intracellularly.
- It also contributes to collateral tissue damage that occurs during inflammation.

Macrophages

- Tissue macrophages and newly recruited monocytes which differentiate into macrophages, also function in phagocytosis and intracellular killing of microorganisms.
- In addition, macrophages are capable of extracellular killing of infected or altered self target cells.
- Furthermore, macrophages contribute to tissue repair and act as antigen-presenting cells, which are required for the induction of specific immune responses.

The phagocytes are inflammatory cells in nature like macrophages and neutrophils. They have the ability of recognition of infectious agent via receptors on them which are termed as pattern recognition receptors (PRRs).

Followings are the different kinds of PRRs

- **Fc receptors**

Bacteria with IgG antibody on their surface have the Fc region exposed and this part of the Ig molecule can bind to the receptor on phagocytes. Binding to the Fc receptor requires prior interaction of the antibody with an antigen. Binding of IgG-coated bacteria to Fc receptors results in enhanced phagocytosis and activation of the metabolic activity of phagocytes (respiratory burst).

- **Complement receptors**

Phagocytic cells have a receptor for the 3rd component of complement, C3b. Binding of C3b-coated bacteria to this receptor also results in enhanced phagocytosis and stimulation of the respiratory burst.

- **Scavenger receptors**

Scavenger receptors bind a wide variety of polyanions on bacterial surfaces resulting in phagocytosis of bacteria.

- **Toll-like receptors**

Phagocytes have a variety of Toll-like receptors (Pattern Recognition Receptors or PRRs) which recognize broad molecular patterns called PAMPs (pathogen associated molecular patterns) on infectious agents. Binding of infectious agents via Toll-like receptors results in phagocytosis and the release of inflammatory cytokines (IL-1, TNF-alpha and IL-6) by the phagocytes.

PRRs recognize various molecular patterns on the surface of pathogens which are called as pattern associated molecular patterns (PAMPs). These are in the following forms

- Lipopolysaccharides (LPS) - These are recognized by TLRs
- Flagellins - These are also recognized by TLRs
- Cell wall component

These are recognized through complement receptor after binding with complement

Phagocytes ingest infectious agent by the process of phagocytosis (Fig.1)

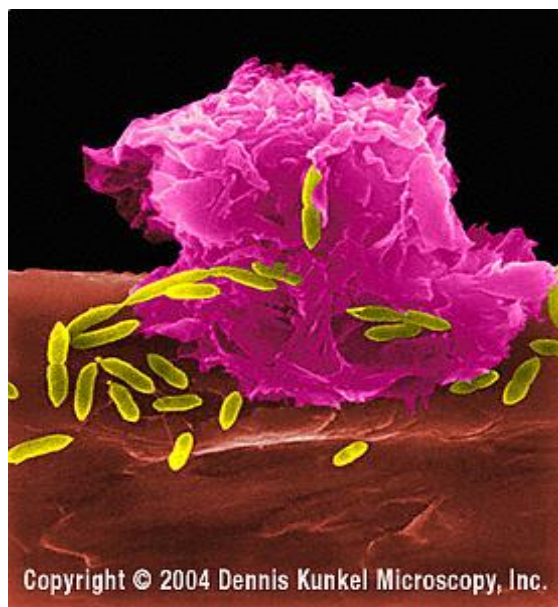


Figure 1: The process of phagocytosis of bacteria by phagocytes

Neutrophils ingest the pathogen first and then kill them intra-cellularly. During inflammation there is collateral tissue damage. Macrophages also have the ability to ingest and kill the infectious agent. Macrophages are Tissue macrophages which generate inflammatory response in tissue. They also contribute in tissue repair and the process of antigen presentation.

Lesson 126 : Chemotaxis & Diapedesis of Leucocytes in Inflammation:

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Chemotaxis & Diapedesis of Leucocytes in Inflammation

Leucocytes provide response against infectious agents by

- **Chemotaxis**

This is the movement of circulating phagocytes towards the site of infection by various chemicals also called as chemotactic agents for example save our soul (SOS) signals from bacteria like Nformylmethionine containing peptides. The clotting system peptides and complement proteins also serve as chemotactic agents. For macrophages, cytokines are considered as good chemotactic agents (Fig. 1).

- **Adherence of phagocytes on endothelial surface**

SOS signals also activate endothelial cells of blood vessels. As a result, there is an increased expression of adhesion molecules like ICAM-1 and selectins. This increased expression facilitates the binding of phagocyte to adhesion molecules (Fig. 1).

- **Rolling of phagocytes**

The release of vasodilators causes the loosening of the gap between endothelial cells. After loosening of endothelial cells there is squeezing of phagocytes occur between endothelial cells (Fig. 1).

- **Diapedesis**

This is the process of crossing the endothelial barrier after squeezing & movement extravascularly towards the site of infection. Infected tissues sites cause the attraction of phagocytes called as chemotaxis and the activation of phagocytes (Fig. 1).

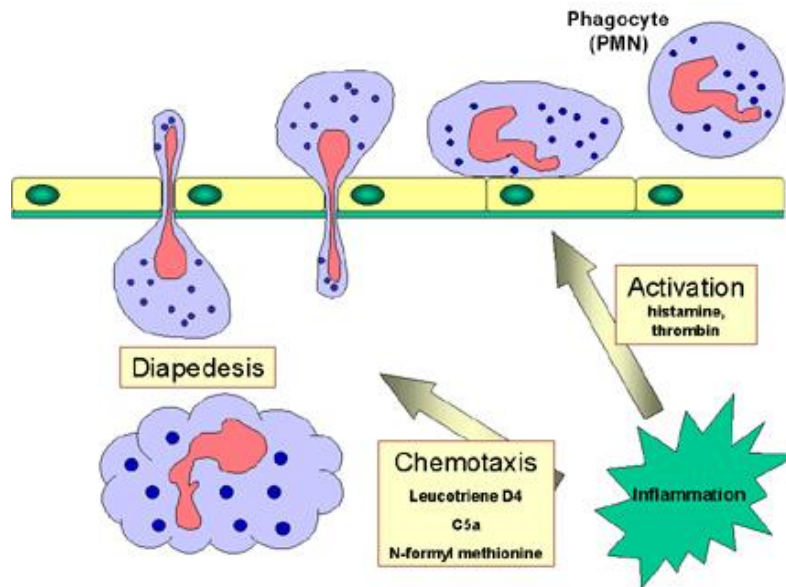


Figure 1: The events for the process of phagocytosis

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Lesson 127 : Opsonization of Bacteria:

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Opsonization of Bacteria

The enhancement of phagocytic process is called as opsonization. In this process, the pathogen is marked for ingestion by phagocytes. The marking of bacteria by a molecule which enhance the phagocytosis is termed as opsonins like immunoglobulin (IgG & IgM) and complement proteins.

In this process of opsonization, the opsonins bind with bacterial surface. However, under normal inflammatory circumstances, PAMPs of bacteria bind with PRRs of phagocytes which mediate neutrophil or macrophage phagocytosis. PRRs also cause the expression of opsonin receptor on phagocytes like Fc receptors and complement receptors

There is interaction of opsonin receptor of phagocytes with opsonin on bacterial surface which cause the binding. After binding, the bacteria cell is internalized in the vesicle called as phagosome where killing of bacteria occur through intracellular killing.

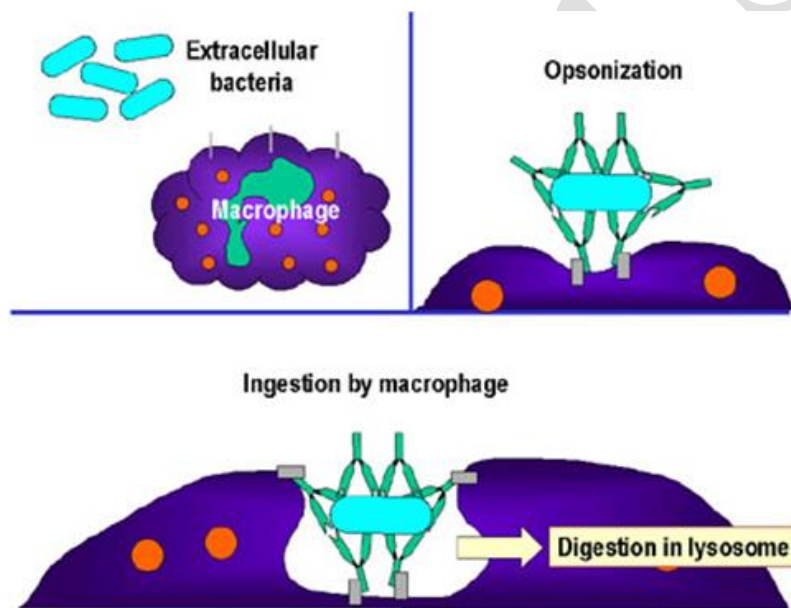


Figure 1: Opsonization of bacteria by opsonins

Lesson 128 : Intracellular Killing by Leucocytes during Inflammation:

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Intracellular Killing by Leucocytes during Inflammation

During inflammation, the ingested bacteria are killed by phagocytes. After phagocytosis, the ingested bacteria is being killed by a process called as intracellular killing

There are following two ways of intracellular killing

- Oxygen independent
- Oxygen dependent

1) Oxygen Independent

In this process of intracellular killing of bacteria, there is no need of oxygen. The granules and vesicles of phagocytes secrete various enzymes which are hydrolytic proteins. Those proteins are bacteriocidal in nature according to their modes of action (Table 1)

Table 1: List of hydrolytic proteins with their functions

Effector Molecule	Function
Cationic proteins (including cathepsin)	Damage to microbial membranes
Lysozyme	Splits mucopeptide in bacterial cell wall
Lactoferrin	Deprives proliferating bacteria of iron
Proteolytic and hydrolytic enzymes	Digestion of killed organisms

2) Oxygen Dependent

There is requirement of oxygen for such kind of intracellular killing of bacteria. It is also termed as “Respiratory Burst” as requirement of glucose and oxygen get increased after phagocytosis. As a result, oxygen containing bacteriocidal radicals are produced. This process can be of two types

Oxygen dependent Myeloperoxidase (MPO) dependent MPO from granules of phagocytes are involved in this process as a result halide ions (OCl^-) are formed as a result of the following reactions (Fig. 1). These are bacteriocidal in nature

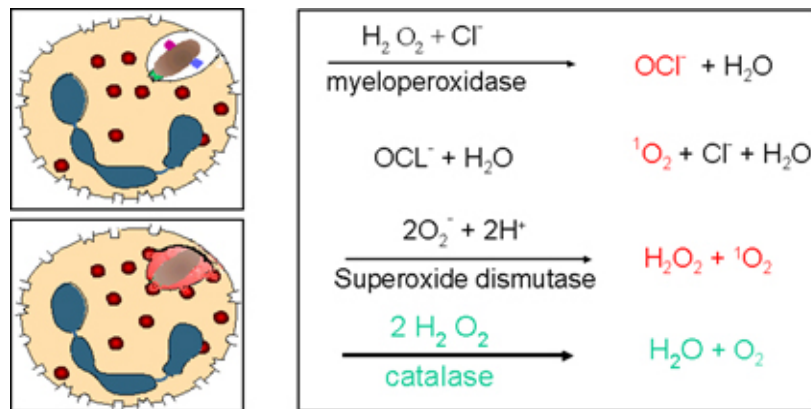


Figure 1: Oxygen dependent Myeloperoxidase (MPO) dependent

Oxygen Dependent Myeloperoxidase (MPO) independent

It is myeloperoxidase independent in which there is involvement of Hexose monophosphate shunt. The reactive oxygen species (ROS) for example superoxide radicals, hydrogen peroxide and singlet oxygen as a result of following reactions (Fig.2)

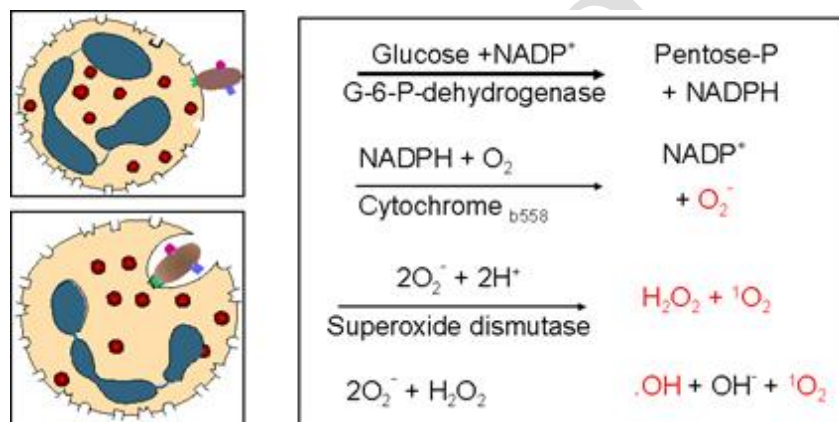


Figure 2: The process of Oxygen dependent myeloperoxidase (MPO) independent

Lesson 129 : Role of Cytokines in Inflammation:

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Role of Cytokines in Inflammation

Cytokines are considered as mediators of inflammation. A complex variety of mediators are involved in acute inflammatory response. Some of cytokines directly act on smooth muscles wall surrounding the arterioles in order to alter the blood flow. On the other hand, some act on venules to cause contraction of endothelial cells for the opening of junctions as a result there is migration of leucocytes from bloodstream.

Moreover, cytokines also up regulate the expression of adherence molecules on endothelial cells. Likewise, the adherence molecules on the surface of leucocytes are also up regulated. Cytokines also lead the leucocytes towards the inflamed site through chemotaxis.

For inflammation cytokines can be grouped into two groups

• Pro-Inflammatory cytokines

These are involved in up regulation of inflammatory process. These are predominantly produced from activated macrophages and Th cells like IL-1 β , IL-6 & TNF- α , IL-12 and INF- γ . These have role in mediating innate immune response. Furthermore, various inflammatory diseases like atherosclerosis, cancer and depression also occur due to these cytokines.

• Anti-Inflammatory cytokines

These cytokines are involved in down regulation of inflammatory process. These also control proinflammatory cytokine response. In this category of cytokines IL-4, IL-10, IL-11 and IL-13 are involved. These have pathological role in systemic inflammatory states. A balance between Pro & anti-inflammatory cytokines is required.

Lesson 130 : Inflammasomes in Inflammation:

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Inflammasomes in Inflammation

A multi-protein oligomer which is responsible for activation of inflammatory responses is called as **inflammasomes**.

These promote the maturation and secretion of pro-inflammatory cytokines like IL-1 β and IL-18.

Inflammasomes are expressed in myeloid cells which is the important component of innate immune system. There is secretion of cytokines upon inflammasomes activation cells leading towards death process. The kind of cell death induced after inflammasomes is called as **pyroptosis**.

It mediates the inflammatory process by killing infected cells in physiological state while pathologically increased pyroptosis leads towards inflammatory diseases like Rheumatoid arthritis, inflammatory bowel disease, Inflammatory disorder, metabolic disorders and cancer. The germline encoded PRRs of immune cells like TLRs, Nod like receptors (NLRs) recognize PAMPs on pathogens lead the assembling of inflammasomes. As a result, caspase-1 gets matured which further cleaves the pro or inactive forms of pro-inflammatory cytokines like IL-1 β & IL-18 into mature ones (Fig.1)

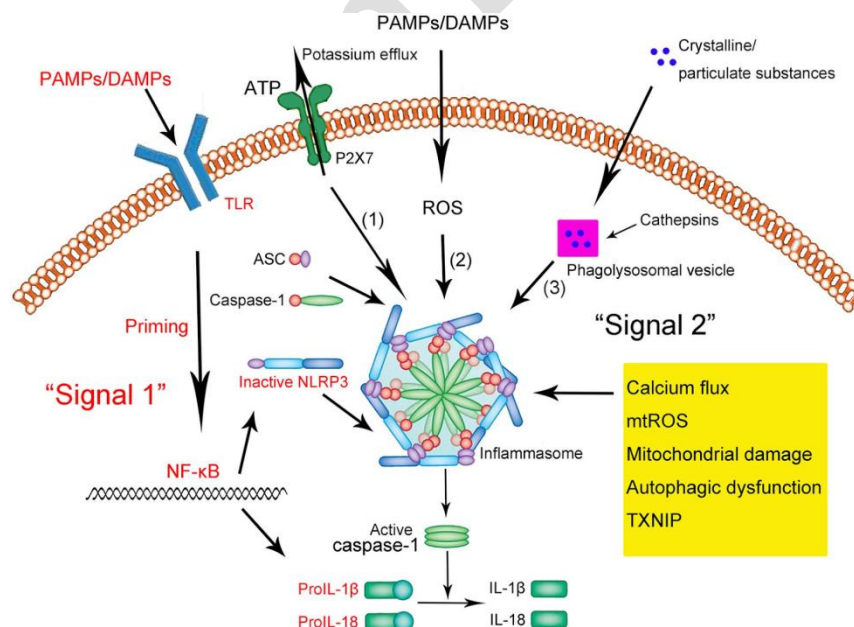


Figure 1: Activation of inflammasomes by pathogens

Inflammasomes are activated either by microbial attack or sterile attack (Fig.2)

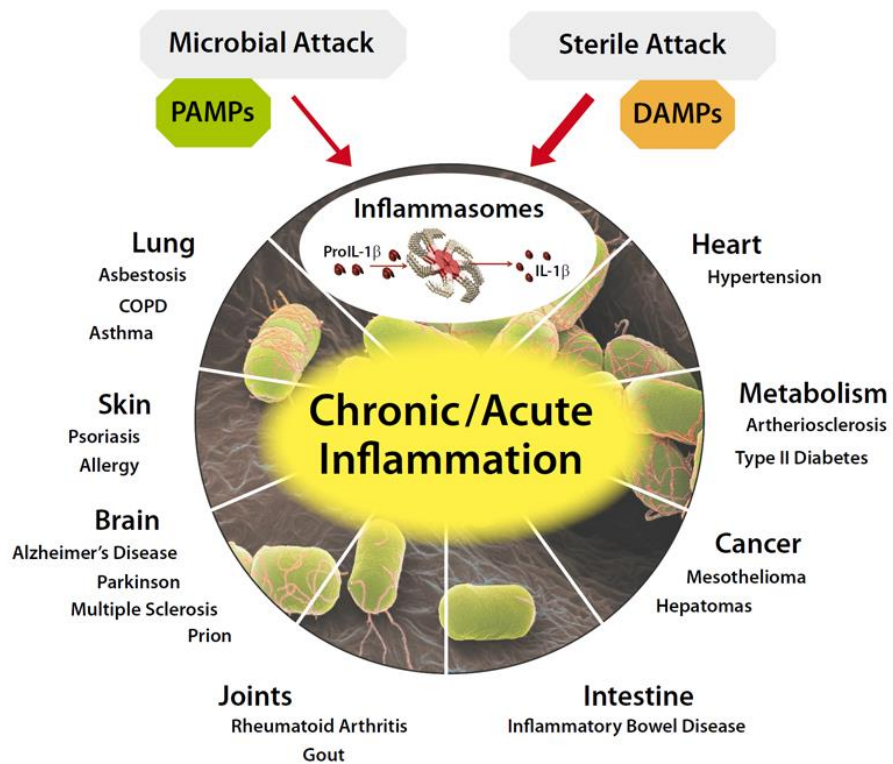


Figure 2: Stimulus for inflammasome activation in various tissues of host

Lesson 131 : Acute vs Chronic Inflammation:

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Acute vs Chronic Inflammation

Acute Inflammation

This form of inflammation is a short term process occurring in response to tissue injury. It appears in minutes to hours. It is also involved purely in physical damage and then healing process. There is also activation of immune system in this sort of inflammation.

Three main processes involved in acute inflammation which are as follows

1. Increased blood flow
2. Increased permeability
3. Migration of Leucocytes

Increased blood flow

It happens due to dilatation of blood vessels in the effected tissue. Redness appears in the affected area with increase in temperature of effected area

Increased permeability

There is also increase in vascular permeability which causes the leakage of plasma into tissue interstetium. As a result, there is accumulation of fluid in tissue leading towards edema which is the swelling of the affected area due to additional accumulation of fluid into interstitial space of the region. The endothelial cells also separate due to increased vascular permeability (Fig.1).

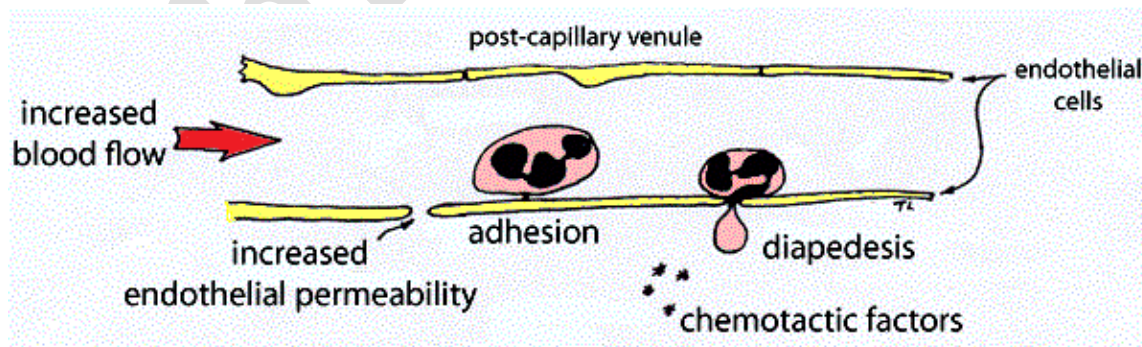


Figure 1: The process of acute inflammation

Migration of Leucocytes

The increased movement of leucocytes towards the site of infection is called as chemotaxis. Cytokines also play their role in migration of leucocytes from circulation to site of infection. Moreover, the expression of adhesion molecules on vessels and leucocytes occur

Chronic Inflammation

This form of inflammation persists for long time. It is a slow process occurring in response to tissue injury. This process lasts for months to even years. The hallmark of chronic inflammation is macrophages and lymphocytes as neutrophils are short lived so those are replaced by lymphocytes and macrophages.

There are two types of chronic inflammation

- Non-Specific proliferative inflammation
- Granulomatous inflammation

Non-Specific Proliferative Inflammation

It is characterized by the presence of non-specific granulation tissue. There is also infiltration of mononuclear cells like Lymphocytes, monocytes and plasma cells. In this process, proliferation of fibroblasts also occur which is called as fibrosis. Moreover, the proliferation of connective tissues and vessels occur for example in case of lung abscess (fig. 2).

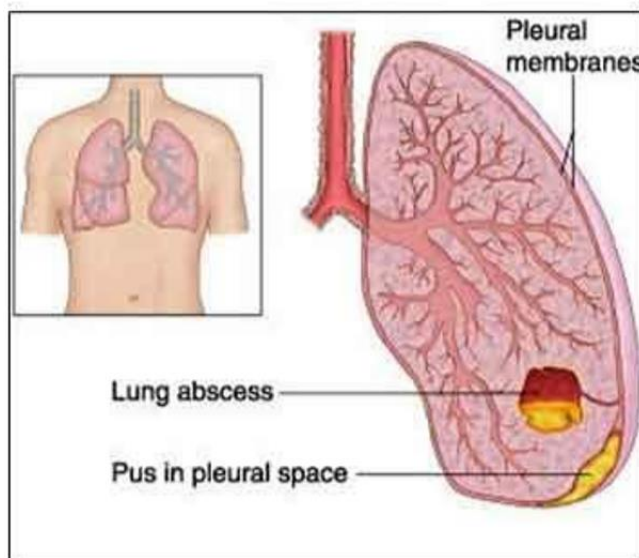


Figure 2: Non-Specific Proliferative Inflammation of lung

Granulomatous Inflammation

It is characterized by the presence of distinct nodular tissue called as granuloma which is the aggregation of activated macrophages with their derived cells called epithelioid cells. Infectious granuloma occur due to infections for example tuberculosis and leprosy (Fig. 3).

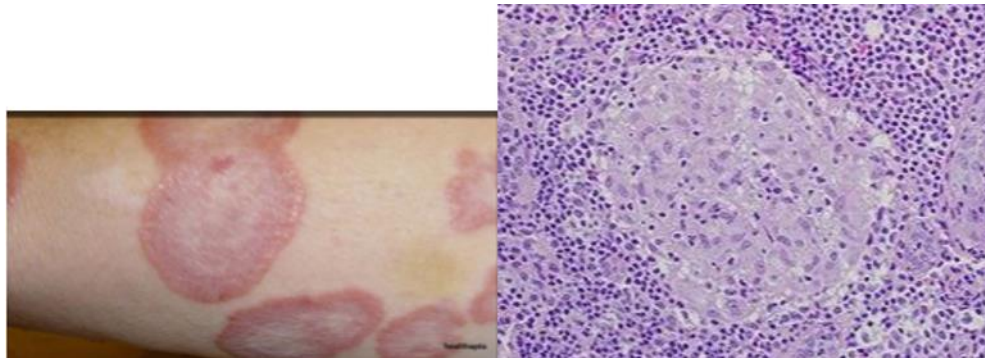


Figure 3: The Granulomatous Inflammation

Adverse Effects of Inflammation is associated with general flu like symptoms including

- Fever
- Chills
- Fatigue/Lethargy/Loss of energy
- Headaches
- Loss of appetite
- Body pains
- Muscle stiffness

Lesson 132 : Vaccines:

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Vaccines

Introduction to Vaccination (Immunization)

The ways of providing specific protection against many common and damaging pathogens after stimulation of organism's or individual's immune system is called as vaccination or immunization.

For vaccination, there is stimulation of humoral immunity for production of antibodies against pathogen. These antibodies has the ability to neutralize specific antigens of pathogens. This is depending upon the nature of pathogenesis and the site of infection by pathogen. These antibodies are also called as Antitoxin which neutralize the toxin production from pathogens. Those toxins get masked by antitoxins and not able to bind its specific receptor on target cell. Moreover, antibodies against pathogens also bind complement and lead to lysis of pathogen which is also termed as complement mediated intracellular killing of pathogen. However, intracellular pathogens cannot be neutralized by antibodies for which cell mediated immunity is activated by providing specific T-cells to host. Immunization against intracellular pathogen is provided through cell-mediated immunity

BT302 - Immunology

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1 to 167 Lectures merged

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Lesson 133 : Types of Vaccination:

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Types of Vaccination

There are two modes of providing specific protection against mostly common & damaging pathogens. These modes of immunization can be natural or artificial in their nature. Following are the types of immunization

1. Active Immunization
2. Passive Immunization

• Active Immunization

It is the induction of immunity after exposure of host to antigen. In this case, antigenic exposure is mandatory. Active immunization can occur naturally by exposing to microbe or other antigen when there is no prior exposure before the entry of antigen. Similarly there is also no pre made antibodies in the host's serum. In this form of immunization, immune system develops antibodies against the microbe after its exposure. This process of antibodies generation is considered slow. Moreover a memory response is also generated in which antibodies remained in use for longer time. This form of immunization can be artificial in nature which is achieved by either injection of microbe before natural exposure or treated microbes or toxins.

• Passive Immunization

It is the induction of active humoral immunity. In this kind of immunization, antibodies are mandatory. Passive immunization can occur naturally by transfer of maternal antibodies to fetus through placenta. However, the artificial induction can be done by injecting gamma globulins from other individuals or animals to host.

Lesson 134 : Natural vs Artificial Immunization:

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Natural vs Artificial Immunization

Natural active immunization

Natural immunization is acquired immunization which is achieved by natural exposure to antigen or acquiring infection by an infectious agent or microorganism e.g measles virus. This is also termed as natural active immunization. This form of natural exposure to antigen stimulate the humoral immunity as a result specific antibodies are generated against exposed microbe. Moreover, there is also activation of cell mediated immunity for intracellular pathogen.

Artificial Active Immunization

Artificial immunization is the kind of immunization which is acquired by artificial exposure to antigen or microorganism which is administered to body in less virulent or attenuated form like heat killed forms of microbes. This artificial exposure with antigen stimulate the humoral immunity in the form antibodies generation. Moreover, artificial activation of cell mediated immunity for intracellular pathogen for example BCG for Mycobacterium tuberculosis. Artificial immunization: acquired immunization by artificial exposure. This artificial exposure to antigen or

microbe which is by administration of less virulent (attenuated), heat killed microbes to organism. Artificial exposure to antigen stimulate the humoral immunity for antibodies

Natural Passive Immunization

On the other hand, natural immunization can be acquired immunization which can be obtained by isolating gamma globulins (antibodies) from other's individual. This is called as natural passive immunization. Furthermore, the transfer of gamma globulins (IgG) from mother's blood to fetus via placenta. The placental cells contain FcR γ for IgG transfer to fetal circulation. Similarly, the natural transfer of IgA can be occurring through breast milk to new-born baby.

Artificial Passive immunization

This form of artificial immunization is the acquired immunization which is achieved by administering gamma globulins or antibodies from others individual's or immune animal's serum for example antibodies against tetanus.

Lesson 135 : Artificial Passive Immunization:

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Artificial Passive Immunization

This form immunization is done by transferring or injecting of specific gamma-globulins from other individual or from other immune animals .Artificial passive immunity is used against various following acute bacterial infections

- Diphtheria
- Tetanus

Moreover, artificial passive immunity is also used against various poisoning conditions like

- Insects or sting biting e.g immunization against various strains of Malaria producing Plasmodium
- Reptiles e.g anti-venom administration in case of snake biting
- Food poisoning (Botulism)

Artificial passive immunity is also used as prophylactic (preventive) measure against various viral infections like Influenza, Poliomyelitis, Measles and Rabies.

For artificial passive immunization, antibodies which are used usually from human origin which are also termed as homologous antisera. Homologous antisera has potential risk of transmitting HIV and Hepatitis among individuals. However, these are also raised in other species or animals after immunization called as Heterologous antisera. Heterologous antibodies provide immediate protection but also a source of allergies like serum sickness and anaphylaxis.

Likewise, passive transfer of cell mediated immunity is also used for treating cancer and immunodeficiency

Lesson 136 : Artificial Active Immunization:

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Artificial Active Immunization

This form of immunization is achieved by transferring or injecting live, dead or other components of microbes

Followings are the important vaccines which are used as artificial active immunization

1. Live (attenuated) organisms
2. Killed organisms
3. Sub-unit vaccines

1) Live (attenuated) Vaccines

These vaccines are produced after inactivation of organism by heat for the purpose of loss of pathogen's virulence. These are used against various viral infections like Smallpox, measles, mumps, hepatitis A virus Moreover, these are used as live bacterial vaccine against tuberculosis: for example Bacille Calmette-Guerin vaccine (BCG)

2) Killed Vaccines

Such vaccines are produced after killing of organism by heat, chemical treatment or UV irradiations. These are used against various viral infections like Polio, rabies & influenza. Also, most bacterial vaccines are also killed like typhoid fever, cholera, plague and pertussis.

3) Subunit Vaccines

These vaccines consist of various components of microbes like polysaccharides from capsule and surface proteins. The polysaccharides are T-independent antigens while proteins are T-dependent antigens. These vaccines are used for reducing the toxicity for example Pneumococcus vaccines.

Lesson 137 : Novel Vaccines:

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Novel Vaccines

There are also various novel ways of designing vaccines in order to reduce in toxicity of vaccines. These have the potential to provoke both humoral & cell-mediated immunity. These are mainly used in experimentally for research purpose. However, these would be available for clinical use in future

1) DNA vaccines

These are cloned viral peptides which are transfected into host cell. These vaccines can generate both humoral and cell-mediated response like live attenuated vaccines. Similarly, Anti HIV-DNA vaccines are used in experimental stage which have very low efficacy so far.

2) Immunodominant Peptides

These are simple and easy to prepare. Moreover, these peptides are incorporated with MHC to induce both humoral & cell-mediated responses.

3) Anti-Idiotypic antibodies

These antibodies are used against polysaccharide antibodies. These have long lasting immune response with memory

Lesson 138 : Role of Adjuvant in Vaccination:

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Role of Adjuvant in Vaccination

These are the substances which increase the antigenicity of weak antigens. These can be of two forms

1. Chemicals
2. Biological

1) Chemical

These are basically aluminum salt or Alum in its structure. This is the only chemical suitable for human use. These are used in DTP (Diphtheria, Tetanus & Pertussis) vaccine. Alum causes slow release of antigen in order to increase in TLR interaction, activation of mononuclear phagocytes. As a result, increased cytokine secretion is induced for appropriate immune response generation

2) Biological adjuvants

Various bacteria are used as biological adjuvants. These are basically the bacterial products

3) B. pertussis used as adjuvant

4) M. bovis (BCG and others)

These are used in combination with oil and detergent for better activation of mononuclear phagocytes and cytokine secretion.

Lesson 139 : Adverse Effects of Vaccination:

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Adverse Effects of Vaccination

As live microbe are used for vaccination so these have some adverse effects on host. These effects can be severe, rare, mild and moderate in its nature. For example, active immunization causes fever, malaise and discomfort. Followings are the adverse effect in respective kind of vaccination

- Joint pains (Arthritis): Rubella vaccination
- Can be fatal: in case of Pertussis
- Neurological disorders: Influenza vaccination
- DPT vaccination has local effects like Redness, swelling and pain
- Mild/Moderate effects: fever, drowsiness. Vomiting and anorexia
- Severe effects: persistent crying, fever, convulsions, collapse, acute encephalopathy, permanent neurological deficit due to pertussis components

Lesson 140 : Tolerance, Diseases of Immune System-Autoimmunity:

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Tolerance, Diseases of Immune System-Autoimmunity

Introduction to Tolerance

The specific immunological unresponsiveness or non-reactivity to an antigen resulting from a previous exposure to same antigen is called as immunological tolerance. This kind of unresponsiveness is mostly against self-antigens. This form of non-reactivity can be induced for non-self (foreign) antigen. The antigens which induce tolerance are called as tolerogen. Physiologically, there is no immune response against self-antigens which is also termed as selftolerance. On the other hand, if immune response is generated against self-antigens is called as autoimmunity (Pathology). Immune system recognize self-antigens and mount strong immune response. The discrimination between self and non-self-antigens is achieved by a process called as self MHC recognition. The tolerance can be generated against non-self-antigens. However, immune response should be against non-self-antigens. The modification of antigens leads towards immune tolerance. Most bacteria and viruses develop tolerance which is the way to exploit or evading host immune system for example in Lepromatus type of leprosy there is tolerance against M. leprae as a result no immune response is generated. Tolerance to tissue and cell antigens can be induced artificially for example by injecting hemopoietic stem cells in neonatal or severely immunocompromised animals. Likewise, the generation of chimeras animals through transferring of allogeneic primary lymphoid tissues at early developmental life of animal. These are the ways of inducing tolerance against allogeneic tissues.

Lesson 141 : Immunological features of Tolerance:

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Immunological features of Tolerance

Tolerance is a phenomenon of specific immune unresponsiveness which is different from nonspecific immunosuppression & immunodeficiency. This is an active antigen dependent process in response to antigen. Like specific immune response, tolerance is also specific and has immunological memory. Tolerance to T-cells is longer as compare to B-cells. Similarly, the induction of tolerance to T-cells is easier and require small amount of antigen (Tolerogen). However, for B-cell tolerance requires larger amount of tolerogen. For the maintenance of tolerance the persistence of antigen in the body is necessary. The lack of persistence leads to breach of tolerance. Immuno-Tolerance can be break in following two ways

1) Naturally

In case of autoimmune disorders e.g Rheumatoid Arthritis, SLE etc.

2) Artificially

Exposure to immunosuppressive drugs or X-ray irradiation e.g in case of experimental animals for bone marrow transplantation

Lesson 142 : Mechanisms of Tolerance Induction:

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Mechanisms of Tolerance Induction

There are following two forms of immunological tolerance

- Central Tolerance - - It occurs in primary lymphoid organs e.g bone marrow & thymus
- Peripheral Tolerance - - It occurs at secondary lymphoid organs e.g spleen, lymph nodes, tonsils etc

Central Tolerance

In this kind of tolerance the mechanism involved is Clonal deletion. It is also called as negative selection by which Self-reactive B & T-lymphocytes are deleted in bone marrow & thymus respectively. The Clones of auto reactive cells are deleted by programmed cell death is called as apoptosis. B-cells during development in bone marrow encounter with self-soluble or cell surface associated antigen. Such kind of self-reactive B-cells are deleted from bone marrow through negative selection via apoptosis process (Fig. 1)

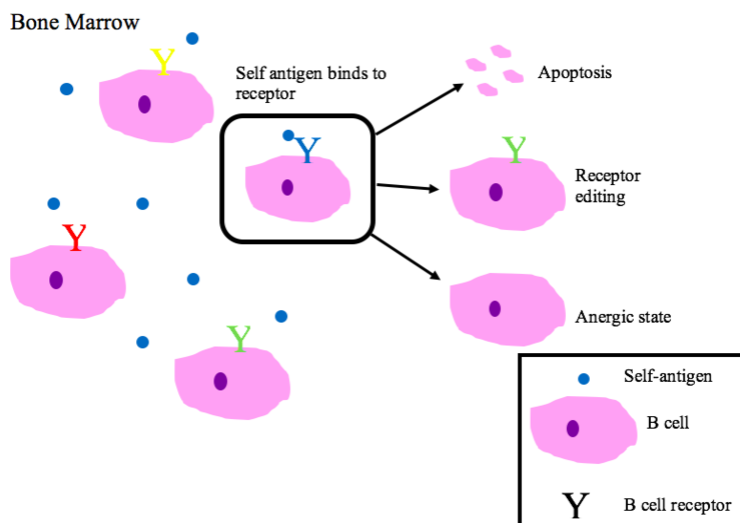


Figure 1: The process of central tolerance for B-lymphocytes in bone marrow

Similarly, T-cells develop in thymus after expressing CD8 & CD4 molecules on its surface. These cells also acquire $\alpha\beta$ TCR. The positive selection of T-cells occur after interacting with self MHC molecule. However, self-reactive T-cells with either class I or II are negatively selected. In this way auto-reactive T-cells are controlled to escape from central lymphoid tissues

Peripheral Tolerance

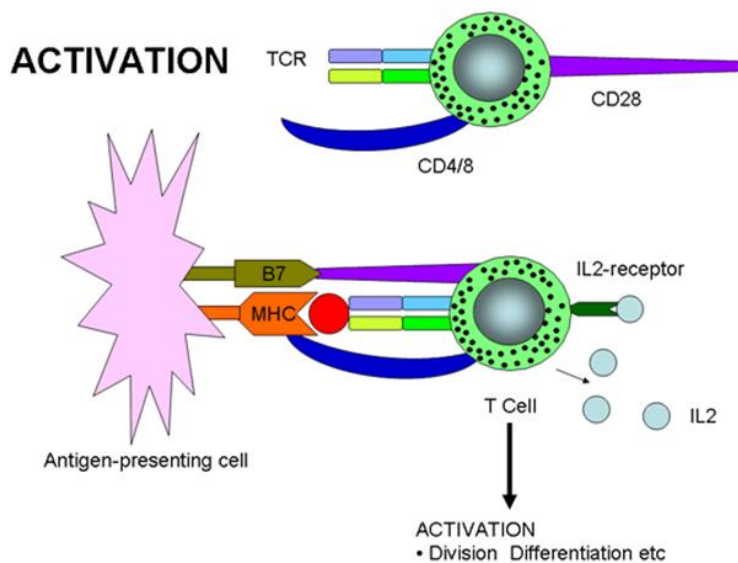
In central tolerance, the clonal deletion is considered as one of the major mechanism for controlling auto reactive T-cells. Sometimes it is not considered as fool proof system as B & T cells fail to undergo deletion and escape from central tolerance and auto-reactive immune cells reach peripheral lymphoid organs. In this case specific un-responsiveness occur in peripheral lymphoid tissues which is called as peripheral tolerance. Followings are the important mechanisms of peripheral tolerance

Activation induced cell death

This is the process by which death of auto reactive T-cell occur upon their activation. Various secretory cytokines from activated T-cells cause the expression of molecules like Fas ligand on Tcells. The self-reactive T-cells in periphery are deleted by the process of apoptosis after engagement of Fas Ligand (FasL) with Fas

Clonal Anergy

Exposure of T-cells to self-antigens lead to functional inactivation which is termed as anergy. The loss of co-stimulation upon interacting with self-antigen on antigen presenting cells (APC). There is no interaction of C28 on T-cells with CD80 (B7-1) or CD86 (B7-2) on APC which lead towards functional unresponsiveness of auto-reactive T-cells (Fig. 2)



ANERGY: Engagement of TCR and Ag/MHC in absence of co-stimulation can lead to anergy

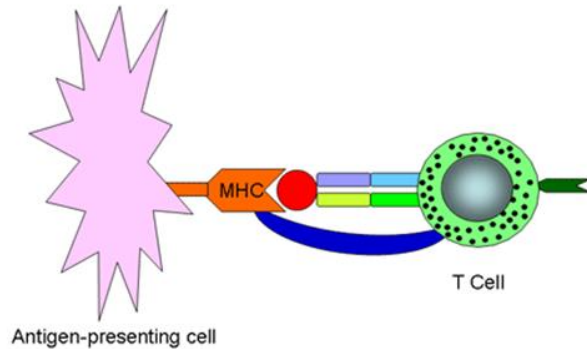


Figure 2: The clonal anergy of auto-reactive T-cells

Clonal Ignorance

In this mechanism, there is lack of interaction of T-cells with appropriate antigen. After maturation in thymus, auto reactive T-cells reach the periphery where there is sequestration of these selfreactive T-cells in inaccessible tissues for ignorance. The death of such clones happen due to continuous ignorance termed as clonal ignorance.

By Regulatory T-cells (Suppressor T-cells)

These are T-lymphocytes which are CD4 and CD25 +ve. These cells cause the secretion of immunosuppressive cytokines like TGF- β & IL-10. These cytokines cause the inactivation or suppression of auto-reactive cells

Lesson 143 : Tolerance to Tissues & Cells:

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Tolerance to Tissues & Cells

This specific unresponsiveness against various cells & tissues cause inhibition of immune response against antigens of cells and tissues. These antigens are foreign to immune system which are also called allogeneic antigens. It occurs in tissue graft like allogenic graft. In this case tolerance is induced against tissues and cells in the following ways

1. By injecting hematopoietic stem cells at neonatal stage as during early development stage there is no development of fully matured lymphoid tissues.
2. By immunosuppression e.g lethal irradiation for killing of host's own stem cells in primary lymphoid organs.
3. By using immunosuppressive drugs in order to inactivate immune cells of host
4. By Chimeras

These are animals with hybrid nature of immune cells with host's own cells and donor's cells. In this case both transplantation of donor's bone marrow and thymus in early age or by immunosuppression which leads towards functional inactivation of host immune cells.

BT302 - Immunology

**Teacher: Dr Numan Javed (Phd Holder at SHanghai
Institute of Sciences, China)**

1 to 167 Lectures merged

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Lesson 144 : Introduction to Autoimmunity:

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Introduction to Autoimmunity

All mechanisms which are responsible for breakdown of self-tolerance against self-antigens are called as autoimmunity or auto immune reactions. In this condition, there is generation of immune responses against components of self-tissues. These are harmful or aberrant immune responses against self-tissues. Various products of immune system damage the host tissues like both antibodies and T-cells are involved in autoimmunity. Moreover, genetic predisposition like certain genes of immunoglobulins, TCR & MHC are also associated with various autoimmune diseases. Various environmental factors are also responsible for autoimmunity e.g drugs & infections.

BT302 - Immunology

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Lesson 145 : Etiology of Autoimmunity:

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Etiology of Autoimmunity

The exact mechanism for autoimmunity is still unknown. However, various theories have been proposed for better understanding the mechanism of autoimmunity. These are as follows

- Sequestered antigens
- Escape of auto reactive cellular clones
- Lack of regulatory T-cells
- Cross reactive antigens

Sequestered antigens

In this mechanism, lymphoid (immune) cells may not be exposed to certain self-antigens during development and differentiation. Certain self-antigens are confined to specialized organs for example testis, brain & eyes etc. The release of such antigens from tissues due to any injury or accident initiate various autoimmune diseases of these organs.

Escape of auto reactive

clones The loss of central tolerance leads auto-reactive T-cells to escape from thymus to periphery. As not all of self-antigens are presented to T-cells in thymus for clonal deletion which make them auto-reactive in peripheral lymphoid tissues. Likewise, auto-reactive B-cells also escape from clonal deletion or negative selection

Lack of regulatory T-cells

There are few regulatory T-cells in autoimmune diseases which leads towards absence of effective suppressive against T-cells. Furthermore, absence of immunosuppressive cytokines like TGF- β & IL-10 also augment this condition.

Cross Reactive antigens

Antigens on certain pathogens have determinants which can cross react with self-antigens. There is generation of antibodies against those determinants which can also cross react with selfantigens. This happens during Post-streptococcal nephritis & carditis where M-proteins of Streptococcus are involved in this mechanism

Lesson 146 : General Classification of Autoimmunity:

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General Classification of Autoimmunity

Autoimmunity is classified based on the type of tissues or organs involved There are following two categories of autoimmunity

- Organ-specific autoimmunity
- Non-organ Specific autoimmunity

Organ Specific autoimmunity

In this kind of autoimmunity, the immune response is generated against organ associated specific antigens in the form of auto-antibodies. These antibodies are against organ associated antigens which damage the organ. For organ specific autoimmunity, following organs can be target organs with particular auto-immune disease and the antibody involved (Table.1)

- Skin
- Thyroid gland
- Muscles

Table 1: Various Diseases of Organ Specific Autoimmunity

Disease	Organs	Antibody to
Hashimoto's thyroiditis	Thyroid	Thyroglobulin, thyroid peroxidase (microsomal)
Primary Myxedema	Thyroid	Cytoplasmic TSH receptors
All hemolytic anemia	RBC	RBC antigens
Good Pasteur's	Kidney,	Renal & Lung basement membrane

Syndrome	Lung	
Ulcerative colitis	Colon	Colon Lipopolysaccharide

Non-Organ Specific autoimmunity

In this kind of autoimmunity, the immune response is not against organ associated specific antigens. Similarly, the auto-antibodies which are generated in response to immune response are not against organ associated antigens. This kind auto-immunity effect mainly all those tissues which are present throughout the body like skin, joints, soft tissues etc. That's why these disease are also termed as systemic autoimmune diseases (Table 2)

Table 2: Various diseases of Non-organ Specific Autoimmunity

Disease	Organs	Antibody to
Rheumatoid Arthritis	Lung, kidney, Joints etc	Self
Systemic Lupus erythematosus	Lupus in, joints	NA, RNA, cleoproteins
Graves's syndrome	Endocrine-producing glands	Basement membrane
Scleroderma	Lung, blood vessels, muscles & internal organs	NA, RNA, cleoproteins
Paraneoplasia	Lung, skin, heart, nervous system	Antibody to reactive T-cells

Lesson 147 : Diagnosis of Autoimmune Diseases:

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Diagnosis of Autoimmune Diseases

The autoimmune diseases are diagnosed based on symptoms and the nature of autoantibodies produced. Autoantibodies can be against two kinds of antigens

1. Self-cell or cell associated antigens
2. Soluble antigens
3. Biochemical test

Autoantibodies against self-cell/Cell associated antigens

These auto-antibodies are detected using tissues section like kidney, skin etc through immunofluorescence technique. These auto-antibodies can produce following two kind of patterns in immunofluorescence based on the kind of auto-immunity

- Linear Pattern (Goodpastuer Syndrome)
- Granular Pattern (Systemic Lupus Erythramatous)

Autoantibodies against Soluble Antigens

Autoantibodies against soluble antigens are detected using patient's serum for auto-antibodies by following techniques

- Agglutination e.g Anti-nuclear antibodies (ANA) for SLE & RA
- Enzyme linked Immunosorbent Assays (ELISA) e.g Rheumatoid factor for Rheumatoid Arthritis
- Radioimmunoassay (RIA) for detection of anti-thyroid antibodies

Biochemical Assays

Following biochemical assays in the serum of autoimmune patients can be performed for their diagnosis of disease

- a. For Intrinsic factor (IF) in case of Pernicious anemia
- b. Competition for TSH receptors in case of Grave's disease
- c. Auto-reactive T-cells can also be detected by flow-cytometry
- d.

Lesson 148 : Treatment of Autoimmune Diseases:

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Treatment of Autoimmune Diseases

Autoimmune diseases are treated with the goal of reducing symptoms or the better control of autoimmune response in order to increase the quality life of patient. Moreover, treatment strategy is for increasing the ability of immune system to fight against infections. Treatment varies based on specific disease and symptoms for reducing inflammatory process & immune response. Followings are the drugs used for treating auto-immune diseases

• Anti-Inflammatory drug therapy

Corticosteroid is used which particularly inhibit the cellular signaling for production of proinflammatory cytokines. Moreover, it relieves from the symptoms of inflammatory process like pain, redness and fever and reduce inflammatory process

• Immunosuppressive drug therapy

Following immune-suppressive drugs can be used for treating auto-immune diseases

- i. Cyclosporine
- ii. Cyclophosphamide
- iii. Azathioprine

These drugs reduce immune response against self-antigens. Also provide relieving from the symptoms of immune response

• Specific approaches

These kind of strategies using specific antibodies against receptor blocking the effects of immune response generated during auto-immunity. Currently, these approaches are used mainly in research. Followings are the antibodies which are used against their corresponding antigens on the surface of immune cells

- Anti-TNF α receptor antibodies
- Anti IL-2 receptor antibodies
- Anti CD4 antibodies
- Anti TCR antibodies
- Anti-Idiotypic antibodies against autoantibodies

Lesson 149 : Transplantation Immunology:

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Transplantation Immunology

Immune Response against Transplants

The process of moving cells, tissues and organs from one site to another among individuals is termed as transplantation. In this process of transplantation, one person called as donor who donates tissue to other person termed as recipient. Followings are the important kinds of tissue transplantation based on the source of donated tissue

Isograft

This is a form of tissue which is transplanted between individuals of same genetic makeup like identical twins.

Autograft

It may be from the same person tissue to another site within same individual.

Allograft

This form of transplantation which can be from one person to other non-identical person

Xenograft

This kind of transplantation is across the species for example from animal to human (Fig 1)

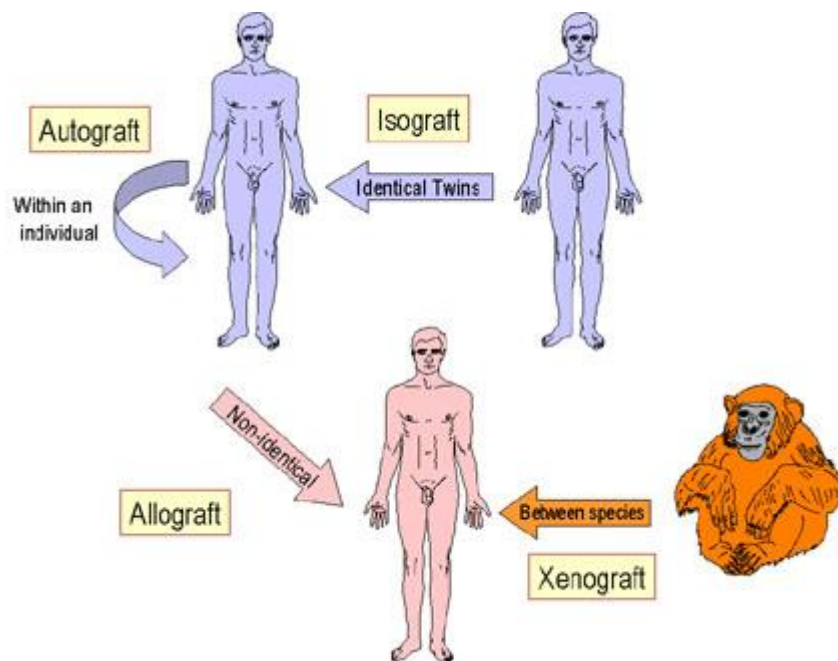


Figure 1: Kinds of tissue transplants

Immune system plays an important role in tissue transplantation as it acts as a barrier for transplantation.

Immune system identifies tissue transplant as foreign and mount an immune response leading destruction or damage of transplanted tissue. This process through which recipient's immune system activation occur against donor's tissue is termed as **transplantation rejection**.

The immunocompetent host recognize foreign antigens on grafted tissues as a result immune response is generated against grafted tissue is called as **host vs graft rejection**.

Similarly, tissue transplants in immunocompromised individuals lead to immune response by immunocompetent cells in graft is termed as **graft vs host rejection**.

In this case, host antigens serve as foreign. However, if there is no immune response against donated tissue by the recipient's immune system due to similar genetic makeup then it is called as **graft acceptance**.

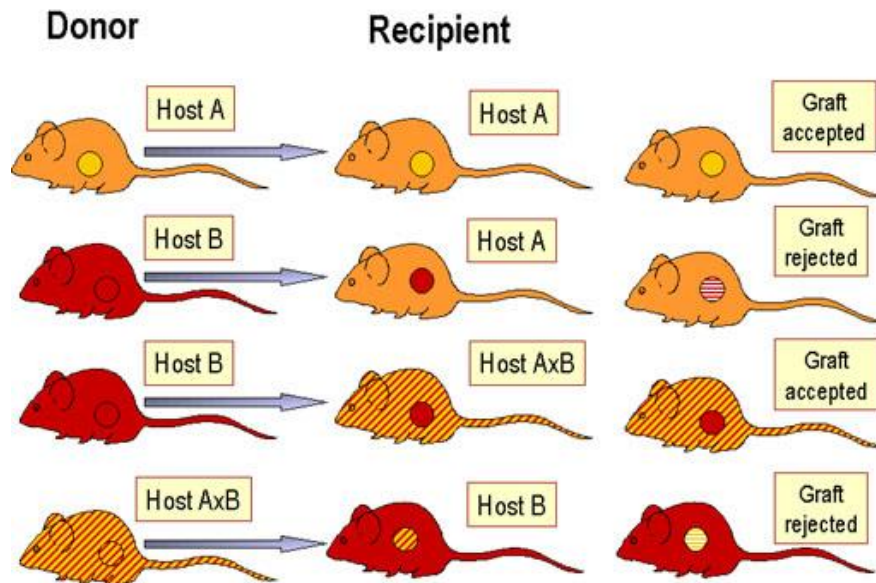


Figure 2: Tissue transplantation acceptance and rejection by host

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Lesson 150 : Transplantation Antigens:

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Transplantation Antigens

The transplantation antigens which are also termed as Major Histocompatibility Complex (MHC) or Human Leucocytes Antigens (HLA). The MHC exist as a complex which is encoded by group of genes on a same chromosome called as haplotype. These genes are in fact responsible to influence allograft rejection There are following two major types of MHC

1. Class I MHC
2. Class II MHC

Class I MHC

These genes are for Human MHC which are located at chromosome no. 6. These contain following three major loci

1. Locus B
2. Locus C
3. Locus A

Each major locus encode for polypeptide for MHC structure. In which there is a α -chain that contains antigenic determinants

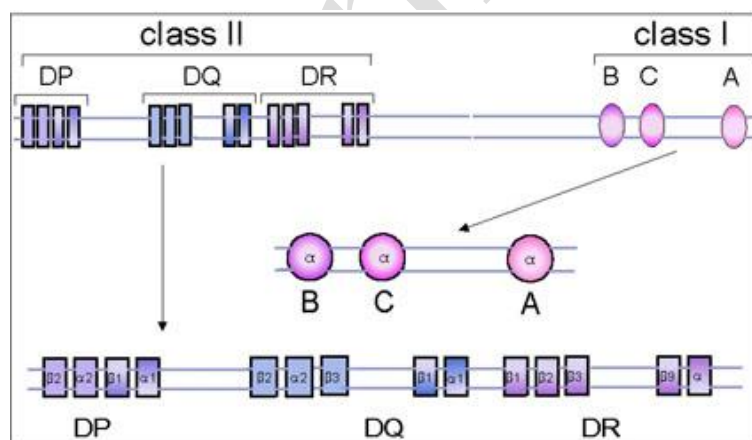


Figure 1: Structure of MHC locus

There are many alleles of α -chains which make it polymorphic in nature. Another chain of MHC is β 2-microglobulin (β -chain) which is encoded by outside of MHC I haplotype. This β 2 chain has role in expression of class I MHC on cell surface (Fig.19.3). If there is any defect in β -2 chain then there is no expression of Class I MHC on cell surface. This condition leads towards the deficiency of cytotoxic T-lymphocytes (CTLs) which have to interact with class I MHC on target cell

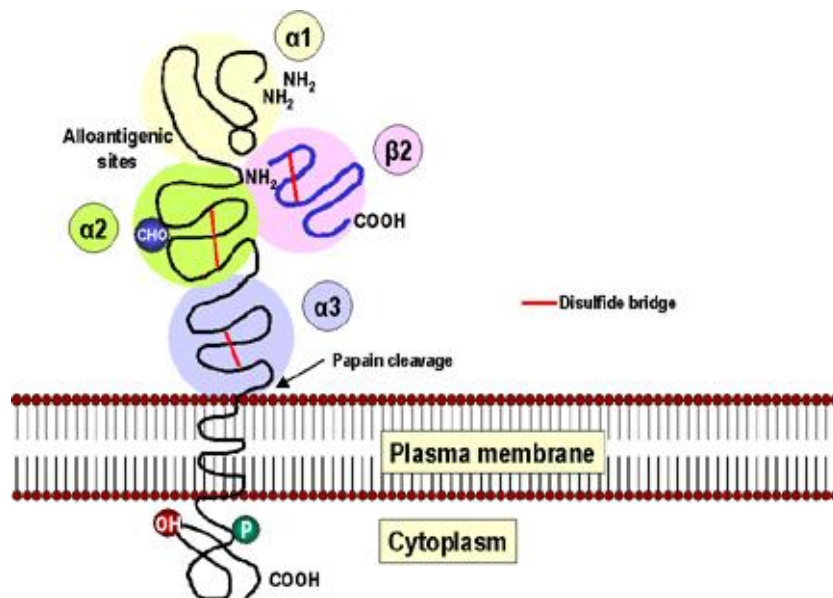


Figure 2: Structure of Class I MHC molecule on cell surface

Class II MHC

The class II MHC is encoded by genes for Human MHC which are located at chromosome 6. Structurally, the Class II MHC complex is also composed of three following major loci (Fig. 1)

1. DP
2. DQ
3. DR

Each of these loci code for one alpha & one beta chain. Both of alpha and beta chains associate together to form class II MHC. Likewise to class I, class II antigens are also polymorphic in nature which means these contain many alleles of α -chains. The DR locus contains more than one beta chain genes and possibly four in various combinations. The Class II MHC is expressed on B-lymphocytes and antigen presenting cells (APC).

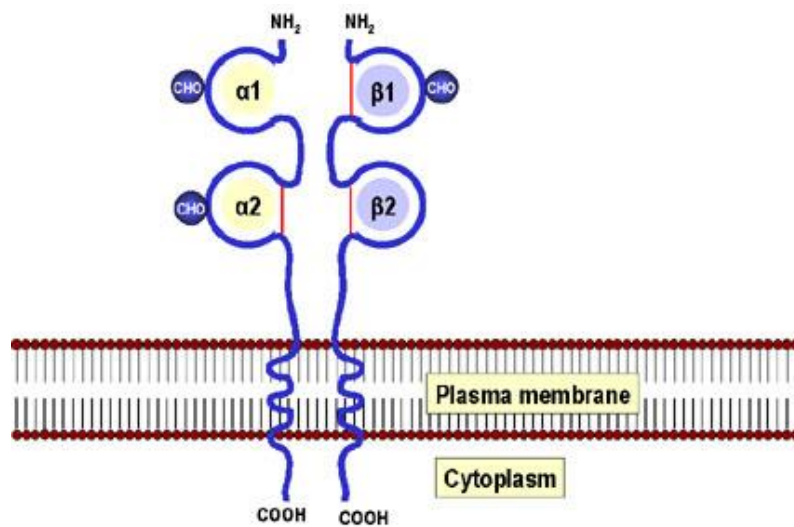


Figure 3: Structure of class II molecule on cell surface

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Lesson 151 : Induction of Immune Responses against Transplants:

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Induction of Immune Responses against Transplants

Clinically the significance of MHC is in tissue transplantation. The cells and tissues are transplanted for the treatment of various diseases. The immune response is generated against transplanted tissue which is called as rejection or destruction of transplant. Immune response can be of following two types based on the kind of tissue rejection

1. Host vs Graft Rejection

In this kind of transplant rejection, there are antigens on the surface of graft which are being recognized as foreign by host immune system

2. Graft vs Host Rejection

In this form of transplant rejection, the lymphoid (immune) tissues in graft recognize host immune system as foreign as a result immune response is mounted

Both of these immune responses lead towards graft rejection which is in fact the loss of grafted tissue. The rejection is based on antigenic nature of graft and host immune status. Induction of immune response is mediated by following ways

- I. Inflammatory process through inflammatory cells
- II. Antibodies against antigens of graft which lead towards complement mediated lysis of graft
- III. Antibody mediated Cellular cytotoxicity (ADCC) of graft tissue
 - a. T-lymphocytes mediated lysis of graft tissue

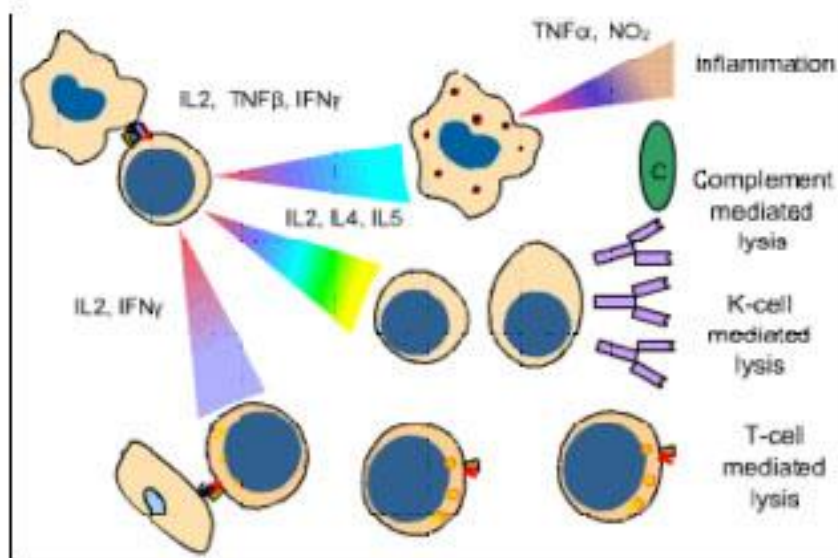


Figure 1: Mechanisms of destruction of transplanted tissues

Lesson 152 : Immune Mechanisms of Graft Rejection:

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Immune Mechanisms of Graft Rejection

The reaction of host against allo-antigens of graft which is also called as host vs graft (HVG) rejection. This is considered as a main obstacle in organ transplantation. The involvement of immune mechanisms in graft rejection is based on

- a. Time of rejection
- b. Nature of allo-antigens of graft
- c. Immune status of host

According to the time of rejection, the graft rejection can be of following types with distinct immune mechanisms

1. Hyper acute rejection
2. Accelerated rejection
3. Acute Rejection
4. Chronic Rejection

Hyper-acute Rejection

This kind of transplant rejection has very quick onset for tissue rejection. It occurs within minutes to hours. The high titer of pre-formed antibodies against antigens of graft are responsible for such form of reactions. Antigen/antibody (immune) reaction occurs on the tissue surface. Afterward the fixation of complement leads towards graft destruction

Accelerated Rejection

This kind of rejection is also called as secondary or 2nd set rejection. It occurs after transplantation of second graft. It is due to sharing of antigenic determinants with the first transplant. Time required for this form of rejection is within 2-5 days. The sensitized T-cells during first graft are responsible for this kind of rejection. Moreover, the Lymphokines and cytotoxic T-lymphocytes (CTLs) augment it.

Acute Rejection

It is also called as primary or 1st set of tissue rejection. It occurs during first graft with allo-antigen on grafted tissue. The time span required for such form of rejection is 1-3 weeks. These

reactions are mediated by sensitized T-cells to class I & II of allo-graft. Furthermore the secretion of Lymphokines leads towards the activation of monocyte and macrophages.

Chronic Rejection

This form of transplantation rejection is called as delayed rejection which occur within months to years. After transplantation the graft remains normal for months to years but sudden rejection occur due to still unknown mechanisms. These are many hypotheses reading such form of rejection like the involvement of various infections which are responsible for loss of immunological tolerance by grafted tissue in host.

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Lesson 153 : Prevention & Treatment of Graft Rejection:

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Prevention & Treatment of Graft Rejection

If there is decrease in tissue rejection, then there is relative increase in survival of graft. There are successful grafts occur for example mostly in case of kidney & cornea. For successful transplantation there is need of better understanding of immune response and MHC

The success of tissue graft is based on following important factors

1. Donor selection
2. Recipient preparation
3. Immunosuppression

Donor Selection

For appropriate donor selection, there should be MHC compatibility with recipient. Identical twins are considered as the ideal donor which are also termed as Isograft. HLA matched siblings have 95-100% chance of graft success. One haplotype of parent or sibling must be HLA-D matched for successful graft. Moreover, ABO compatibility is also essential for high success rate of transplant

Recipient Preparation

For successful tissue graft, the recipient should be in good health with no current infection. Similarly the absence of active malignancy would also increase the success rate. Moreover, in recipient the absence of any systemic diseases would lead towards better rehabilitation. Likewise, the recipient should not be hypertensive. One to five transfusions of 100-200 ml of donor's blood at 1-2 weeks interval would also increase the compatibility level with donor.

Immunosuppression

This is the way through which host immune system can be suppressed and is most essential component of allo-transplantation. For this there is usage of following immunosuppressive drugs with their specific mode of actions

- **Cyclosporin A**

It inhibit IL-2 synthesis following antigen exposure to host as a result no immune response is generated for allo-antigens.

- **Rapamycin**

It inhibits signal transduction for the activation and differentiation of immune cells. For instance inhibition of T-cells proliferation and activation is achieved by this.

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Lesson 154 : Transplantation of Blood Cells & Bone Marrow:

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Transplantation of Blood Cells & Bone Marrow

Clinically, transplantation of blood and bone marrow is used for treating various hematological disorders. The success of these graft is achieved by having good compatibility between donor and recipient. For this purpose following procedures are used

1. ABO blood grouping
2. Cross matching
3. HLA typing

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Lesson 155 : Tumor Immunology:

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Tumor Immunology

Evidence of Immune Reactivity to Tumor

- The mass containing un-controlled proliferating cells is called as tumor.
- There are a lot of evidences which indicate that tumors elicit immune response. Like in case of age, the young & old populations have increased incidence of tumors because due to insufficiency of immune response at these ages.
- Similarly, tumors having mononuclear infiltration have better recovery as compare to those which lack mononuclear cells which indicate the involvement of immune cells during clearance of tumors.
- Tumor regression also occur due to appropriate immune response for example certain tumors regress spontaneously like melanomas (tumor of melanin producing cells of skin), neuroblastomas (tumor of nerve cells).
- Moreover, immune system facilitate the regression of metastatic tumor which has the tendency to move from tissue of origin to other tissues. For example in case of metastatic tumors, the removal of primary tumor regress metastatic tumor due to decrease in tumor load.
- Additionally, there is increased incidence of tumor in immune deficient patients like patients suffering from AIDS are susceptible to Kaposi Sarcoma. Likewise, the patients receiving transplants also get Epstein-Barr virus (EBV) induced lymphoma due to immune suppression. The presence of tumor specific antibodies & T-lymphocytes in patient's serum is also the hallmark of immune reactivity against tumor.

Lesson 156 : Tumor Associated Antigens:

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Tumor Associated Antigens

In order to generate effective immune response by immune system, tumor must have certain antigens on its surface. These antigens are generated through alteration in number of gene expression by a process called as tumorigenesis. The expression of new antigens on the surface of tumor are called as neo-antigens. Moreover, there is also alteration in existing antigens which are present on normal cells. These antigens are membrane bounded receptors, regulators of cell cycle & apoptosis and molecules of signal transduction

There are following two main types of tumor associated antigens

1. Tumor specific transplantation antigens
2. Tumor Associated transplantation antigens

1) Tumor specific transplantation antigens

These antigens are unique to tumor cells. These are not expressed on normal cells. These antigens are responsible for rejection of tumor by host immune system. In most cases, these antigens cannot be easily identified

2) Tumor Associated transplantation antigens

These antigens are expressed by both tumor & normal cells in contrast to tumor specific transplantation antigens. Various chemicals, UV radiations & viruses are responsible for expression of these neo-antigens. Majority of these tumors are weakly immunogenic or onimmunogenic

Lesson 157 : Tumor Associated Transplantation Antigens:

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Tumor Associated Transplantation Antigens

The tumor antigens which are also expressed by normal cells. These are expressed in high levels by tumor cells as compare to normal cells. These kind of antigens are also called as onco-fetal antigens as these are expressed during early development & lost during adult life. However these antigens re-expressed on tumor cells. These antigens also serve as tumor markers due to their importance in the diagnosis & prognosis of cancers

- There are following two types of onco-fetal antigens

1. Alpha-fetoproteins (AFP)
2. Carcino-embryonic antigens (CEA)

1) Alpha-fetoproteins (AFP)

These antigens are found as secretory protein in serum. Their level get raised during hepatocellular carcinoma.

2) Carcino-embryonic antigens (CEA)

These antigens are found both in secretory & cell associated form. Their level get raised in colon cancer

Lesson 158 : Immunity Against Tumors:

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Immunity against Tumors

- The immune system provides anti-tumor activity in humans, however, the evidence for immunity against malignancy mainly come from experimental studies with animals.
- Laboratory animals like mice can be immunized with irradiated tumor cells for providing defense against tumors.
- Similarly, after removal of primary tumor if mice are challenged with the same live tumor then there would be resistance against same tumor upon re-challenge.
- Moreover, antibodies also play important role against various cancers for their neutralization.
- Likewise, the cell-mediated immunity also play pivotal role in tumor rejection by host immune system.
 - Helper T-lymphocyte process the tumor antigen like shed from tumor and present with Class II MHC.
 - Additionally, Helper T-lymphocyte also help B-cells to produce antibodies against tumor antigens.
 - The role of CTL is also very critical for tumor regression. Helper T-lymphocyte also help in activation & differentiation of cytotoxic T-lymphocytes (CTLs).
- Cells of innate immune system like natural killer (NK) cells also kill the tumor cells due to lack of class I MHC. Furthermore, the cytokines like Interferon -gamma (IFN- γ) for tumorocidal activity.

Lesson 159 : Escape of Tumor from Immuno-Surveillance:

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Escape of Tumor from Immuno-Surveillance

According to the Immune Surveillance Theory, cancer cells that arise in the body are eliminated by the immune system. However, due to impaired immune reactivity, cancer cells may escape destruction. Tumors evade immune recognition by several mechanisms. Tumors may not express neo-antigens that are immunogenic or they may fail to express co-stimulatory molecules required for the activation of T cells. In addition, certain tumors are known to lack or be poor expressers of MHC antigen. Another reason for failure of immune surveillance may be the fact that in the early development of a tumor, the amount of antigen may be too small to stimulate the immune system (low dose tolerance) or, due to the rapid proliferation of malignant cells (high dose tolerance), the immune system is quickly overwhelmed. In addition, some tumors may evade the immune system by secreting immunosuppressive molecules and others may induce regulatory cells particularly the CD4+CD25+ FoxP3+ T regulatory cells. Also, some tumors may shed their antigens which in turn may interact and block antibodies and T cells from reacting with the tumor cells.

BT302 - Immunology

**Teacher: Dr Numan Javed (Phd Holder at SHanghai
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Lesson 160 : Use of Tumor Neo-antigens in Immuno-diagnosis & Immunotherapy:

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Use of Tumor Neo-antigens in Immuno-diagnosis & Immunotherapy

Tumor neo-antigens on tumor cells can be used for both immune-diagnosis & immunotherapy
Immune-diagnosis

Tumor neo-antigens are used for In-vivo detection of relatively small tumor foci (masses) through radiolabelled monoclonal antibodies against tumor antigens. Likewise the In-vitro use for finding the cell origin of undifferentiated tumor particularly lymphocytic origin in the blood & bone-marrow. Furthermore, Immuno-histochemical use of monoclonal antibodies against metastatic foci of tumors can help in the diagnosis of such tumors (Fig. 1)

Immunotherapy

The tumor neo-antigens can also be used for Immunotherapy for treating various tumors in different following ways

1. Active immunotherapy

In this form of immunotherapy, host actively participates against tumors. Active immunotherapy is achieved by inducing immune response by injecting irradiated tumor cells having neo-antigens on them. The active form of immunotherapy has following two forms

a. Non-specific active immunotherapy

This is generated for activation of immune response against neo-antigens using various vaccines e.g BCG for activation of macrophages against tumors

b. Specific active immunotherapy

In this form, the killed tumor cells or their antigens are used for killing of tumor cells

2. Passive immunotherapy

This form of immune-therapy uses the pre-formed antibodies against specific neo-antigens of tumor through following ways. The specific monoclonal antibodies can be used against tumor antigen can be served for

a) As a vehicle for delivering anti-cancer drug to tumor or cancer

b) Activation of components of innate immune system like complement proteins

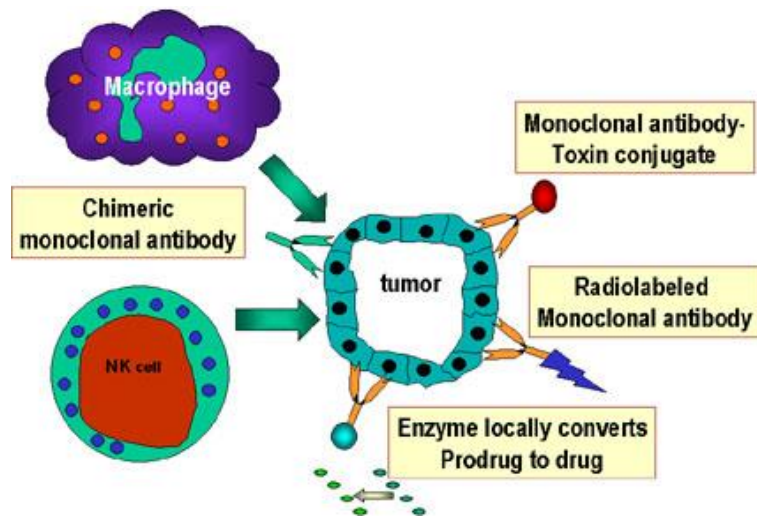


Figure 1: Use of Tumor neo-antigens for immuno-diagnosis & immunotherapy

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Lesson 161 : Immunodeficiency:

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Immunodeficiency

Introduction to Immunodeficiency

The deficiency or failure of immune system is called as immunodeficiency. It is a state of complete absence of immunity or effective immune response. There are defects regarding generation of immune reactivity against infectious agents like bacteria & virus etc. Furthermore, there is no immune response against transformed cells like tumors or malignancy. The persons having immunodeficiency are also called as immunocompromised patients because there is increased vulnerability for opportunistic infections in addition to other forms of infections. As there is decrease surveillance against tumors so more chances of getting tumors are there in immunodeficient patients.

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Lesson 162 : Classification of Immunodeficiency:

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Classification of Immunodeficiency

Immunodeficiency is classified into following two main categories based on mode of its acquisition

1. Primary Immunodeficiency
2. Secondary Immunodeficiency

Primary Immunodeficiency

Primary immunodeficiency is basically the inherited or congenital defects of the immune system. These defects can be either in specific or non-specific immune systems. These are classified on the basis of the site of lesion in the developmental or differential pathway of immune system. The chances of susceptibility to variety of infections get increased in this kind of immunodeficiency. The nature of infection depends on the kind of immunodeficiency

Primary immunodeficiency can be of two kinds

1. Immunodeficiency in specific immune system
2. Immunodeficiency in non-specific immune system

Immunodeficiency in specific immune system

This kind of immunodeficiency is due to defects in stem cell differentiation. Followings are the important disease of immunodeficiency in specific immune system

Reticular dysgenesis

It is either complete absence or severe deficiency of lymphocytes & granulocytes in blood due to decrease differentiation of bone marrow stem cells.

Severe combined immunodeficiency (SCID)

This disorder is due to the involvement of lymphoid stem cells. There is complete absence of T & B cell immunity. The patients suffering from SCID are susceptible to variety of bacterial, viral, mycotic & protozoan infections.

DiGeorge Syndrome

This is the disorder of T-lymphocytes due to congenital thymic aplasia or hypoplasia. This condition is highly immunodeficient as live vaccines also cause infections in the patients suffering from DiGeorge syndrome.

X-linked hypo gammaglobulinemia

This kind of immunodeficiency is due to functional disorder of B-Lymphocytes. Such patients are highly susceptible to recurrent bacterial infections.

IgA deficiency

There is deficiency of IgA in this disorder as a result individuals are susceptible to mucosal surface infections like GIT, eye & nasopharynx.

Selective IgG deficiency

In this kind of immunodeficiency, there is selective deficiency of IgG which leads towards hyper IgM immunodeficiency.

Immunodeficiency in Non-Specific immune system

Followings are the important conditions which are linked with immunodeficiency in non-specific immune system

Cyclic Neutropenia

There is low number of circulating neutrophils in the blood which cause recurrent bacterial infections among such patients.

Chronic Granulomatous disease (CGD)

This kind of immunodeficiency is due to defects in phagocyte functions.

Complement deficiency

There is deficiency of complement proteins in this kind of immunodeficiency. Patients suffering from such disorder are susceptible to infections particularly Neisseria

Secondary Immunodeficiency

This kind of immunodeficiency is not congenital in nature in contrast to primary immunodeficiency. This state of immunodeficiency is associated with infections for example AIDS. There is also association of aging factor like hypo-cellularity in bone marrow. Malignancies for example Leukemia and Myeloma are also linked with secondary immunodeficiency. Furthermore, other metabolic disorder like diabetes and renal malfunction etc are also linked with secondary immunodeficiency.

Lesson 163 : Acquired Immunodeficiency Syndrome (AIDS):

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Acquired Immunodeficiency Syndrome (AIDS)

This condition is caused by a virus called as human immunodeficiency virus (HIV). This condition is initiated through an initial infections with influenza like illness. Further, the disease progress with other defects in immune system leading towards severe immunodeficiency. The patients get severe opportunistic infections like tuberculosis. Moreover, the chance of getting tumor also increased. AIDS is linked with abnormalities in circular lymphocytes. There is reduction in the number of helper-T cells which are also called as CD4+ ve cells which consequently cause reversal in CD4+ /CD8+ T-cell ratio. Furthermore, there is normal NK cells number but with reduced activity to kill virally infected as well as transformed cells. AIDS patients have increased susceptibility to infections with opportunistic pathogens like Cryptococcus, herpes simplex, herpes zoster, and mycobacterium etc. During initial or primary infection there is systemic fever, lethargy & malaise with localized tissue involvement (Fig. 1). AIDS patients have also increased susceptibility to various following tumors

- Kaposi sarcoma
- Burkitt's lymphoma
- Primary central nervous system lymphoma

Main symptoms of Acute HIV infection

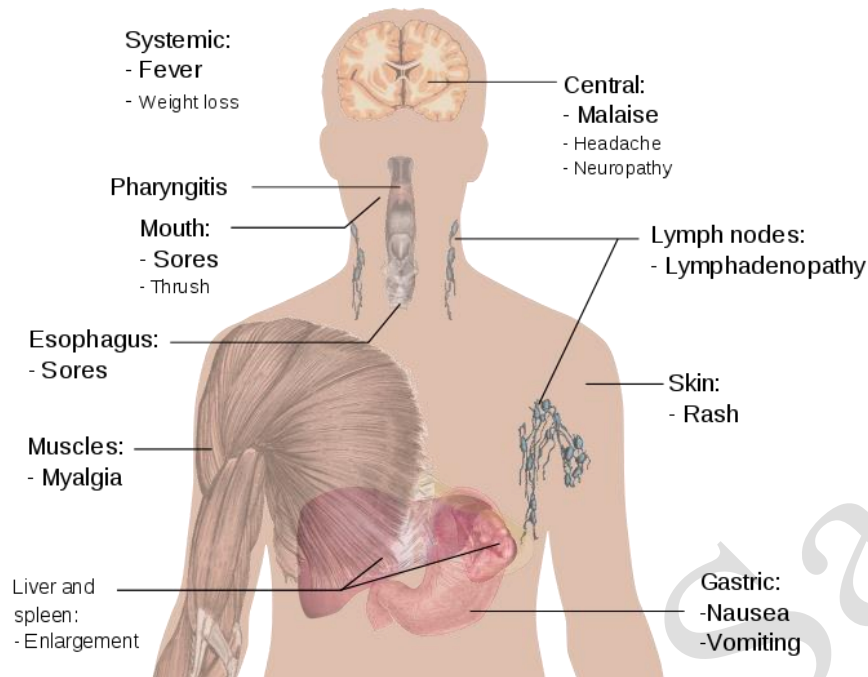


Figure 1: Involvement of various tissues during AIDS

For AIDS, HIV infection is transmitted by following two modes

1. **Horizontal transmission**

This kind of transmission occurs usually by person to person contact either sexually or through body fluids like blood and semen

2. **Vertical transmission**

This form of transmission is from mother to fetus during the process of pregnancy, delivery & breast feeding

Lesson 164 : Immunodeficiency Disorders of T- Cells:

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Immunodeficiency: Disorders of T- Cells

As T-lymphocytes have very significant role in providing cell mediated and humoral immunity, that's why immunodeficiency associated with T-cells disorders effects both cell mediated & humoral immunity. There is complete absence & functional abnormality in T-cells which increase the chances of getting multiple infections like viral, protozoal & fungal infections. In immunodeficiency associated with T-cells the incidence of viral infections like cytomegalovirus and measles get increased.

DiGeorge Syndrome

It is one of the primary immunodeficiency in which there is complete absence of T-lymphocytes. This condition is also called as thymic aplasia or hypoplasia. This form of immunodeficiency is also linked with hypoparathyroidism due to abnormal development of fetus. There is poor development of heart, thymus & parathyroid due to which there is abnormal development of heart, thymus & parathyroid. All patients suffer from DiGeorge syndrome have thymic aplasia. This disease is autosomal dominant caused by deletion in chromosome number 22. This deletion is of variable size which doesn't correlate with severity of disease. The treatment of DiGeorge syndrome occur through thymic graft only.

T-cells deficiency with variable degree of B-cells deficiency

This form of immunodeficiency is linked with T-lymphocytes deficiency which also effect the number of B-lymphocytes with a variable degree. In this condition, both cell mediated as well as humoral immunity get effected leading to multiple form of infections.

Ataxia-telangiectasia

This form of T-cells associated immunodeficiency has deficient T-cells with less movement of blood vessels which also effects the function of T-lymphocytes.

Wiskott-Aldrich Syndrome

In this condition, there is normal number of T-cells with reduced functions as a result chances of getting multiple infections become increased.

Lesson 165 : Immunodeficiency Disorders of B- Cells:

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Immunodeficiency: Disorders of B- Cells

This form immunodeficiency is associated with normal T-cells, however number of Blymphocytes may be low or normal. As a result, the serum Immunoglobulin levels are low. There is more chances of getting pyogenic infections like bacterial infections

X-linked hypo-gammaglobulinemia

In this form of B-cells associated immunodeficiency, the numbers of B-cells are very low. Due to which the immunoglobulin levels are also very low. This condition is linked with defect in B-cell maturation. Patients suffer from this immunodeficiency are prone to recurrent bacterial infections

IgA deficiency

This is the commonest form of immunoglobulin deficiency. There is defect in class switching phenomenon of immunoglobulin synthesis. These patients are prone to mucosal surface infections like GIT, eye & nasopharyngeal infections. This form of immunodeficiency is diagnosed through measurement of serum IgA levels by various immunological methods

X-linked hyper IgM -Immunodeficiency

This form of B-cell associated immunodeficiency has very high levels of IgM. As a result, the levels of IgG & IgA get reduced. The underlying defect in this condition is in class switching due to defect in CD40 ligand on CD4 T-cells. The patients are susceptible to pyogenic infections like bacterial infections. This condition can be treated with intravenous gamma globulins injections.

Lesson 166 : Immunodeficiency Defects of Phagocytic Cells:

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Immunodeficiency: Defects of Phagocytic Cells

The primary immunodeficiency associated with cells of non-specific immune system including phagocytic cells like neutrophils, monocytes and macrophages. There is also defect in killer cells like NK cells

Congenital Agranulomatosis

This form of congenital immunodeficiency has decreased neutrophil count. There is defect in myeloid progenitor differentiation into neutrophils. These patients are prone to pyogenic infections like bacterial.

Chronic Granulomatous disease (CGD)

This form of primary immunodeficiency has decreased neutrophil function while their number remains normal. The underlying defect is due to poor intracellular killing ability of neutrophils. This defect of neutrophils is due to deficiency of NADPH oxidase & other co-factors which are required for respiratory burst. CGD patients have increased susceptibility to bacterial infections.

Leukocyte Adhesion deficiency

As the name suggests this form of immunodeficiency is linked with defect in integrin molecules as a result there is decreased process of diapedesis & defective neutrophils movement towards chemotactic signals. This defective phagocytic function leading towards recurrent bacterial infections among such patients

Lesson 167 : Immunodeficiency Defects of Complement System:

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Immunodeficiency: Defects of Complement System

In this form of immunodeficiency of non-specific immune system, there are abnormalities in complement proteins which are also called as Hypo-complementemia. These defects are genetic in nature due to inherited defects in the synthesis of various complement proteins. There is major defect in the synthesis of C3 due to defective C3 synthase. Moreover, defects are also due to various regulatory proteins like Factor H & I. Majority of complement deficiency is autosomal recessive in nature except properdin deficiency which has X-linked inheritance. However, MBL deficiency can be both. The patients suffering from such disorders have recurrent bacterial infections particularly Neisseria & Streptococcus infections. Furthermore, these conditions also cause autoimmune disorders like SLE, vasculitis etc.

The -----End

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